BofA SECURITIES

Technicals Explained

In 2026, get to know technical strategy

Primer

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Technical Strategy
Global
FICC and Equity

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Note: For abbreviations and acronyms, see the Appendix: Acronyms & abbreviations section.

A resolution to know more about Technical Analysis

This primer summarizes some of the indicators and process discussed in our technical research, on trading desks and at investment meetings. Get to know technical analysis to refine an investment approach, time trades and estimate risk/reward. Whether you're a novice or a seasoned investor, we're confident there is something in here for you.

Fifteen ways to get to know technical strategy in 2026

1. Put the data, narratives, tweets and risks aside. Start by getting into the Technical Analysis Mindset. After all, price is a fact, unrevised and truth.
2. Technical process is top down and bottom up. The fractal nature of price is key to find trends, Support & Resistance and Terrific trend lines.
3. Wondering which indicator(s) to use? Discover Evidence of the most popular indicators and then continue reading about each of them...
4. Is a new trend forming? Consider multiple time frame analysis with Moving Averages, Mighty MACD, Ichimoku Analysis and DMI + ADX.
5. Find short term trends with the Triple Moving Average Cross strategy and medium-term trends with The Golden Cross & Death Cross signal.
6. How much momentum is too much and for how long? Use RSI to measure momentum and/or Stochastics, Williams %R or Bollinger Bands.
7. Something goes parabolic every year, but do you hold on long enough? Consider using Trender, a trailing-stop-like line and tracking futures positioning.
8. Is a trend at risk of exhausting, pausing or changing? TD Sequential: Green 9s and red 13s and TrendStall can warn of such risks.
9. Neckache? Get to know A well massaged head and shoulders. Coffee or tea drinker? Sip from a Cup and handle. World traveler? Visit some Island reversals.
10. Bullish in the morning but then a headline flips things bearish in the afternoon? Outside reversals and Candle patterns can signal change in investor sentiment.
11. Markets tend to repeat history. Get to know Seasonality, the Decennial Cycle, U.S. Presidential Cycle, Benner Cycle, January Trifecta and Don't sell in May!
12. Start investing with the tide, riding the waves and persevering through the ripples by getting to know Elliott Wave patterns and counts.
13. This is one of the most important Fibonacci numbers. If we could fit 21 ways on one page, we would. Get to know the Fibonacci ratios and measures.
14. Confirm uptrends with Dow Theory and Market Breadth. Track sector rotations with the Relative Rotation Graph.
15. What tends to happen to macro markets after the VIX spikes, a Breadth Thrust, a NYSE 90% up days or when Margin Debt rises or falls too quickly?

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Refer to important disclosures on page 172 to 174. **12925273**

Technical Analysis Mindset

- The mindset and process of a technical analyst begins with learning about the many aspects of technical analysis, and there are many.
- Get to know the theories, tools and assumptions, and begin applying them to financial markets.
- Create a process to evaluate markets and a checklist of your preferred indicators. Consult and learn from them over time.

Technical approach explained

To engage in technical analysis is to assess market trends and find high conviction trade and investment ideas with attractive risk/reward. This can be accomplished by applying a diverse set of technical tools, strategies and theories. An aspiring technician should define technical analysis, understand and learn about its tools, and then create, implement and refine a process.

Technical analysis defined, in our view

Technical analysis is the extraction of information from market data into objective visualizations primarily through the use of charts and mathematics with an emphasis on investor behavior and supply and demand to explain the current and anticipate the future path of markets (Ciana, P. New Frontiers in Technical Analysis. John Wiley & Sons Inc. 2011). This definition has five specific parts, including:

- **Market data** – price is by far the most important, then volume, open interest, breadth, spreads, ratios, yields, volatility, etc.
- **Objective visualizations** – lead with a clear and concise chart to show a view. Consider using other tables and figures to support it. Don't use too many indicators, lines, methods all at once.
- **Mathematics** – apply measurements, calculations and statistics to help define trends, targets, levels and to explain your view. Every technical indicator we use has a mathematical formula and purpose behind it.
- **Investor behavior and supply/demand** – identify rational and irrational market movements with price patterns, positioning and sentiment analysis.
- **Anticipate the future** – apply a process of technical tools to forecast markets, assess risk/reward, and make buy and sell decisions.

The body of "technical" knowledge is vast

There are many tools and strategies available to the technician. Taking the time to get to know them is an important step. The following table lists some of the technician's tools in six categories.

Exhibit 1: The Technician's Toolbox
There are many areas of technical analysis to learn about and apply

Chart types	Indicators	Theories	Data	Testing	Management
Line	Trending	Intermarket	Price & volume	Rules based	Underlying or derivatives
Bar	Oscillators	Sector rotation	Accurate history	Define probabilities	Risk/Reward
Candle	Statistical	Cycles	Sentiment	Optimize	Position size
Market profile	Exhaustion	Seasonality	Breadth	Test a universe	Confirmations
KASE bars	Intraday	Dow Theory	Macro / Economic		Disciplined
Heikin-ashi	Fibonacci	Elliott Wave	Multi-time frame		
Point & figure					

Source: BofA Global Research

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Technical analysis has three assumptions

1. Market actions discount everything

Price should reflect all available information. To a technician, how markets are moving is more important than why. By observing price and other tools, a technician can assess the trend and estimate future price action.

2. Prices move in trends

Like Newton's law, it is widely believed a trend in motion remains in motion until acted upon by a large enough counterforce. Technicians follow the trend and attempt to enter and exit markets at the most opportune times.

3. History repeats itself

Whether history repeats or rhymes, there is an echo, and technicians believe market patterns and signals will repeat based on human nature such as fear and greed.

We propose a fourth: First discipline, then conviction

The tools and rules of a process should be prioritized over a current conviction and evaluated regularly. This, we think, should allow for more timely position changes, limit losses by not sticking with a losing trade for too long, and limit "gut" decisions.

Start the process with a question

Is the market in a trend? Is it range bound? Is it leading or lagging? Or has it possibly moved too far, too fast? Many technical indicators are meant to be used in a certain type of market or interpreted differently, depending on the state of the market.

Work through a checklist

- Apply your preferred indicators to multiple time frame charts. Consider those that are more applicable than the others. For example, if trending, apply moving averages, Ichimoku, MACD (Moving Average Convergence/Divergence Indicator) and ADX (Average Directional Index). If range bound, apply oscillators such as RSI (Relative Strength Index) or Stochastics. If parabolic, estimate an end with candle patterns, TD Sequential, long-term trend lines, divergences, and log scaled charts.
- Identify key levels for support, resistance and trend continuation with relevant moving averages, trend lines and Fibonacci retracements and projections.
- Review multiple time periods and study the trend to find strong trend lines, significant highs, lows and consolidation areas and key Fibonacci levels.
- Run some tests. Test what strategies are working and decide if they may continue. Find what odds or probabilities exist to support or contradict a view.
- Determine what other technical charts and processes suggest. Such as Elliott wave, bar and candle charts, market profile or another technical charts for added color.
- Review intermarket relationships to better understand the market environment. Compare trends and relationships to other markets using charts, ratios, correlations, regressions and distributions.
- Review market breadth, volumes, flows. Use such data to assess market internals and compare that to market trends.
- Review the sentiment in the market. Look at CFTC/COT (Commodity Futures Trading Commission/Commitments of Traders) data, surveys, put/call ratios and compare that to market trends.
- Find alignments and compare conclusions. Review conclusions with the fundamental, quant and economic views but always prioritize technical.

Make the call

Present high conviction views with a convincing chart or visual and a brief explanation of the call including key levels and potential risk/reward.

The fractal nature of price

- Fractals are patterns that recur on different scales. Each smaller portion of time is a scaled-down version of the whole.
- Price is fractal and hierarchical. Market behavior repeats across time frames, but higher time frames tend to carry greater more influence. Lower time frames capture the same price action as higher time frames, but with more detail.
- Because markets are fractal, technical analysis encourages the use of multiple time frames. Alignment across time frames can increase confidence in a setup or signal. For example, four charts, such as a 60 minute, daily, weekly and monthly.

Technical analysis operates across multiple time horizons

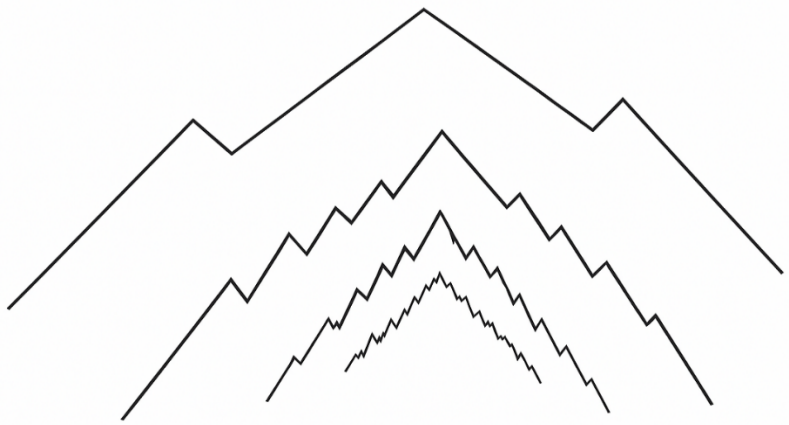
Price does not trend in a single time frame. At a given point in time, a market may be advancing on a long-term basis, consolidating on an intermediate-term basis and correcting short-term. These movements are not contradictory. Instead, they reflect the varied objectives and expectations of market participants operating across different horizons. Multi-time frame analysis (MTA) is the process of studying price with the goal of placing shorter-term movements into longer-term context. MTA can help determine where a market sits within its broader trend, and whether lower time frame (LTF) action aligns with or opposes higher time frame (HTF) structure.

Fractals are everywhere, including financial markets

Fractals are patterns that recur at different scales. Like Fibonacci sequences, they appear throughout nature – snowflakes, river networks, plant structures and even the human vascular system are examples of fractals. The same basic patterns can be seen whether something is viewed up close or from far away.

Financial markets are similar. Patterns and trends appear and repeat across time frames. This is because markets, like natural systems, reflect the collective behavior of individual participants with different objectives and horizons. Because the mechanics of supply and demand are consistent, the same technical principles can be applied on all time frames. However the longer-term charts tend to exert greater influence over the trend and direction of their shorter-term counterparts.

Exhibit 2: Price action is fractal. Charts reflect the same picture with more or less detail. Higher time frames capture big-picture trends and patterns while lower time frames provide a granular view.
Example of degrees of detail provided by higher and lower time frame charts



Source: BofA Global Research

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Time frame hierarchy and analytical importance

In technical analysis, HTFs reflect secular trends and the behavior of longer-term participants and institutions. LTFs respond quickly to news and near-term sentiment changes but carry less structural weight. We consider fractals through three time frames:

- Monthly charts (primary trend)
- Weekly charts (refined primary / intermediate trend)
- Daily charts (tactical / short-term trend)

Monthly charts define the primary trend

Monthly charts provide the clearest view of long-term direction. Each bar represents a month of trading, so short-term volatility ("noise") is removed. This helps make major trends and inflection points more visible. Trend changes on monthly charts occur infrequently but carry meaningful long-term implications. Monthly charts serve as the anchor for MTF analysis.

Weekly charts refine primary trend structure

Weekly charts provide a zoomed in view of the monthly chart. Instead of capturing trading activity with a single bar, four bars reflect the same activity. Weekly charts balance noise and color and are good for identifying patterns and trend changes within a larger context.

Daily charts help reduce risk and improve timing

A daily chart captures trading activity for a single session or day. A single weekly bar is expanded and shown as five bars. Daily charts add granularity but are typically not used to determine primary direction unless one has a tactical bias. Daily charts are good for monitoring short-term momentum, confirming HTF signals, making entries and exits more precise and reducing risk. When daily charts conflict with weekly and monthly charts, trades may be less effective because long-term investors and institutions tend to prioritize HTFs for decision making and positioning.

Traditional top-down analytical process

Multi-time frame analysis typically follows a top-down approach. Analysis begins with the highest time frame and progresses lower: **1)** the monthly chart establishes primary trend structure and major levels, **2)** the weekly chart identifies intermediate patterns and trend maturity, and **3)** the daily chart is used to refine execution and monitor trade behavior. This approach helps keep the main thing the main thing.

Time frame alignment increases conviction

When multiple time frames suggest the same thing, a technical analyst can have more confidence in a signal or setup. MTF alignment can also reduce risk. When time frames diverge, outcomes become less predictable. Conflicting time frames may also be indicative of transition between trend phases. This doesn't mean that an idea or trade cannot work or be pursued. However MTF context should be kept in mind.

Patterns within patterns

Because markets are fractal, chart patterns can exist within larger patterns. A daily breakout pattern may develop inside a weekly trend continuation. A weekly trend continuation pattern may develop ahead of a major monthly turning point. This nesting behavior can help explain why technical patterns sometimes fail. A daily bullish breakout near weekly resistance is more likely to be rejected than a daily bullish breakout after weekly consolidation at monthly support. When in doubt, higher time frames receive more weight than lower timeframes.

Multiple time frame technical analysis applied to the S&P 500 bottom in 2022

The use of three-time frames, moving averages and pattern recognition during the cyclical bear market of 2022 helped determine when the market was bottoming.

Exhibit 3: Daily chart of S&P 500 index shows detailed price action of 2022 cyclical bear market and 2023 recovery.
SPX daily chart, 50d SMA, 200d SMA



Source: BofA Global Research, Bloomberg

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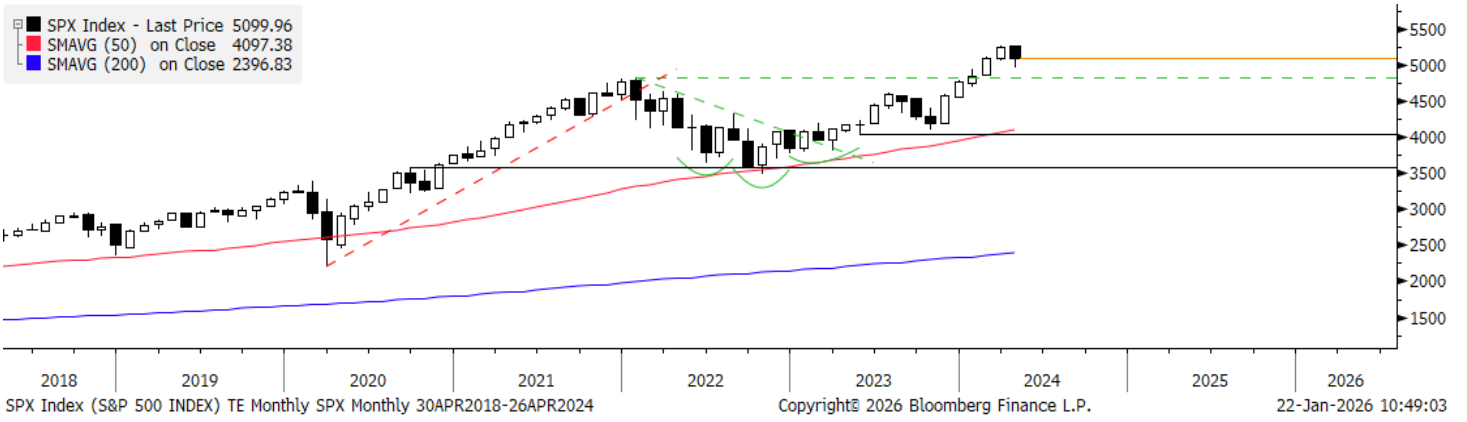
Exhibit 4: Weekly chart of S&P 500 index also shows 2022 cyclical bear market and recovery but with less noise. 200wk SMA provided support.
SPX weekly chart, 50wk SMA, 200wk SMA



Source: BofA Global Research, Bloomberg

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Exhibit 5: Monthly chart of S&P 500 kept 2022 cyclical bear market in perspective with support from the 50m SMA.
SPX monthly chart, 50m SMA, 200m SMA



Source: BofA Global Research, Bloomberg

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Support & Resistance

- Support (demand for an asset) halts a decline ahead of a move higher. Resistance (supply of an asset) halts a rally ahead of a move lower.
- Support & resistance are important for identifying trends (up, down or trading range) as well as chart patterns (reversal or continuation).
- Support & resistance can reverse roles. Use time, tests, depth & volume to assess the importance of a support or resistance level.

Support & resistance = Supply & demand

When technical analysts refer to support and resistance, they are referring to areas on a chart of demand (support) and supply (resistance). It is economics 101.

Buyers win at support & sellers win at resistance

Buyers equal demand and win when support holds on a decline. Sellers equal supply and win when resistance contains a rally.

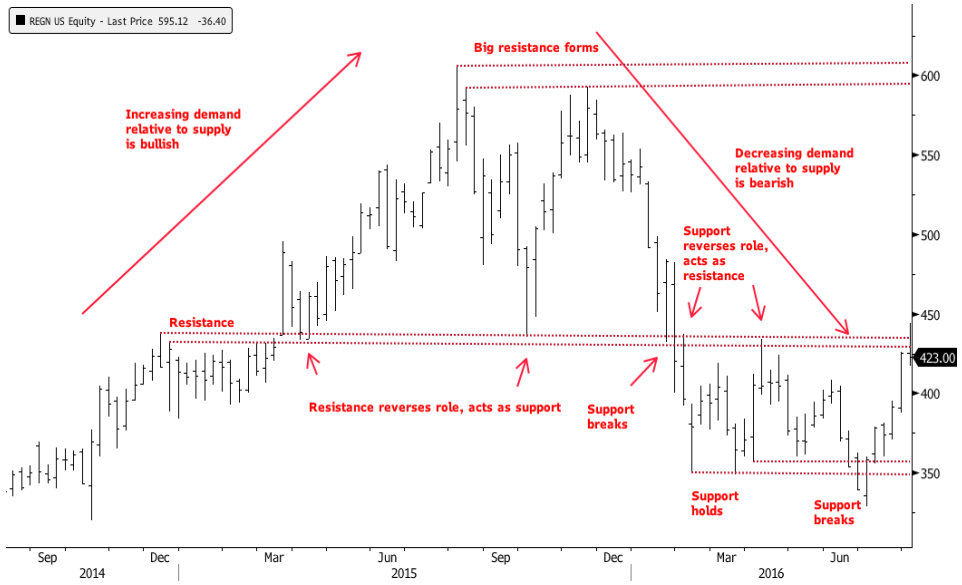
Support halts a decline ahead of a move higher

Support is a price level or zone where buyers (or demand) are strong enough to overpower sellers (or supply) and halt a decline ahead of a move higher. Holding support is bullish. From a fundamental perspective, a decline to a support may have made a security cheap enough to entice investors to buy and sellers to step aside.

Resistance halts a rally ahead of a move lower

Resistance is a price level or zone where sellers are strong enough to overpower buyers and halt a rally ahead of a move lower. Holding resistance is bearish. From a fundamental perspective, a rally into resistance may have made a security expensive enough to entice investors to sell and buyers to step aside.

Exhibit 6: Illustrative example of support & resistance on Regeneron (REGN)
When technical analysts refer to support and resistance, they are referring to areas on an Exhibit of demand (support) and supply (resistance). It is economics 101.



Source: BofA Global Research, Bloomberg

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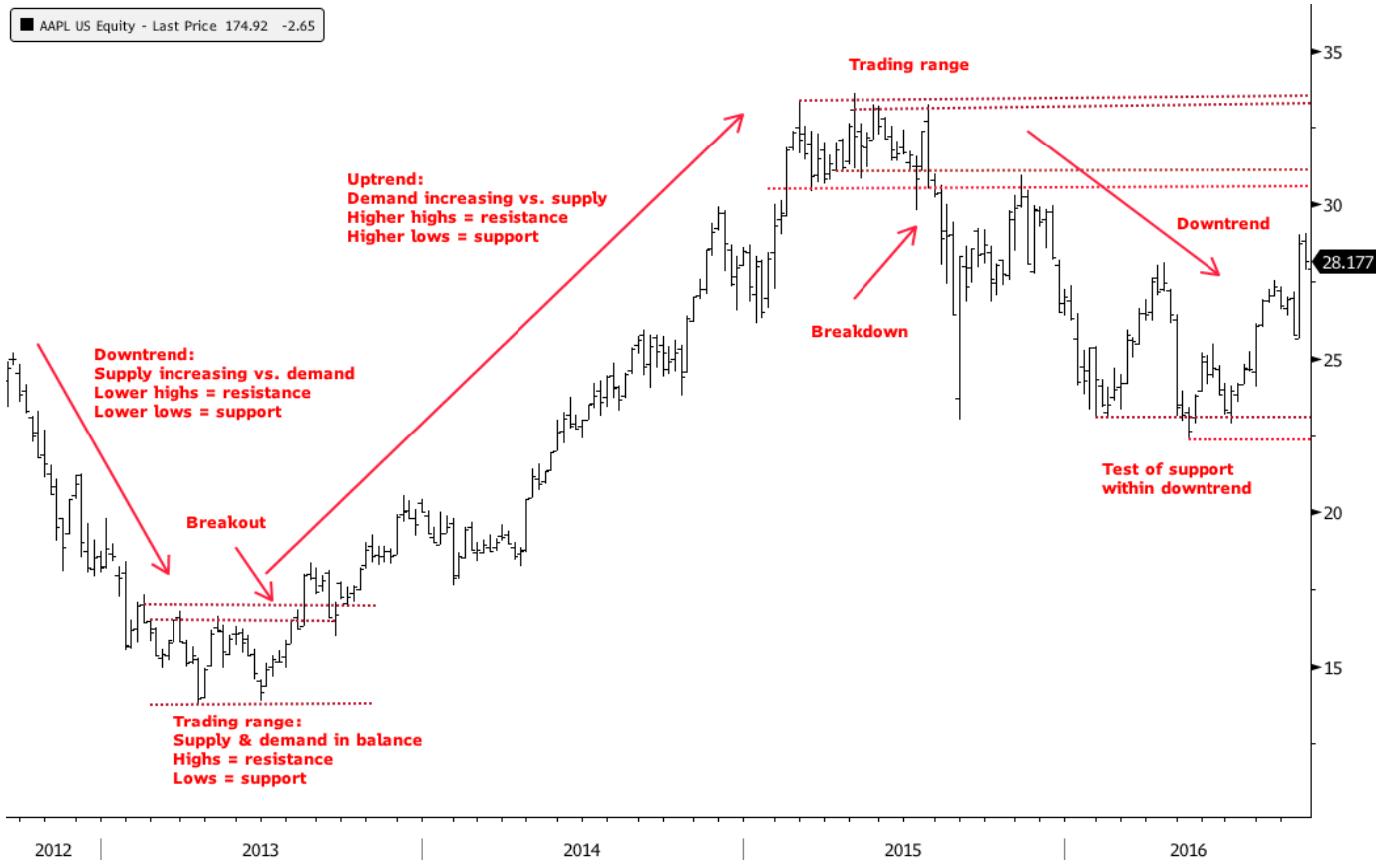
Lows & highs define support & resistance

Price lows or troughs are potential support (demand) and price highs or peaks are potential resistance (supply). Buyers tend to become more interested near lows and sellers tend to become more interested near highs.

Supports & resistances define trends

- **Uptrend**: When demand is increasing relative to supply, investors are willing to pay higher prices for a security. This often leads to higher support and resistance levels (higher lows and higher highs).
- **Downtrend**: When supply is increasing relative to demand, investors are willing to sell a security at lower prices. This often leads to lower resistance and support levels (lower highs and lower lows).
- **Trading range**: Supports form at or near the same price below the market and resistances form at or near the same level above the market. There is a supply and demand balance between these levels that forms a trading range.
- **Range breakouts and breakdowns** negate the supply/demand balance of a trading range. An upside break or breakout from a trading range above resistance is bullish and suggests that demand has become strong enough to overcome supply. A downside break or breakdown from the range below support would indicate that supply has become strong enough to overcome demand.
- **Trendless**: Supports and resistances do not have a well-defined pattern.

Exhibit 7: Support & resistance on Apple (AAPL) - illustrative example
Price lows or troughs are potential support (demand) and price highs or peaks are potential resistance (supply).



Source: BofA Global Research, Bloomberg

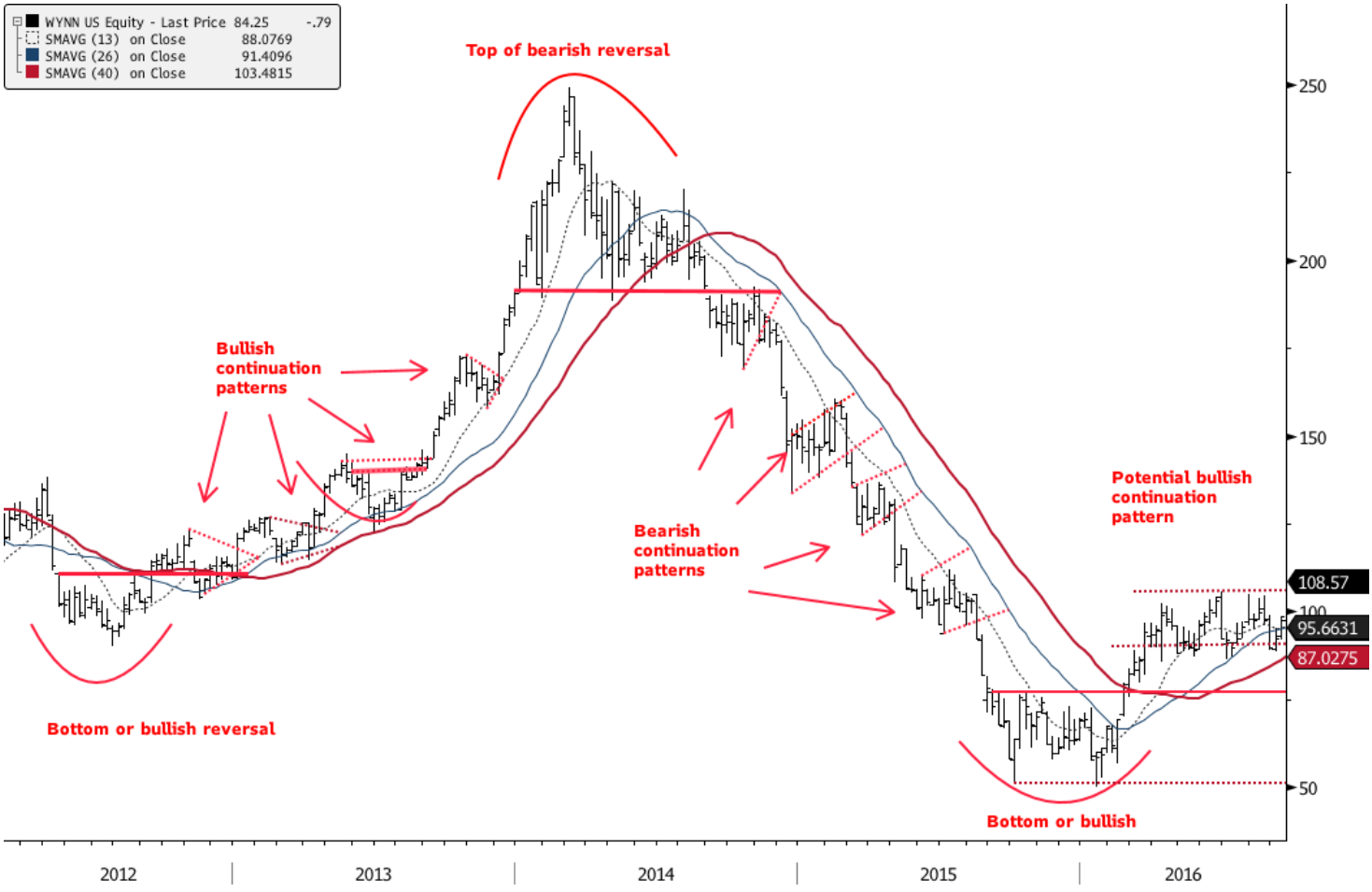
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Supports & resistances define price patterns

There are two types of price patterns: reversal and continuation. Support and resistance levels play important roles in defining these patterns.

- **Reversal patterns**, such as double bottoms (tops) and head and shoulders bottoms (tops), are characterized by changes in supply and demand. Bottoms occur when supply is getting exhausted and demand is improving to the point where the trend changes from down to up. Top patterns show diminishing demand and increasing supply that is sufficient enough to change the trend from up to down.
- **Bullish continuation patterns**, such as flags, pennants, and triangles, occur within uptrends. These patterns reflect a loss of demand and/or increase in supply that is not strong enough to reverse the uptrend but rather leads to a consolidation or a pullback. This pause in the uptrend moves prices down to a level where the security finds renewed demand or support that is strong enough to continue the uptrend.
- **Bearish continuation patterns** occur in downtrends when there is a pick-up in demand and/or decrease in supply that is not strong enough to reverse the downtrend but rather that leads to a rebound or an upside consolidation. This pause in the downtrend moves prices up to a level where the security finds increased selling pressure or resistance that is strong enough to continue the downtrend.

Exhibit 8: Support & resistance & price patterns on Wynn Resorts (WYNN) - illustrative example
Supports and resistances define, as well as confirm, price patterns.



Source: BofA Global Research, Bloomberg

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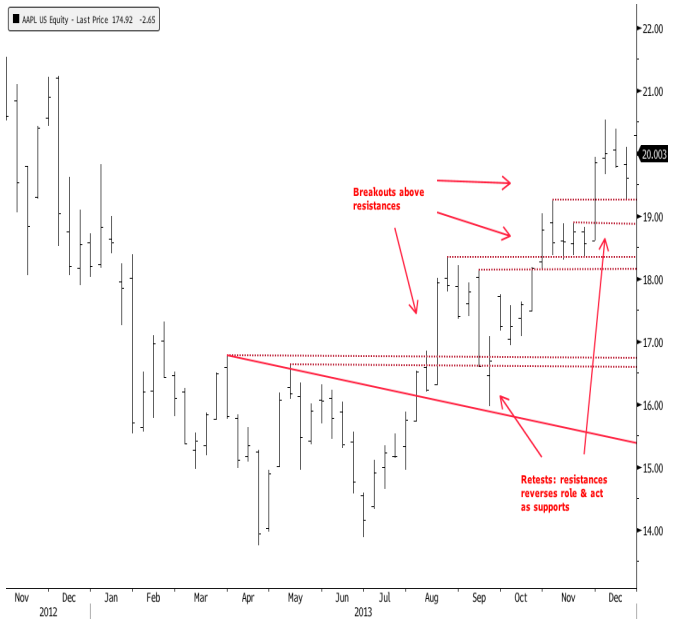
Supports & resistances can & do reverse roles

When a support or resistance is breached, it can and often does reverse roles. This is a signal that the supply and demand relationship has changed and occurs within uptrends and downtrends as well as after prices break out or break down from price patterns.

- **Breakouts**: When price breaks out above resistance and the resistance reverses roles to act as support, the security or market shifts from "sell the rallies" to "buy the dips." This role reversal happens because buyers are happy and looking to buy more and sellers (or shorts) are not happy and looking to get out even on dips.

- **Breakdowns**: When price breaks below support and the support reverses roles to act as resistance, it means that the security or market shifts from "buy the dips" to "sell the rallies." This happens because sellers are happy and looking to sell (or short) more and buyers are not happy and looking to get out even on rallies.

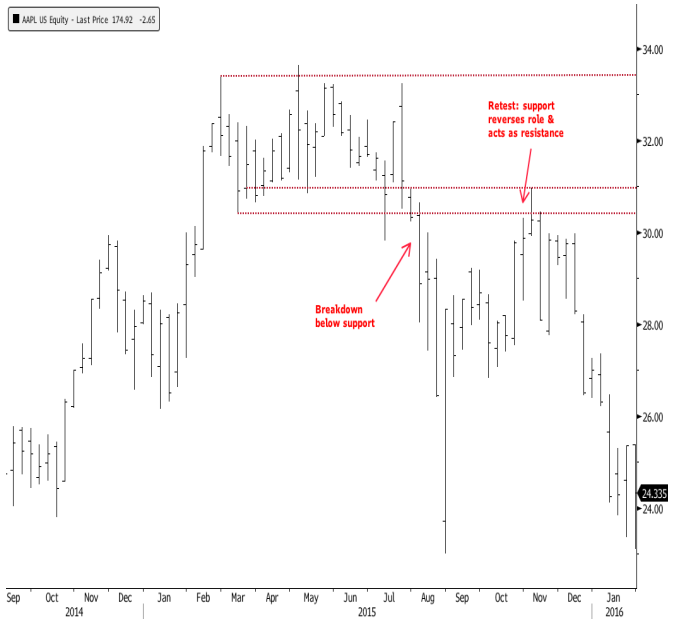
Exhibit 9: Resistances break & then reverse role to act as supports
A shift from greater supply (sell strength) to greater demand (buy dips).



Source: BofA Global Research, Bloomberg

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Exhibit 10: Support breaks & then reverses role to act as resistance
A shift from greater demand (buy dips) to greater supply (sell strength).



Source: BofA Global Research, Bloomberg

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Terrific trend lines

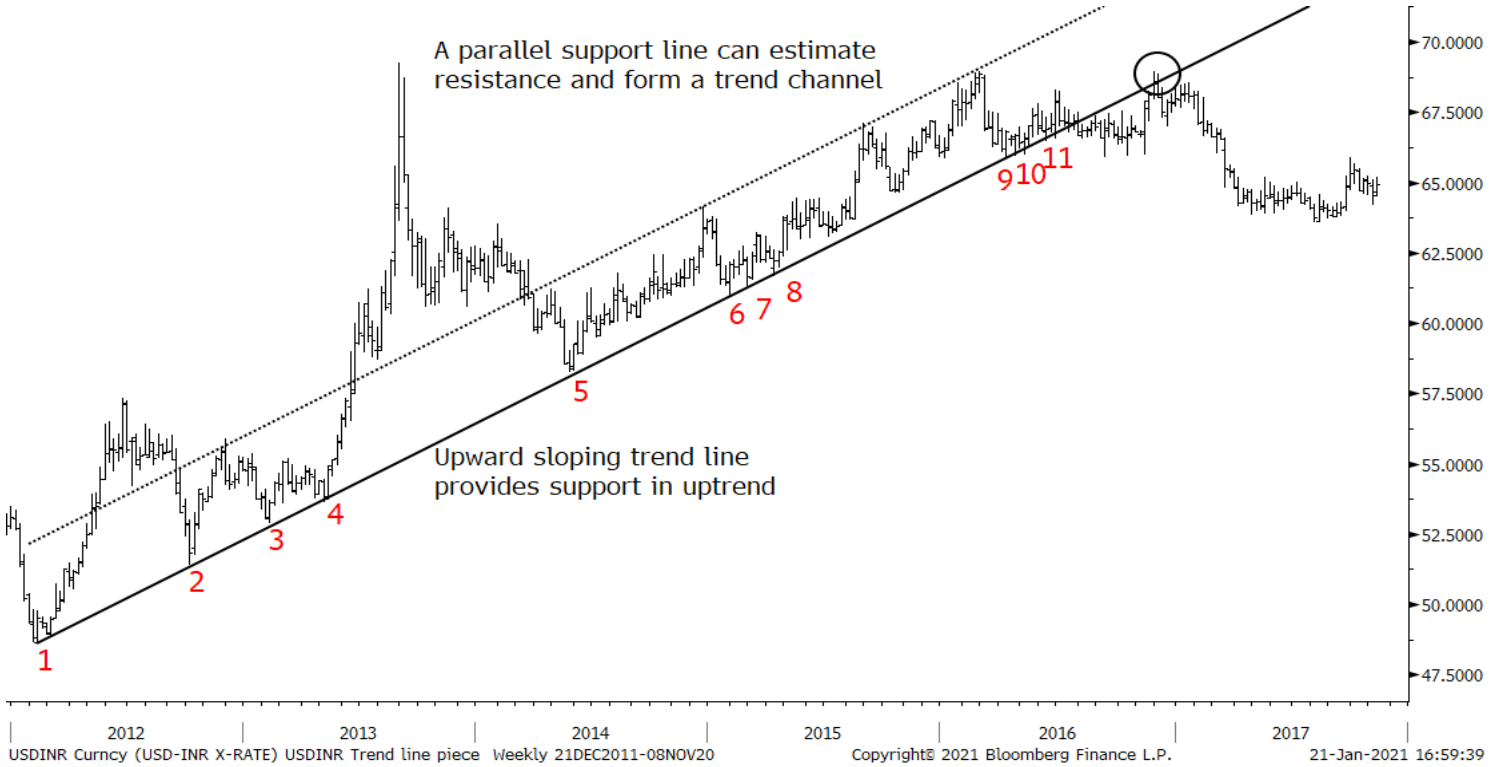
- A trend line is a directionally sloped line that provides support in uptrends and resistance in downtrends.
- Multiple points and time make a trend line more important and useful to identify support, resistance and changes in trend.
- We provide rules that define a trend line as broken and how to create targets using Fibonacci extensions and measured moves.

Trend lines imply trend direction and can signal change

A trend line is a guide to the overall direction of the market in an upward, sideways or downward direction. It is there for four reasons:

1. Estimate support in an uptrend as price temporarily corrects the larger trend. Support is a level where buyers outnumber sellers. It is a location where a dip is bought.
2. Estimate resistance in downtrends during relief rallies. Resistance is a level where sellers begin to outweigh buyers, which can push prices lower. It is a location in a rally where sellers add to shorts.
3. Create a parallel channel line based on 1 and 2 above. A support line in an uptrend can be used to estimate the top of a channel. The resistance line in a downtrend can be used to estimate the bottom of a channel.
4. When a trend line is broken, it adds to the body of evidence that the trend has changed.
5. A broken trend line can chance roles, such as turning to resistance.

Exhibit 11: USD/INR weekly Exhibit with long term support trend line and parallel channel line - illustrative example
This is an example of a long-term trend line that was tested 11 times, broken and then the line acted as resistance.



Source: BofA Global Research, Bloomberg

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Not all trend lines are created equal

A variety of characteristics help to differentiate good trend lines from not so good ones. A few ways to differentiate them are listed below.

1. Multiple touch points with time between each test.

2. Duration or age of the line increases significance. This trend line in Exhibit 11 is over four years old.

3. Not too steep, not to flat. Steep trend lines are often broken early and flatter trend lines provide confirmation signals. A 30 to 50 degree slope is ideal.

4. If the distance from the start of the trend line to the current price level exceeds 30%, it is recommended to also consider a log scale chart in addition to an arithmetic scale. The log scale chart can result in an earlier trend line break. However, in this case we prefer the arithmetic scale because the 50wk moving average (red) aligns with the support line.

What constitutes a broken trend line?

If price were to break down through a supporting trend line, it would be ideal if at least one or more of the following conditions occurred:

1. Price closes at least 1% below the trend line.

2. Volume, if available, is greater than the average volume.

3. Consecutive bars close below the line.

4. Post break, intraday rallies test and fail to move back above the line. Such tests are more convincing if they occur on less than average volume.

5. The size of the break down is greater than the ATR value.

Average true range (ATR) measures the volatility of prices over the past x bars. In Exhibit 12, the five week ATR has a value that rounds to .90. A close .90 below the trend line could also be used to confirm a trend line break. ATR is calculated by taking the average of the largest absolute value of the following three calculations for each included period: the high minus low, high minus prior bars close and low minus prior bars close.

Exhibit 12: USD/INR with Average True Range indicator (ATR)
Consider the current size of the ATR as an amount to break through a trend line.



Source: BofA Global Research, Bloomberg

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Projecting price after a qualified trend line break

Suppose price broke down through a trend line. A measured move calculation or a Fibonacci Extension can be used to estimate targets. Calculate a target by taking the height of prior moves above the trend line and subtracting that from the confirmed breakdown point.

The distance from the peak in 2016 down to the trend line was about -5.05%. This is a reasonable technical target to subtract from the trend line break down point. Choose the next largest height such as that period in 2012 for a more medium- or long-term possibility.

Also pay close attention to horizontal levels that were once resistance such as the area at about 64.00 and 61.00. They have a tendency to turn into support during declines. As more levels break without technical signals to refute the decline, the deeper targets become more achievable.

Exhibit 13: Measured move example
Measure the height of a channel and subtract it from the breakdown point to derive a target for the trend when it changes.



Source: BofA Global Research, Bloomberg

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Fibonacci Extensions can also be used to estimate targets

By measuring the largest distance from the trend line we can project lower from the break out point including key Fibonacci ratios. This projection method can be more aggressive so it is not to say they will all be reached. Rather the point is to isolate levels that might matter and to have a roadmap should the trend accelerate. In Exhibit 14, the largest height above the trend line was in 2013. A 38.2% target comes close to the 61.00 area discussed in Exhibit 13. The 50% target aligns with the 2014 lows.

Exhibit 14: Projecting levels using Fibonacci Extensions

The height of a trend above a trend line can be used for downside targets. In this case, a Fibonacci extension offers multiple levels.



Source: BofA Global Research, Bloomberg

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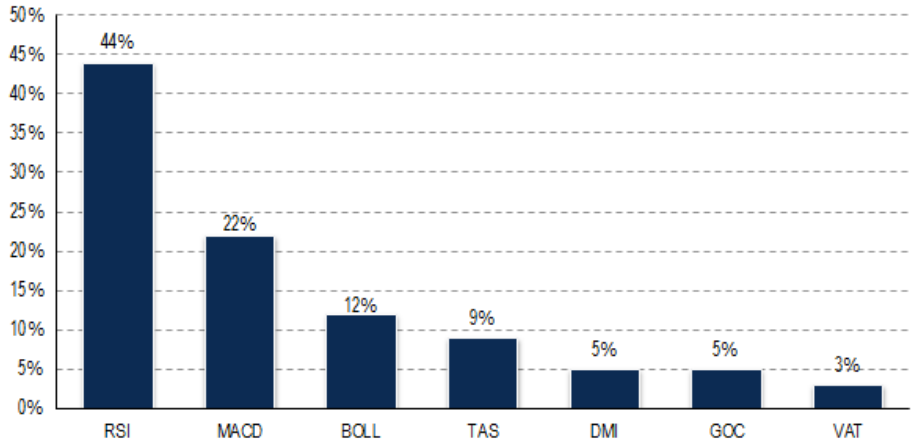
Evidence of the most popular indicators

- RSI is the most popular technical indicator, ahead of MACD and Bollinger Bands. It is used to measure and trade momentum.
- We review the essence of the RSI formula as well as the facts and myths of overbought and oversold trends.
- We also discuss our most preferred signals including a three-troughed bullish divergence and three-peaked bearish divergence.

Relative Strength Index is the most popular

Data published in 2011 suggests the Relative Strength Index (RSI) is the most popular technical study used on the Bloomberg terminal. The Moving Average Convergence Divergence (MACD) is the second most popular, though half as popular as RSI. Third place goes to Bollinger Bands (BOLL). The top three indicators represented 78% of the seven most used indicators. The others include Stochastics (TAS), the Directional Movement Index (DMI), and an increasingly popular study called Ichimoku (GOC). The last is Volume at Time (VAT). Notably missing and assumed in the top 10 are simple moving averages and Fibonacci price retracements. In our view, knowing the different ways the popular indicators are used is an important step in learning and using technical analysis.

Exhibit 15: Indicator use as a percentage of the top seven most used
The Relative Strength Index (RSI) is the most popular technical indicator (excludes moving averages).



Source: Ciana, P. New Frontiers in Technical Analysis. John Wiley & Sons Inc. 2011

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Markets use different types of indicators

The top three most popular indicators suggest market participants use a diverse group of indictors. RSI is a momentum oscillator offering confirmation of price breakouts, short-term buy and sell signals, and longer-term, trend-ending implications from price and momentum divergences. MACD is a trending indicator that attempts to improve upon moving averages by offering an earlier trend start and end signal. It too can provide divergence signals. Bollinger bands is a statistical indicator that attempts to identify when markets may be overdue for a volatile move when the bands are relatively narrow or when the market has moved to far too fast as price exits the bands.

Get a chart with the popular indicators

There is a sample chart on Bloomberg that is accessible by typing {G BBTA 10<GO>} with all of the popular indicators preloaded. Click the red “Actions’ button, choose “Save a copy,” title it popular indicators, and click the gray “Save” button. The right side of the red toolbar should say G # and the title. Type, for example, EURUSD Curncy G#<GO>, to open a chart of the euro with the popular indicators.

Moving Averages

- Moving averages are one of the most popular trend-following, support/resistance and quoted indicators in technical analysis.
- With many uses, in this report we review the simple moving average, generally accepted signals, popular periods and examples.
- We correctly define the "golden cross" & "death cross." We also recap the TMAC and MACD moving average-based strategies.

Moving average cross strategies

The 50-day and 200-day simple moving averages (SMA) are the most popular and quoted moving average (MA) signals and levels. There are many ways to leverage their value. In this primer we introduce the MA, different types of averages, look-back periods and multiple time frame uses. We recap the golden cross and death cross, a Triple Moving Average Cross (TMAC) and Moving Average Convergence/Divergence (MACD).

Why 50 & 200? What about the 100, 150 or Fibonacci #s?

History suggests the 50- and 200-day SMA, and to an extent the 100- and 150-day SMA, are the most popular because they were simple to calculate by hand prior to computers. Now they tend to be the default on charting software. They may work at times because others are looking at them however plenty of other conditions exist when technical traders, trend followers and systematic firms act. A trader, whose horizon is shorter, consider smaller SMA periods such as the 5-, 10-, 15-, 20-, 30- and 50-day MAs. Futures traders tend to remove one day for the roll. A Fibonacci enthusiast may follow a few in the Fibonacci sequence such as 5, 8, 13, 21, 34, 55, 89, 144 and/or 233.

Generally accepted bullish and bearish signals

Moving average signals are bullish (bearish) when the faster moving elastic time series moves above (below) a slower moving inelastic line, such as 1) Price crosses above (below) a moving average, 2) When the slope of an average turns positive (negative), 3) When a shorter period average crosses above (below) a longer period average, 4) When price trends too far below (above) the average relative to history.

Golden Cross & Death Cross is reserved for 50 & 200-day

The golden cross and death cross names are generally reserved for the crossing of the 50-day and 200-day simple moving averages. A golden cross occurs when the 50-day simple moving average crosses above the 200-day. The signal suggests the time series will rise. The death cross is the opposite of the golden cross and occurs when the 50-day moving average crosses below the 200-day. This signal suggests the time series will fall. We recommend being mindful/explicit of the time series the signal occurs on because a golden cross on the VIX, a bond future or JPY/KRW may infer a bearish risk tone.

TMAC and MACD moving average strategies

We discuss the TMAC and MACD strategies which are two popular ways to leverage a combination of MAs to generate signals for a qualitative and quantitative approach.

Trend following with a set of moving averages

- In this section we summarize and review examples of using the 20-, 50-, 100-, 150- and 200-day simple moving averages.

Below we present a price chart with five moving averages on it including the 20-, 50-, 100-, 150- and 200-day simple moving averages (Exhibit 16). As a rule of thumb, it's visually simple to differentiate the short from the long based on how much it moves. The more elastic the line the shorter the period. Price is the most elastic time series while the 200-day moving average is relatively inelastic. Given the generally accepted rules for using moving averages, there are a total of 20 conditions and signals generated with price and five moving averages on a chart.

1. When price is above a moving average it is considered bullish so higher prices should follow. When price is below a moving average it is considered bearish so lower prices should follow. In this example there are a total of five conditions.

2. When the slope of a moving average is positive it is considered bullish. When the slope of a moving average is negative it is considered bearish. In this example there are a total of five conditions. When a long-term average looks flat, we assume there is little trend.

3. When a shorter term moving average crosses above a longer term moving average it is considered bullish. When a shorter term moving average crosses below a longer term moving average it is considered bearish. In this example there are a total of ten bullish and bearish conditions.

Using a set of moving averages is referred to as a moving average fan. It is used to determine short, medium and long term trends, estimate targets and estimate support and resistance. Given the number of bullish and bearish conditions it is fairly simple to conclude how strong or weak the trend is by aggregating the conditions (more on this later in the TMAC section). Or, when looking at the chart and seeing the averages are dispersed (fanned out) like in December 2019, the stronger the trend is assumed to be.

Exhibit 16: An example of five simple moving averages on a daily Exhibit
Price vs a 20, 50, 100, 150 and 200-day simple moving average



Source: BofA Global Research, Bloomberg

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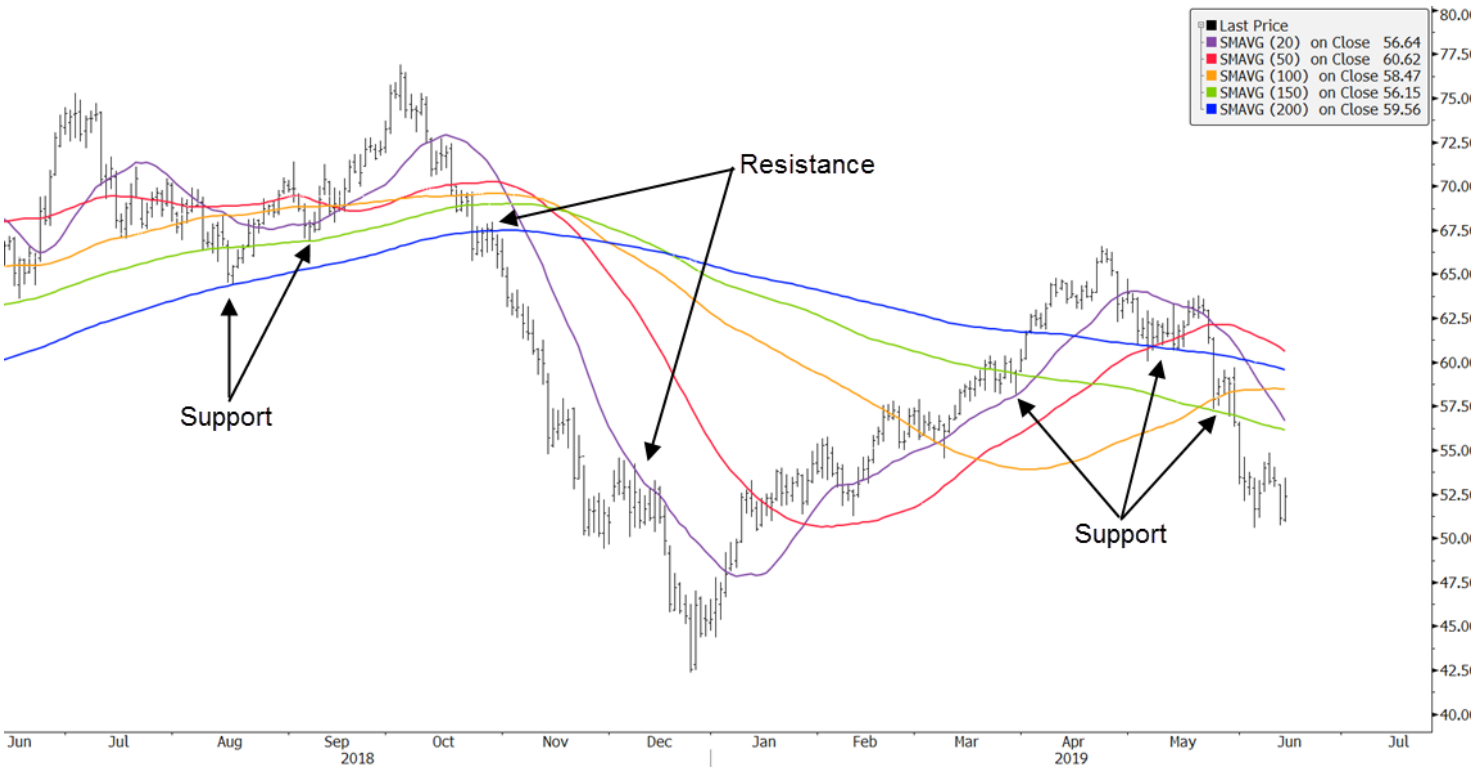
Use moving averages for support and resistance

Moving averages can provide support to price in uptrends and resistance to price in downtrends. We define support as a level the market will not allow price to move below and so the market begins to buy the security. In other words, demand is greater than supply. Similarly, resistance is a level that the market won't let price go above and so supply becomes greater than demand leading to a turn lower in prices.

In August 2018 the 200-day simple moving average (blue) provided support to price. The 100, 150 and 200-day moving averages still had a positive slope. The test of the 150-day moving average (green) in September resulted in a higher low with support holding. A rally to a new high followed.

A downtrend from October through January presents many examples of resistance from a moving average. In October price closed below the 200-day moving average and could not get back above it. In December price action failed to close back above the 20 day moving average. It wasn't until January that it did and then struggled at the 50d SMA. Eventually an uptrend persisted guided higher by the 20 day moving average.

Exhibit 17: An example of five simple moving averages on a daily Exhibit
Moving averages can act as support and resistance to price trends.



Source: BofA Global Research, Bloomberg

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Many types of moving averages...keep it simple

- A simple moving average is an equal weight average of a specified look back.
- An exponential moving average is a form of a weighted average that gives more consideration to newer price data.

There are many MA types to consider. The simple moving average is often used the most followed by the exponential moving average. There are many other types such as weighted, variable, adaptive and triangular. Then there are indicators such as MACD, Envelopes, Bollinger Bands, TRIX and Trender that use moving averages as the basis of their calculation. We recommend keeping it simple and, over time, experimenting with

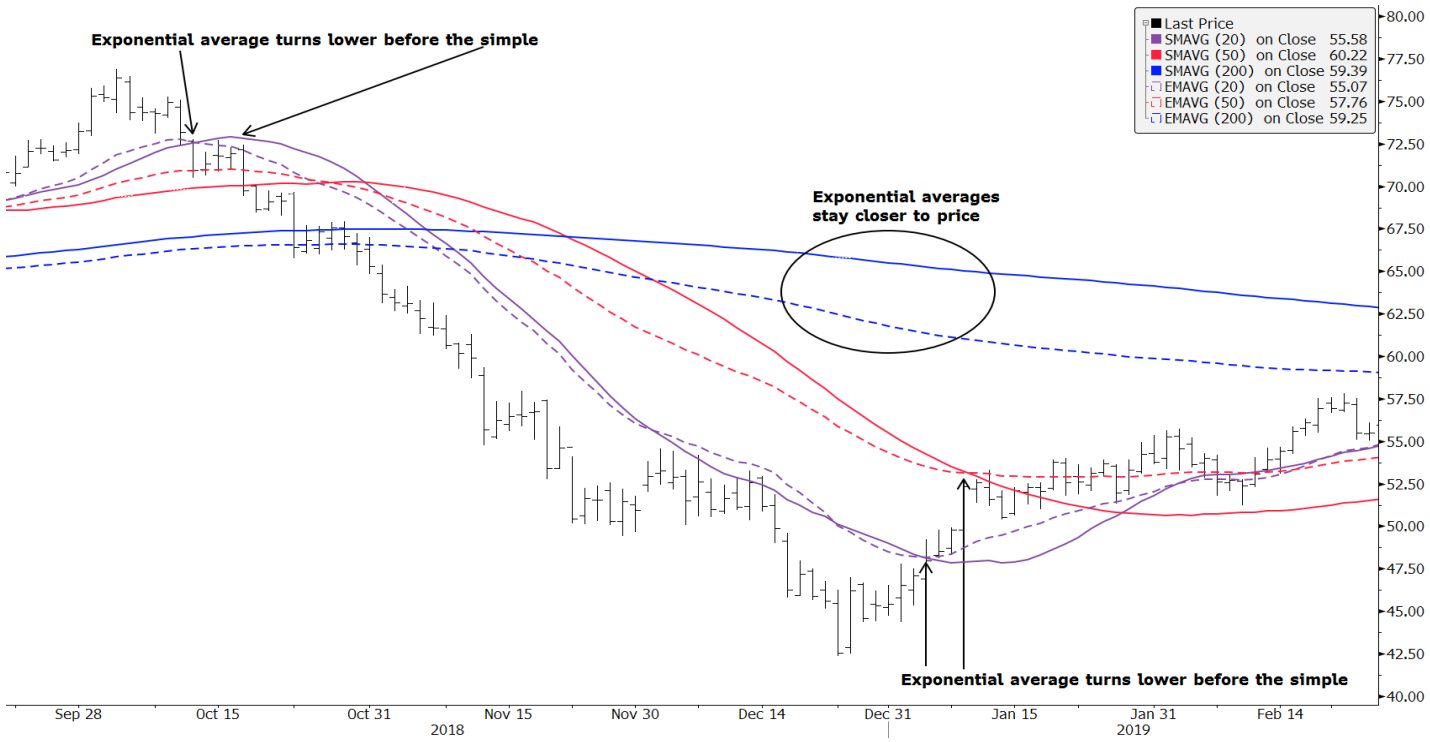
other averages especially on different asset classes. For example, a rule of thumb says rates tend to mean revert more often, commodities tend to trend longer and stock indices tend to go up. Therefore, an exponential moving average on a commodity may work better in your process than a simple average because the average will trend closer to price as it reaches new extremes. This will allow it to trigger a reversal signal sooner.

Simple vs. Exponential moving averages

A simple moving average is an equal weight average. It takes the sum of x data points and divides by x to calculate the current value. An exponential moving average is a form of a weighted average that gives more consideration to newer price data. However, in contrast to a weighted moving average it maintains a longer memory of historical price action in its calculation. In other words, there is a slower roll off of old data points while emphasizing new data while a weighted average has a fixed roll off. This exponential average is the basis of the MACD indicator.

Besides its memory of historical price action, exponential moving averages are viewed as appealing because they tend to stay closer to price. This means their slopes tend to change sooner, price tends to break them sooner and a faster average tends to cross a slower average sooner. All together this could mean earlier signals with the same period input as a simple MA. Whether they offer the correct signal more than simple averages is debatable. It more likely boils down to process and a disciplined application.

Exhibit 18: Example of simple moving averages (solid) vs. exponential moving averages (dashed) on a daily Exhibit
Exponential moving averages tend to stay closer to price trends than simple moving averages.



Source: BofA Global Research, Bloomberg

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Not just daily moving averages

- Looking at the 50, 100 and 200 period moving averages on many time frames can show where short-, medium- and long-term support and resistance levels lie.

Multiple time frame technical strategy is important

The markets are comprised of short, medium and long term traders and investors. These market participants will choose to look at different moving averages but they most certainly look at different time frame charts. The integration of popular moving averages

across time frames can often tell a participant what others may be looking at or thinking with regard to trend conditions, support/resistance and targets. The short term oriented may look at a 5 minute, 60 minute, 240 minute and daily set of charts with 200 period moving averages on each. A medium term participant may look at a 240, daily and weekly. A long term participant may only concern themselves with a weekly, monthly and even a quarterly chart. Here are some examples where different time frame 200 periods SMA's were highly relevant.

US 10 year treasury yield peaked at the 200 month SMA

The bond bear market of 2016-2018 was capped by the 200 month simple moving average. This averaged had not been tested since 1989 however it became increasingly relevant on the weekly chart below as yield began to double top at 3.25% (Exhibit 19).

Exhibit 19: US 10 year treasury yield – weekly Exhibit with 200 month simple moving average
The 200 month simple moving averaged resisted yield from going any higher in 4Q18



Source: BofA Global Research, Bloomberg

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The S&P 500 tests the 200-day SMA and the 200 week SMA in 2018

During 2018, heightened volatility, geopolitical risks and macro risks caused the S&P 500 to decline to the supportive 200-day moving average in March. Later that year a new high was made and then another decline followed. This time the 200-day was less supportive rather the market traded around the average a few times as it hesitated before its next decline. The next decline tested support at the 200 week moving average and marked a significant low before a strong rally in 2019 tested the 200d SMA, again.

Exhibit 20: S&P 500 Daily Exhibit with 200-day and 200 week simple moving average plotted
The S&P 500 finds support and resistance at the 200-day SMA and 200 week SMA.



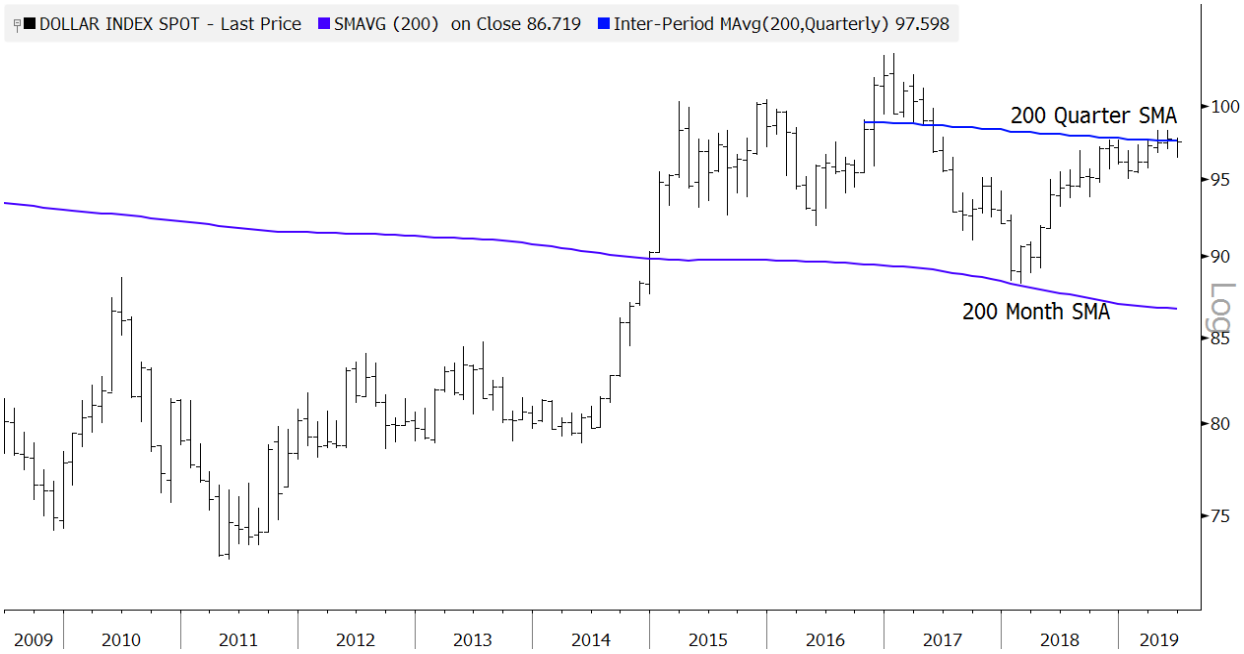
Source: BofA Global Research, Bloomberg

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DXY bounces between the 200 month and 200 quarter moving averages

The long term trend, or a monthly chart, of the DXY shows the index trading down to the 200 month moving average in 2018 and struggling for three quarters to close above the 200 quarter moving average in 2018-2019.

Exhibit 21: DXY Index – Monthly Exhibit with 200 month SMA and 200 quarter SMA plotted
The DXY finds support at the 200 month SMA and resistance at the 200 quarter SMA.



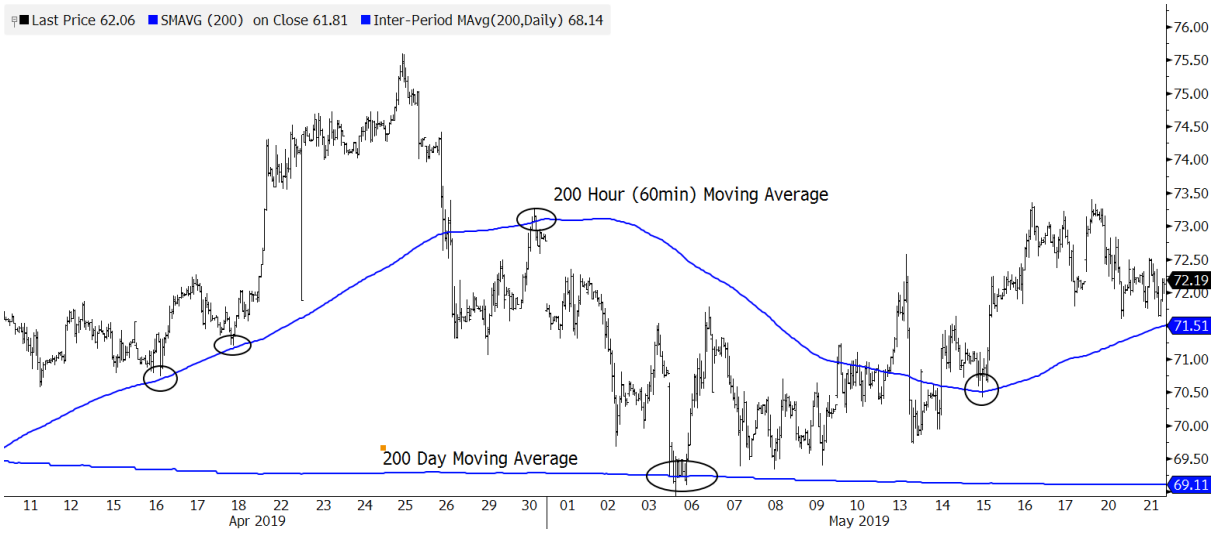
Source: BofA Global Research, Bloomberg

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Brent trends up along the 200 hour SMA and then tests the 200-day

The uptrend in Brent oil futures tested the 200 hour (60 minute) moving average twice during April to then peak in the 75s. Then it declined below the 200h SMA and failed to move back above it as it became resistance. Later, price tested the 200d SMA for support and held above it for four trading days forming a base.

Exhibit 22: Brent oil vs the 200 hour (60 min bar Exhibit) and 200-day moving average
Brent oil future was supported by the 200-day SMA and both supported and resisted by the 200 period (60min) SMA.



Source: BofA Global Research, Bloomberg

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The Golden Cross & Death Cross

- A popular moving average cross signal is the 50-day and 200-day moving average.
- When the 50d crosses above the 200d, a golden cross is said to have occurred. When the 50d crosses below the 200d, a death cross is said to have occurred.
- We look at historical forward returns for the golden cross and death cross on the DXY index and incorporate the 200d SMA slope, too.

Golden Cross & Death Cross is reserved for 50 and 200

The golden cross and death cross names are generally reserved for the crossing of the 50-day and 200-day moving averages. A golden cross occurs when the 50-day moving average crosses above the 200-day moving average. The signal suggests the time series it is calculated on will rise. The death cross is the opposite of the golden cross and occurs when the 50-day moving average crosses below the 200-day moving average. This signal suggests the time series it is calculated on will fall. We recommend being mindful/explicit of the time series the signal occurs on because a golden cross on the VIX, a bond future or JPY/KRW may infer a bearish risk tone.

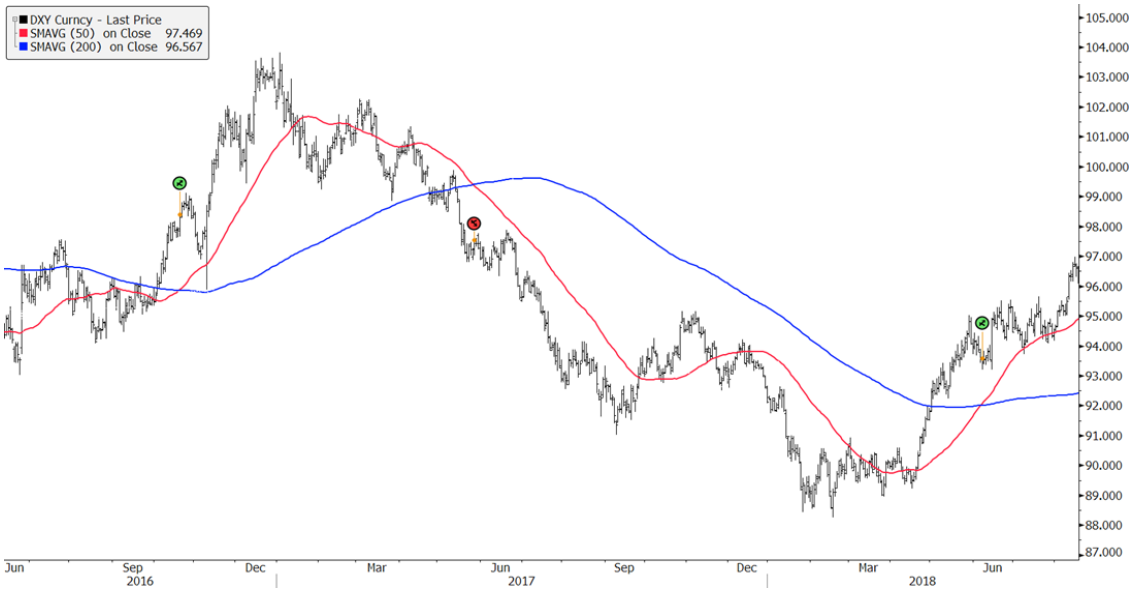
Golden Cross = 50d crossing above the 200d

In Exhibit 23 a golden cross occurred on October 20th 2016 because the 50-day moving average crosses above the 200-day moving average (red line crossed above the blue line as indicated by the green circle). This is considered a bullish signal suggesting price should rise. The closing value on this day was 98.31. The trend persisted higher to the 103s however then turned down. Eventually a death cross occurred.

Death cross = 50d crossing below 200d

On May 26th 2017 the 50-day moving average crossed below the 200-day moving average (Red line crossed below the blue line as indicated by the red circle). This is considered a bearish signal and suggests price should fall. The closing value on this day was 97.44. A downtrend persisted into 2018 and on June 7th 2018 another golden cross occurred with a closing value of 93.43.

Exhibit 23: Example of two golden crosses and one death cross
A golden cross occurred in October 2016 and June 2018. A death cross occurred in June 2017.



Source: BofA Global Research, Bloomberg

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Good for context, not necessarily entry AND exit strategy

A trend can persist after a golden cross or death cross occurs. By the time the opposing signal occurs, however, much of the trend may have already reversed. Relying on it for both entry and exit like a "black box" system may mean much of the trend based on the prior signal has already been given back. Using the golden cross and death cross may be ideal for context within a trend that is already underway and for understanding historical forward returns.

Below we show the percent change from the close on the golden cross in October 2016 to the high as +5.6%. By the time the opposite signal occurred, the death cross, price was -.89% below the entry point. From this short point, the trend was down as much as -9.43% in February 2018. The next golden cross didn't occur until June 2018. The bullish signal was -4.10% below the short entry point leaving about half the trend behind.

Exhibit 24: Example golden cross and death cross signals on the DXY index – Daily Exhibit
Percent change after a golden cross and death cross signal.



Source: BofA Global Research, Bloomberg

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DXY after a golden cross since 1979

When the 50d SMA crossed above the 200d SMA, in the past the US Dollar index (DXY) was higher 20-80 trading days later. Since 1979, the DXY was higher 68-84% of the time (Exhibit 25). The best "% Up" ratio shown in Exhibit 25 is 45 days later up 84% of the time with positive average and median gains and max gain greater than man loss. If entering at the close of the golden cross signal or within the first fifteen trading days that are less conclusive, history suggests the sweet spot to target a higher DXY is in 35-60 trading days as the up ratios are 74-84% of the time.

A golden cross with rising 200d SMA more bullish

When the 50-day SMA crossed above the 200-day SMA and the 200-day SMA had a positive slope (rising), the DXY was mostly higher 20-80 days later with the strongest point being also being 45 days later. It was up 21 of 23 times or 91% of the time. (Exhibit 26).

Exhibit 25: Price trend in the DXY after the 50-day simple moving average crosses above the 200-day simple moving average
The DXY tends to rise 25-80 trading days after a golden cross signal.

Ticker: DXY			Start Year: 1979		# Signals:	31 (32nd on going)		# < 80 days: 1					
Days after	**5 Day**	**10 Day**	**15 Day**	**20 Day**	**25 Day**	**30 Day**	**35 Day**	**40 Day**	**45 Day**	**50-day**	**60 Day**	**70 Day**	**80 Day**
% Up Ratio	44%	53%	53%	69%	68%	68%	74%	77%	84%	74%	77%	68%	68%
Up	14	17	17	22	21	21	23	24	26	23	24	21	21
Down	18	15	15	10	10	10	8	7	5	8	7	10	10
Average	0.10%	0.03%	0.17%	0.60%	0.69%	0.95%	1.03%	1.27%	1.39%	1.52%	1.50%	1.40%	1.64%
Median	-0.16%	0.06%	0.37%	0.87%	1.15%	1.19%	1.14%	1.67%	1.78%	1.75%	1.78%	2.35%	2.28%
Min	-2.03%	-4.60%	-2.82%	-3.23%	-3.92%	-4.74%	-5.35%	-6.04%	-7.06%	-8.25%	-8.96%	-9.74%	-10.13%
Max	2.50%	3.09%	2.70%	3.15%	3.89%	5.44%	8.91%	7.96%	9.53%	10.08%	10.49%	8.35%	8.69%

Source: BofA Global Research

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Exhibit 26: Price trend in the DXY after the 50-day average crosses above the 200-day average and the 200-day average was rising
The DXY tends to rise after a golden cross signal with a 200d SMA that has a positive slope as much as 21 of 23 times at 45 trading days after.

Ticker: DXY			Start Year: 1979		# Signals:	23 (24th ongoing)		# < 80 days: 0					
Days after	**5 Day**	**10 Day**	**15 Day**	**20 Day**	**25 Day**	**30 Day**	**35 Day**	**40 Day**	**45 Day**	**50-day**	**60 Day**	**70 Day**	**80 Day**
% Up Ratio	54%	58%	58%	75%	70%	74%	87%	83%	91%	78%	83%	70%	70%
Up	13	14	14	18	16	17	20	19	21	18	19	16	16
Down	11	10	10	6	7	6	3	4	2	5	4	7	7
Average	0.28%	0.12%	0.36%	0.89%	1.00%	1.29%	1.60%	1.80%	1.95%	1.99%	1.95%	2.00%	2.25%
Median	0.36%	0.13%	0.43%	1.25%	1.39%	1.33%	1.76%	1.76%	2.39%	1.75%	1.78%	2.96%	2.24%
Min	-2.03%	-4.60%	-1.68%	-3.23%	-3.92%	-4.74%	-5.35%	-6.04%	-6.55%	-7.06%	-7.35%	-8.60%	-8.31%
Max	2.50%	3.07%	2.70%	3.15%	3.89%	5.44%	8.91%	7.96%	9.53%	10.08%	10.49%	8.35%	8.69%

Source: BofA Global Research

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DXY after a death cross

Since 1979 there have been 31 instances when the 50-day SMA crossed below the 200-day SMA. In the past, this signal has not been indicative of a sustained downtrend (Exhibit 27). In Exhibit 28 we show a death cross with a negatively sloped 200d SMA. There are only 15 signals since 1979 vs. the 31 in table 3. There is a stronger tendency for the DXY to be lower after a death cross with a negatively sloped 200d SMA, but less consistency. After 20 and 45 trading days, the DXY was down 11 of 15 times or 73% of the time (Exhibit 28).

Exhibit 27: Price trend in the DXY after the 50-day simple moving average crosses below the 200-day simple moving average
The DXY tended to decline 5-10 trading days later 66-69% of the time.

Ticker: DXY			Start Year: 1979		# Signals: 31			# < 80 days: 1					
Days after	**5 Day**	**10 Day**	**15 Day**	**20 Day**	**25 Day**	**30 Day**	**35 Day**	**40 Day**	**45 Day**	**50-day**	**60 Day**	**70 Day**	**80 Day**
% Down Ratio	58%	58%	45%	55%	52%	48%	45%	45%	55%	52%	42%	42%	45%
Up	13	13	17	14	15	16	17	17	14	15	18	18	17
Down	18	18	14	17	16	15	14	14	17	16	13	13	14
Average	-0.20%	-0.36%	-0.19%	-0.40%	-0.42%	-0.21%	-0.21%	-0.33%	-0.38%	-0.44%	-0.16%	0.49%	0.33%
Median	-0.28%	-0.16%	0.04%	-0.91%	-0.68%	0.32%	0.24%	0.44%	-0.26%	-0.27%	0.22%	0.59%	0.80%
Min	-2.38%	-4.32%	-4.51%	-5.24%	-5.74%	-6.55%	-7.65%	-7.01%	-6.51%	-8.44%	-10.00%	-6.12%	-11.39%
Max	2.70%	2.51%	4.78%	5.06%	5.42%	5.41%	5.96%	5.33%	5.48%	5.29%	7.43%	8.37%	7.77%

Source: BofA Global Research

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Exhibit 28: Price trend in the DXY after the 50-day average crossed below the 200-day average and the 200-day average was falling
The DXY tended to decline after a death cross with falling 200d SMA

Ticker: DXY			Start Year: 1979		# Signals: 15		# < 80 days:	0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Down Ratio	60%	67%	60%	73%	67%	53%	47%	47%	73%	67%	47%	53%	53%
Up	6	5	6	4	5	7	8	8	4	5	8	7	7
Down	9	10	9	11	10	8	7	7	11	10	7	8	8
Average	-0.40%	-0.79%	-1.00%	-1.54%	-1.38%	-1.02%	-0.95%	-1.07%	-1.45%	-1.30%	-1.11%	-0.88%	-0.82%
Median	-0.29%	-0.63%	-1.06%	-1.29%	-1.05%	-0.01%	0.24%	0.32%	-0.76%	-0.87%	0.07%	-1.50%	-1.10%
Min	-2.38%	-4.32%	-4.51%	-5.24%	-5.74%	-6.55%	-7.65%	-6.39%	-6.03%	-7.75%	-10.00%	-6.12%	-6.92%
Max	0.80%	1.12%	2.56%	2.06%	2.04%	2.88%	3.65%	3.43%	3.35%	3.86%	6.73%	5.41%	7.18%

Source: BofA Global Research

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US long bond future after a golden cross

Since 1977, the US long bond treasury future (front month US1) saw 34 golden cross signals implying higher prices (lower yields). 40-50 trading days later bond prices rose 68% of the time or 23 out of 31 times. There was also some tendency to see prices higher 15 trading days after the signal (Exhibit 29). When a golden cross signal occurred and the 200d SMA was rising, the front month continuous long bond treasury future price was up 12 of 14 times 45-50-days later, or 86% of the time. (Exhibit 30)

Exhibit 29: Price trend after the 50d SMA crossed above the 200d SMA
The front month-long bond treasury futures tends to rise after a golden cross 40-50 trading days later.

Ticker: US1			Start Year: 1977		# Signals: 34			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	56%	56%	68%	62%	59%	55%	59%	68%	68%	68%	59%	56%	53%
Up	19	19	23	21	20	18	20	23	23	23	20	19	18
Down	15	15	11	13	14	15	14	11	11	11	14	15	16
Average	0.16%	0.23%	0.66%	0.35%	0.67%	0.60%	0.81%	1.04%	1.44%	1.40%	0.88%	0.57%	0.65%
Median	0.09%	0.15%	0.57%	0.67%	0.85%	0.29%	1.06%	0.96%	1.56%	1.62%	0.30%	1.24%	0.60%
Min	-3.25%	-5.76%	-6.26%	-7.58%	-7.94%	-9.92%	-12.44%	-13.48%	-15.24%	-15.96%	-14.44%	-18.77%	-17.12%
Max	3.26%	4.09%	5.72%	6.21%	7.04%	10.68%	10.48%	13.22%	13.42%	12.05%	16.20%	15.17%	12.00%

Source: BofA Global Research

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Exhibit 30: Price trend after the 50d SMA crossed above the 200d SMA and the 200d SMA was rising
When a golden cross occurred and the 200d SMA was rising, the US1 bond future was up as much as 12 of 14 times 45-50-days later.

Ticker: US1			Start Year: 1977		# Signals: 14			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	57%	50%	64%	50%	57%	50%	64%	79%	86%	86%	64%	57%	50%
Up	8	7	9	7	8	7	9	11	12	12	9	8	7
Down	6	7	5	7	6	7	5	3	2	2	5	6	7
Average	0.16%	-0.01%	0.27%	0.13%	0.58%	0.65%	0.70%	1.73%	2.49%	2.54%	2.22%	1.66%	0.69%
Median	0.33%	-0.18%	0.44%	0.06%	0.85%	0.09%	1.09%	1.23%	1.83%	1.97%	1.04%	1.24%	-0.09%
Min	-3.25%	-1.90%	-3.32%	-5.53%	-5.18%	-4.16%	-5.02%	-5.41%	-5.10%	-5.83%	-7.66%	-8.56%	-10.22%
Max	2.43%	2.09%	2.38%	3.36%	4.80%	5.43%	6.33%	13.22%	13.42%	11.33%	13.34%	15.17%	11.95%

Source: BofA Global Research

BofA GLOBAL RESEARCH

US long bond future after a death cross

There have been 34 death cross signals for the US long bond treasury future suggesting lower prices and higher yield since 1977. History shows these signals have little relevance on the future trend. (Exhibit 31)

Exhibit 31: Price trend after the 50d SMA crossed below the 200d SMA
The death cross presents little trend bias for the long bond future.

Ticker: US1			Start Year: 1977		# Signals: 34			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Down Ratio	53%	41%	53%	41%	47%	36%	45%	42%	52%	48%	42%	42%	45%
Up	16	20	16	20	17	21	18	19	16	17	19	19	18
Down	18	14	18	14	15	12	15	14	17	16	14	14	15
Average	-0.08%	0.02%	-0.43%	-0.04%	-0.18%	0.22%	-0.13%	-0.25%	0.07%	0.49%	0.44%	0.41%	0.29%
Median	-0.04%	0.17%	-0.22%	0.27%	0.03%	0.84%	0.81%	0.25%	-0.34%	0.82%	0.95%	0.87%	0.51%
Min	-5.06%	-6.06%	-5.71%	-7.64%	-9.79%	-10.38%	-11.44%	-10.28%	-11.24%	-9.62%	-12.07%	-13.93%	-10.88%
Max	3.48%	9.59%	6.04%	8.50%	7.95%	9.32%	11.66%	11.75%	20.81%	21.25%	22.61%	22.09%	19.91%

Source: BofA Global Research

BofA GLOBAL RESEARCH

Oil after a golden cross

Since 1983, there have been 32 golden cross signals. In the past, oil prices were higher 45-50 trading days after the signal 69% of the time or 22 out of 32 (Exhibit 32). When the 200d SMA had a positive slope and a golden cross occurred, the signal was bearish for the first ten trading days and bullish at 45-50-days (Exhibit 33). However when the 200d SMA had a negative slope, the golden cross signals tended to be better. From the signal through 10 days oil was up 69-77% of the time and from 45-50-days later it was up 69% of the time (Exhibit 34).

Exhibit 32: Price trend after the 50day MA crossed above the 200-day SMA
Oil prices tended to be higher 50-days later 67% of the time of 20 of 30 times.

Ticker: CL1 R:05_0_N Comdty			Start Year: 1983		# Signals: 32			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	53%	47%	50%	59%	53%	50%	53%	50%	69%	69%	48%	50%	47%
Up	17	15	16	19	17	16	17	16	22	22	15	16	15
Down	15	17	16	13	15	16	15	16	10	10	16	16	17
Average	0.53%	0.43%	0.00%	0.72%	0.64%	0.31%	1.16%	0.06%	-0.06%	0.81%	-1.12%	-1.11%	-4.06%
Median	0.71%	-0.48%	0.01%	2.61%	0.96%	0.14%	0.41%	-0.41%	2.49%	2.55%	-0.04%	-0.27%	-2.88%
Min	-10.45%	-11.00%	-18.29%	-17.23%	-19.54%	-16.16%	-17.53%	-26.90%	-67.56%	-74.65%	-111.36%	-111.26%	-160.10%
Max	11.92%	14.42%	13.82%	17.61%	27.21%	17.18%	30.25%	25.03%	15.23%	20.88%	25.38%	29.23%	30.62%

Source: BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 33: Price trend after the 50day MA crossed above the 200-day SMA and the 200-day SMA had a positive slope
Beware of a positively sloped 200d SMA when a golden cross occurs. Oil tends to be lower through ten days later, higher 45-50 and lower at about 80 days.

CL1 R:05_0_N Comdty			Start Year: 1983		# Signals: 15			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	20%	33%	40%	47%	47%	47%	47%	47%	67%	67%	53%	47%	33%
Up	3	5	6	7	7	7	7	7	10	10	8	7	5
Down	12	10	9	8	8	8	8	8	5	5	7	8	10
Average	-2.04%	-2.02%	-1.15%	-0.71%	-0.43%	-0.56%	1.07%	-0.39%	-2.50%	-0.23%	-4.48%	-5.39%	-10.50%
Median	-1.45%	-2.70%	-0.80%	-2.28%	-1.37%	-0.49%	-0.99%	-1.16%	1.85%	4.17%	6.22%	-1.92%	-4.29%
Min	-10.45%	-11.00%	-14.93%	-17.11%	-19.54%	-16.16%	-17.36%	-26.90%	-67.56%	-74.65%	-111.36%	-111.26%	-160.10%
Max	3.31%	4.74%	10.28%	13.34%	27.21%	17.18%	30.25%	25.03%	10.54%	20.88%	11.37%	19.69%	19.21%

Source: BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 34: Price trend after the 50day MA crossed above the 200-day SMA and the 200d SMA had a negative slope
When the 200d SMA had a negative slope and a golden cross occurred, oil prices tended to be higher through 10, 20 and 45-50-days later.

CL1 R:05_0_NComdty	Start Year: 1983			# Signals: 17				# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	81%	63%	56%	69%	63%	56%	63%	56%	69%	69%	47%	56%	56%
Up	13	10	9	11	10	9	10	9	11	11	7	9	9
Down	3	6	7	5	6	7	6	7	5	5	8	7	7
Average	2.92%	2.81%	1.02%	1.99%	1.78%	1.16%	1.41%	0.59%	2.20%	1.83%	1.97%	2.88%	1.55%
Median	2.43%	2.44%	1.02%	4.59%	4.43%	1.79%	1.45%	1.66%	3.38%	2.17%	-0.78%	3.54%	3.60%
Min	-2.36%	-7.84%	-18.29%	-17.23%	-18.28%	-12.90%	-17.53%	-22.23%	-22.95%	-21.09%	-22.23%	-30.50%	-37.61%
Max	11.92%	14.42%	13.82%	17.61%	14.57%	10.46%	11.05%	14.27%	15.23%	20.03%	25.38%	29.23%	30.62%

Source: BofA Global Research

BofA GLOBAL RESEARCH

Oil after a death cross

There have been 32 signals of when the 50-day SMA crossed below the 200-day SMA on oil prices since 1983. This signal has not been indicative of a new trend (Exhibit 27). When the 50-day SMA crossed below the 200-day SMA and the 200-day SMA had a negative slope, oil prices tended to rise (the opposite of the signals intention). The % Down ratio was just 29% about 45 days later, or up 71% of the time.

Exhibit 35: Price trend after the 50-day SMA crossed below the 200-day
Oil prices did not find a repeating trend after the death cross.

CL1 R:05_0_NComdty	Start Year: 1983			# Signals: 32				# < 80 days: 2					
Days after	**5 Day**	**10 Day**	**15 Day**	**20 Day**	**25 Day**	**30 Day**	**35 Day**	**40 Day**	**45 Day**	**50-day**	**60 Day**	**70 Day**	**80 Day**
% Down Ratio	56%	48%	50%	47%	53%	50%	50%	47%	38%	44%	44%	47%	38%
Up	14	16	16	17	15	16	16	17	20	18	18	17	20
Down	18	15	16	15	17	16	16	15	12	14	14	15	12
Average	-0.87%	-4.21%	-4.22%	-5.00%	-5.94%	-6.87%	-6.86%	-7.70%	-7.52%	-6.93%	-5.64%	-4.45%	-2.27%
Median	-1.75%	0.16%	0.04%	0.18%	-1.92%	-0.15%	0.06%	0.85%	0.89%	0.59%	0.95%	1.41%	5.01%
Min	-10.23%	-37.31%	-60.20%	-73.16%	-89.06%	-74.75%	-92.08%	-128.68%	-119.80%	-73.24%	-92.60%	-97.19%	-86.61%
Max	7.68%	7.06%	11.46%	14.85%	11.80%	8.82%	10.57%	12.22%	14.59%	15.29%	15.45%	27.19%	41.58%

Source: BofA Global Research

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Exhibit 36: Price trend after the 50-day SMA crossed below the 200-day SMA when the 200-day SMA had a negative slope
Oil prices, opposite of the signal intention, went higher 45 days later 71% of the time (Down ratio 29%)

CL1 R:05_0_NComdty	Start Year: 1983			# Signals: 17				# < 80 days: 1					
Days after	**5 Day**	**10 Day**	**15 Day**	**20 Day**	**25 Day**	**30 Day**	**35 Day**	**40 Day**	**45 Day**	**50-day**	**60 Day**	**70 Day**	**80 Day**
% Down Ratio	47%	44%	53%	41%	59%	47%	41%	41%	29%	41%	35%	47%	35%
Up	9	9	8	10	7	9	10	10	12	10	11	9	11
Down	8	7	9	7	10	8	7	7	5	7	6	8	6
Average	-0.06%	-4.34%	-5.29%	-7.09%	-8.17%	-8.38%	-7.18%	-9.60%	-9.61%	-7.69%	-5.03%	-2.51%	0.32%
Median	0.11%	0.94%	-0.19%	0.70%	-2.34%	0.37%	1.55%	1.21%	1.76%	1.94%	1.14%	0.72%	5.29%
Min	-5.61%	-37.31%	-60.20%	-73.16%	-89.06%	-74.75%	-92.08%	-128.68%	-119.80%	-73.24%	-56.48%	-41.00%	-28.30%
Max	7.17%	5.17%	5.42%	6.39%	11.80%	8.11%	9.08%	12.22%	14.59%	15.29%	15.45%	27.19%	41.58%

Source: BofA Global Research

BofA GLOBAL RESEARCH

S&P 500 after a golden cross

A golden cross for the S&P 500 has tended to be a bullish event. Out of 49 signals since 1927, the S&P 500 was up 59-73% of the time 10-80 days later with the strongest points being 25-40 days and 50-80 days later. A golden cross when the 200d SMA had a positive slope resulted in a higher S&P 500 as much as 26 of 34 times 25 days later or up 76% of the time (Exhibit 38).

Exhibit 37: Price trend after the 50d SMA crossed above the 200d SMA
The S&P 500 tended to be up 25-80 trading days later.

Ticker: SPX			Start Year: 1927		# Signals: 49			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	55%	59%	59%	58%	73%	69%	63%	65%	59%	69%	65%	73%	73%
Up	27	29	29	28	36	34	31	32	29	34	32	36	36
Down	22	20	20	20	13	15	18	17	20	15	17	13	13
Average	0.45%	1.11%	0.59%	1.19%	1.74%	1.85%	2.28%	2.48%	2.01%	2.23%	3.36%	3.90%	4.23%
Median	0.39%	0.97%	0.48%	1.09%	1.85%	2.63%	2.87%	2.74%	2.08%	2.91%	4.00%	4.52%	4.65%
Min	-4.99%	-7.14%	-13.39%	-9.51%	-8.23%	-10.77%	-8.53%	-13.51%	-11.14%	-10.08%	-9.68%	-11.05%	-14.87%
Max	9.11%	13.06%	15.78%	9.85%	17.21%	20.48%	28.82%	30.08%	16.64%	12.44%	17.04%	23.63%	21.84%

Source: BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 38: Price trend after the 50d SMA crossed above the 200d SMA and the 200d SMA had a positive slope
The S&P 500 tended to be 25 trading days later 76% of the time or 26 of 34 times.

Ticker: SPX			Start Year: 1927		# Signals: 34			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	59%	62%	68%	59%	76%	71%	62%	65%	62%	71%	71%	74%	71%
Up	20	21	23	20	26	24	21	22	21	24	24	25	24
Down	14	13	11	14	8	10	13	12	13	10	10	9	10
Average	0.44%	1.24%	1.15%	1.48%	1.88%	2.16%	2.63%	2.88%	2.00%	2.17%	3.07%	3.69%	3.92%
Median	0.40%	1.49%	0.67%	1.68%	1.82%	2.81%	4.06%	3.88%	2.71%	3.17%	3.76%	4.44%	4.77%
Min	-4.99%	-4.96%	-6.01%	-7.21%	-6.55%	-7.89%	-8.39%	-13.51%	-11.14%	-10.08%	-9.68%	-11.05%	-14.87%
Max	6.96%	13.06%	15.78%	9.85%	17.21%	20.48%	28.82%	30.08%	16.64%	11.26%	16.26%	23.63%	21.84%

Source: BofA Global Research

BofA GLOBAL RESEARCH

S&P 500 after a death cross

A death cross for the S&P 500 has typically not been a significant bearish signal (Exhibit 39). Price had a slight tendency to be lower 15 days later 28 of 28 times (or 58% of the time). However, by 30 days later price was up 27 of 48 times (or 63% of the time). When the death cross occurred and the 200d SMA was declining (negative slope) then 15-20 days later the S&P 500 tended to be lower (Exhibit 40). It then appears to rebound about 60 trading days after.

Exhibit 39: Price trend after the 50d SMA crossed below the 200d SMA
Not a significant trend signal.

Ticker: SPX			Start Year: 1927		# Signals: 50			# < 80 days: 0					
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Up Ratio	47%	44%	50%	52%	42%	40%	50%	42%	46%	50%	38%	50%	46%
Up	26	28	25	24	29	30	25	29	27	25	31	25	27
Down	23	22	25	26	21	20	25	21	23	25	19	25	23
Average	-0.05%	0.24%	-0.76%	-0.53%	0.43%	0.79%	0.62%	0.85%	1.03%	1.18%	2.25%	2.65%	2.62%
Median	0.34%	0.66%	-0.07%	-0.65%	0.98%	0.55%	0.33%	1.33%	0.36%	-0.08%	2.19%	0.17%	1.06%
Min	-6.34%	-11.50%	-15.48%	-11.83%	-18.19%	-19.56%	-18.05%	-14.86%	-27.73%	-26.97%	-19.99%	-19.59%	-19.59%
Max	8.09%	8.67%	8.95%	23.72%	32.87%	32.16%	37.38%	38.61%	45.93%	51.77%	55.23%	66.57%	55.14%

Source: BofA Global Research

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Exhibit 40: Price trend after the 50d SMA crossed below the 200d SMA and the 200d SMA had a negative slope
A death cross with a declining 200d SMA tended to lead prices lower 15-20 days later and then say a rebound at about 60 days after.

Ticker: SPX			Start Year: 1927		# Signals: 24			# < 80 days:	0				
Days after	5 Day	10 Day	15 Day	20 Day	25 Day	30 Day	35 Day	40 Day	45 Day	50-day	60 Day	70 Day	80 Day
% Down Ratio	55%	50%	63%	67%	53%	47%	60%	47%	50%	50%	33%	50%	40%
Up	13	15	11	10	14	16	12	16	15	15	20	15	18
Down	16	15	19	20	16	14	18	14	15	15	10	15	12
Average	0.03%	-0.03%	-1.70%	-1.77%	-1.07%	-0.27%	-0.89%	-0.25%	-0.14%	0.24%	1.17%	1.39%	1.89%
Median	-0.11%	0.19%	-1.09%	-1.94%	-0.46%	0.25%	-0.57%	1.42%	-0.04%	0.13%	2.19%	0.86%	2.44%
Min	-4.80%	-11.50%	-15.48%	-10.97%	-18.19%	-19.56%	-18.05%	-14.86%	-27.73%	-26.97%	-19.99%	-19.59%	-19.59%
Max	8.09%	8.67%	8.01%	8.63%	8.81%	8.86%	11.49%	14.49%	17.30%	19.44%	16.05%	18.23%	20.85%

Source: BofA Global Research

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Triple Moving Average Cross

- As the name implies, the triple moving average cross (TMAC) system uses three moving averages to define trend and buy/sell signals.
- Typically it is a shorter term set up using the 10, 20 and 30 day moving average.
- TMAC Sum aggregates the 9 trend conditions available in TMAC to easily see trend and create a rule set for trading it.

Triple moving average cross as a trading system

The triple moving average cross trading system, or TMAC, utilizes three moving averages on a price chart. It generates signals when price closes through the moving averages and when the shorter period moving averages cross the longer period moving averages. It can consider changes in average slope, too.

In Exhibit 41 we show multiple sell signals in the middle of October and a down trend that followed. In December price consolidated and flirted with triggering buy signals but never confirmed. It wasn't until January that price rallied enough to trigger TMAC buy signals.

Exhibit 41: Price vs. three moving averages. Green = 10d, Purple = 20d, Red = 30d
Price versus three moving averages with buy and sell signals noted



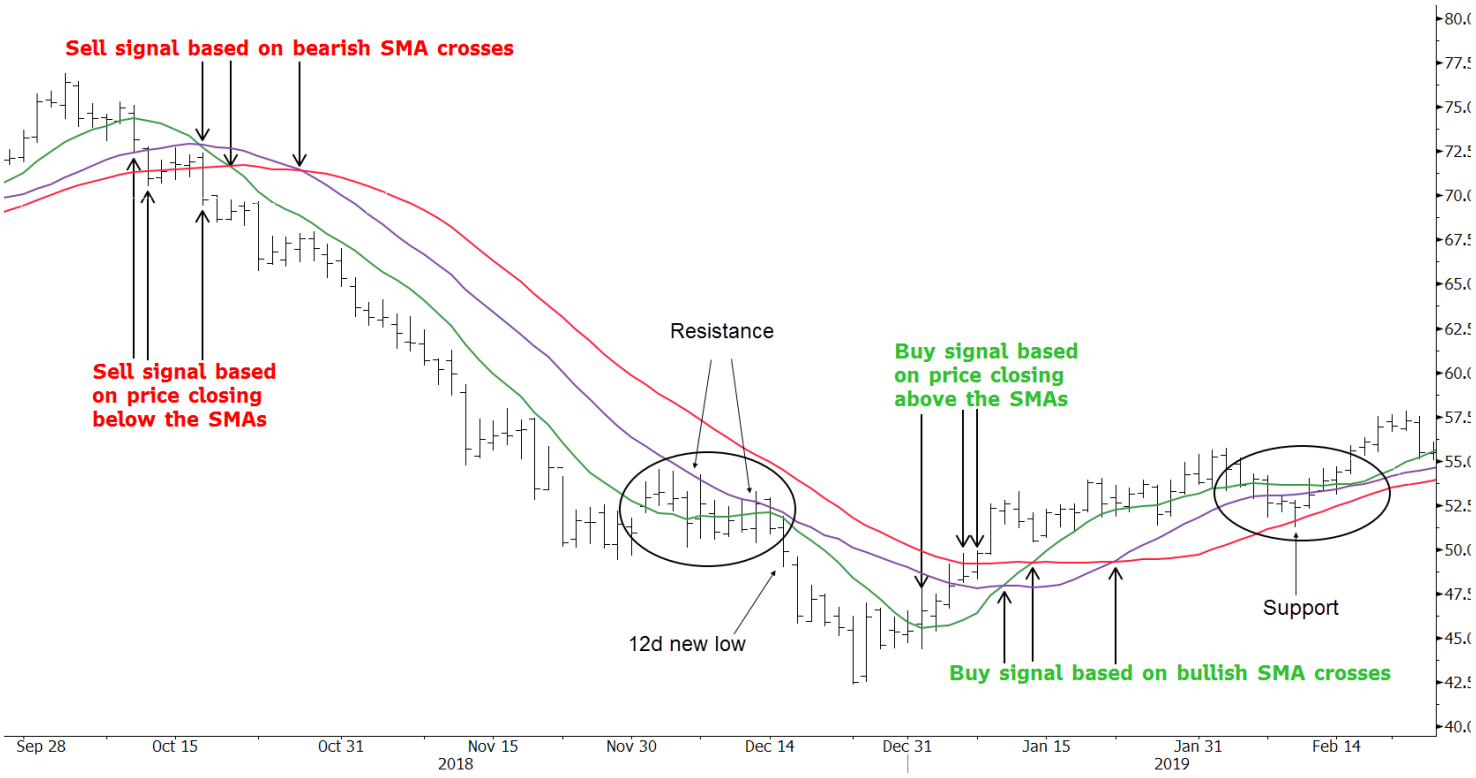
Source: BofA Global Research, Bloomberg

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TMAC down trend observations in 2018

In late November price action began breaking above the shortest period moving average when price closed above the green moving average (left black oval in Exhibit 42). Price never closed above the purple moving average which acted as closing resistance. It also didn't close above the red moving average and the green moving average never crossed above the purple. The slope on the purple and red average remained negative, or bearish, while the green turned sideways. In conclusion, the trend remained mostly bearish according to the failure of price action to break more resistance levels and the moving average conditions. During such a period some may have pared back their shorts in the event the trend changed to lock in gains from the downtrend already realized. It is also possible they began to sell again when price closed at a 12 day new low in mid-December.

Exhibit 42: Triple Moving Average Cross example (TMAC)
Support and resistance from the moving averages



Source: BofA Global Research, Bloomberg

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TMAC up trend observations in 2019

As price was rising during January 2019 it closed above all three moving averages in just five trading days. Such a decisive break above all levels speaks to the change in trend that may be underway and the large capitulate range days at the December low. The moving averages crossed in January to confirm the price breaks. In February, price declined below two of the three moving averages and the green average almost crossed the purple one. Longs may have begun trimming their position due to some of these signals occurring however the red average held on a test and closing basis as support. When price rallied back above the averages and the green avoided a cross below the purple average, the uptrend is thought to have resumed.

TMAC Sum tracks trend strength, possible reversals

Using three moving averages and generally accepted moving average rules means there are nine conditions to evaluate. This includes where price is vs. each moving average (3), where each moving average is vs. the other (3) and the slope of each average (3). The below chart set up makes it simple to see how bullish or bearish the moving average trend conditions are and can be useful for creating a system, for example:

5. Be long when TMAC is greater than zero and be short when it is less than zero.

6. Buy when the TMAC sum = 9 (Max) and add to longs on any down day that does not reduce the sum. Reduce long if TMAC equals 6. Exit if it equals 3. Sell when the TMAC sum = -9 (Min) and add to shorts on any up day that does not reduce the sum. Reduce shorts if TMAC equals -6 and exit if equals 3.

7. Buy 1/3 when TMAC >= 3, another 1/3 when it >= 6 and a final 1/3 when it = 9. Exit part of longs on moves back to 5, 2 and 0. Sell 1/3 when TMAC <= -3, another 1/3 when it <= -6 and a final 1/3 when it = -9. Cover part of shorts on moves back to -5, -2 and 0.

As with any system, there are many considerations…

There are many other considerations however this is an introduction to the methodology behind trend following with moving averages. Other considerations may include filtering signals base on a long-term moving average. For example, be long when TMAC is greater than zero and the 200-day moving average has a positive slope. This may better incorporate the long-term trend with the shorter term signals. The above rules are written in a way to try and capture trends; however some markets trend more than others. For markets that tend to trend less, they may need a rule to exit when a certain gain is reached. And like we said on the front page "plenty of other conditions exist when technical traders, trend followers and systematic firms take action."

Exhibit 43: Triple Moving Average Cross example (TMAC) with sum of signals and paint bars for max condition
Color coded price bars and a histogram showing bullish and bearish conditions accumulated.



Source: BofA Global Research, Bloomberg

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RSI to measure momentum

- RSI is the most popular technical indicator, ahead of MACD and Bollinger Bands, and is used to measure and trade momentum.
- We review the essence of the RSI formula as well as the facts and myths of overbought and oversold trends.
- We also discuss our most preferred signals including a three-troughed bullish divergence and three-peaked bearish divergence.

Measure and trade with momentum

The Relative Strength Index (RSI), which is the most popular technical indicator used on Bloomberg, measures the momentum of a trend by comparing the average size of the up periods to the average size of the down periods over a specified timeframe, such as 14 days. Common thought suggests if momentum reaches an extreme, a reversal, or at least a consolidation, can occur. While this may be the short-term outcome, sometimes the trend resumes until momentum finally depletes as the RSI flashes a bullish or bearish divergence. In this primer, we will review RSI and the signals it can generate.

The essence of the RSI calculation and levels

If price went up $1.00 for 10 of the last 14 days and down $0.25 four of the last 14 days, then we know price went up on average much more than down. However, in financial markets, the up/down relationship is rarely this clear cut. This type of market may be described as overbought and the opposite as oversold. Both mean the market moved too far too fast. RSI is normalized to fall between 0 – 100. So, when it is above 70, the trend is described as overbought. When it is below 30, it is described as oversold.

The facts and myths of overbought and oversold

The terms overbought and oversold mean the market may have risen or fallen too much, especially in the short term. Exiting or reversing a position solely because the market is deemed overbought or oversold may be the right trade for a short period of time but could be the wrong trade for the intermediate-term trend. One way of filtering this is to incorporate support and resistance. In theory, momentum complements a breakout similar to greater-than-average volume. If price and momentum break up through resistance, the expected overbought dip could be buyable. Similarly, if price and momentum break down through support, a bounce may be sold.

Momentum trades vs momentum trends

In a trendless market, also referred to as a sideways or rangebound trend, RSI rarely moves through overbought or oversold readings. When price reaches the high-end of the range and RSI is equal to or less than its prior peaks, price can be sold. If price were to breakout to new highs and RSI is still not overbought, then the breakout lacks confirmation from RSI and may soon fall. However, when price and RSI break out together, a new uptrend is underway, and longs could be held or added to on a dip.

Price vs RSI divergences signal a trend is changing

When prices make lower lows and the RSI indicator makes higher lows, a bullish divergence is forming. Conversely, when price makes higher highs and RSI makes lower highs, a bearish divergence is forming. Think of price as a car and RSI as the foot on the accelerator. If a car is going uphill and the foot is pushing the accelerator, then the car is moving uphill. However, if the pressure on the accelerator starts to decrease, the car will be climbing the hill at a slower speed (with less momentum). Eventually the car makes it to the top of the hill, or the peak of the uptrend, and the foot comes completely off the pedal. If the bearish divergence is correct, all that's left is for the car to go down the hill.

Exhibit 44: RSI is the most popular technical indicator (excluding moving averages) - Example 1
Understanding overbought, oversold and RSI divergences.



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

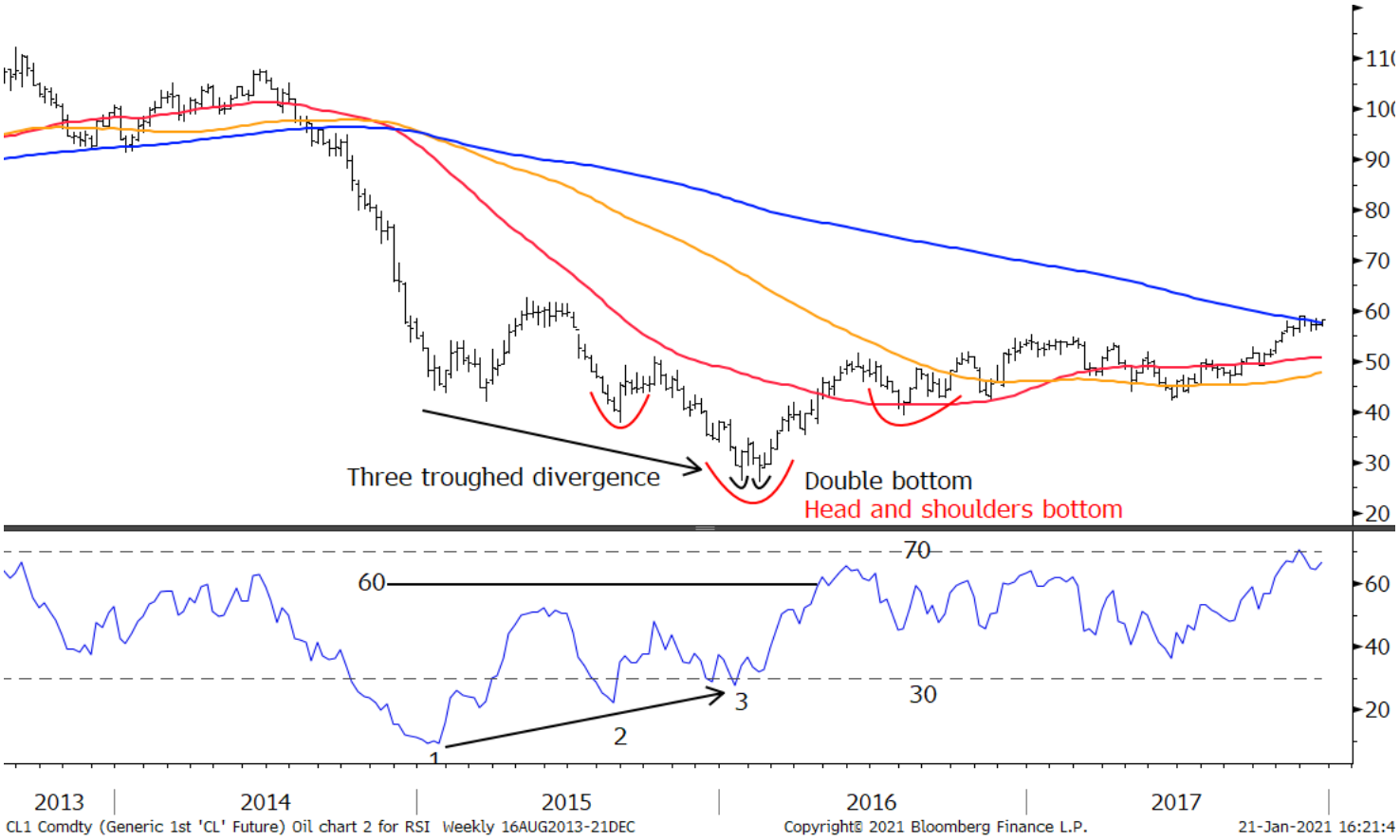
Adjust overbought and oversold levels for trend direction

In the above chart, another important aspect of using RSI is shown. When markets are in a clear downtrend, they should be oversold more often than not. Conversely, markets in uptrends should be overbought most of the time. When a well-defined trend is present, it usually makes sense to adjust the overbought and oversold levels, traditionally 70 and 30, respectively. In uptrends, consider using 40-80 and in downtrends 20-60. If price is in a downtrend and sees a bullish bounce, watch the RSI as price meets resistance. For the downtrend to continue, price should make a lower high relative to its last peak (pivot), which is resistant. If RSI is at or below 60 upon arriving at resistance, the level is more likely to hold and the downtrend may continue. If RSI is breaking above or already above 60 at resistance, there's a higher likelihood buyers may overwhelm sellers, and the downtrend may be nearing an end. The inverse application can be applied to RSI 40 in an uptrend. An adage to help you remember this trend-following application is, "If RSI is above 60, supply (resistance) is a lie. If RSI is below 40, demand (support) will not stand."

Three-troughed divergences are strong reversal signals

When prices undergo three momentous moves lower and the rate at which those moves occur is less severe each time, a strong "bullish divergence" signal is forming. In Chart 2 below, we show the difference between a small divergence, which can still be valid and is tradable, and a longer-term bottoming process. Price levels at points 1, 2 and 3 are lower while the RSI lows are higher. This is a three-troughed bullish divergence. By the third point, a double bottom formed, which produced a large rally. This rally led RSI to break up through the 60-level, confirming the end of the downtrend signaled by the three-troughed divergence. However, RSI has since failed to reach the standard overbought threshold of 70 since early 2011, so price still lacks the requisite momentum for an uptrend. Price has also yet to break trend line resistance. If price were to move above the trend line and RSI reaches 70, then technically a new uptrend may be forming.

Exhibit 45: RSI example 2
A bullish RSI divergence as part of a head and shoulders bottom pattern



Source: BofA Global Research, Bloomberg

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Three-peaked divergences are strong reversal signals

When prices undergo three momentous rallies and the rate at which those moves occur is less overbought each time, a strong "bearish divergence" signal is forming. In Exhibit 46 below, we show a three-peaked divergence in addition to some other signals and topics discussed in this larger report.

In this example, price was rising for much of 2013-2017. However, a bearish divergence began building in late 2015 and into 2016. A TD Sequential '13' sell signal formed at the third peak of the divergence and what turned out to be the head of a head and shoulders top. The neckline and trend lines were later broken, suggesting the trend was down.

Exhibit 46: RSI Example 3 including TD Sequential, Head and Shoulders top and trend lines
A bearish RSI divergence as part of a head and shoulders top.



Source: BofA Global Research, Bloomberg

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Mighty MACD

- No one likes to be late to a trend. Waiting for moving average crosses takes time, and price may be moving away from a high/low by the time a cross occurs.
- MACD provides earlier signals by comparing the spread between two moving averages to an average of the spread.
- Get to know MACD and you may be earlier to trend reversals. Signals occur when the lines cross and when they diverge from the current trend.

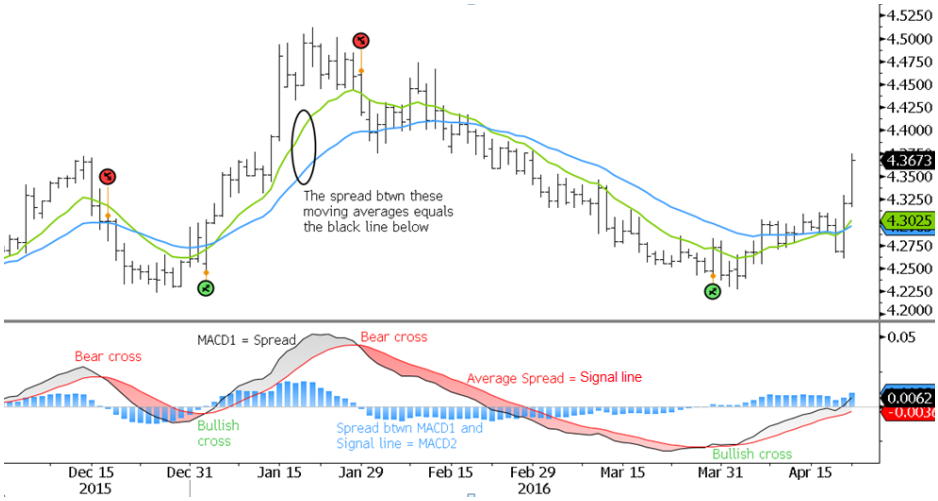
MACD provides earlier signals than moving averages

Waiting for a moving average cross can take a while. Often price moves away from the highs or lows, creating a less opportune time to trade and the feeling of being late to the move. The Moving Average Convergence Divergence indicator (MACD) attempts to provide an earlier signal by tracking the spread between two moving averages (MA).

MACD = Spread of two MA vs an MA of the spread

The MACD line is the spread between two exponential moving averages of price. Typically it is the spread between the 12 and 26 period averages. A second line, called the Signal Line, is the 9 period moving average of the MACD line. Plotted on the same panel below price, the MACD1 line is in black and the Signal line is in red; crossings create reversal signals.

Exhibit 47: Example of two exponential moving averages and MACD signals
The MACD line is the spread between two moving averages of price. The signal line is a moving average of the MACD line.



Source: BofA Global Research, Bloomberg

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Bullish and bearish MACD cross signals

The chart above highlights bullish cross signals with green circles and bearish signals with red circles. It is bullish when the MACD1 line crosses above the Signal line while less than zero. It is bearish when the MACD1 line crosses below the Signal line while greater than zero. Bullish or bearish crosses that occur when positive/negative can complement continuation patterns, such as a flag or pennant pattern. It is also considered bullish when the MACD line crosses above zero and bearish when it crosses below. This is the same as the underlying MACD1 moving averages crossing on the price chart.

Price and MACD divergences warn of a change in trend

The overall trend of price and MACD can lead to trend changing conditions that a MACD cross confirms. When price continues to trend to new highs or lows and MACD fails to do the same, a divergence is occurring.

Bearish divergence can lead to a market decline

Below is an example of a two peaked bearish divergence. Price was in an uptrend guided by two trend lines of short and long duration. A small head and shoulders top pattern developed with two TD Sequential trend exhaustion signals on the head of the pattern. During the formation of the head, price and MACD were bearishly diverging. Price was at a higher high while MACD was at a lower high. Then the MACD1 line crossed below the signal line. The neckline support broke projecting price down to trend line support. After grinding slightly higher between the trend lines, price could not break horizontal resistance, so it broke down below the long term trend line and declined.

Exhibit 48: Example of a bearish divergence
MACD bearish divergence and bearish cross as part of a head and shoulders top pattern.



Source: BofA Global Research, Bloomberg

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A bullish divergence can lead to a rally

Below is an example of a three troughed bullish divergence. Price trended to lower lows three times while the MACD indicator graduated to higher lows. It wasn't until a third low in price and third higher low in MACD that the downtrend turned into an uptrend. The break of resistance in March added confirmation of an uptrend where in 4Q16 it continued to fail at resistance. Later, the MACD line crossed above zero, also indicating an uptrend.

Two vs three trough divergences

Going long in September or October 2015 near the second higher low or bullish cross would have been a reasonable attempt to take advantage of a two troughed divergence. However price action after this point did little to confirm an uptrend would begin, such as breakout higher through resistance. Due to the lack of confirmation, new longs may have chosen to exit. The decision to go long again would be considered when the third low occurred in January 2016.

Exhibit 49: Example of a MACD bullish divergence
Bullish MACD divergence



Source: BofA Global Research, Bloomberg

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Timing a MACD cross with the MACD histogram

The prior chart also had a signal for a counter trend rally according to the MACD histogram. The MACD histogram (dark blue) measures the distance between the MACD and Signal line. When prices are declining to new lows and the MACD histogram is making higher lows, then a bullish MACD cross is approaching. This can be seen in the MACD histogram trend from September 2014 through March 2015. The MACD lines eventually crossed bullish and a rally followed. However, in this example, the rally ended with the market selling into the highs of three consecutive trading days and no break of resistance. So in this example, the MACD histogram divergence signaled a short term trend reversals where the larger trend may or may not be ending. A three troughed divergence was the final end to the downtrend.

Exhibit 50: Example of MACD histogram signal
Bullish MACD divergence, candle patterns and resistance



Source: BofA Global Research, Bloomberg

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Example top, reversal and continuation

An uptrend was under way into the end of December 2016. A bearish price vs MACD divergence began to form and price was failing to continue its rally higher. During the first week of January, the MACD line crossed below the Signal line (Bearish cross). One more intraday rally occurred with a new high that was faded by the market (the third high of the triple top). This formed a small triple top as price plunged lower two days in a row. The MACD line also declined to a new low adding confirmation to the top and decline. A downtrend followed until a bullish MACD histogram divergence warned of a bullish MACD cross. The MACD lines did not diverge; however, a cross occurred and a rally followed. Then another bearish MACD histogram divergence formed, warning of a bearish MACD cross. The MACD line then crossed below the signal line suggesting a deeper decline to follow. Price closed at short term support and traders may use a break of that support as addition confirmation of a decline.

Exhibit 51: Bullish and bearish MACD crosses, divergences and histogram divergences
An uptrend that topped with a bearish MACD divergence, cross and then turned up again with a MACD histogram signal



Source: BofA Global Research, Bloomberg

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Bollinger Bands

- The Bollinger Band (BB) indicator is the third most popular indicator after the Moving Average Convergence Divergence (MACD) and Relative Strength Index (RSI).
- It plots a moving average and two standard deviation bands on the price chart and two oscillators below the price chart called bandwidth and %B.
- While price tends to stay within the standard deviation bands, excessively narrow or wide bands can generate trade signals. Bandwidth and %B support trade ideas.

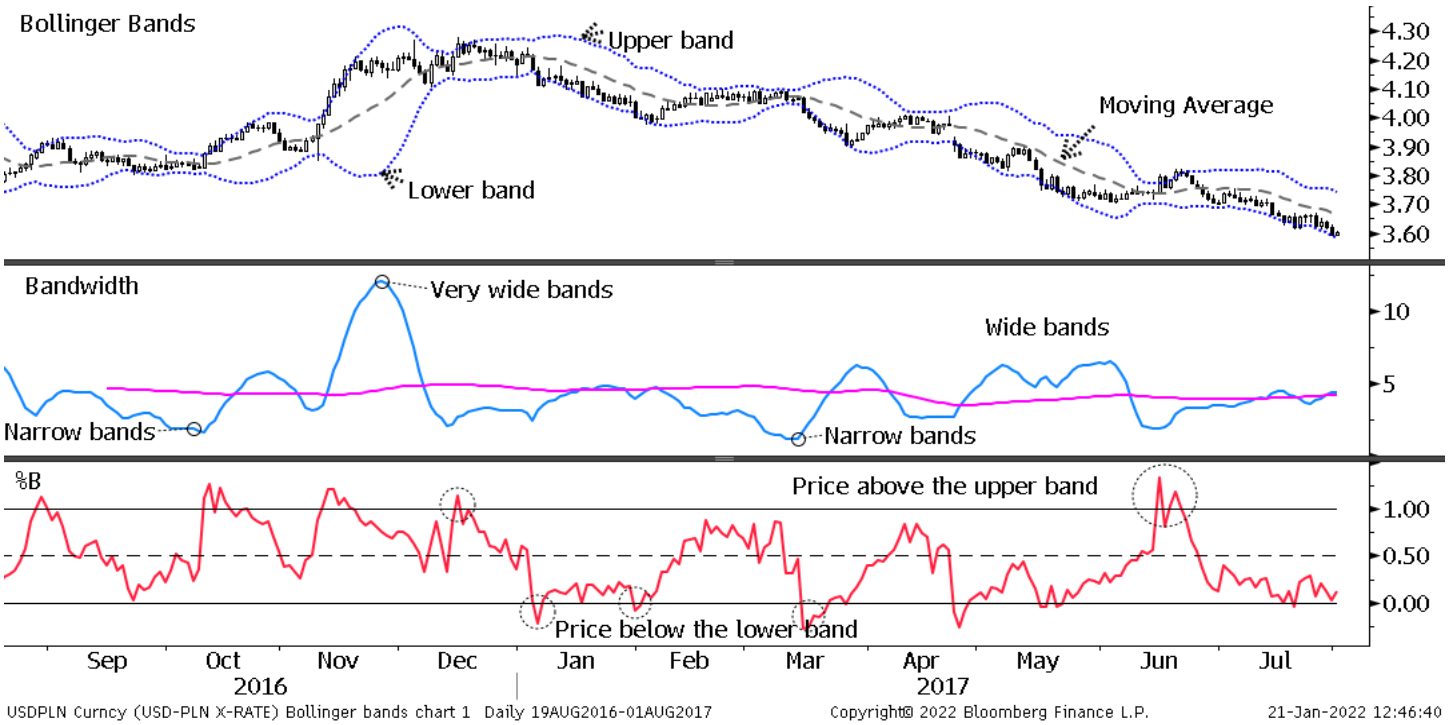
The Bands are +/- 2 std dev from the 20-period average

The Bollinger bands are usually calculated with a 20-period moving average. The higher band shows the current value of two standard deviations above average. The lower band is two standard deviations below the average. The middle line is the average. The bandwidth oscillator measures the difference between the standard deviation bands. In Exhibit 52, we have also calculated a moving average on the bandwidth to define a narrow and wide band bias. The %B oscillator measures where price is relative to the bands. Price is above the upper band when %B is greater than 1, price is between the upper band and average when between .50-1, it is between the average and lower band when between .50-0, and below the lower band when below 0.

BBs are a mean reversion and trend breakout indicator

The BB indicator signals for mean reversion trades and breakout trades. The bands often serve as an estimate of support and resistance levels. When markets are range bound prices tend to trade between the bands. When the bands are narrow relative to the past and prices break out through a band then a new trend and trade in the direction of the breakout is signaled. Price trends with widening bands until the bands become too wide. Trades can be exited when bands are excessively wide, when price comes back inside the bands or when price reaches the moving average band.

Exhibit 52: Bollinger Bands example 1
The Bollinger bands are comprised of a moving average, a +2 std dev line, a -2 std dev line, a bandwidth line and % bandwidth line.



Source: BofA Global Research, Bloomberg

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Trading with Bollinger bands

The chart below shows an example of four bullish breaks and three bearish breaks. By following the generally accepted rules, five of them were winning trades and two were losing trades. We connect some examples with the candle patterns discussed previously.

Bullish breaks 1 and 2

At "Bullish break 1" the bandwidth was considered narrow as it was below the bandwidth average. Price closed above the upper band and trended higher for about nine trading days. Then it started moving sideways as the bandwidth widened. Two doji candles formed suggesting the prior trend might be ending. Price began to revert to the mean. The bandwidth narrowed again and another bullish breakout occurred. A larger rally followed, and the bands became very wide, which is often unsustainable. Price action began to move sideways and eventually reached the moving average.

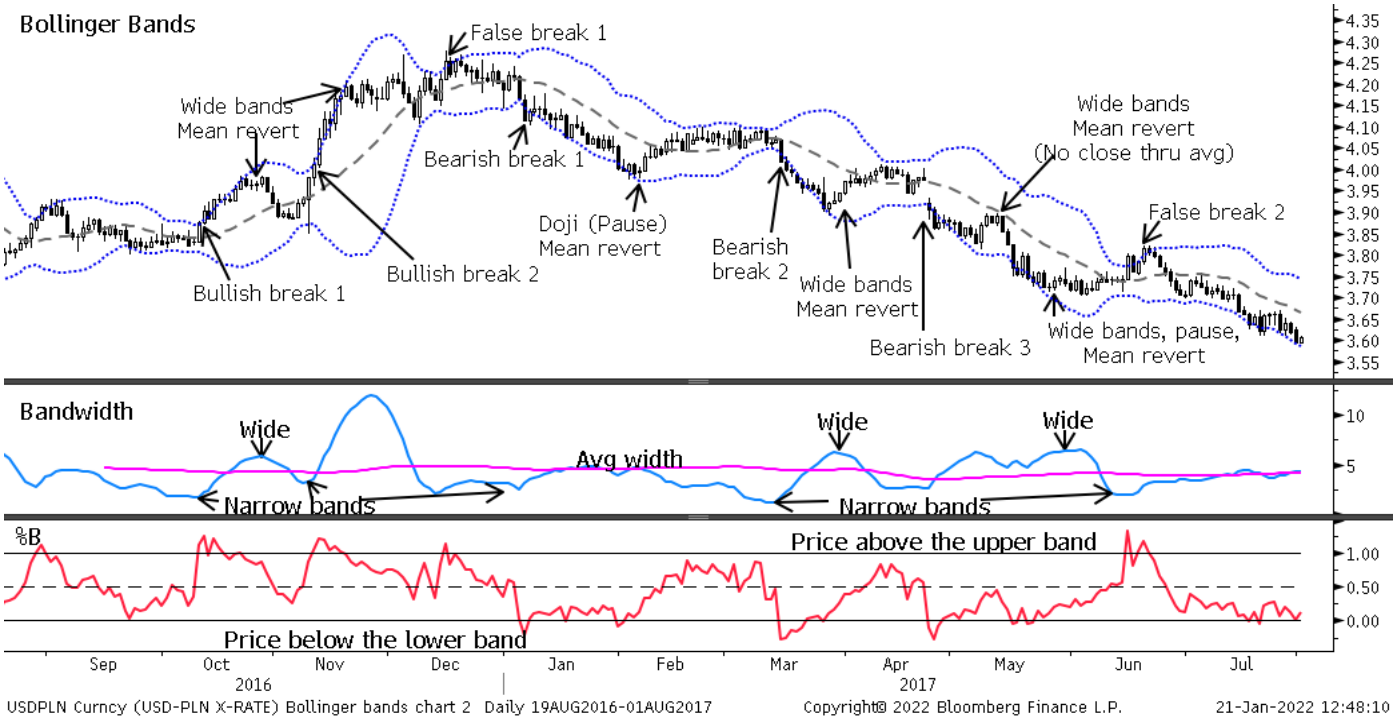
Bearish break 1, 2 and 3

In January 2017, bearish break down 1 occurred with a narrow bandwidth. This trade worked as price "rode the band" lower for about a month until the bandwidth reached average width and a doji candle formed. Price then reverted to the mean. The downtrend continued at "Bearish break 2" when price closed below the lower band with the bandwidth at its narrowest point all year. Prices declined and the bandwidth widened to its highest level since November 2016. Price then reverted to the mean again. This example shows that exiting a trade at the mean often results in a late exit. Bearish break 3 worked particularly well because price never moved through the mean when it bounced in May. A rules-based technical trader or system may not have exited until early June when price reached the average, however this is subjective.

False break 1 and 2

In December 2016, a close up through the upper band with a narrow band reading occurred. This was a false break as price quickly reverted back to its mean. Similarly, in June 2017, price closed above the upper Bollinger band twice while they were narrow. A short-lived rally occurred, and then it quickly reverted to its mean.

Exhibit 53: Bollinger Bands example 2
Bullish and bearish Bollinger band signals with complementary Bandwidth and %B indicators



Source: BofA Global Research, Bloomberg

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Stochastics

- The stochastics indicator is an oscillator that measures the momentum of a trend similar to the Relative Strength Index (RSI).
- In comparison to RSI, the stochastic indicator has multiple lines where the momentum measure is represented by %K and the rest are moving averages.
- The indicator measures if price is closing closer to the highs or the lows over time. When one direction persists, the trend becomes overbought or oversold.

Stochastics explained

The Stochastics indicator is a momentum study similar to the Relative Strength Index (RSI). It measures whether price is consistently closing each trading session nearer to the highs or lows of the session. Depending on the speed at which one prefers to trade, there are two stochastic oscillators, a fast and a slow. The %K line is considered the stochastic indicator line while the %D line is a moving average of the %K.

Bullish and bearish stochastic signals

The indicator rises and falls on a scale between 0 and 100 where overbought is typically set at 75 and oversold at 25. When the stochastic %K and %D lines are in overbought or oversold territory, it means the prior trend is getting stretched and becoming more vulnerable to a consolidation or reversal whether it be tactical or medium term. When the %K line crosses the %D line when stretched a reversal signal occurs. When these lines exit stretched territory the tactical trend change is underway as price and momentum revert.

Another way to view the stochastic lines is to seek divergences from the price trend. When prices are rising and the stochastic is making higher highs, then the trend is strong. When the stochastic starts to make lower highs while the price trend is still making higher highs then a bearish divergence has formed and price might peak soon. One example of a bearish divergence can be seen in July-August 2020 in the chart below.

Exhibit 54: Stochastics on the US long bond treasury future in 2020 (G 1493)
An example of a bearish divergence between rising prices and lower overbought highs in slow stochastics.



Source: BofA Global Research, Bloomberg

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DMI + ADX

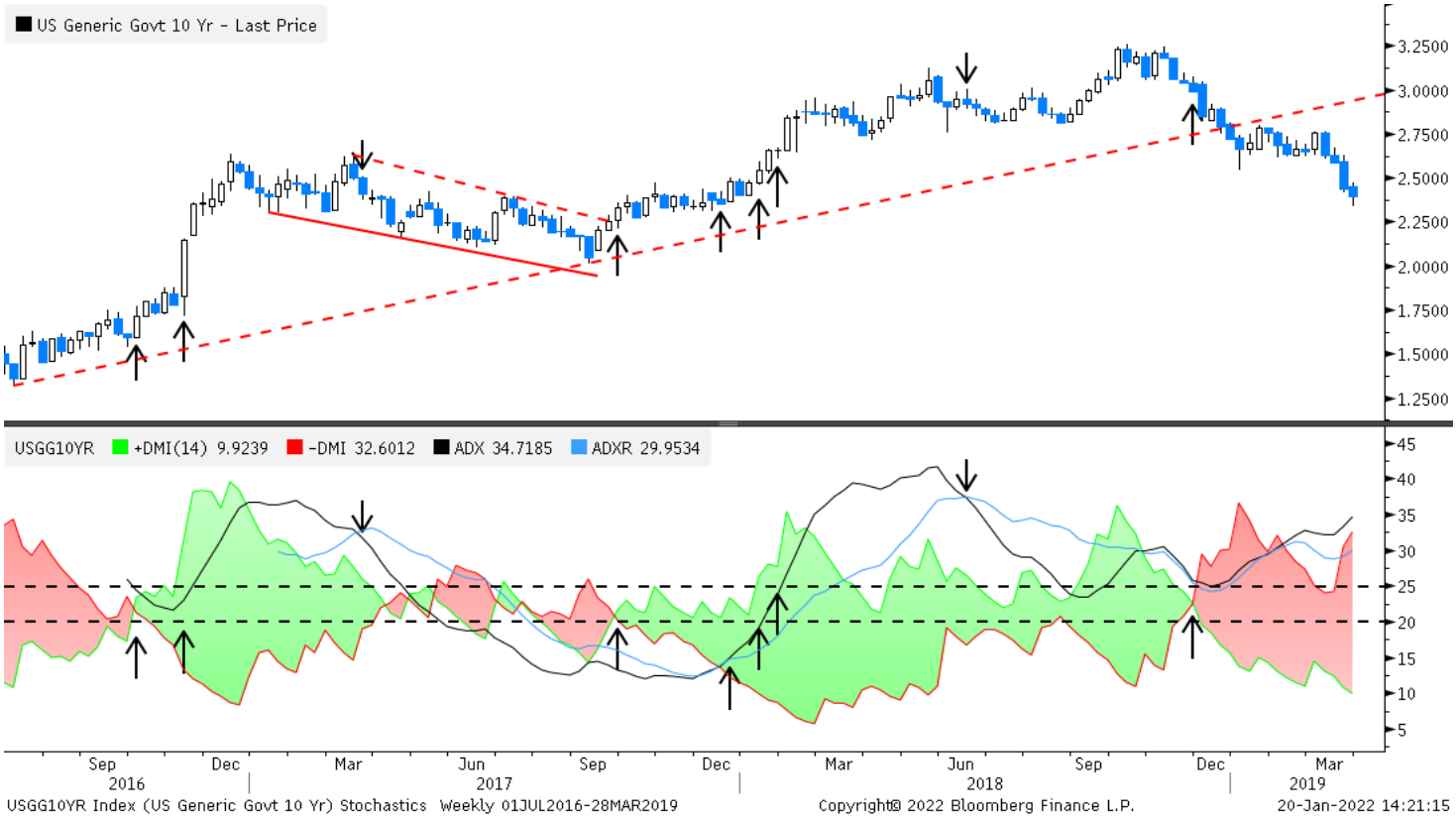
- The Directional Movement Indicator (DMI) and Average Directional Movement indicator (ADX) define a trend as up, down or range bound.
- The DMI has two lines: The +DMI line tracks higher highs and a –DMI tracks lower highs. When the +DMI is above the –DMI, the time series is rising.
- The ADX line measures trend strength, or lack thereof. When the ADX is above 25, price is described as trending, below 20 range bound, and between is transition.

DMI and ADX Explained

The DMI indicator is comprised of two lines, a +DMI (Green) and –DMI (Red). The green line rises when prices are making higher highs, and the red line rises when prices are making lower lows. When the green crosses above the red, it suggests the time series is rising (October 2016, Sept 2017), and when the red crosses above the green, it suggests the time series is falling (December 2018).

These lines are often used in conjunction with the ADX and ADXR lines. The ADX is a combination of the DMI lines and is used to define the path of prices as range bound or trending. When the ADX is below 20, it is said to be in a range (2H17). When the ADX rises above 20, price may be transitioning to a trend. If the +DMI is above the –DMI then the ADX crossing above 20 would suggest a transition toward an uptrend (January 2018). When it rises above 25 then price is trending (January 2018). The ADXR is a moving average of the ADX. An early indication of a trend starting is when the ADX is below 20 and crosses above the ADXR (December 2017). An indication of a trend ending is when the ADX is higher, such as 40, and crosses below the ADXR (March 2017, June 2018).

Exhibit 55: Example of the DMI and ADX on crude oil in 2017-2018 (G 1494)
DMI and ADX signals on US 10Y yield from 2016-2019



Source: BofA Global Research, Bloomberg

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Ichimoku Analysis

- Ichimoku analysis defines trend direction, support, resistance and offers many trend-changing signals.
- The cloud is the most referenced part. Signals occur when price breaks through the cloud and when the cloud lines cross.
- Additional signals are derived when the conversion line crosses the base line and based on the position of the lagging line.

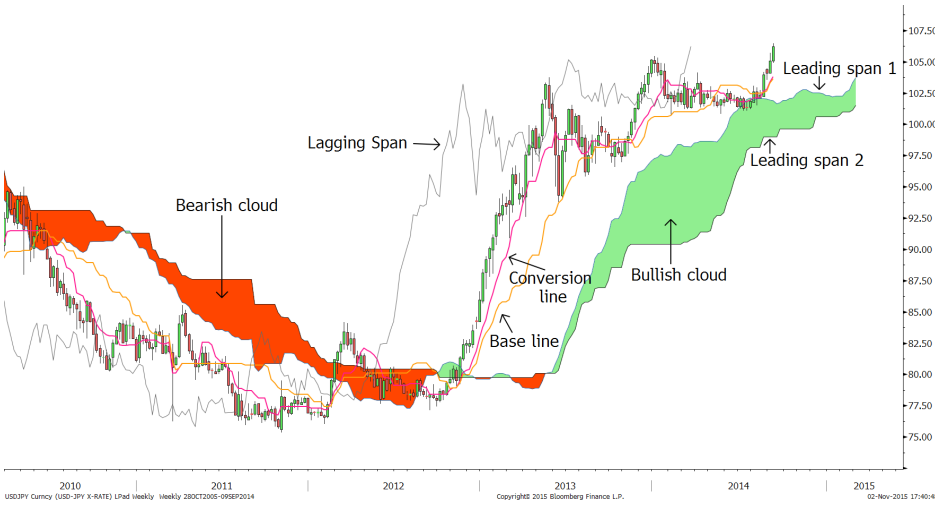
Ichimoku defines trend direction, support and resistance

This indicator consists of five lines overlaid on the price chart that define the trend as up, down or range-bound, and provide estimated levels of support and resistance. It is often combined with candle pattern signals and a momentum oscillator, such as RSI. The more technically savvy may apply Heikin-ashi candles with Ichimoku.

The five lines: two sets of two and a lagging one

The five lines that comprise the Ichimoku indicator include the conversion line, base line, leading span one, leading span two, and the lagging line. The conversion line and base line are used together. Leading span one and leading span two are used together and create the "cloud." The lagging line is often compared to price and the cloud.

Exhibit 56: USD/JPY - Weekly Ichimoku
The five lines that comprise the Ichimoku cloud indicator



Source: BofA Global Research, Bloomberg

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Ichimoku is a series of midpoint calculations

The conversion line is a rolling midpoint of the last nine days. The base line is a rolling midpoint of the last 26 days. Leading span one is the average of the conversion line and base line. Leading span two is the midpoint of the high and low over a look-back of twice the base line – in this case, 52 days. The leading span lines are shifted forward in the chart by the look-back period minus one (25) and form the cloud. The lagging line is shifted back in time by the same amount the leading span lines are shifted forward.

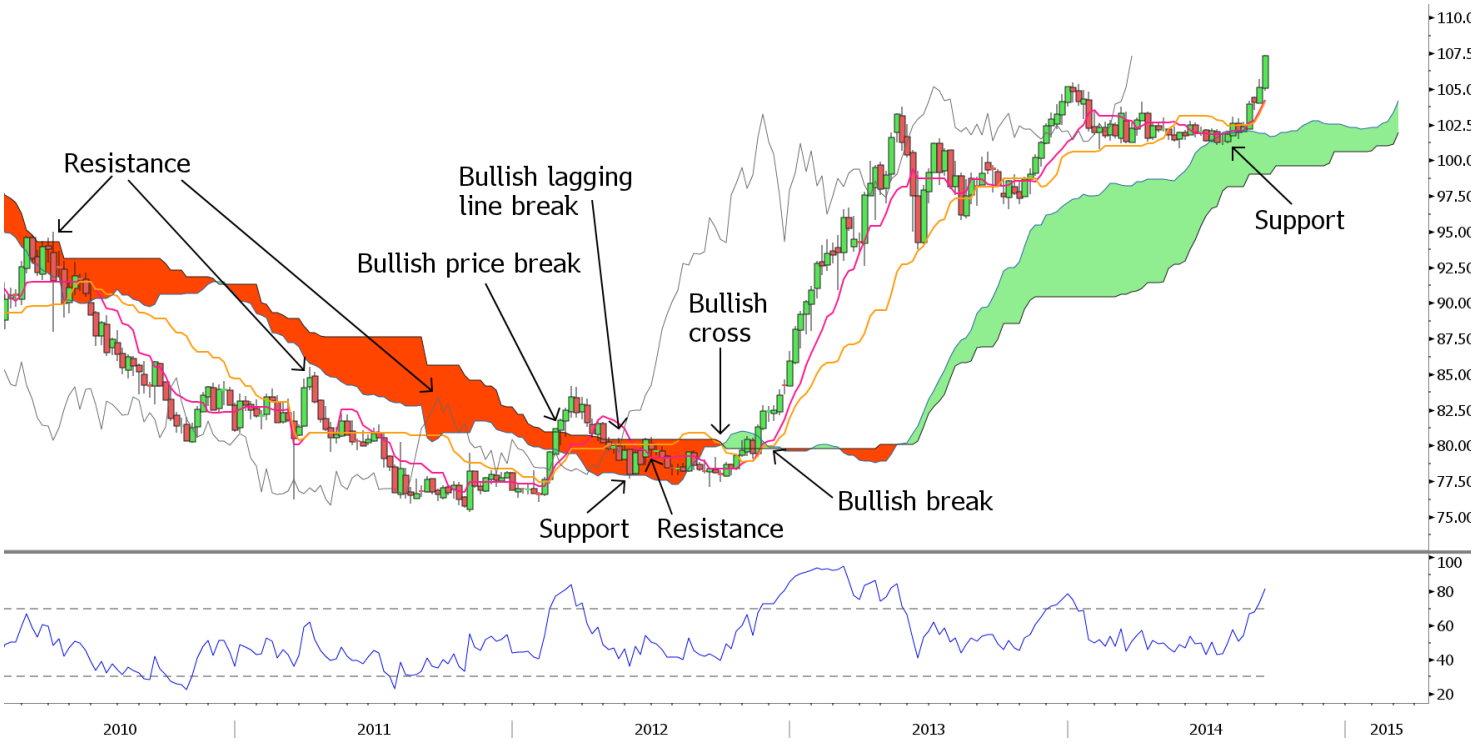
Bullish signals from Ichimoku

A variety of signals can be derived when using Ichimoku. Many market participants focus on price vs. the cloud, the cloud crosses and the thickness of the cloud. A summary of bullish signals are as follows. The opposite would be considered bearish signals.

Ichimoku cloud signals

- The cloud, which is the area between leading span one and leading span two, provides support in uptrends and resistance in downtrends.
- The cloud is shifted forward on the chart to estimate future levels of support and resistance and their strength. The thicker the cloud, the more support or resistance is present.
- When leading span one crosses above leading span two, it is bullish.
- If price enters the cloud, the trend is viewed as range-bound or trendless. But when price exits up through the cloud, it suggests the new trend is higher.

Exhibit 57: Examples of signals derived from Ichimoku
Examples of bullish and bearish signals and of support and resistance from the Ichimoku cloud



Source: BofA Global Research, Bloomberg

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The conversion line and base line signals

- Used for short-term trend signals, just like moving averages.
- It is bullish when the conversion line crosses above the base line while *above* leading span one and two. It is bearish when the conversion line crosses below the base line while *below* leading span one and two.
- When these lines exit the cloud, it confirms a new trend.

The lagging line signals

- It is bullish when the lagging line is above the price compared at that point.
- It is bullish when the lagging line crosses up through the cloud. The cloud also can provide support or resistance to the lagging line.

Williams %R

- Williams %R (percent range) is a price momentum indicator the moves between 0 and -100. A reading above -20 is overbought and a reading below -80 is oversold.
- It measures the relationship between the most recent closing price and the range between the high and low over a specific period. This period is most often 14 days, but we have preferred to use a 28-day period.
- Although Williams %R is considered a momentum indicator, it also can provide guidance for price trends on persistent oversold and overbought readings, especially when using longer timeframes.

Interpreting Williams %R

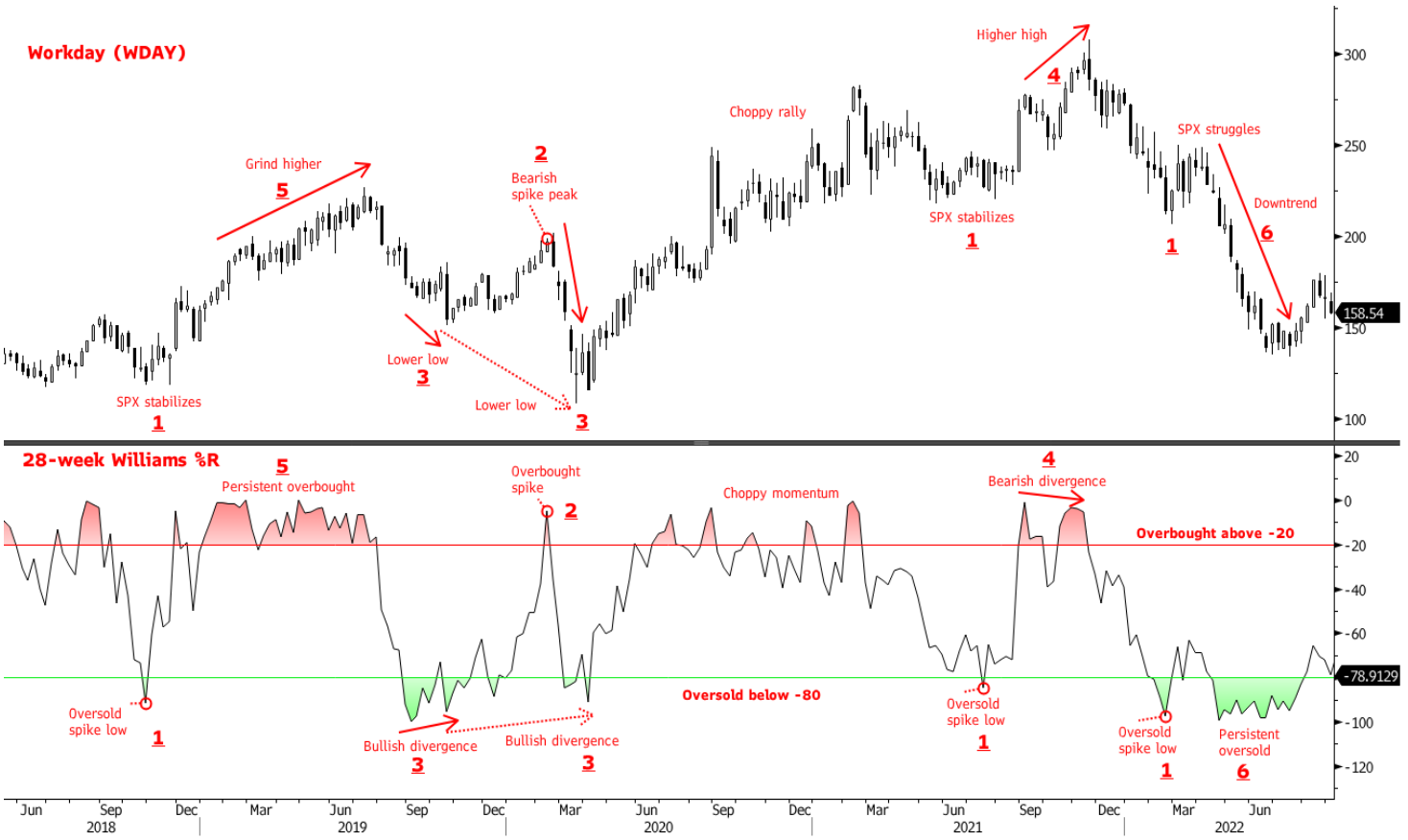
Williams %R is a price momentum indicator the moves between 0 and -100. A reading above -20 is overbought and a reading below -80 is oversold. It measures the relationship between the most recent closing price and the range between the high and low over a specific period. This period is most often 14 days, but we have preferred to use a 28-day period. Although Williams %R is considered a momentum indicator, it also can provide guidance for price trends.

Using signals from Williams %R

1. **Oversold spike**: A quick downward spike into oversold territory can help confirm lows and/or stability for price after a decline. Action: Look to buy or cover shorts on the move out of oversold.
2. **Overbought spike**: A quick upward spike into overbought territory can help confirm a peak and/or a stalling out in price after a rally. Action: Look to short or sell out of a long on a move out of overbought.
3. **Bullish divergence**: Williams %R shows improving price momentum, forming a higher low when price declines to a lower low. Action: Look to buy or cover shorts on the move up from the higher low, especially on a move out of oversold.
4. **Bearish divergence**: Williams %R shows deteriorating price momentum, forming a lower high when price rallies to a higher high. Action: Look to short or sell out of a long, especially on a move out of overbought.
5. **Persistent overbought**: Williams %R stays at or near overbought for an extended period, especially on the weekly and monthly charts, triggering a grind higher with an uptrend in price. Action: Don't fight trend. Stay long, avoid shorts if overbought.
6. **Persistent oversold**: Williams %R stays at or near oversold for an extended period, especially on the weekly and monthly charts, triggering a grind lower within a downtrend in price. Action: Don't fight trend. Stay short, avoid longs if oversold.

Example of price trend and Williams %R rules

Exhibit 58: S&P 500 (top) and the 28-week Williams %R (bottom): Weekly chart
Williams %R is a price momentum indicator the moves between 0 and -100. A reading above -20 is overbought and a reading below -80 is oversold.



Source: BofA Global Research, Bloomberg

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Average True Range (ATR)

- The range of a security is often considered the high minus the low. However, this fails to capture shifts in sentiment and volatility when markets are closed.
- In addition to the high minus low, True Range (TR) takes the absolute difference of the current session high and prior day close as well as the low and prior day close.
- The largest of the three values is considered the True Range. A simple or exponential moving average is taken to define the Average True Range (ATR).
- Historically high or diverging ATR trend (from price) may indicate a turning point.

Going beyond the range to find the truth

The range of a security during a daily or weekly trading session is considered the spread between the high and the low. But sometimes markets gap higher or lower at the open of trading the next day relative to the prior day close. True range captures this closed-market shift in sentiment by taking the largest value of the three measures below. It also represents a more complete measure of market volatility.

1. High minus low of current session.
2. Absolute high of current session minus yesterday's close.
3. Absolute low of current session minus yesterday's close.

ATR spikes or divergences can signal a turn in prices

The chart below shows the weekly prices of the front month continuous brent oil future in the top panel and the Average True Range with two standard deviation bands in the bottom panel (Exhibit 59). Five signal examples are noted and viewed as follows.

Signal 1: Bullish divergence. A gap down in prices led to the ATR rising above the upper band. Seven trading weeks after the gap a bullish divergence formed as the price of oil made new intraday lows and the ATR a lower high.

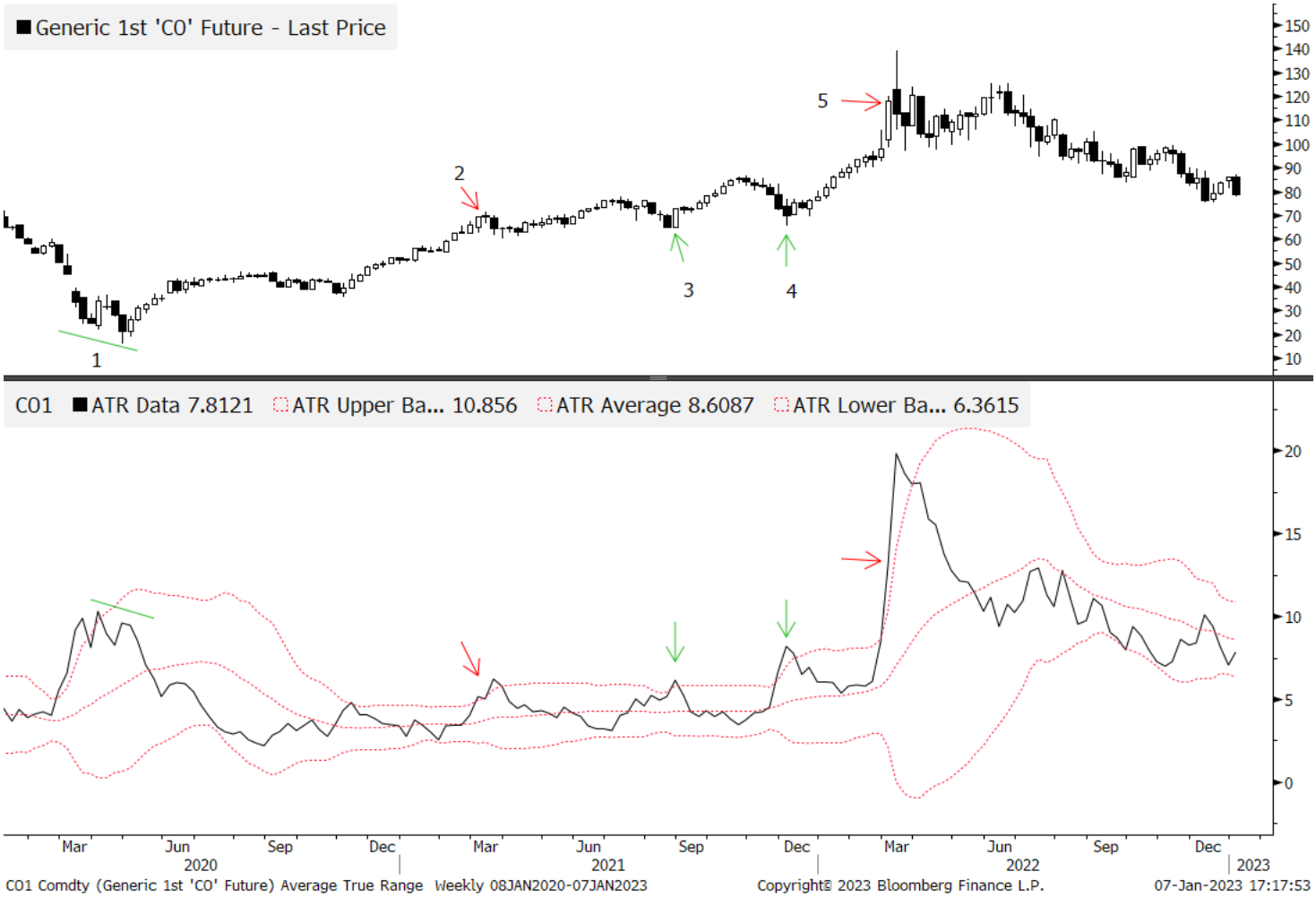
Signal 2: Oil prices were in an uptrend and the ATR broke above the upper band line. The next day a doji candle formed, suggesting possible trend exhaustion (as discussed in the "Candles light the way" section of this primer). Price dipped for two weeks and didn't make a new high for twelve weeks. The uptrend peaked and turned sideways.

Signal 3: Oil prices were correcting for a six-week period and the ATR line spiked higher above the upper band when prices had a big up week. This type of signal indicated the decline was over as the correction lower was turning up and volatility was expanding.

Signal 4: Like signal 3, a spike in the ATR line above the upper band implied the six-week correction may be ending.

Signal 5: The ATR broke out the week prior to the blow-off spike in oil prices in March 2022. The combination of an excessively high ATR and spike in prices favored a peak.

Exhibit 59: Weekly chart of Brent oil future front month continuous contract
Five examples of signal analysis from the Average True Range indicator with standard deviation bands



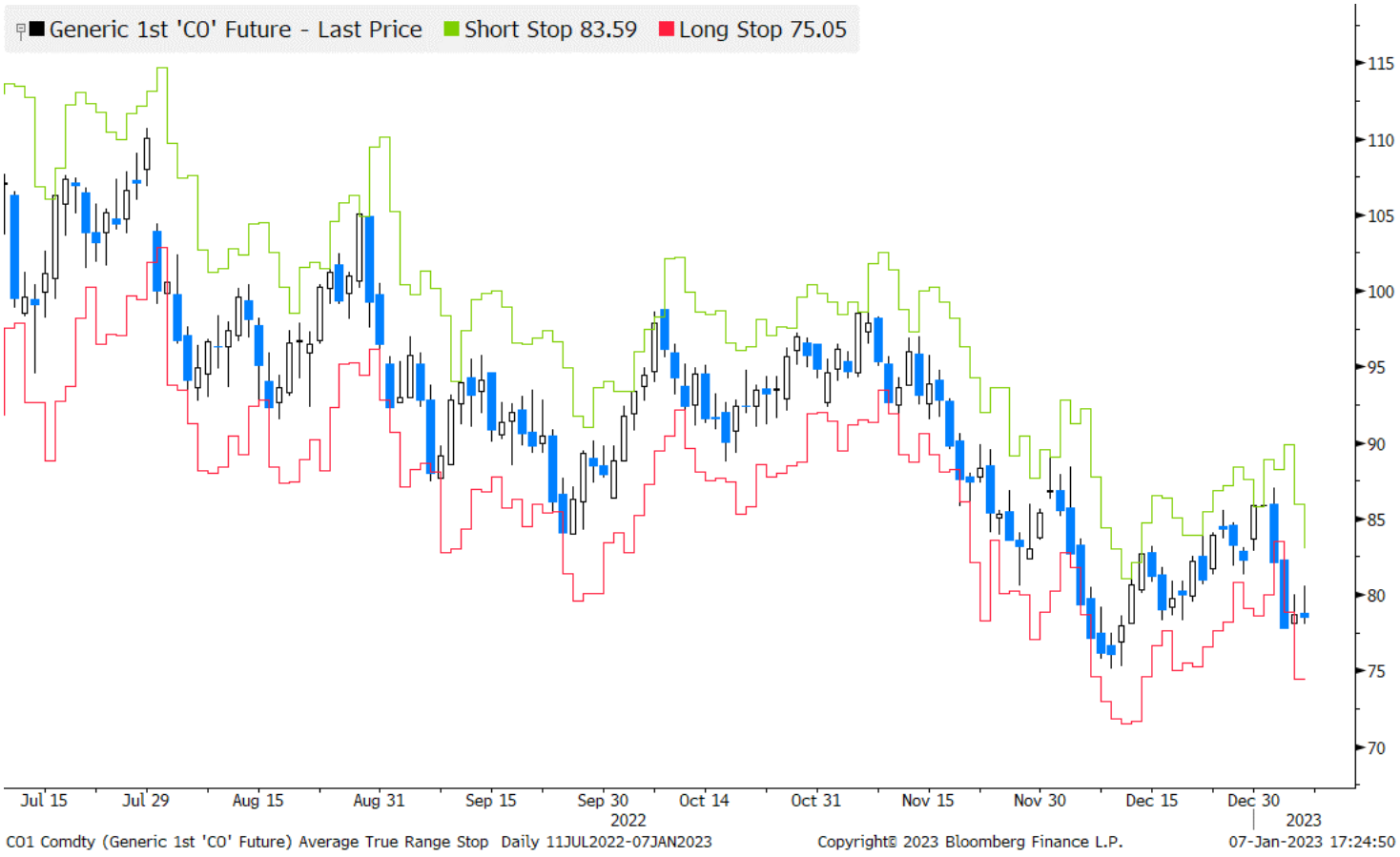
Source: BofA Global Research, Bloomberg (G1644)

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Use ATR values as a trailing stop when trading

Since the ATR is a measure of volatility as well as sentiment, the value can also be used to approximate risk threshold. In the chart below we add the ATR value to the prior day's high to plot a short stop for each new day. This line is in green and above price. A market participant with a bearish view might consider entering short when prices near the green line and stopping if it broke through it. We also subtract the ATR value from the prior day's low to plot a stop line for longs. A market participant with a bullish view might consider entering long when prices near the red line, which is the long stop line.

Exhibit 60: Brent oil future front month continuous – Daily chart (G1645)
Basic example of ATR trend-following lines, which can be used as stops for trades or entry points for bigger picture trend views. Breaking above or below a line would likely stop short-term traders. Longer-term market participants might use the reaching of a line as a place to add to a core position or trim a core position.



Source: BofA Global Research, Bloomberg

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TD Sequential: Green 9s and red 13s

- TD Sequential® is comprised of two parts, the TD Setup® and TD Countdown®.
- TD Setup signals short term trend exhaustion with a green 9 on the chart. It points to a consolidation or correction for one to four periods.
- TD Countdown signals indicate trend exhaustion with a red 13 on the chart. It warns of a trend change within the next 12 periods.

TD Sequential indicator signals trend exhaustion points

The primary goal of the TD Sequential indicator is to identify a short-term and long-term trend exhaustion points. It is comprised of two parts: TD Setup and TD Countdown. For short-term trend exhaustion signals, it will paint a green 9 on the chart. For long-term trend exhaustion signals, it will paint a red 13. When it prints a 9 or 13 above the price bars, it implies the prior up-move may be ending and turning sideways or down. When it prints a 9 or 13 below the price bars, it implies the prior down-move may be ending and turning sideways or up such as that shown in Exhibit 61.

The green 9 pattern is the first part = TD Setup

The TD Setup is the proper name for a green 9 signal. It is the first part of the TD Sequential indicator. It is finalized on the chart only when nine consecutive price bars have a close greater than or less than the close four bars prior. Nine consecutive closes above the close four bars prior is a sell 9 signal. Nine consecutive closes below the close four bars prior is a buy signal, like in Exhibit 61. The ninth bar tends to be a point where the prior trend pauses or reverses for up to four price bars

The red 13 pattern is the second part = TD Countdown

The TD Countdown is the proper name for the red 13. It is the second part of the TD Sequential pattern and can only occur after a TD Setup completes. In a downtrend like in Chart 1, it seeks 12 bars that have a close lower than the low two bars prior. The pattern does not have to occur consecutively like the TD Setup, but it could. The 13th bar must have a low below the low of the red 8 in the countdown. A red 13 suggests the prior is exhausted and may turn. It is recommended to find confirmation of trend exhaustion over the next 12 bars alongside other indicators. The chart below is an example of a bullish TD Sequential pattern where the TD setup 9 signaled a bounce and the TD countdown 13 signaled a timely trend change to up.

Exhibit 61: Example bullish TD Sequential signals
TD Setup 9 buy signal and TD Countdown 13 buy signal comprise the TD Sequential indicator.



Source: BofA Global Research, Bloomberg, DeMark Analytics

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Horizontal lines

When a TD Setup completes, a red dotted line is drawn from the highest true high or a green dotted line from the lowest true low of the setup bars. This price level represents the most the trend can reverse to void the following TD Countdown from occurring. If price breaks through that line satisfying some specific rules then the line turns solid and the trend has reversed. These lines are called TDST Lines®.

A dotted magenta line is drawn on the chart when a TD sequential countdown completes. This estimates risk, or how much price may continue in the current direction. It is viewed as support in a downtrend and resistance in an uptrend. If price breaks through this level and satisfies some specific rules, then the line turns solid suggesting the prior trend is not exhausted after all. It is referred to as the TD Risk Level®. This level is calculated by taking the largest true range from the countdown and adding it to or subtracting it from the highest high or lowest low of the countdown.

Multiple setups and countdowns

Multiple TD Sequential counts can occur simultaneously. When a green 9 occurs on or near a red 13, it is thought to be a stronger signal. Also, when a 9 or a 13 occur on the daily and weekly chart it is thought to be a stronger signal.

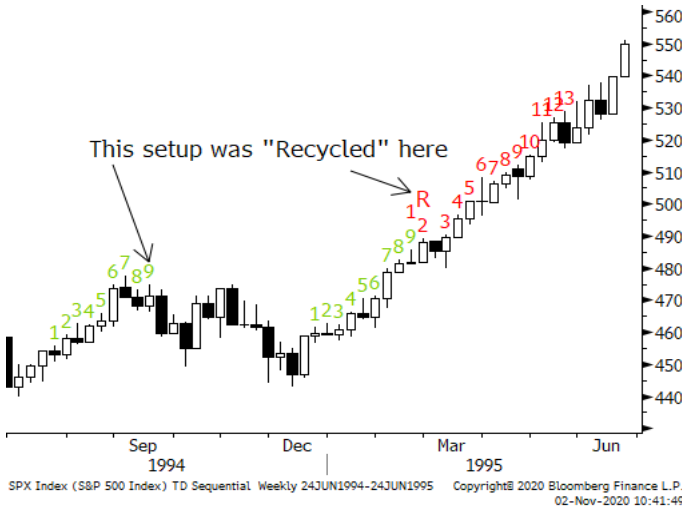
Setup perfection arrows, recycling and plus signs

There are three other good-to-know outputs to TD Sequential. Green and red arrows apply to the setup. They suggest the setup is perfected or more complete with an arrow than without. A perfected sell setup occurs with a green 9 and when the high of bar eight or nine is greater than or equal to the high of bar six and seven. If it is not perfected, the arrow will appear later when a high exceeds bar six and seven. In Exhibit 61 the perfected buy setup arrow occurred nine days after the 9 buy signal.

An R symbol may occur during the countdown phase. R stands for recycle and it suggests the momentum of the trend as measured by an extended TD setup of 22 bars is too much to continue the current countdown. So it places an R on the chart and starts the countdown from the most recently completed TD Setup. (Exhibit 62)

A plus sign will show on the chart after a red 12 and before a red 13. It delays a 13 signal to allow for the 13 to have a high (or low) greater than or less than the close of the red 8 in the countdown. (Exhibit 63)

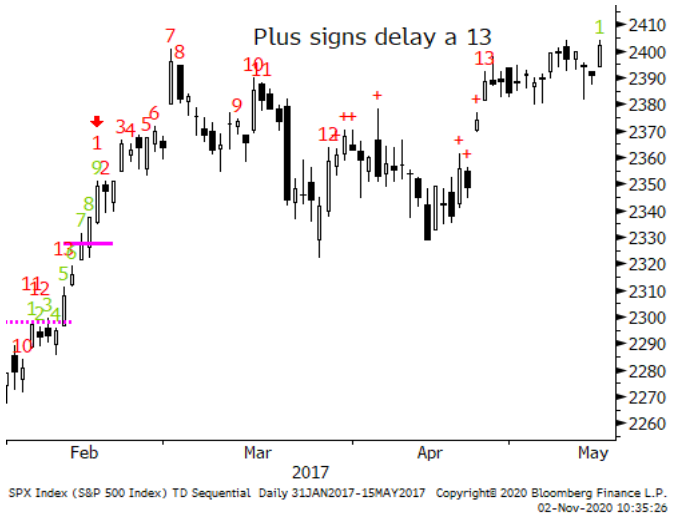
Exhibit 62: Recycle example
When a countdown recycles due an "R" shows on the Exhibit.



Source: BofA Global Research, Bloomberg, DeMark Analytics

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Exhibit 63: Plus sign example
A + sign on the Exhibit means an additional rule needs to be true for a 13.



Source: BofA Global Research, Bloomberg, DeMark Analytics

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Bloomberg Indicators

- The Bloomberg professional service offers a small suite of Bloomberg technical indicators to measure momentum, trend direction, signal reversal and more.
- Three helpful and interesting indicators include Fear and Greed, Trender and TrendStall. We describe the gist of these indicators.

Fear and Greed: A momentum and divergence indicator

The Fear and Greed indicator (FG) is a momentum measure that tracks the speed of price trend to suggest when trend strength is changing, may have gone too far or is weakening. The indicator calculates the spread between two weighted averages of true range (TR). For more on TR please see the Average True Range (ATR) section. Depending on your selected inputs, the formula may apply a sensitivity factor to the lookback period. A larger sensitivity factor and/or larger lookback periods make the indicator slower and perhaps more ideal for the medium/long-term investor. Conversely, smaller input values make the indicator more appealing for short-term trading.

Below we chart the S&P 500 on a weekly basis with the defaulted setting of 5 for the Fear and Greed indicator. When prices are trending to higher highs and "greed" makes a lower high, then it infers a peak / turn down in prices. Similarly, lower lows in price and a higher low in "fear" implies a bottom/turn up in prices. Historically large Fear or Greed levels can support a tactical reversal point.

Exhibit 64: S&P 500 (top) and Fear and Greed oscillator (Bottom)
A bullish and bearish divergence from the fear and greed oscillator.



Source: BofA Global Research, Bloomberg

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Trender: A trailing-stop-like line for trend following

The Trender indicator is designed to trail prices from below in an uptrend and from above in a downtrend. When price is trending the line attempts to stay close enough to keep a position open to follow the trend until price reverses and breaks through the line to stop a position and signal trend change. The gist of the calculation for a Trender Up line (green) begins with an exponential moving average of the midpoint whose value is adjusted (in this case decreased) by a fraction of the Average True Range (ATR). For more on ATR please see the Average True Range (ATR) section. This value is again adjusted through a sensitivity input which is said to be standard deviation based. Inputs range from 1-10. A Trender Up line never goes down while prices are rising.

Below is an example of a Trender Down line (Red – TrndrDn) with a default input of 3. As euro declined in 2021-2022 it followed the downtrend very well. Price action in February 2022 tested it but did not close above it to break the line and trigger a short stop. In November 2022 euro rallied above the short stop line and the Trender up line began as prices began trending up.

Exhibit 65: EURUSD weekly chart with the Trender indicator
The Trender indicator on Bloomberg followed euro's decline well in 2021-2022.



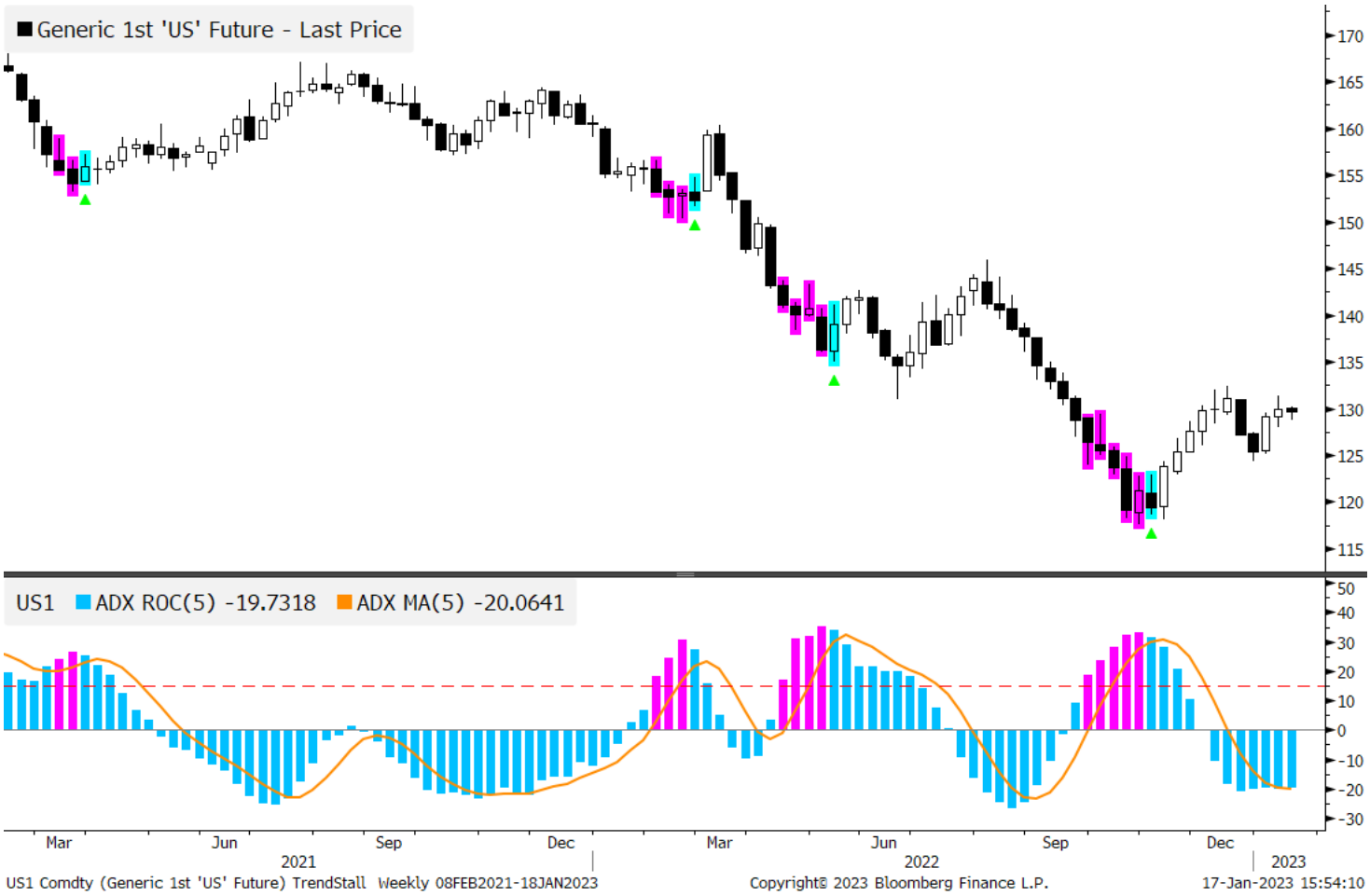
Source: BofA Global Research, Bloomberg

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TrendStall

The TrendStall indicator attempts to signal when the current trend is stalling and may see a short/long-term directional change. The indicator is calculated by taking the rate of change (ROC) of the Average Direction Movement (ADX). For more on this, see the DMI + ADX section. This data is shown as the histogram below price. A moving average is applied to the ADX ROC. A threshold on the histogram (typically 15) is required to generate signals. When the ADX ROC rises above 15, the indicator turns magenta and paints the price sessions. This suggests a strong trend is underway. When the ADX ROC decreases (while above 15) a green or red triangle is plotted on the chart and the paint on price turns cyan. This means the strong trend may be stalling or reversing.

Exhibit 66: Example of the TrendStall indicator on the US long bond treasury future
While bond prices were falling in 2021-2022, the TrendStall indicator signal identified four points where price may consolidate or reverse.



Source: BofA Global Research, Bloomberg

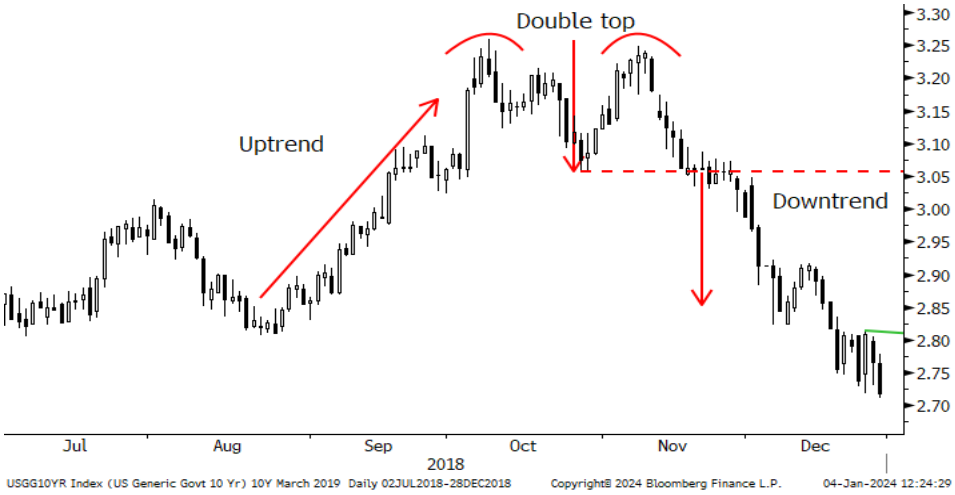
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Double tops and bottoms

- A double top forms after an uptrend, when price tests the same high level twice and the declines below the trough between the highs. Subtract the height of the pattern from the breakdown point (trough low) to determine the initial downside target.
- A double bottom forms after a downtrend, when price tests the same low level twice and the rises above the peak between the two lows. Add the height of the pattern to the breakout point (peak high) to determine the initial upside target.
- Positions can be entered in thirds, for example, a speculative 1/3 after the second turn down from an equal high, another 1/3 when the breakdown confirms the pattern, and a remaining 1/3 if the breakdown point is retested a few days later.

Visualizing a double top and double bottom pattern

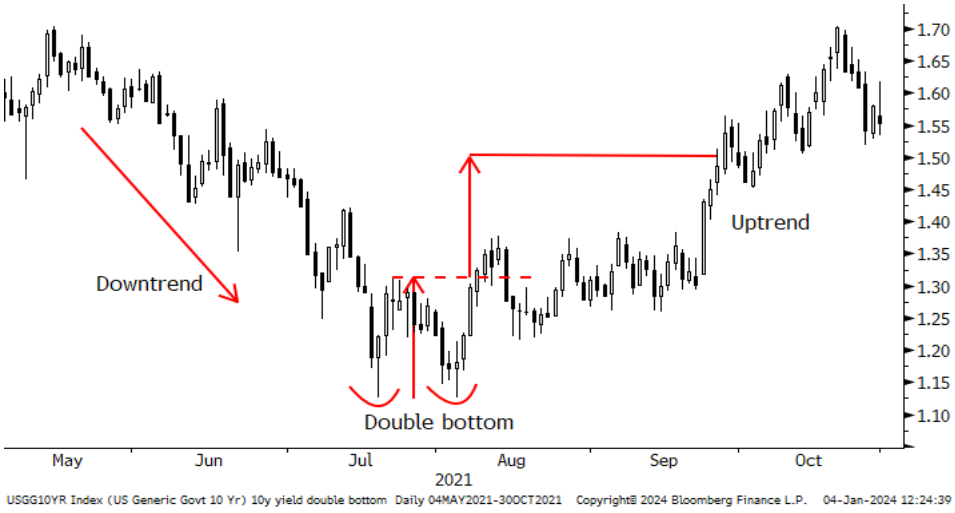
Exhibit 67: A double top on US 10Y yield
US 10Y yield was in an uptrend from August-October, began forming a double top in early November, and then confirmed the DT in mid-November.



Source: BofA Global Research, Bloomberg

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Exhibit 68: A double bottom on US 10Y yield
US 10Y yield declined from May-August, tested 1.15% twice in summer of 2021, then rose above the peak between the two troughs. The late August-Sept range required some patience, but an uptrend followed.



Source: BofA Global Research, Bloomberg

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A well massaged head and shoulders

- The head and shoulders pattern is a popular chart pattern that usually implies the prior trend is changing direction.
- We explain what to expect as the pattern forms and how to use other indicators to increase conviction in the formation.
- We show how to create targets using Fibonacci and discuss multiple examples when indicators aligned to strengthen the pattern.

Spotting a head and shoulders formation

The head and shoulders formation is useful in identifying market tops and bottoms. It can even be a continuation pattern if it appears in the opposite direction of the trend. In this primer we review the necessary inputs for a top pattern and summarize additional technical criteria to increase conviction when trading it.

The basics of a head and shoulders top

A prior uptrend to reverse. If there is not a trend prior to the formation, then there is not a trend to correct or change.

1. Three peaks and two troughs. The first peak is the left shoulder, the second the head and the third the right shoulder. The head is always the highest high. The troughs separate the peaks from one another and help create the neckline.
2. Estimate a neckline by drawing a trend line after the second trough forms. In a top pattern, the neckline should have a slightly positive slope or at least be flat. This is important because price will be in the earlier phase of a downtrend when the neckline breaks, making the sell entry point higher.
3. The right shoulder should make a lower high than the head. It is also ideal for the right shoulder to be similar or smaller than the left.
4. A decisive move down through the neckline with greater-than-average volume should occur to solidify the pattern.
5. The neckline may be tested as resistance on a light-volume rally, which offers a second opportunity to sell the security. An appropriate stop is often just above the high of the right shoulder.

Volume can enhance the pattern

When an uptrend is coming to an end, it is preferred to see lighter volume in up moves and heavier volume in down moves. When volume is available it is better if it corresponds to a head and shoulders top pattern in the following ways:

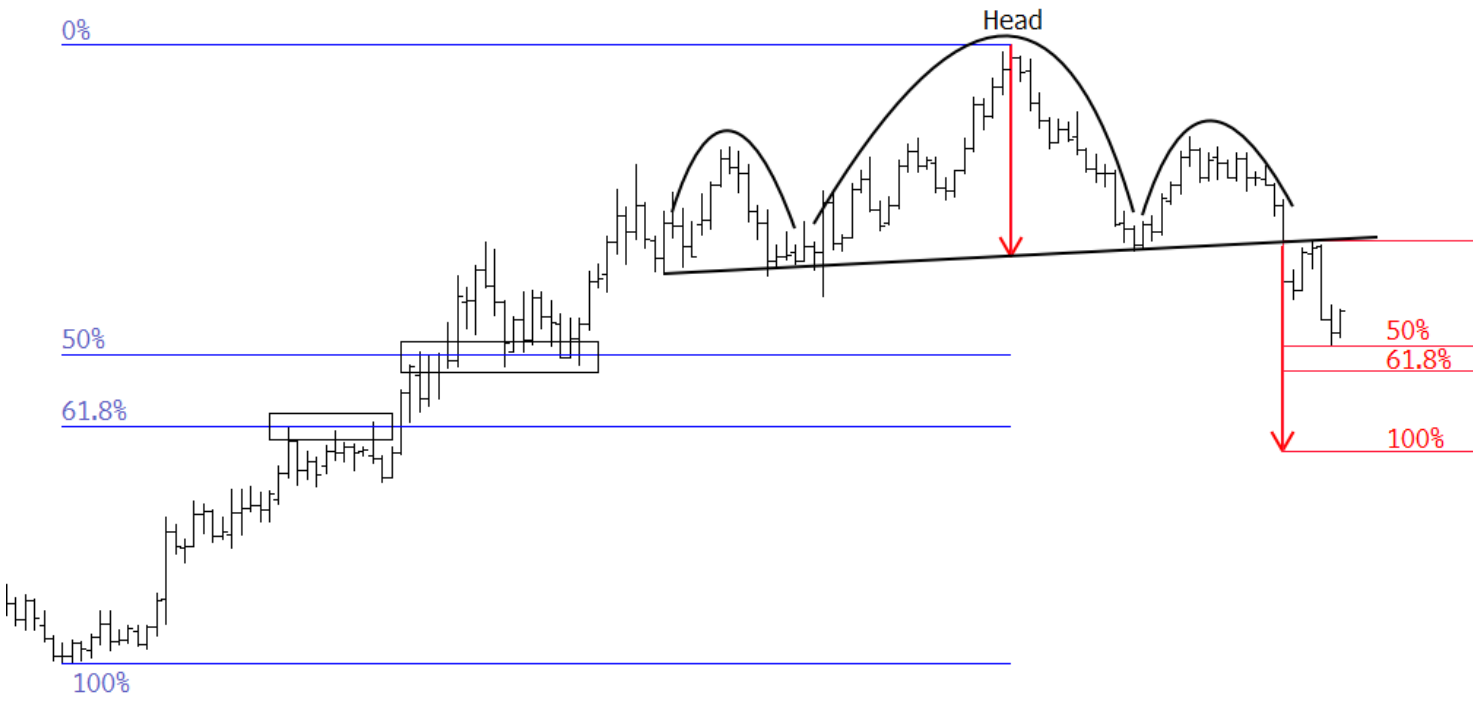
- The rally to the head occurs on light volume. It could be compared to the volume to the top of the left shoulder or to a moving average.
- The decline down to the second trough occurs on rising volume suggesting an increase in supply or selling pressure.
- The rally to the top of the right shoulder occurs on light volume suggesting a lack of demand or little buying pressure.
- The decline from the top of the right shoulder through the neckline should occur on heavy volume.

Calculate targets with Fibonacci

There are two ways to calculate downside targets for a head and shoulders top. The first is to use a measured move projection with a Fibonacci Extension annotation (red). Measure the distance from the top of the head to the neckline and subtract that from the breakdown point. This will also create levels where price may find some support.

The second is to apply a Fibonacci retracement (blue) of the prior uptrend to find levels of confluence. In this example the 50% provided resistance and then support in the uptrend. The 61.8% retracement also acted as resistance. They also come close to aligning with some of the red projections. The confluence of levels suggests a high-conviction view that the blue 50% and red 61.8% levels are reached. A probable conviction would foresee price reaching the blue 61.8% and red 100% projections.

Exhibit 69: Calculating H&S top targets with Fibonacci retracements and extensions. Areas of confluence (Fibonacci retracement-extension levels overlap or cluster together) can strengthen conviction in a level. Downside H&S top targets can act also act as support upon arrival.
A Fibonacci retracement and Fibonacci extension can be used to estimate downside levels for this head and shoulders top pattern.



Source: BofA Global Research, Bloomberg

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Use additional indicators to confirm the pattern

When a head and shoulders pattern is developing, check other technical indicators for complimentary signals at turning points and breaks.

- Check if the head or shoulder failed at a moving average or trend line on the current or a higher time frame. Often a key trend line or moving average toggles between resistance and support when developing the pattern, which is called polarity.
- Check for bar or candle pattern formations at the peaks and troughs. Bearish patterns such as an outside day, doji candle or bearish engulfing pattern may form at the peaks emphasizing the technical trend in the market.
- Momentum divergence – It is considered bearish when a new price high occurs forming the head of a pattern and a lower high in momentum also occurs (such as RSI, Stochastics or MACD histogram).
- Check to see if trending indicators such as MACD or Ichimoku are approaching or already crossed bearish.
- For an index, review the breadth of index constituents and sentiment indicators while the pattern develops.

Examples with additional indicators

Multiple time frame resistance and trend lines

In this daily chart, the 200wk moving average (blue) was pivotal. It provided resistance during the formation of the left and right shoulder. It was also important during the rally to the head and on the decline to the second trough. The failure to break back above it while forming the right shoulder was an early indication the pattern would form. The trend line (magenta) showed polarity. It was support in the uptrend until the left shoulder broke it. Then it became resistance when forming the head.

Exhibit 70: Multiple time frame indicators and trend lines help to frame out pattern
The 200WK SMA and trend line helped identify resistance points while the head and shoulders top formed.



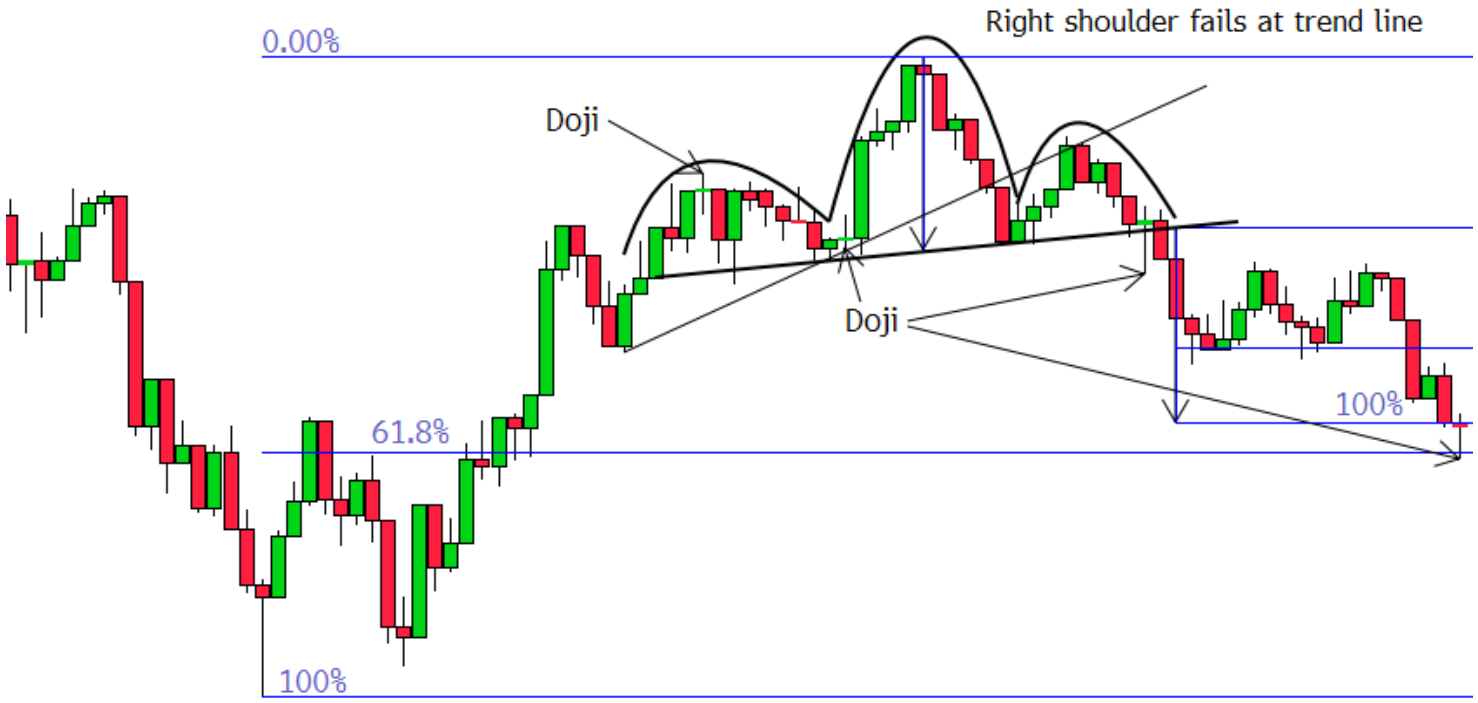
Source: BofA Global Research, Bloomberg

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Doji candles form at key points including Fibonacci targets

Doji candles are known for representing indecision in continuing the prior trend. It forms when price opens and closes at nearly the same level and trades higher and lower during the period. In this example a doji candle formed at the top of the left shoulder, the first trough and just before the neckline break. When price reached the 100% Fibonacci measured move target it also formed a doji where the low touched the 61.8% retracement.

Exhibit 71: Doji candles form at key points in forming top pattern and reaching target
Candle patterns can be used to identify tactical turning points when a head and shoulders pattern is forming.



Source: BofA Global Research, Bloomberg

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Trend exhaustion signals help to form a bottom

Two technical indicators suggested a bottom was forming. A TD Setup green 9 'buy' signal occurred at the lows. Also, a bullish momentum divergence occurred when price made new lows but momentum made a higher low. When price broke above the neckline it was also able to break above the 50-week and 12-month moving average. Later, price reached the aligned Fibonacci targets of the measured move projection and the 61.8% retracement.

Exhibit 72: Head and shoulder bottom formed with bullish momentum signals
The TD Sequential indicator can be used to identify turning points when a head and shoulders pattern is forming.



Source: BofA Global Research, Bloomberg, DeMark Analytics

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Double trend exhaustion, momentum divergence, gaps

In this bottoming example, four technical signals pointed to a bottom pattern. There were two trend exhaustion signals near the lows of the head; a TD Setup 9 and TD Countdown 13. Momentum as defined by RSI presented a bullish divergence. The decline to new lows did not even exhibit enough momentum to reach oversold. A short-term trend line was broken when forming the right side of the head and provided support to the right shoulder. The trend also gapped higher a few times when forming the right side of the head and right side of right shoulder.

Exhibit 73: Head and shoulders bottom with multiple technical signals
An example of a head and shoulders base with RSI divergence, a TD Sequential reversal signal, trend line break and measured move targets.



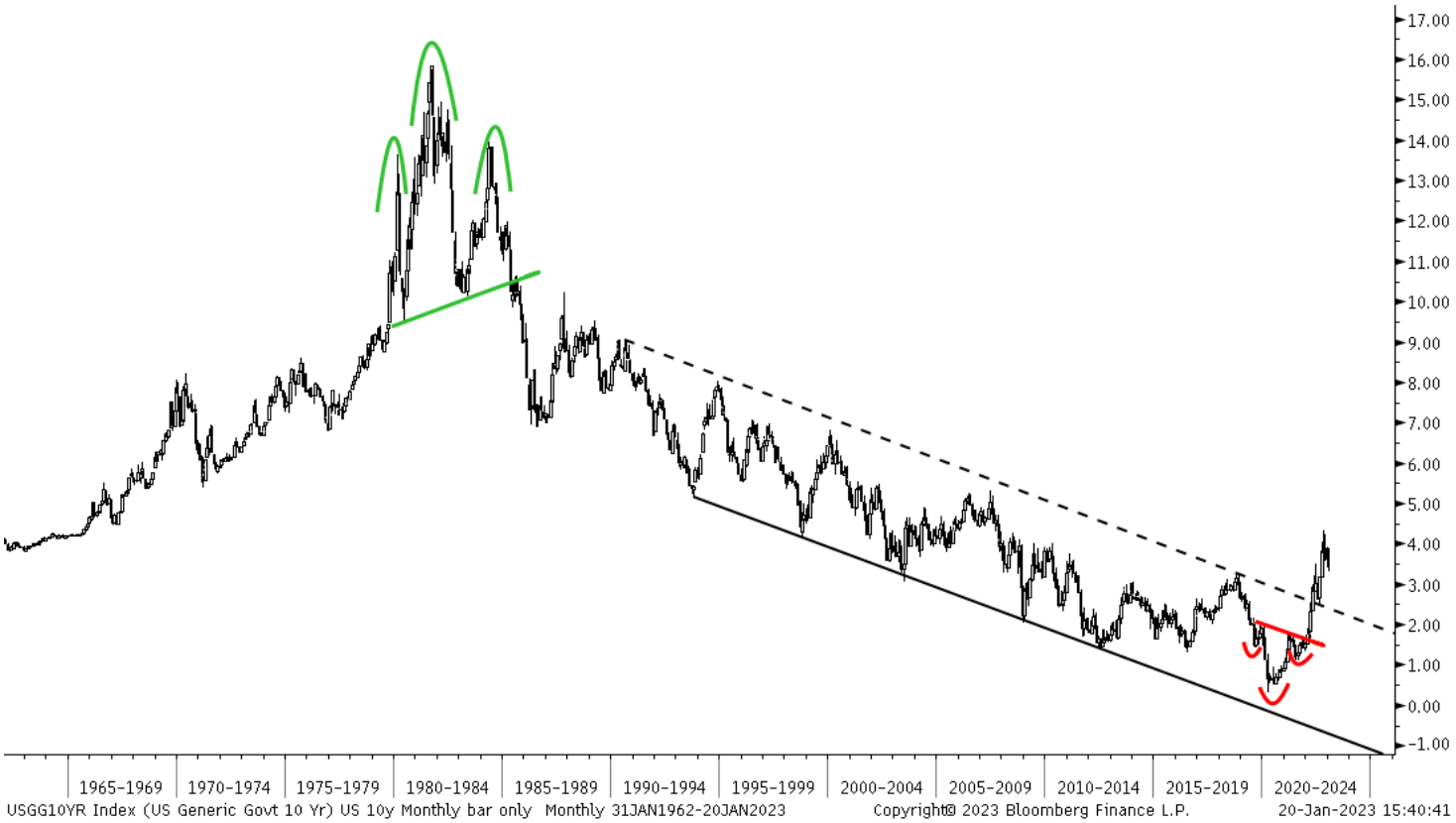
Source: BofA Global Research, Bloomberg, DeMark Analytics

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The US 10Y yield secular trend changes with head and shoulder patterns

In the early 1980s a head and shoulder top formed on the monthly chart of US 10Y yield. This signaled the end of a secular bear market and start of a secular bull market in US treasuries. In 2019-2022, a head and shoulders bottom in 10Y yield formed to signal the end of the secular bull trend and start of a secular bear trend.

Exhibit 74: US 10Y yield – Monthly chart's secular trend changes signaled by head and shoulder patterns
In the early 1980s a head and shoulders top defined the start of a secular bull market in bonds. A poetic end to the secular bull market occurred with a head and shoulders bottom being confirmed in 2022.



Source: BofA Global Research, Bloomberg

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Cup and handle

- A cup and handle is usually a bullish continuation pattern in the shape of – you guessed it – a cup and handle. It can also form at the end of a downtrend.
- The "cup" is a deeper "U"-shaped corrective phase, and the "handle" is a smaller "u"-shaped retracement of the cup.
- A break above the price high that represents the end of the cup and start of the handle confirms the pattern and establishes the upside target.

The cup & handle refreshes an uptrend, ends a downtrend

A bullish continuation pattern restarts an existing uptrend. This pattern takes the form of a "U"-shaped "cup" followed by "handle" that retraces a portion of the rally from the cup low to the cup high. The handle often takes the shape of a smaller "u". As a rule of thumb, the handle pullback should not retrace more than 38.2% to 50% of the cup rally. The price that marks the end of the cup and the start of the handle is the breakout point. A decisive rally above this level completes the cup and handle, establishing an upside target for the bulls. This upside target is the difference from the breakout point to the low of the cup projected higher from the breakout point. As a bottoming pattern, the handle represents the first "higher low" with an uptrend to follow.

Volume and/or breadth should confirm the pattern

Volume should decrease as the cup and handle forms and then increase to confirm the breakout. When compared to volume on the rally from the cup low to the cup high, a decrease in volume on the pullback into the handle low would be viewed as bullish. When a cup and handle forms on an equity index, watch important market breadth indicators, like advance-decline lines, which can provide a leading to confirming indicator for bullish continuation patterns at the index level.

Time in pattern

Some cup and handles are shorter-term patterns that form in a couple of months. Others also form big bases that can span two or three years, or even up to a decade or more.

How to trade the cup and handle

Buy a bullish handle pullback

Volume should decline on the handle when compared to volume on the right side of the cup to support a bullish pullback. Look for bullish reversal signals to help confirm a tradable low between the 38.2% and 50% Fibonacci retracement levels of the rally from the cup low to the cup high, which is where the handle low should find support ahead of a rally. Place a stop below the handle low or under a retracement level. This allows an investor or trader to establish a long position prior to the cup and handle breakout. Sometimes a shallow handle forms, holding above the 38.2% retracement, while other times a deeper handle could reach the 61.8% retracement.

Buy an upside breakout above the handle downtrend line

Handles can form flags or pennants, which are bullish consolidation patterns. Rallying above these patterns' downtrend lines completes the formations and suggests that higher prices are to come.

Use moving averages as a guide for when to buy

Since a cup and handle pattern is a bullish continuation pattern, it should form within the context of rising longer-term moving averages (MAs), such as the 40-week MA. A close below a rising 40-week MA followed by a close back above could help confirm the handle low and provide a buy signal. A stop would go below the price low that preceded the cross back above the MA, or below the low of the week during which price crosses and closes back above the 40-week MA.

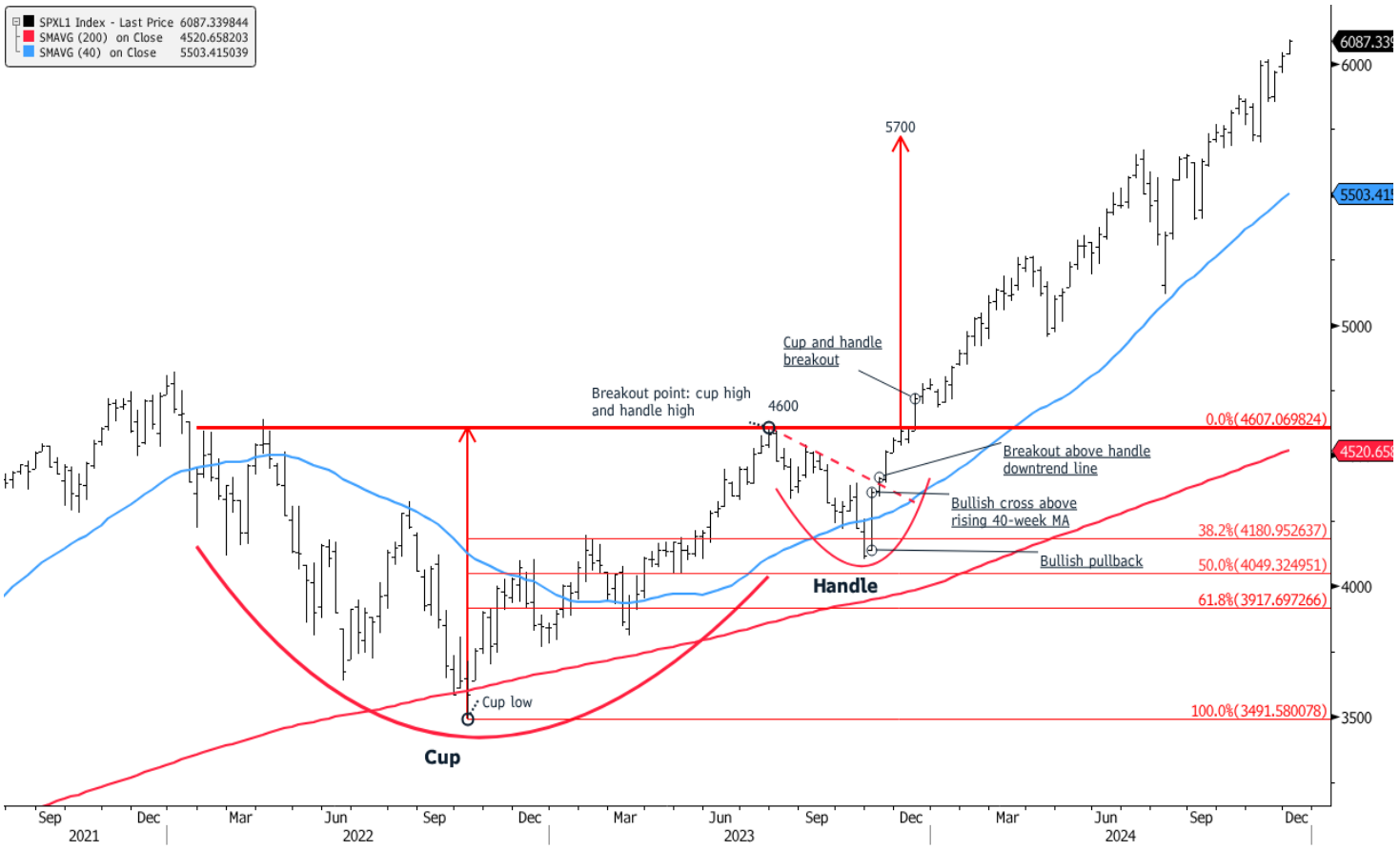
A breakout above the cup rally high

The three signals highlighted above suggest establishing a long position prior to the "official" cup and handle breakout, which occurs on a rally above the price that marks the end of the cup and the start of the handle.

Cup and handle introduced by William J. O'Neil in 1988

William J. O'Neil introduced the cup and handle pattern in his 1988 book, *How to Make Money in Stocks*.

Exhibit 75: S&P 500 weekly chart with 40-week and 200-week moving averages
The S&P 500 completed an early 2022 into late 2023 cup and handle base. The breakout above 4,600 projected upside potential toward 5,700.



Source: BofA Global Research, Bloomberg

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Candles patterns light the way

Candle patterns

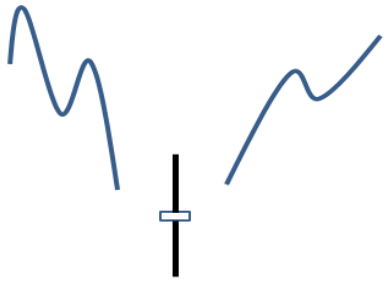
- Japanese candle stick charts are a way to visualize price action to show who is in control of the trend and if there is potential for a change in trend.
- In the prior section we introduced the bar and candle chart and the popular bullish and bearish engulfing patterns, including the doji candle.
- In this section we expand our candle pattern coverage to the primary candle stick terms including one, two and three candle stick patterns.

A doji candle suggest balance between buyers and sellers

A Doji candle forms when price opens and closes at about the same level on the trading day having traded reasonably higher and lower during the session. By closing near the open it suggests the market found balance between buyers and sellers. Balance suggests the prior trend may be at risk of reversal. So, if a downtrend was occurring and a doji forms then it may turn up, such as Exhibit 76. Should price gap down to open the doji day, it is considered an even stronger pattern because the pattern rejected follow through during the session. If an uptrend was occurring and a doji forms, the trend may turn down. There are other types of doji candles with unique characteristics, including:

- **Dragonfly Doji:** This is a bullish candle that occurs in a downtrend to signal a reversal higher. The open and close are nearly equal at the top of the trading range. A long lower tail defines the pattern as it represents sellers that were overtaken by buyers with price closing at the open and high. (Exhibit 77)
- **Gravestone Doji:** This is a bearish candle that occurs in an uptrend to signal a reversal lower. The open and close are nearly equal at the bottom of the trading range. A long upper tail defines the pattern as it represents buyers that lost control to sellers as price closes at the open and low of the day. (Exhibit 78)
- **Long Legged Doji:** This candle is viewed as a more extreme doji candle because the trading range on the day was much wider than a typical day. Like a doji candle, it suggests the prior trend is set to reverse but the battle that ensued during the trading session was larger than usual. (Exhibit 79)
- **Doji Star:** This candle is viewed as a strong signal of market balance at current price levels. It has a very small trading range on the day and price still opens and closes at about the same level. This candle is often stronger if there was a gap to open the day, such as down in Exhibit 80. This candle is often combined in other patterns to increase its value which we'll cover later.

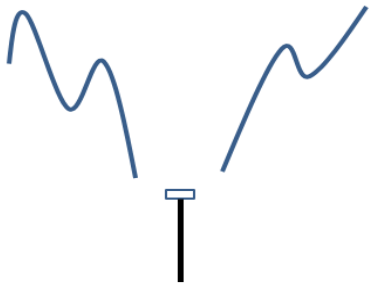
Exhibit 76: Doji candle
Doji candle in a downtrend may turn price up



Source: BofA Global Research

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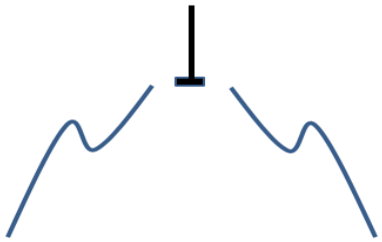
Exhibit 77: Dragonfly Doji
A bottoming candle



Source: BofA Global Research

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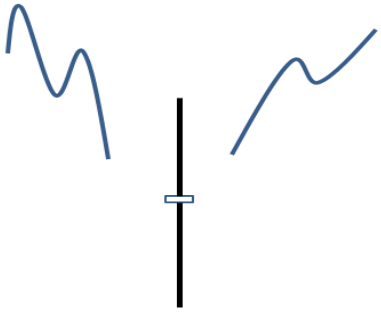
Exhibit 78: Gravestone Doji
A top candle



Source: BofA Global Research

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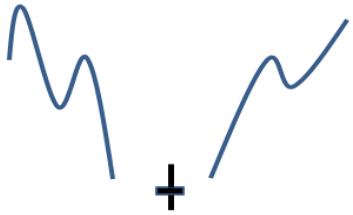
Exhibit 79: Long legged Doji
A more volatile range and reversal candle



Source: BofA Global Research

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Exhibit 80: Doji star
A simple change in trend



Source: BofA Global Research

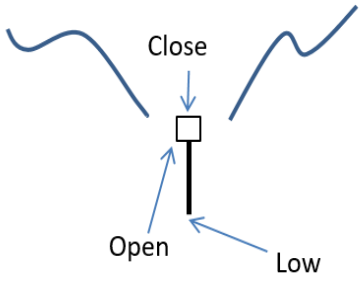
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Hammer out a bottom in a downtrend

The hammer candle looks distinctly different than many candles. As the name implies, it looks like a hammer. It only forms in a downtrend to signal the end of the decline. It has a small body at the top of the trading range with a long lower trading range. The lower range is usually 2-3 times the size of the candle body. The long lower range shows buyers stepped in during the trading session resulting in price closing above the open (Exhibit 81). It is considered more bullish than a dragonfly doji because price closes above the open as opposed to equal to it.

An inverted hammer can only form at a peak. It forms when price is in an uptrend, trades to new highs and then closes below the open and at/near the low of the day (Exhibit 82). The upper candle wick is about 2-3 times the size of the body. It represents sellers overwhelmingly entering at higher prices of the trading session. Supply therefore outweighs demand and price falls.

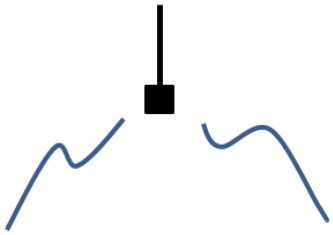
Exhibit 81: Hammer out a bottom
A hammer candle is a bottom signal in a downtrend.



Source: BofA Global Research

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Exhibit 82: Inverted hammer
An inverted hammer is a top signal in an uptrend.



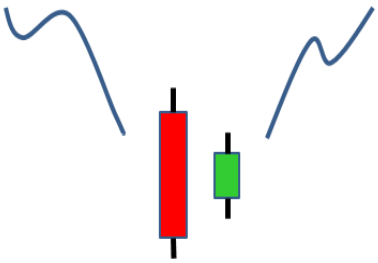
Source: BofA Global Research

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Bullish and bearish harami patterns

A harami pattern switches the candles in a bullish and bearish engulfing pattern. By reversing the candles, it is thought that the second trading day is rejecting the large directional day that preceded it. A bullish harami forms in a downtrend. The first candle is a down day and the second candle a small up day within the candle body (open and close) of the first session. The second candle tends to gap up and close a little higher on the session. These characteristics suggests price will turn higher rejecting or trapping the sellers on the first down session like in Exhibit 83. A bearish harami is the opposite of this. It forms in an uptrend and warns of a turn lower. The first candle is a strong up session and the second candle gaps down at the open, closes slightly lower and within the body of the first (Exhibit 84).

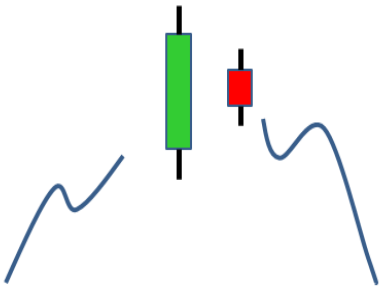
Exhibit 83: Bullish Harami
A bottom signal for a downtrend



Source: BofA Global Research

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Exhibit 84: Bearish Harami
A top signal for an uptrend



Source: BofA Global Research

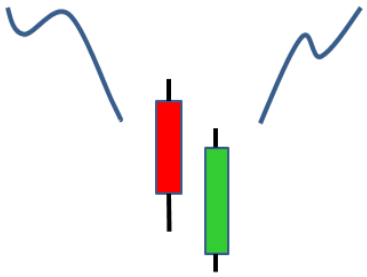
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Piercing lines and dark cloud cover

The piercing line candle pattern is another variation of a bullish engulfing pattern and only occurs in a downtrend to signal a turn higher. The primary difference is price opens lower on the second day, trades to a new low and then closes up about 50% of the way into the first candle. This suggests buyers entered late in the session on the second day shifting the supply/demand relationship in favor of the bulls. (Exhibit 85)

A dark cloud cover pattern occurs in an uptrend and is bearish. Price action on the first day is up. The second day begins bullish by gapping up at the open. However, price then reverses to close lower, about 50% of the way into the first candle (Exhibit 86).

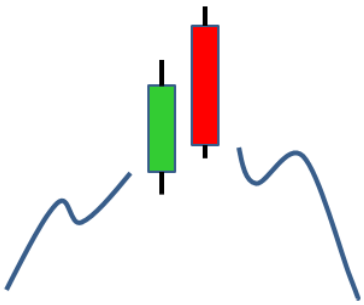
Exhibit 85: Piercing line (Bullish)
A bottom signal for a downtrend



Source: BofA Global Research

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Exhibit 86: Dark cloud cover (Bearish)
A top signal for an uptrend



Source: BofA Global Research

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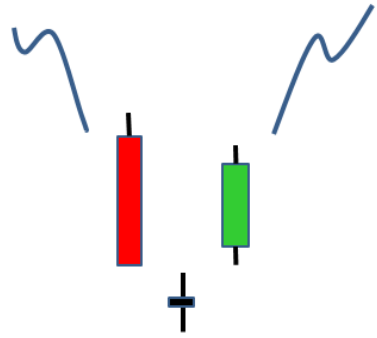
Shooting stars are abandoned candles, trapped positions

Shooting stars or abandoned candles refer to trading days that sit out in the air of a chart. There are no other trading days nearby in recent history. A gap up and then down or down and then up occurs leaving it in some empty space.

To infer an abandoned candle as bullish there are a few criteria. The setup is referred to as a morning star pattern. First a downtrend, then a strong down day, then a gap down forming a doji-like candle and then a gap up forming a bullish up day at least 50% into the first down day. These tree candles in this order is read as a shift in bearish sentiment to bullish. Many times, bears get caught in a position on the first and second day and are later forced to exit their trade in the coming days as prices rise. (Exhibit 87)

An evening star pattern forms in an uptrend and is bearish. Sometimes it is called a shooting star pattern in relation to those found in outer space. Prices in space meaning prices are too high and eventually a shooting star loses speed, burns out and rolls over. A strong up day, gap up to a shooting doji star and then a gap down for a down day occur. The down day confirms the doji star and that a reversal lower is underway. (Exhibit 88)

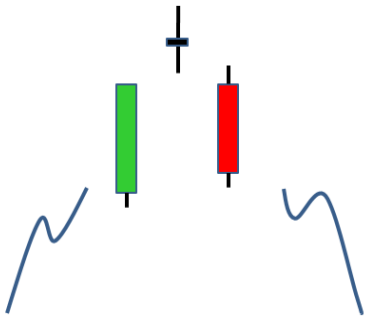
Exhibit 87: Morning star pattern
A bottom pattern in a downtrend.



Source: BofA Global Research

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Exhibit 88: Evening star pattern
A top pattern in an uptrend,



Source: BofA Global Research

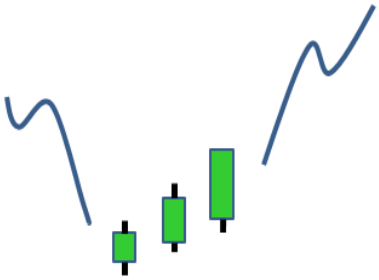
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The three soldiers are setting up for a big move

The three soldiers pattern is a trend reversal set up that represents gradually increasing demand to change the prior trend. Consider Exhibit 89, a bullish setup where a new low is made but closes a little up on the day. Then the market gaps down to open the second day and succeeds in closing up on the day above the high of the first. Then on the third

day the market gaps down again, maybe halfway into the second day's range, to then close above the second days high for a third up day of even larger size (Exhibit 89). This sequence of events suggests the bears tried and failed three days in a row to push price lower. These failures suggest bulls are taking control and price is set to move higher. Three bearish soldiers would see three down days of increasing size beginning with a new high and a down day. Then with gaps up on the second and third day rejected by bears as the market closes lower each day. (Exhibit 90)

Exhibit 89: Three bullish soldiers
A bullish pattern gaining momentum.



Source: BofA Global Research

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Exhibit 90: Three bearish soldiers
A bearish pattern gaining momentum.



Source: BofA Global Research

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Engulfed by greed, reversed by fear

Outside reversals

- We review the characteristics and application of outside reversal signals, also known as engulfing patterns, which are often quoted in strategy reports.
- These valuable reversal patterns formed at the 2017 bottom in GBP/USD, the 2008 peak in oil and 2016 low in US 30Y yield.
- In our examples we incorporate other technical indicators and signals from other sections in this primer series.

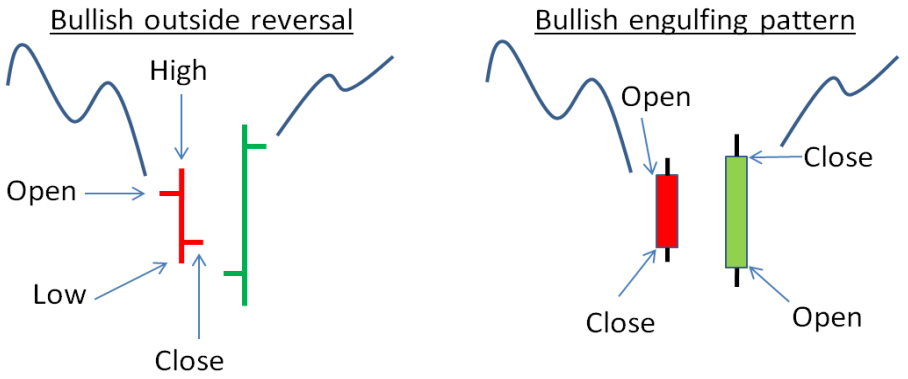
What is an outside reversal and engulfing pattern?

Whether you prefer to use bar or candle charts, the technical significance of an outside bar or engulfing candle is the same. Both signal a trend reversal point if and only if there is a trend to reverse. In this report we summarize what an outside reversal and engulfing pattern is and discuss qualitative technical examples of where they may work well.

Recognize a bullish outside reversal / engulfing pattern

The following graphic is an example of a bullish outside reversal pattern and bullish engulfing pattern. The following conditions are important to define them. First there must be an established downtrend to reverse. If this pattern formed at new highs, it wouldn't be called a reversal pattern because price just closed at a new high. The pattern consists of two trading ranges. The first period of the pattern (left) should be a smaller trading range with a down close. The second period of the pattern (right) should open lower, trade down to a new low and then reverse to a higher high. The second period and market sentiment should look, feel and be a bullish up period overall.

Exhibit 91: Bullish outside reversal pattern and engulfing pattern (Red = down. Green = up)
After a downtrend, the second candle entirely engulfs the first to suggest higher.



Source: BofA Global Research

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Technical conditions that may enhance a reversal pattern

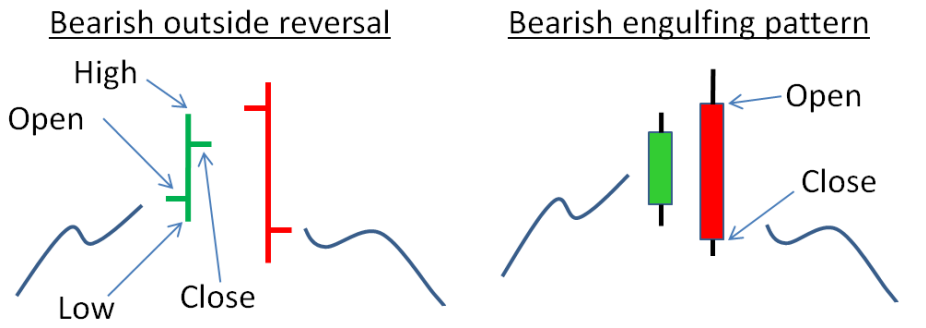
The following conditions in a technical chart may make a bullish reversal more effective:

1. If price is in a tactical correction while above a positively sloped 200-period SMA or if the 50-period SMA is already above the 200-period SMA.
2. If price is declining to test trend line, moving average or Fibonacci support and a bullish reversal pattern forms on or around support suggesting, it will hold.
3. If the downtrend prior to the reversal pattern results in RSI reaching recent lows or oversold levels, then a rally may follow a reversal pattern.
4. Low volume/flows on the first period and high volume/flows on the second. This may suggest hesitation to go lower and agreement on the outside.

Bearish outside reversals and engulfing patterns

The opposite holds true for bearish reversal patterns. First there must be an established uptrend to reverse. If this pattern formed at new lows, it wouldn't be called a reversal pattern because price just closed at a new low. The first period of the pattern (left) should be a smaller trading range and close higher. The second period of the pattern (right) should open higher, trade to a new high and then reverse to a lower low. Ideally, price should close below the prior range low or at least the prior range open.

Exhibit 92: Bearish outside reversal and engulfing patterns
A bearish reversal occurs in an uptrend when the second bar entirely engulfs the prior bar.



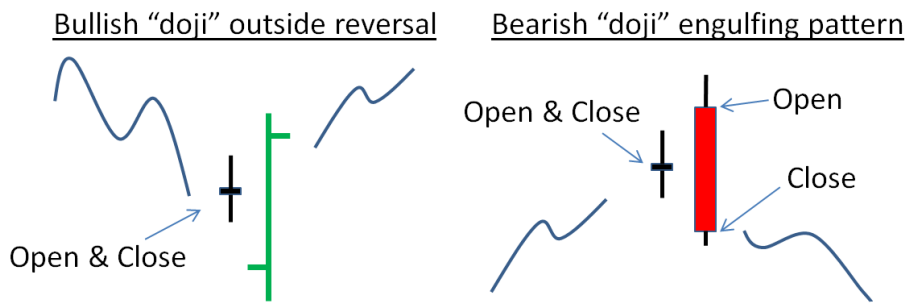
Source: BofA Global Research

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Doji applied to bullish and bearish reversal patterns

A Doji candle forms when price opens and closes at about the same level after trading higher and lower on the period. This type of price action after a trend is viewed as a change to neutral because buyers and sellers have agreed that prices should not move higher or lower at that time. In other words, supply and demand is balanced. When a doji forms in an established downtrend it could be a hint a rally may occur. Similarly, if it occurs in an established uptrend, it could be a hint a decline may follow. These patterns are most useful near important support and resistance levels or after parabolic moves. When the period after the doji engulfs the entire doji candle, that too signals a reversal. It confirms the doji candle sentiment and forms an outside reversal / engulfing pattern.

Exhibit 93: Doji candle as part of a bullish outside reversal or bearish engulfing pattern
A doji means the a balanced or neutral trend When engulfed it is a change in trend signal.



Source: BofA Global Research

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Reversal patterns in practice

- In this section we review three charts with bullish and bearish outside/engulfing reversal patterns and pair them with other technical indicators.

- We review the bullish reversal patterns in GBP/USD during the 1Q17 bottom, the bearish reversal in WTI crude oil prices at the all-time highs in 2008 and US 30Y yield's outside reversal after making an all-time low in 2016.

Bullish reversal present during Cable's post Brexit bottom

The downtrend in GBP/USD leading up to and after Brexit resulted in a technical base forming near 1.20. Two bullish outside week patterns formed in 1Q2017. The first occurred in January and the second in March. Each bullish reversal was part of a shorter term and longer-term downtrend. But more importantly the price action that immediately preceded the patterns was lower. The psychological support level of 1.20 and a supportive trend line continued to hold while these bullish patterns formed.

Additional technical signals occurred before and after these bullish outside weeks. Before the outside weeks, there was a bullish TD Sequential signal (red 13) at the October 2016 low. The reversal weeks and no new low offered confirmation for this signal and a possible bottom. Rising trend line support and declining trend line resistance formed a triangle pattern. Price resolved higher through trend line resistance (dashed line), confirmed the prior bullish signals and a bottom in GBP/USD.

Exhibit 94: GBP/USD Weekly bar chart with bullish outside reversal weeks
Two bullish reversal patterns helped form a triangle bottom pattern.



Source: BofA Global Research, Bloomberg

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Oil: Reversal month signals end to rally

Below we show a monthly chart of oil prices and notate bullish and bearish engulfing patterns. The largest and most significant bearish engulfing month occurred at oil's peak in 2008. In June 2008 oil prices rallied for the fifth consecutive month and the Relative Strength Index (RSI) reached the most overbought condition since 1996. In July, price action made a new high above June's high and then began to decline. By the end of July, prices declined to a low below June's and closed near the low of the month. This price action combined with RSI is a bearish reversal. A downtrend followed, which found support in early 2009 at the 200-month simple moving average.

In 2011 another bearish engulfing pattern formed. This pattern formed as price closed above and then back below the 50% Fibonacci retracement of the 2008-2009 decline. This implied the midpoint of the prior decline was rejecting price from moving higher. Price trended lower for four more months. After the four-month decline, a bullish engulfing pattern emerged. Oil prices rose for about five months to a lower high.

Exhibit 95: WTI Crude Oil monthly Exhibit - roll adjusted for difference and for active future
A bearish engulfing pattern with overbought RSI is two bearish conditions simultaneously.



Source: BofA Global Research, Bloomberg

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US 30Y treasury yield's outside reversal at all-time lows

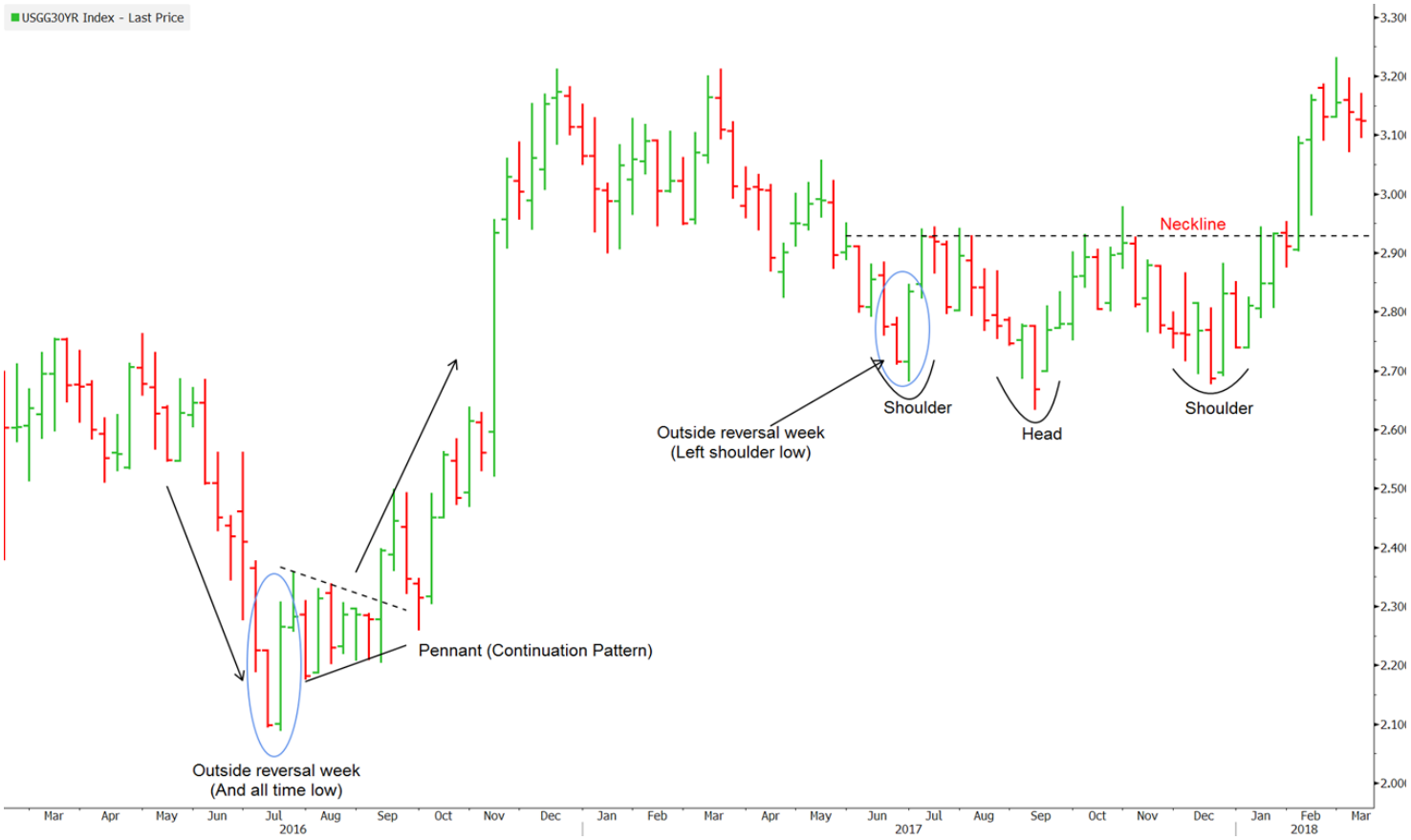
When the bond market peaked in 2016, US 30Y yield formed an outside reversal week at its all-time low. The six-week decline in yield during June-July of 2016 made a closing low of 2.0936%. Entering the second week of July yield traded a few ticks lower to 2.0882% and then began a sharp reversal higher, engulfing the prior week and reversing the decline.

In this example, it would have been more ideal for the open of the outside bar (2.1003) to be below the close of the prior bar (2.0983%) as that is one of the aforementioned criteria. Such action on the weekly chart would suggest the same sentiment persisted over the weekend, which can be an important ingredient to a stronger reversal signal. Given the marginal difference between the close and open (.0020%), and that a new low occurred, we consider this reversal valuable and worthwhile to review. In practice, we would also look for other technical indicators to confirm.

The week after the reversal, yield continued to rise. However, it quickly fell into a holding pattern called a pennant. A pennant is a short-term continuation pattern and looks like a small triangle. The move higher in September and retest/hold in October emphasized the strength of the outside reversal week at the all-time lows.

An outside reversal week can also be an important part of forming a larger technical pattern, such as a head and shoulders. The bond rally during the latter half of 2017 led yield to form a head and shoulders base. The low of the left shoulder shows an outside reversal week.

Exhibit 96: US 30Y yield – weekly
Outside reversal weeks were seen at the July 2016 yield lows and the left shoulder of a head and shoulders base in 2017.



Source: BofA Global Research, Bloomberg

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Island tops and bottoms

Island reversals

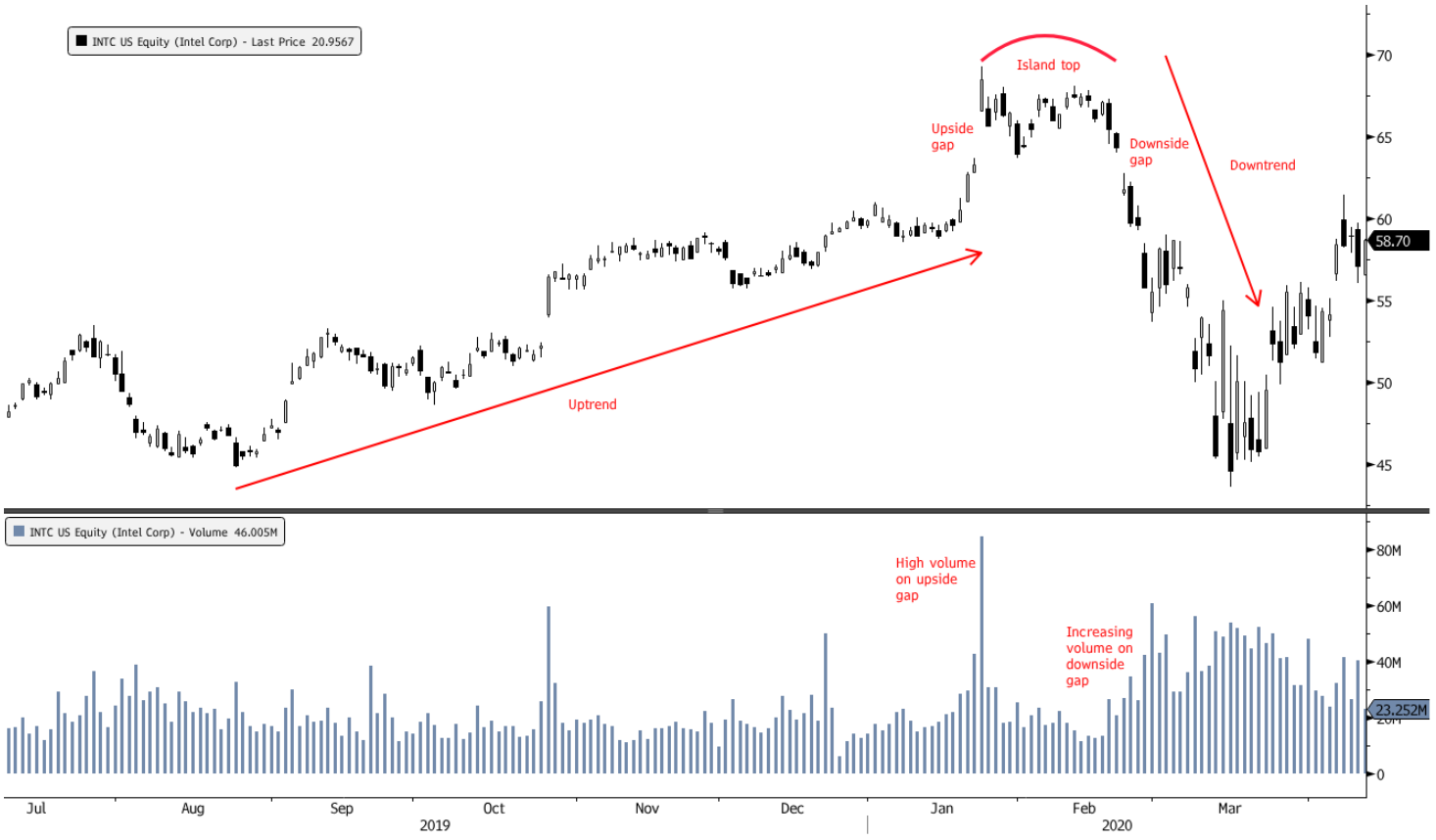
- An island is a reversal pattern that occurs when two gaps create an isolated cluster of trading days.
- As a reversal pattern, an island can signal a potential top within an uptrend or a bottom within a downtrend.
- The isolated cluster of price bars traps the bulls within an island top and offer resistance and traps the bears and offers support within an island bottom.

Island top: Upside gap, price bars and downside gap

The components of an island top are an upside gap followed by a period of trading that is usually within a definable range and then a downside price gap. The upside and downside gaps should overlap and have increased volume. The price bars above the gaps represent the island top. The trend shifts from up to down, trapping the bulls (those who are long or who bought near the highs) on the island top. The area of the gaps and the price action that comprises the island top provide a potential source of resistance on interim rallies within the new downtrend.

The upside gap is an upside exhaustion gap, which is the last gap in an uptrend and the most likely type of gap to fill. The downside gap is a bearish breakaway gap, which confirms the top pattern and is the least likely type of gap to fill anytime soon.

Exhibit 97: Intel (INTC): Daily candlestick chart with volume
A late January into late February 2020 Island Top triggered a deep correction on INTC.



Source: BofA Global Research, Bloomberg

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Island bottom: Downside gap, price bars and upside gap

An island bottom consists of a downside gap followed by a period of trading that is usually within a definable range and then an upside price gap. The downside and upside gaps should overlap and have increased volume. The price bars below the gaps represent the island bottom. The trend shifts from down to up, trapping the bears (those who are short or who sold near the lows) on the island bottom. The area of the gaps and the price action that comprises the island bottom provide a potential source of support on interim pullbacks within the new uptrend.

The downside gap is a downtrend exhaustion gap and most often to be filled by a reversal. The upside gap is a bullish breakaway gap confirming the top pattern and is a gap that is least likely to be entirely filled anytime soon.

Exhibit 98: Nutanix (NTNX): Daily candlestick chart with volume
A late May into late August 2022 Island Bottom confirmed a change in trend from down to up on NTNX.



Source: BofA Global Research, Bloomberg

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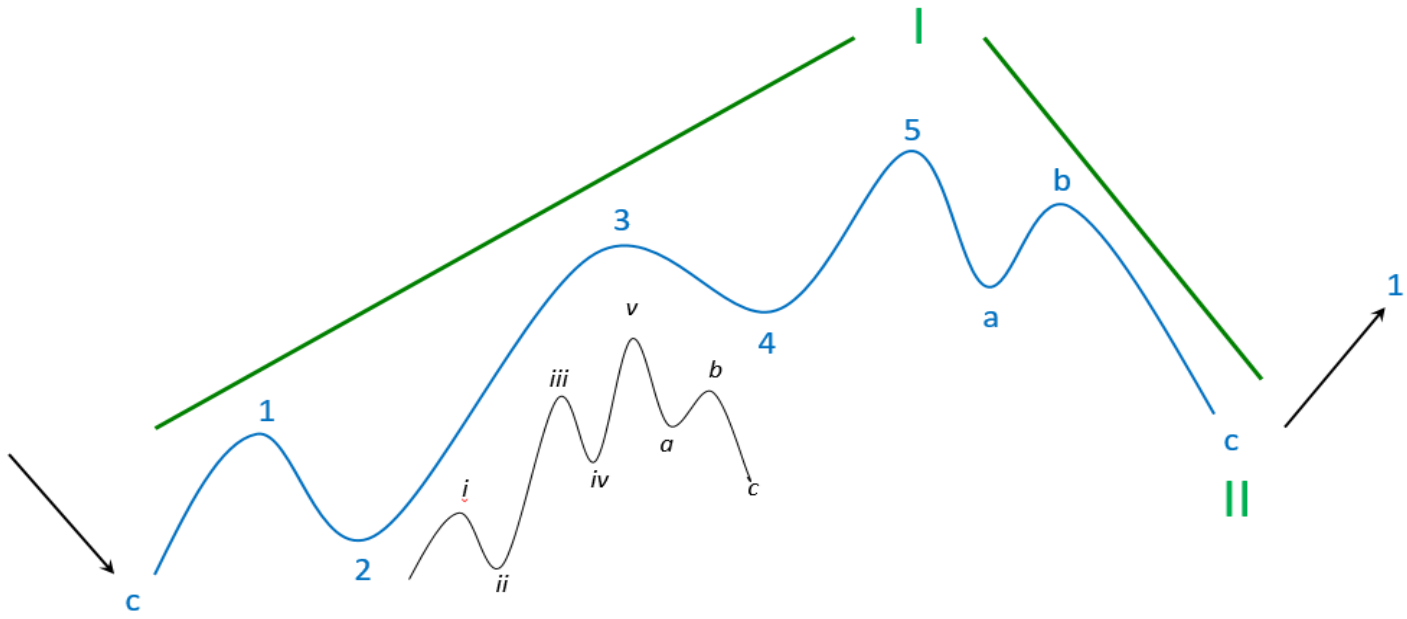
Elliott Wave 101

- Markets trend in cycles of optimism and pessimism both large and small. Elliott wave patterns track these cycles by counting a series of waves.
- An Elliott wave uptrend cycle is a total of eight waves. First a series of five waves higher where three waves up are separated by two corrective waves. Then a three-wave correction (down) period with one wave up separating two down waves.
- In this section we introduce some Elliott wave patterns, rules and general tendencies as a foundation for starting to recognize waves.

Elliott Wave

Elliott wave analysis was founded and began to be publicized in the 1930s. Ralph Nelson Elliott, an accountant and author, started noticing a repeating pattern and psychology behind market trends. The basic pattern he saw time and again was five waves in one direction and then three waves moving counter to it. In an uptrend, five waves up would occur where two of those five waves were corrective. However, by making higher highs in each up wave and higher lows in each corrective wave, prices appreciated. After advancing multiple times, three corrective waves followed lower with one bounce to a lower high separating them. The blue line and "**1 2 3 4 5 a b c**" labels in Exhibit 99 is an example of this.

Exhibit 99: Typical Elliott wave structure
Three Elliott wave time frame counts including the primary wave I up and II down which is comprised of intermediate waves including 12345abc which is comprised of minor waves (lower case roman numerals)



Source: BofA Global Research

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Every wave is part of another wave

In the graphic above there are three wave counts occurring simultaneously representing the short-, medium- and long-term trends (Exhibit 99). The primary trend is labeled with large green roman numerals **I** and **II**. Presumably wave **II** is near an end and wave **III** up is beginning. The intermediate term trend is labeled with blue numerical digits including **12345** up and **abc** down. Presumably wave **c** is near an end (as is the larger degree wave **II**) and wave **1** of **III** up will soon begin. A short-term trend is shown as an example of what wave **3** up may have looked like on zooming in. The small italic lower case roman numerals *i, ii, iii, iv, v, a, b* and *c* count this pattern the same way.

Three rules to Elliott wave counts

Note: For simplicity, this section uses language for counting an uptrend wave pattern.

There are three rules to counting and tracking Elliott wave patterns, including:

Rule 1: Wave 2 cannot decline below the start of wave 1. If it did, price just made a lower low which is how a downtrend is defined.

Rule 2: Wave 3 is not the smallest, or in other words wave 3 is larger than 1 or/and 5.

Rule 3: Wave 4 does not overlap with wave 1. If it did, other corrective patterns may be present or wave 3 may not be complete yet. Over the years, some leniency has been offered to futures/leveraged markets.

The meaning behind each wave

When an up-wave count begins markets are most bearish. Wave **1** up is often a time where smart money is reducing their bearish position to a more neutral position and in some cases starting to build a long. Disbelief remains. Wave **2** tends to be a deep and sometimes sharp correction of wave **1**, such as 61.8% Fibonacci retracement, because markets are still trading with a bearish bias and retail flows press short again as disbelief lingers. However, wave **3** begins with a higher low (end of wave **2** above start of wave **1**) and an improving market backdrop starts to shift investor belief to buy.

Wave **3** tends to be where most investors become bullish and smart money is long. Those that went short in wave **2** are stopping out of their short when a new high above wave **1** is reached. Large investors are piling into longs and small/retail investors are less convinced or concerned about chasing the rally. Wave **3** up tends to be a larger period of appreciation creating a feeling of "missing out" on the move. Wave **3** is probably equal to if not larger than the size of **1** (such as 161.8% of **1**).

When wave **4** begins, it tends to be shallow or a consolidation of wave **3**, such as 26.4-38.2% retracement of wave **3**. It tends to appear more sideways than deeper like a wave **2**. This is because the "buy the dip" mentality is high, and the correction is used to pile in with remaining assets. While wave **4** is often a shallower correction than wave **2**, if wave **2** was shallow than wave **4** is often deeper. Corrective waves tend to alternate.

When wave **5** begins and new highs above wave **4** are reached, so-called smart money starts reducing longs because their gains are significant and sentiment is too bullish. This selling causes wave **5** to peak and a three-wave correction begins. Small/retail flows try to buy more in wave **a** down because they remain bullish. The wave **b** bounce back provides them with initial confidence, but typically a lower/equal high is made in wave **b** as smart money sells the bounce. Wave **c** down washes out all longs and turns positioning neutral.

Corrections inevitable and have more variation

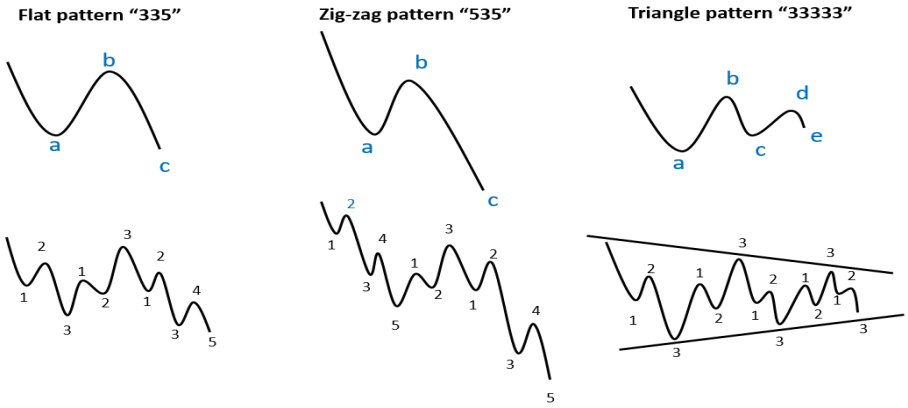
The corrective part of a wave pattern is typically three waves in the oppositive direction. After five waves up in (Exhibit 99), three waves down are labeled with an **abc** where **b** tends to be a lower high and **c** a lower low. Corrections can be shallow or deep and may depend on where the correction falls within the larger wave cycle. For example, the **abc** pattern forming wave **II** down may be deeper, such as a Zig-zag, whereas wave **IV** may be shallower like a flat or a triangle.

There are three types of flat patterns, however we'll discuss a regular flat. Flat patterns are referred to as a 335. This is because the pattern that forms the **abc** has three waves down to form wave **a**, three waves up to form wave **b** and five waves down to form wave **c**. An example of this can be seen on the left side of (Exhibit 100).

A Zig-zag pattern is referred to as a 535. This is because the underlying waves that form the **abc** structure show five waves down, three up and five more down. This pattern tends to be deeper. An example of this can be seen in the center of (Exhibit 100).

Then there is the triangle pattern, a true consolidation of price action that can represent a continuation of the prior trend if it is a wave **2** or **4** or in some cases a top or bottom of a wave. Triangles in Elliott wave have five waves labeled **abcde** with three waves comprising each smaller swing. We tend to trade with the breakout/breakdown. An example of this can be seen on the right side of (Exhibit 100).

Exhibit 100: Elliott wave corrective patterns
Three corrective patterns to know include the flat pattern, zig-zag pattern and triangle pattern.



Source: BofA Global Research

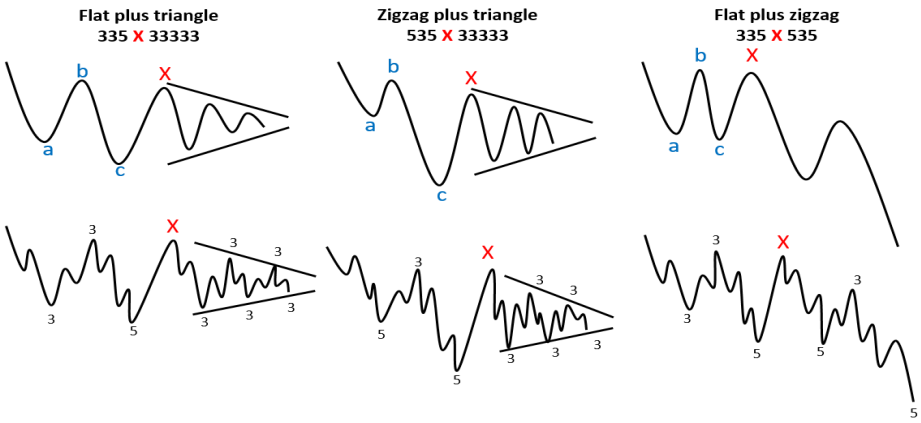
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Corrections can get complex

As fundamentals deteriorate, they can do so in stages with a brief period of hope or false rally initially convincing some market participants the correction is over. These patterns take some practice to identity because the hopeful rally connects two corrective patterns together to form a double correction. This isn't always obvious. In Exhibit 101 we label this wave of so-called hope as an "**X**" wave. It looks bullish and sometimes may even make a high above that of **b**, but ultimately it is too sharp a rally for a bull market and is said to be a "bear market rally" in hindsight.

The wave patterns following an "**X**" wave can be combined in a few ways. Below we show a flat plus a triangle pattern. In this case the market probably saw a 38.2% correction and consolidated into a triangle bottom. Zig-zag plus a triangle pattern probably saw a deeper correction such as 50% or 61.8% and needed to consolidate the emotion of a large decline before it could resume. A flat plus a Zig-zag is a deeper correction which lends itself to a wave 2.

Exhibit 101: Elliott wave double correction patterns
A "bear market rally" tends to create an **X** wave to start a second corrective period.



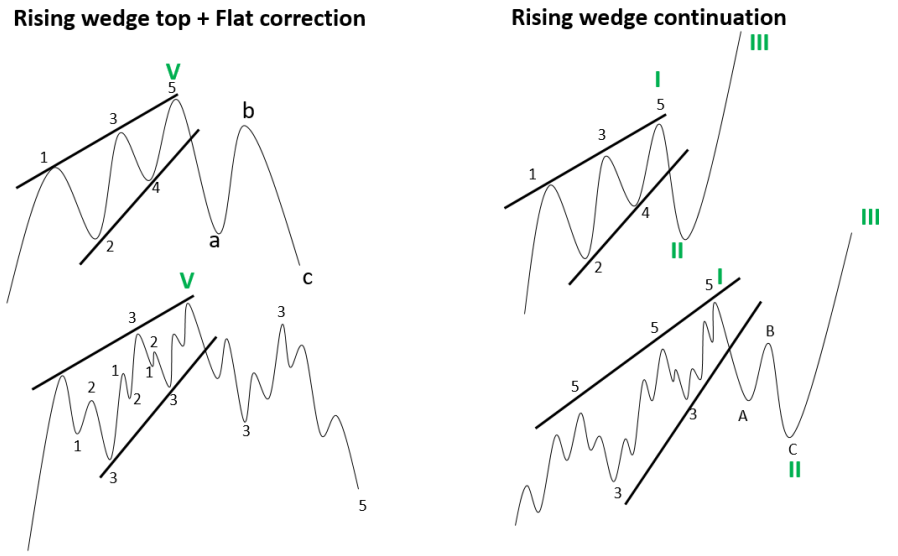
Source: BofA Global Research

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Wedge top and continuation patterns

A wedge pattern is comprised of two converging trend lines that are rising in an uptrend or falling in a downtrend. A rising wedge top, also called an ending diagonal, tends to represent a period of one-way bullish flows or a "melt-up" in prices. The flow is so strong that dips are small and fast until five waves up occur. Then price breaks down. It is typically comprised of five waves higher to end a larger wave count, such as a **V** wave. A good hint of a wedge top is if the underlying wave pattern shows 33333, like the triangle pattern. A continuation wedge pattern tends to occur earlier in a larger wave count to signal a corrective wave is coming. For example, wave **I** up could be a rising wedge leading to II down and III up. For this to be the case, the smaller degree count may present five waves up and three down like a typical up wave pattern. (Exhibit 102)

Exhibit 102: Elliott wave wedge patterns
Two converging trend lines with positive or negative slopes form wedge patterns.



Source: BofA Global Research

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Other topics to mention and review for another time

We think the initial basics of Elliott wave theory have been summarized in the last few pages to provide an introduction. However, we'd be remiss to not mention there are more details and possible wave structures including, but not limited to:

1) Fibonacci ratio measurements are important for estimating support, resistance and target levels. Please see Finding Fibonacci section of this primer for more on this.

2) A triple corrective pattern can occur meaning two "**X**" waves.

3) Flat structure can occur with some variation. In an expanding flat, wave B ends above the start of wave A and C ends below the end of wave A. In a running flat, B ends above the start of wave A and C ends above the end of wave A.

4) A growing tendency for some markets is to trend in three wave patterns.

5) How all the other content in this report can help identify wave counts and reversals. Candle patterns (Candles patterns light the way) may signal the end of a wave near a measured move target. The end of wave 3 may be the head of a A well massaged head and shoulders. RSI divergence is often present to signal the end of 5 or B in an expanding flat (Relative Strength Index is the most popular).

Finding Fibonacci

- The purpose of Fibonacci levels is to estimate support in a downtrend, resistance in an uptrend, estimate price targets and estimate time targets.
- The Fibonacci ratios are derived by the constant relationships found in the number sequence 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, etc where each number is added to the prior to get the next.
- When dividing these numbers by one another the important ratios of 61.8%, 50% and 38.2% are found. Secondary ratios of 76.4% and 23.6% are notable, too. However the most important ratio is 61.8% which ties to the golden mean.

Fibonacci numbers

The Fibonacci number sequence begins with a 0 and 1 and continues should you add the largest to the second largest. In this case, 0+1=1 so the sequence is 0, 1 and 1. Add the two largest together (1+1) and get 2. Then 2+1 is 3, 3+2 is 5 and so on where the never-ending sequence grows to 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, etc. The relationships between these numbers is studied in mathematics, architecture, science, and applied to charts in technical analysis. For example, 89 / 144 = .618 or 61.8%. This is the most significant Fibonacci ratio used in many technical measurements and Elliott wave theory. Similarly, 55 / 144 = .382 or 38.2%. This is another important level.

Why are the ratios relevant? Because they are everywhere.

The Fibonacci number pattern and subsequent ratios are found in many places in the world whether it be formed in nature or man-made. A few examples include many proportions of the human body such as head to waist and waist to feet. In architecture such as the design of the white house and Egyptian pyramids. In high-value art, such as the placement of Mona Lisa on the poplar board and the distance between her facial features. In nature such as the number of petals on different types of flowers, the spirals found in pinecones and seashells, rabbit reproduction, and even outer space. With these number and ratio patterns appearing in so many places, technical analysts use them as a guide to identify support, resistance and estimate targets based on prior price trends. After all, one of the three principles of technical analysis is history repeats itself.

Fibonacci retracements

A Fibonacci retracement is applied to an uptrend when a top is found in order to estimate levels price might retrace down to. When a bottom is found a Fibonacci retracement is applied to estimate how much of the prior trend price might retrace up to. Should a technical top be found, the idea is to estimate levels of support or downside targets for a short trade. If a bottom is found, the idea is to estimate levels of resistance or upside targets for a long trade. It is most important to draw retracements from major turning points (pivots) in trends, as opposed to random points.

For example, the S&P 500 (Exhibit 103) formed a head and shoulders top in 2007 with a high of 1576.09 and a head and shoulders bottom in 2009 with a low of 666.79. These are two distinct turning points in trend. The rally that followed in 2009-2010 was to the 61.8% retracement of 1228.74 of the 2008-2009 decline. This is calculated by taking .618 * (1576.09 - 666.79) + 666.79 = 1228. The April 2010 high was 1220 (close).

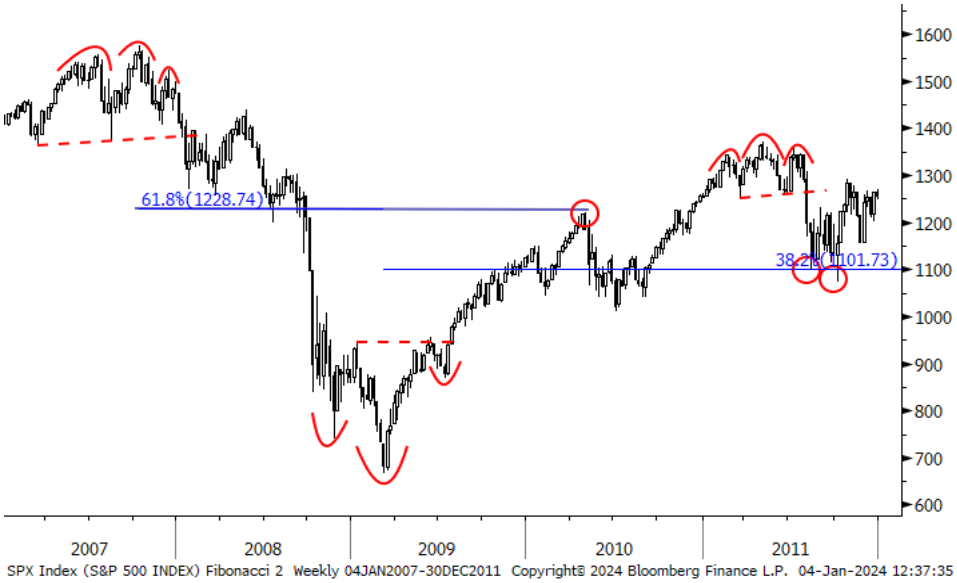
Another example is the head and shoulders bottom that formed in 2009 to the head and shoulders top in 2011. A Fibonacci retracement can be drawn from the low of 2009 to the high of 2011 and we see the 38.2% retracement of 1370.58 - .382 * (1370.58 - 666.79) = 1101.73. The first circled low was 1101.54. The second circled low was 1074 however price closed that week at 1155.

Another example could be the euro shown in Exhibit 104. A double top formed in 2008 leading to a downtrend and a long-range bottom from 2015-2017. The distance from the high in 2008 of 1.6038 to the low in 2017 of 1.0341 is 0.5697. 38.2% of 0.5697 is

0.2176. Add 0.2176 to the low of 1.0341 and an estimated retracement level of the prior downtrend is 1.2517. This is where price peaked in 2018. There was also a double top in 2014 at 1.3993 which led to a downtrend to 1.0341. A 61.8% retracement of this trend is 1.2598. Two of the primary Fibonacci levels point to strong resistance in the 1.25s.

The difficult or speculative part of this example may have been knowing the 2017 low was a big low. The market had just broken down below the 2015-2016 lows, paused with a doji and then a long legged doji for the final low. Once the technical feel was more bullish, some may begin to consider the 1.25s as a longer-term objective. For more on doji patterns, please see the "Candles light the way" section.

Exhibit 103: S&P 500 Fibonacci example 1
Apply Fibonacci from important turning points to estimate future levels of support and resistance.



Source: BofA Global Research, Bloomberg

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Exhibit 104: EUR/USD Fibonacci retracement example
Candle patterns may signal a Fibonacci line will hold.



Source: BofA Global Research, Bloomberg

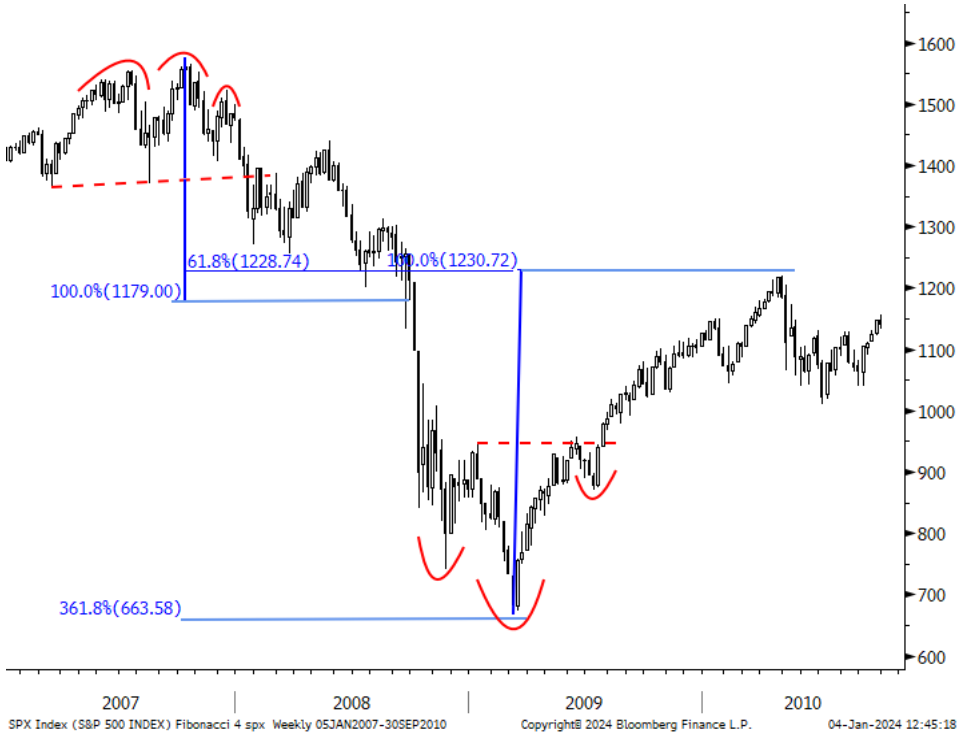
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Fibonacci projections and extensions

Fibonacci projections and extensions are often used to estimate targets for a trend based on the height of a price pattern, range breakout or during price discovery (when price is at new all-time highs or lows). The idea is to take the height of a price pattern, range or prior trend and add that to the breakout point.

In Exhibit 105 below we look at the head and shoulder patterns on the S&P 500 during the 2007 and 2009 period. For the top in 2007 we take the height of the top and subtract it from the breakdown point. For the bottom, we take the height of the pattern and add it to the break-up point. The expectation is to see a trend to the 100% target of 1179 and 1224 respectively. The unknown is if a larger projection factor should be applied. For example, 161.8% or 300% or even 361.8% projection. In the head and shoulders section of this report we go into more detail on this.

Exhibit 105: Fibonacci projections example
Use Fibonacci ratios to estimate multiple targets after a top or bottom pattern forms.



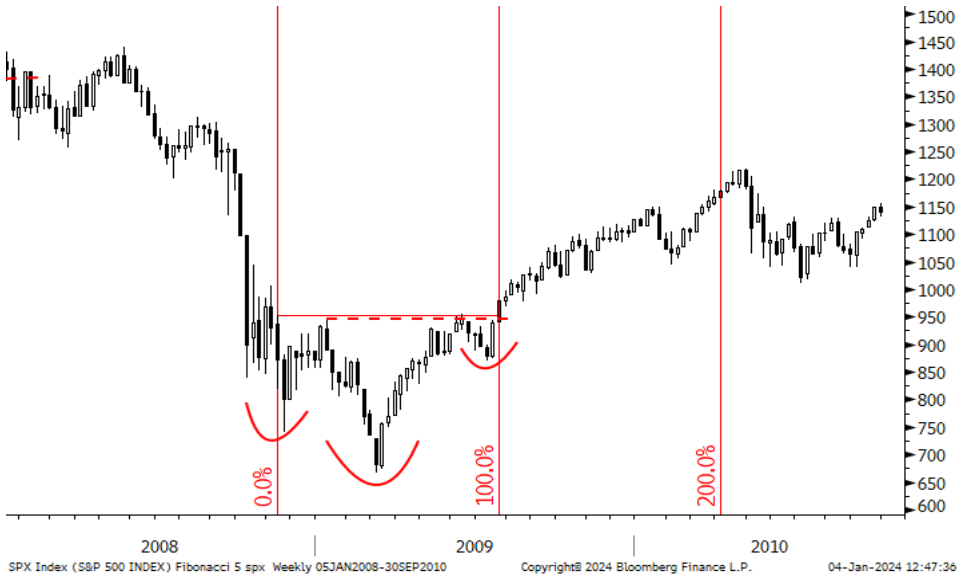
Source: BofA Global Research, Bloomberg

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When may a target be reached? Fibonacci time projections

The Fibonacci sequence and ratios can also be applied to time, not just price, as a guide for when a price target might be reached. In the following example we measure the width (amount of time) the head and shoulders bottom took to form in late 2008 through mid-2009 and found 38 weeks. Adding that distance to the breakout point suggests peaks may occur at the Fibonacci ratios of time, such as 161.8% and 200%, which in this case were pretty close to the actual peaks before the trend turned lower. (Exhibit 106)

Exhibit 106: Fibonacci time projections
Estimate when price might reach a target by measuring the width of a pattern with Fibonacci ratios.



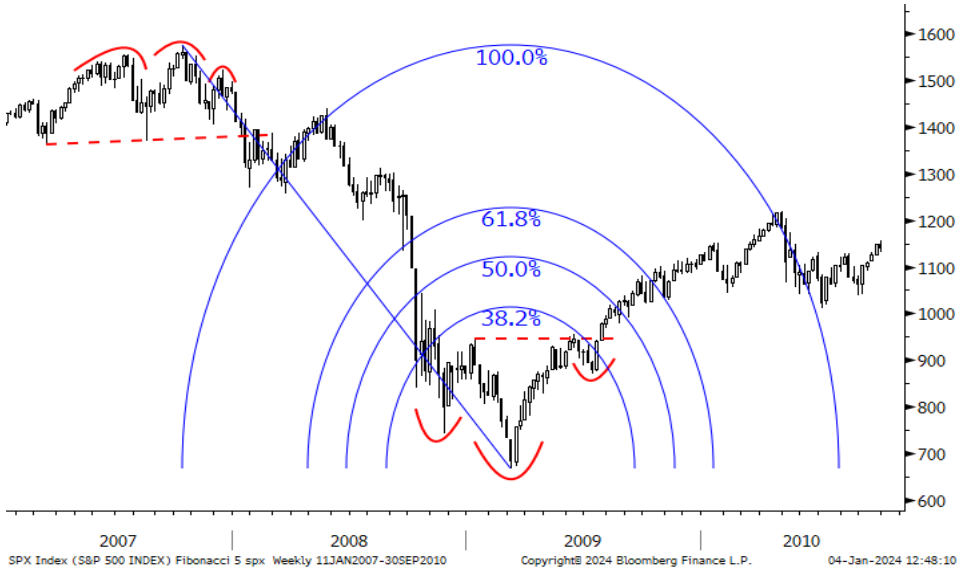
Source: BofA Global Research, Bloomberg

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Fibonacci arcs project price and time

If you want to estimate what price level might be resistance for an uptrend and when that resistance may be found, you can use the Fibonacci arc tool. By choosing the high of the head and shoulders in 2007 and the low of the head and shoulders bottom in 2009 (just like a retracement), the arcs are drawn. The highest point of each arc represents a Fibonacci retracement level. So the 61.8% arc peak shown in Exhibit 107 is at 1228 which is the 61.8% retracement point in Exhibit 103. Time is equally spread on each side and as it turns out the 100% arc time and the 61.8% retracement is where the S&P 500 peaked in June 2010.

Exhibit 107: Fibonacci arc example
Estimate price and time levels by using Fibonacci arcs.



Source: BofA Global Research, Bloomberg

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Use RRG as your relative rotation radar

Sector Rotation

- Relative Rotation Graphs (RRG) visualize relative strength across a group of securities against an index or benchmark.
- RRG gauges relative trend & momentum, showing winners & losers in 4 quadrants: Leading, Weakening, Lagging & Improving.
- Absolute and relative price charts can be used to explore findings in the RRG chart.

Visualize relative rotation through RRG

Relative Rotation Graphs™ (RRG) allow investors to visualize relative strength across a group of securities versus an index or benchmark. RRG gives investors a relative rotation snapshot across a group of securities in just one chart. While most often used to show relative rotation among sectors, RRG can also show rotation for a portfolio of stocks as well as across asset classes.

Relative strength analysis

Before addressing RRG, we need to define relative strength. Relative strength analysis uses a ratio chart to compare the price performance of two securities, such as a stock vs. an index. When the relative ratio rises, the stock outperforms the index, which is bullish for the stock. When it falls, the stock underperforms, which is bearish for the stock.

RRG shows relative winners & losers in 4 quadrants

RRG compares relative strength on a scatter-plot chart using relative trend strength (JdK RS-Ratio) on the x-axis and relative trend momentum (JdK RS-Momentum) on the y-axis. RRG charts also show a trail of history to track relative rotation over time. The RRG graph breaks out relative rotation into four quadrants: Leading, Weakening, Lagging, and Improving. We view the relative trend as more important than relative momentum. This means that regardless of momentum, the right side of the RRG chart shows relative leaders and the left side shows relative laggards.

- **Upper right quadrant = Leadership** (positive relative trend and positive relative momentum) = RS-Ratio and RS-Momentum > 100
- **Lower right quadrant = Weakening Leadership** (positive relative trend and negative relative momentum) = RS-Ratio > 100 and RS-Momentum < 100
- **Lower left quadrant = Laggards** (negative relative trend and negative relative momentum) = RS-Ratio and RS-Momentum < 100
- **Upper left quadrant = Improving Laggards** (negative relative trend and positive relative momentum) = RS-Ratio < 100 and RS-Momentum > 100

Exhibit 108: The four quadrants of the Relative Rotation Graph (RRG)
The four quadrants for the RRG moving in a clockwise direction are Improving, Leading, Weakening and Lagging.

Relative momentum (JdK RS-Momentum)		
	Improving = Negative Trend & Positive Momentum	**Leading = Positive Trend & Positive Momentum**
	Lagging = Negative Trend & Negative Momentum	**Weakening = Positive Trend & Negative Momentum**
	Relative trend (JdK RS Ratio)	

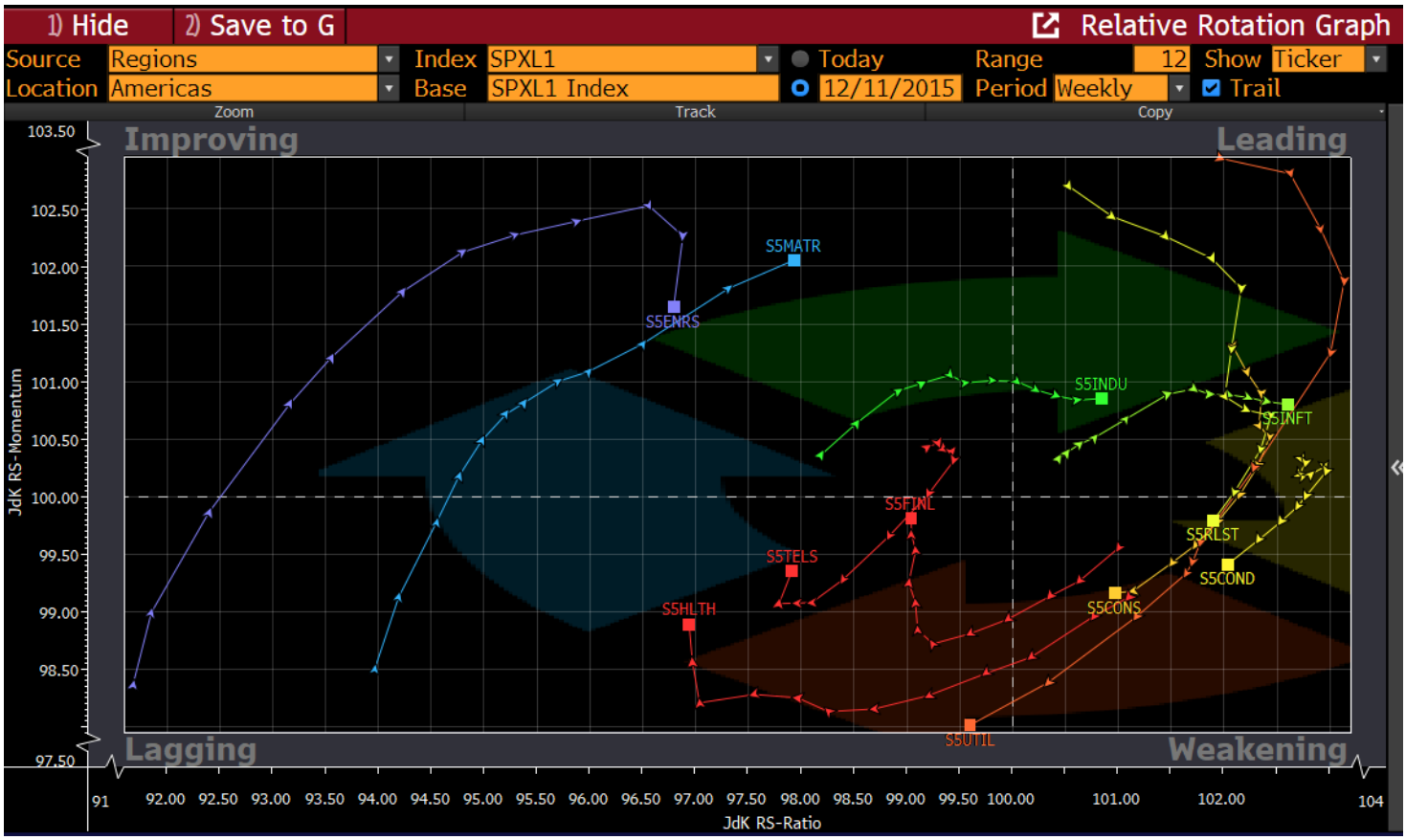
Source: BofA Global Research, Bloomberg, RRG Research

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Relative rotation in action

Exhibit 109 shows a RRG chart for the S&P 500 GICs Level 1 Sectors vs. the S&P 500 benchmark with a 12-week trail. Each point on the trail represents one week and the squares mark the current or most recent week. This allows investors to visualize how the 10 GICs sectors have done over a rolling 12-week period. Moving up and to the right on the RRG graph shows improving relative strength and momentum. This is viewed as bullish. Moving down and to the left shows deteriorating relative strength and momentum and is interpreted as bearish.

Exhibit 109: Bloomberg Relative Rotation Graph (RRG) – weekly as of 12/11/15
RRG Exhibit for the S&P 500 GICs Level 1 Sectors vs. the S&P 500 benchmark with a 12-week trail.



Source: BofA Global Research, Bloomberg, RRG Research. RRGs were created by Julius de Kempenaer in 2004-2005 and Relative Rotation Graphs™ and RRG Charts™ are registered trademarks of RRG Research. For more please see www.relativerotationgraphs.com or New Frontiers in Technical Analysis (by Ciana, P. John Wiley & Sons Inc. 2011).

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Relative rotation graph (RRG) takeaways

- Leading sectors: positive relative trends and momentum (upper right quadrant): Information Technology (S5INFT) and Industrials (S5INDU)
- Lagging sectors: negative relative trends and momentum (lower left quadrant): Health Care (S5HLTH), Telecom (S5TELS), Financials (S5FINL) and Utilities (S5UTIL)
- Improving sectors: negative relative trends but positive relative momentum (upper left quadrant): Energy (S5ENRS) and Materials (S5MATR)
- Weakening sectors: positive relative trends but negative relative momentum (lower right quadrant): Consumer Discretionary (S5COND) and Consumer Staples (S5CONS)
- Bullish rotation (improving relative trend and momentum) over the last 12 weeks in order of relative trend strength from strongest to weakest: Information Technology, Industrials, Materials, and Energy

- Bearish rotation (deteriorating relative trend and momentum) over the last 12 weeks in order of relative trend strength from weakest to strongest: Health Care, Telecom, Utilities, Consumer Staples, and Consumer Discretionary
- Relative trend ranks (strongest to weakest): Information Technology, Consumer Discretionary, Consumer Staples, Industrials, Utilities, Financials, Materials, Telecom, Health Care, and Energy
- Relative momentum ranks (strongest to weakest): Materials, Energy, Industrials, Information Technology, Financials, Consumer Discretionary, Telecom, Consumer Staples, Health Care, and Utilities

Absolute and relative charts = powerful combination

While using the Relative Rotation Graph is helpful to determine shifts in relative performance vs a benchmark, the next step is to look at the ratio in an absolute chart format and apply technical theory like trend lines, moving averages, support, resistance patterns, breakouts etc. to confirm relative trend.

How to read a relative price chart

A ratio chart divides the price of one security by another. Ratio charts identify a stock's relative strength or weakness vs a benchmark. This is also known as comparative relative strength and is not to be confused with the Relative Strength Index or RSI, which is a price momentum indicator. When the relative chart is rising, the stock is outperforming and shows relative strength or leadership. When the relative chart is declining, the stock is underperforming and shows relative weakness or is a laggard. Since a flat relative trend means that a stock is performing in line with the market, it is not bearish for the stock if the market is in a strong uptrend and gained 20% over the last year.

Exhibit 110: Marriot (MAR) relative to the S&P 500
Marriot's relative ratio Exhibit compares the price of MAR relative to a benchmark (S&P 500).



Source: BofA Global Research, Bloomberg

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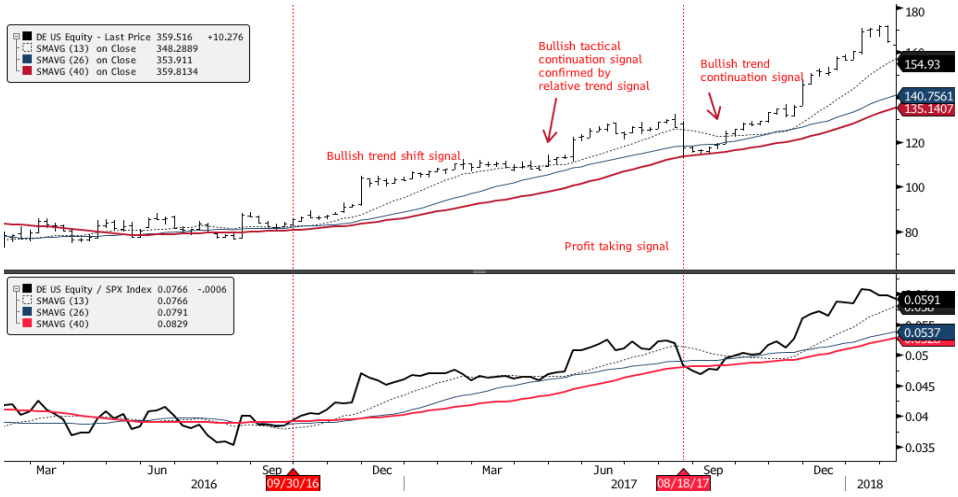
Absolute and relative charts: Trends, leaders and laggards

The strongest technical pattern is a bullish absolute price trend confirmed by a bullish relative (or leadership) trend. The weakest pattern is a bearish absolute price trend confirmed by a bearish relative (or laggard) trend. We use the same techniques on absolute and relative price charts but tend to be more flexible on the relative chart with a greater emphasis on the trends of the relative moving averages.

Bullish absolute and relative set-ups for Deere (DE)

The chart below shows the weekly absolute and relative charts for DE highlighting bullish signals using the absolute and relative price trends. DE had a bullish signal from late 2016 prior to a profit taking signal in August 2017. This profit taking signal did not break the bullish trend (rising 26 and 40-week MAs), and DE embarked on a yearend 2017 into 2018 rally. The late September move back above rising 26 and 40-week MAs provided this bullish signal.

Exhibit 111: Deere (DE) (top) and relative to the S&P 500 (bottom) – weekly Exhibit with MAs
This Exhibit highlights bullish signals using the absolute and relative price trends.



Source: BofA Global Research, Bloomberg

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Bearish absolute and relative set-ups for Qualcomm (QCOM)

Exhibit 112 shows QCOM highlighting bearish signals using the absolute and relative price trends. The relative chart broke down in advance of the absolute chart in mid 2014. A rally for QCOM in late 2014 above the 40-week MA did not get confirmed by a move above the 40-week MA relative to the S&P 500, which was a bearish divergence. After these negative relative signals, QCOM was a bearish momentum stock from late 2014 into 2016, especially relative to the S&P 500.

Exhibit 112: QUALCOMM (QCOM) (top), relative to the S&P 500 (bottom) – weekly Exhibit with MAs
This Exhibit highlights bearish signals using the absolute and relative price trends.



Source: BofA Global Research, Bloomberg

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Relative reversals can precede or lead absolute reversals

Investors like to pick tops and bottoms. Tops or bottoms in the relative price trend often provide a leading indicator of tops (bearish reversals) or bottoms (bullish reversals) in the absolute price trend.

Bearish relative reversal (top) signals in absolute uptrends

Relative price signals that increase the risk for a top in an absolute price uptrend are 1) relative price diverges on lower highs as absolute price hits higher highs, 2) the relative price chart moves lower as the stock trends higher and 3) relative price is below weekly MAs while absolute price is above its corresponding weekly MAs.

Bearish relative signals preceded an absolute top on AutoZone (AZO)

The chart below shows AZO with a bearish relative divergence (lower relative highs vs higher absolute highs), a relative price breakdown in the face of a stronger absolute trend and a move above absolute price MAs without a corresponding move above the relative MAs (also a bearish divergence). This relative deterioration suggested getting more defensive on AZO in mid to late 2016 and preceded an early 2017 top breakdown for AZO. This stock was leadership in 2015.

Exhibit 113: AutoZone (AZO) (top) and relative to the S&P 500 (bottom) – weekly with MAs
A bearish shift on the relative Exhibit was a leading indicator for a top on the absolute Exhibit.



Source: BofA Global Research, Bloomberg

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Bullish relative reversal (bottom) in absolute downtrends

Relative price signals that increase the risk for a bottom in an absolute price downtrend are 1) relative price diverges on higher lows as absolute price hits lower lows, 2) the relative price chart moves higher as the stock trends lower and 3) relative price is above weekly MAs while absolute price is below its corresponding weekly MAs.

Bullish relative signals preceded an absolute bottom on Cisco Systems (CSCO)

The chart below shows CSCO with a bullish relative divergence (higher relative lows vs lower absolute lows), a relative price upside breakout in the face of a weaker absolute trend and a move below absolute price MAs without a corresponding move below the relative MAs (also a bullish divergence). This relative improvement suggested getting more bullish (or less bearish) on CSCO in early 2009 and preceded a breakout and rally for the stock from early 2009 into 2010.

Exhibit 114: Cisco Systems (CSCO) (top) and relative to the S&P 500 (bottom) – weekly with MAs
A bullish shift on the relative Exhibit was a leading indicator for a bottom on the absolute Exhibit.



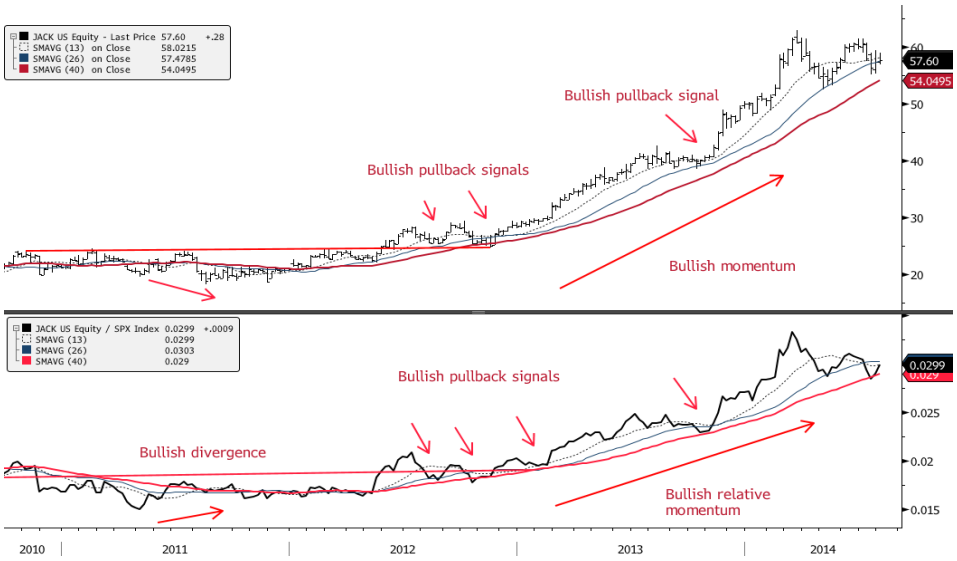
Source: BofA Global Research, Bloomberg

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Bullish momentum stock chart

The chart below shows Jack in the Box (JACK) transitioning from a bearish period to a bullish period. A higher relative low for JACK vs the S&P 500 did not confirm a lower low for JACK in late 2011. This formed a bullish divergence that was part of a 1.5 year bottom or base for JACK. Bullish absolute and relative breakouts in mid 2012 and bullish pullback signals in 2H 2012 preceded a bullish momentum phase for JACK from 2013 and into 2014. JACK had a bullish trend reversion signal in late 2013.

Exhibit 115: Jack in the Box (JACK) (top) and vs the S&P 500 (bottom) – weekly Exhibit with MAs
Bullish momentum phase preceded by bullish absolute and relative breakouts and retests.



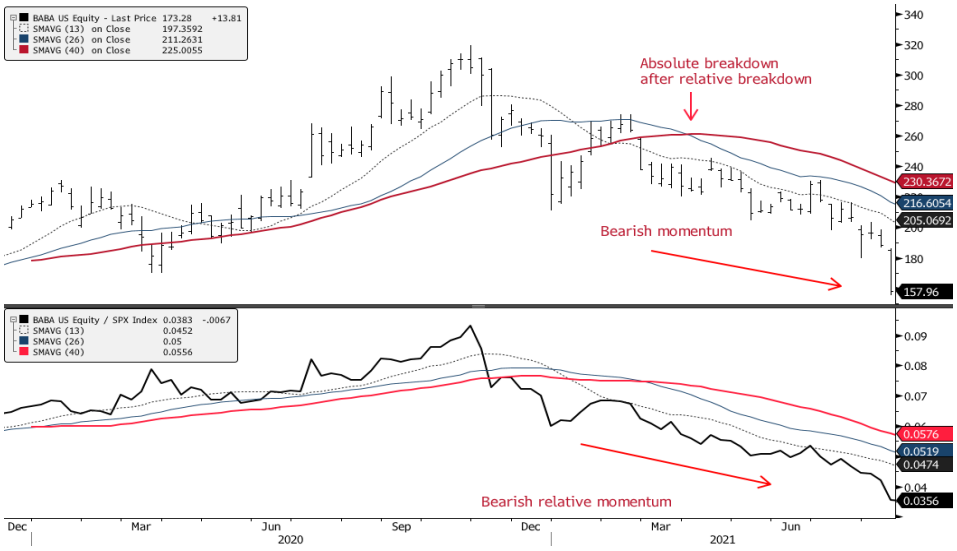
Source: BofA Global Research, Bloomberg

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Bearish momentum stock chart

The chart below shows Alibaba (BABA) going from bullish momentum in 2020 to bearish momentum in 2021. A relative breakdown in late 2020 provided a warning. Bearish momentum has pressured this stock throughout much of 2021 with tactical (and sellable) rallies into declining weekly MAs.

Exhibit 116: Alibaba (BABA) (top) and vs the S&P 500 (bottom) – weekly Exhibit with MAs
Bearish momentum has pressured BABA throughout 2021 with tactical rallies into bearish MAs.



Source: BofA Global Research, Bloomberg

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Weekly MAs and relatives boost chart pattern confidence

Weekly MAs can help increase confidence in reversal patterns (tops and bottoms) and trend continuation patterns. CVS formed a 2015-2016 top. Weakening absolute price MAs, bearish MA signals and a relative price breakdown increased the risk for a top breakdown in 2016. A weak relative trend throughout 2017 suggested that CVS was consolidating within a bearish trend (not forming a bottom) and likely to break lower. Investors short CVS during this period generated alpha relative to the S&P 500.

Exhibit 117: CVS Health Corp (CVS) (top) and vs the S&P 500 (bottom) – weekly Exhibit with MAs
Weekly MAs can help show reversal patterns (tops and bottoms) and trend continuation patterns.



Source: BofA Global Research, Bloomberg

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Seasonality: Average macro market trends

The average trends over various lookback periods

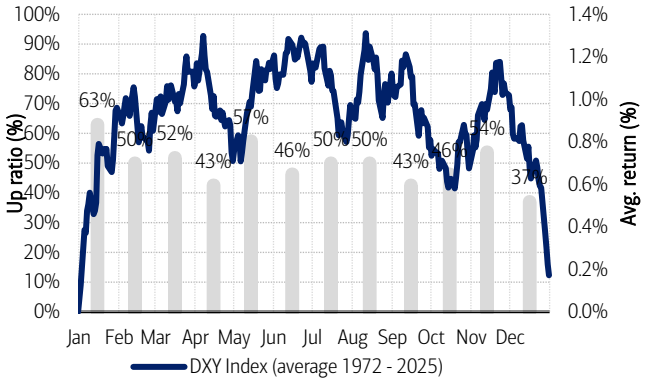
Seasonality trends for the DXY, Gold, Oil, US 10Y and S&P500

In this section we recap the average trend of the top five macro assets in four ways. The first chart looks back over the longest term we find data is reasonable. The second chart looks at the same lookback but only year 2 (Y2) of the US Presidential Cycle (USPC), since 2026 is Y2. The third chart looks back since the euro began trading in 1999. The fourth chart looks back at the last ten years.

DXY: Strong Q1, April dip, May rip, summer weak, Sep-Nov bounce, down Dec

Since 1972, the DXY was up in Q1 with an average gain of 1.06% while in year 2 of the USPC it ended flat. In Y2, April, October and December are weakest months while Jan and May stronger. Over the last 10 years, February, September and October are the strongest months while December tends to be a down month, also since 1999.

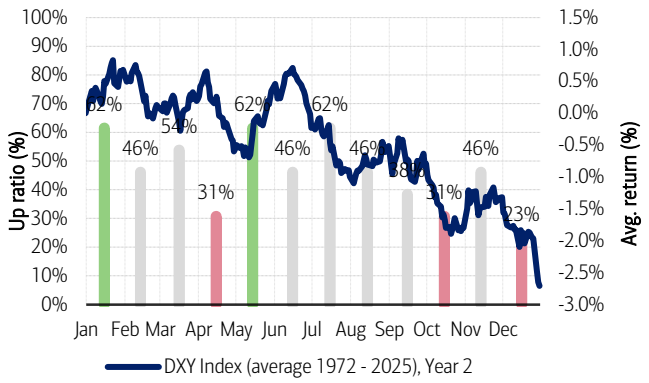
Exhibit 118: DXY trend since 1972
Since 1972, the DXY tended to go up into early April, dip into mid-May, rally into June. A choppy summer, bearish Sept-Oct and weak December.



Source: BofA Global Research, Bloomberg

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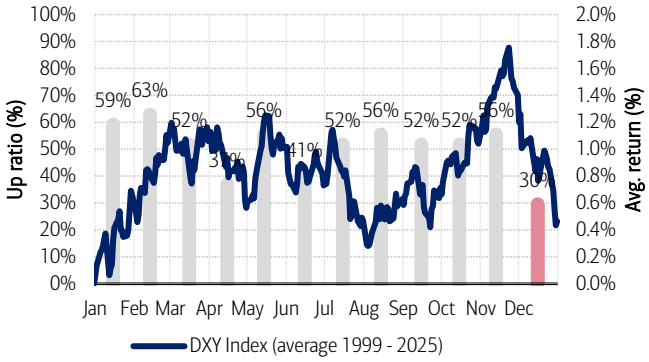
Exhibit 119: DXY average trend in Y2 of USPC was sideways in 1H and down in 2H. April, Oct and Dec stand out as weak. Jan & May stronger.
Average trend in Year 2 of USPC and up ratio.



Source: BofA Global Research, Bloomberg

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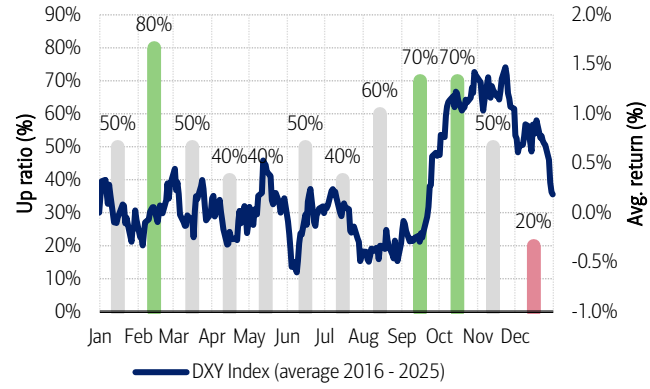
Exhibit 120: DXY trend since euro began in 1999
The DXY has tended to be up in Q1, traded mostly sideways in March-June with a move lower in July then rally back to new high through November and decline into yearend.



Source: BofA Global Research, Bloomberg

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Exhibit 121: DXY trend over the last ten years
Bullish February, September and October. DXY down 8 of 10 times in December.



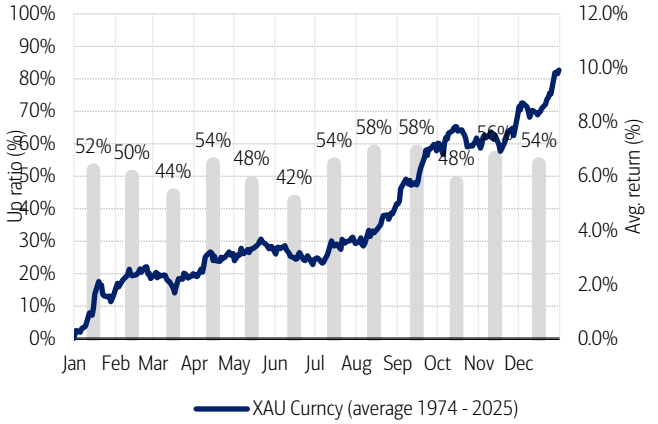
Source: BofA Global Research, Bloomberg

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Gold: Average trend is up with strong H2

Since 1974, gold's average trend tended to go up into December by about 9.9%. Thrusts higher occurred in January, Aug-Oct and Dec. Gold's trend in Y2 of the USPC is bullish. It rallied from Jan to mid-May, Aug-Sep and gain at yearend. It corrected from mid-May until July. Bullish up ratios in January, August, November and December. Since the euro began trading, April has been the most bullish month for gold up 67% of the time. Over the last ten years, bullish months have been January, April, July and December while September tended to be bearish i.e. down 70% of the time.

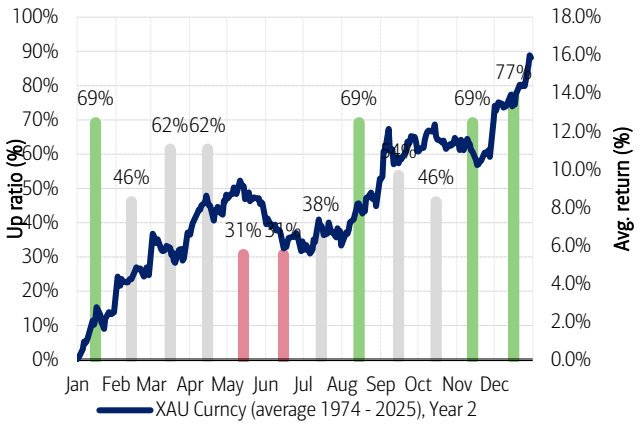
Exhibit 122: Gold average trend since 1974
The average trend tends to be up with the strongest part of the trend from July through October.



Source: BofA Global Research, Bloomberg

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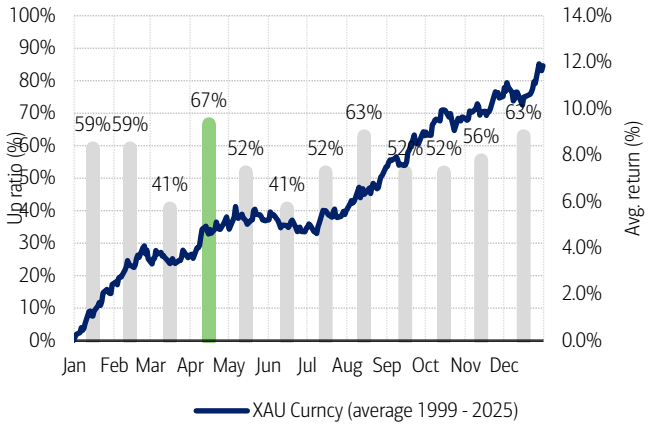
Exhibit 123: Gold's trend in Y2 is bullish. It rallied from Jan to mid-May, Aug-Dec. Gold corrected from mid-May until July.
Average trend in Year 2 of USPC and up ratio since 1974



Source: BofA Global Research, Bloomberg

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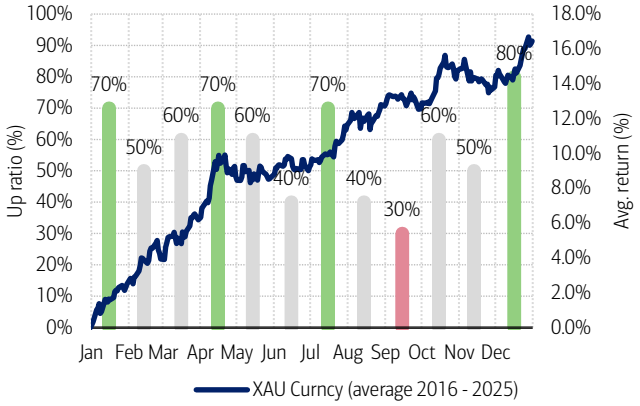
Exhibit 124: Gold average trend since euro began trading
Gold's trend has tended to be up with its strongest up ratio in April of 67%.



Source: BofA Global Research, Bloomberg

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Exhibit 125: Gold trend over the last ten years
Gold prices have tended to rally through April, trade sideways in May & June, rally in July, dip in September and move up into yearend with bullish December



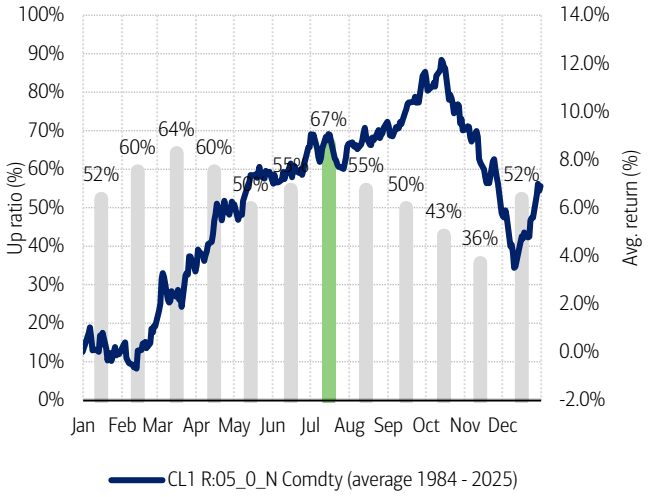
Source: BofA Global Research, Bloomberg

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Oil: Strong from Feb to mid-October then down through Q4

Since 1984, oil prices tended to rise from late February through mid-May and again in July-mid October, then they declined into yearend. In Y2 of the USPC, oil tended to rise in Mar-Apr, trade sideways in the summer, rally in September and decline into YE with bearish October and November. Over the last twenty and ten years, oil prices have peaked earlier in the year (July) with notable up ratios in June then see larger swings sideways in 2H over the last ten years.

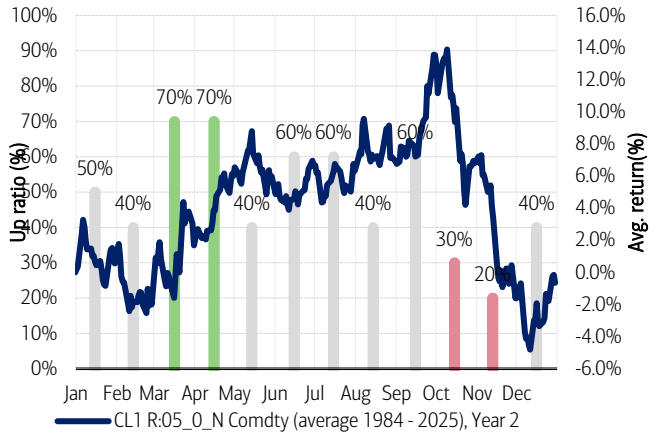
Exhibit 126: WTI crude oil average trend (5d roll adjust)
Up from March to mid-October with bullish July, then down through Q4.



Source: BofA Global Research, Bloomberg

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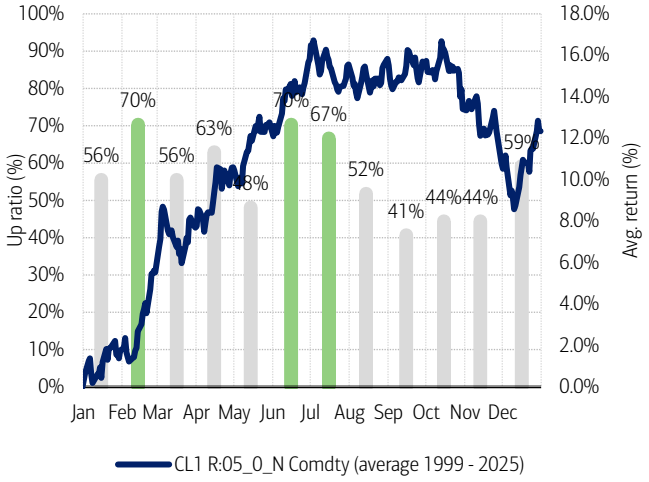
Exhibit 127: Of the last ten Y2 of the USPC, oil tended to rise in Mar-Apr,
trade sideways in the summer, see one more rally in Sept and decline into YE.
Average trend in Year 2 of USPC and up ratio (Ten observations)



Source: BofA Global Research, Bloomberg

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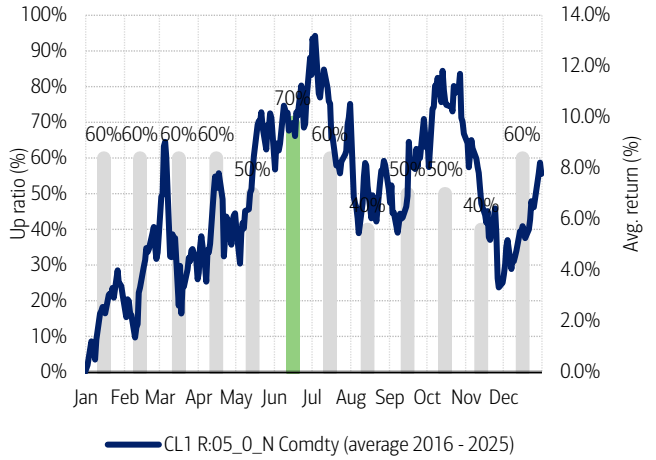
Exhibit 128: Oil price trend since euro began trading in 1999
Oil prices have tended to rally in the 1H, turn sideways in Q3 and decline in Q4. Notable up ratios have been in February, June and July



Source: BofA Global Research, Bloomberg

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Exhibit 129: Oil price trend over the last ten years
Oil prices have tended to rally in the 1H and peak by July. Swings in the 2H are noticeably larger than the last twenty years. The best month for oil has been June i.e. up 7 out of 10 times.



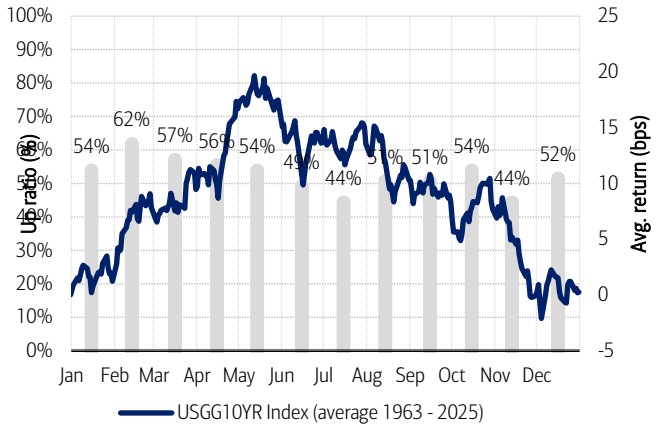
Source: BofA Global Research, Bloomberg

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US 10-year yield: Up into May, down into yearend

Since 1963, the average trend for US 10Y yield is up into May and down into year end. In Y2 of USPC, US 10Y yield rose in Q1, topped in Q2 and declined in 2H. Jan-Mar have been weakest months while June saw favorable statistics i.e. yield down for 67% of the time. Since 1999, the average trend tended to be bullish in June and bearish in October while over last ten years, June was struggling i.e. yield down for 30% of the time.

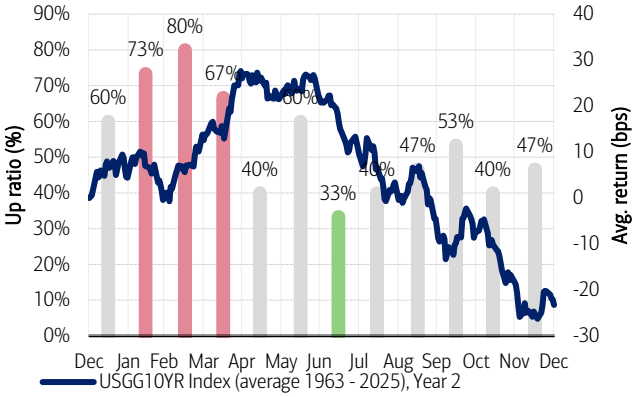
Exhibit 130: US 10Y yield since 1963
Yield tended to rise into May, peak and decline into yearend.



Source: BofA Global Research, Bloomberg

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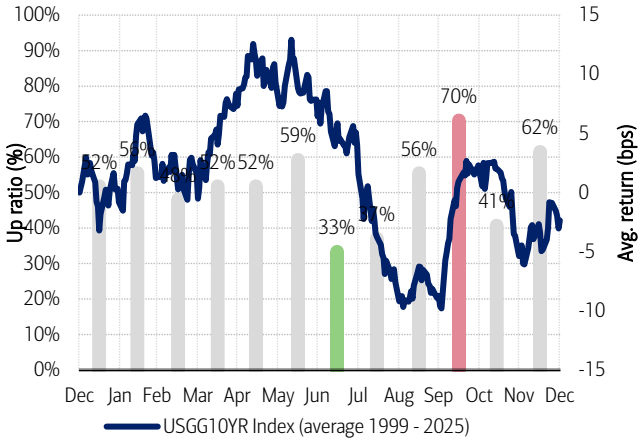
Exhibit 131: In Y2 of USPC, 10Y yield rose in Q1, topped in Q2 and declined through the 2H of the year.
Average trend in Year 2 of USPC and up ratio



Source: BofA Global Research, Bloomberg

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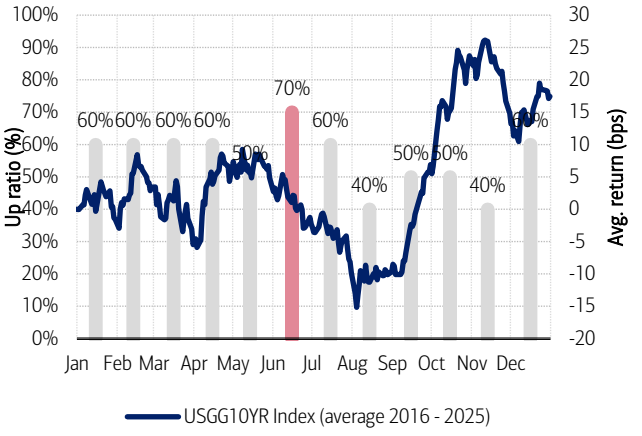
Exhibit 132: US 10Y yield since euro began trading in 1999
Yield has tended to rise into May, decline through September with bullish July, bounce back in Oct-Nov and decline in December. October tended to be bearish



Source: BofA Global Research, Bloomberg

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Exhibit 133: US 10Y yield over last ten years
Yield traded choppy in the 1H, decline from a peak in May to a low in early August, rose into November and dipped into year end. June tended to be a bearish month.



Source: BofA Global Research, Bloomberg

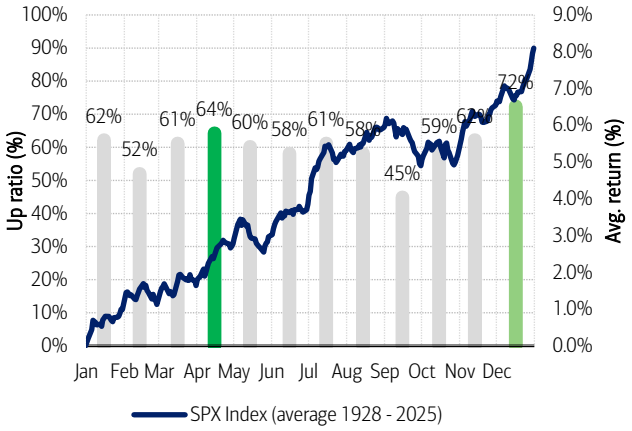
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S&P 500: Average uptrend with dips in Feb, May, Sept and with strong Q4

Since 1928, the S&P 500 tended to rise for the year by about 8%. The months that tended to go up have been April 64% of the time and December 72% of the time. The weakest month tended to be September i.e. up only 45% of the time. In year 2 of the USPC, SPX peaked in April, trended lower until end Q3, then rallied into EOY with bullish October and November. Since the euro began trading, the S&P 500 had a bullish March-May period and strong Q4. Over the last ten years, SPX has been up in July 100% of the time.

Exhibit 134: S&P 500 since 1928

The S&P 500 tends to trend up for the year with corrections in February, May and September-October. The best month tends to December up 72% of the time. The second-best month is April i.e. up 64% of the time

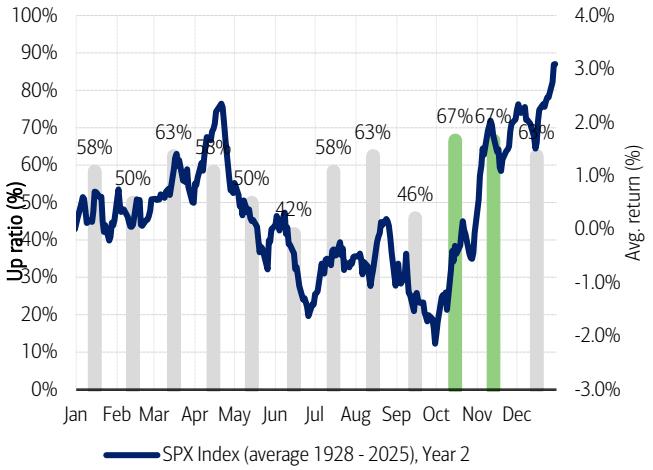


Source: BofA Global Research, Bloomberg

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Exhibit 135: In Y2 of USPC, SPX peaked in April, trended lower until end Q3, then rallied into EOY. Up ratios were bullish in Oct, Nov.

Average trend in Year 2 of USPC and up ratio

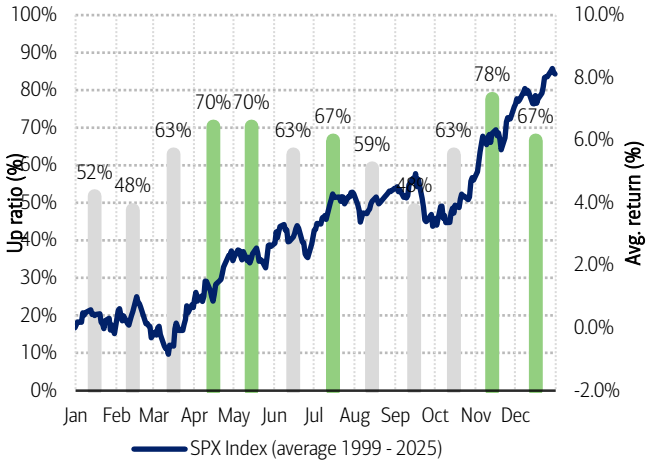


Source: BofA Global Research, Bloomberg

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Exhibit 136: S&P 500 since euro began in 1999

Strong trend and up ratios in March-May. Strong Q4 led by November and December.

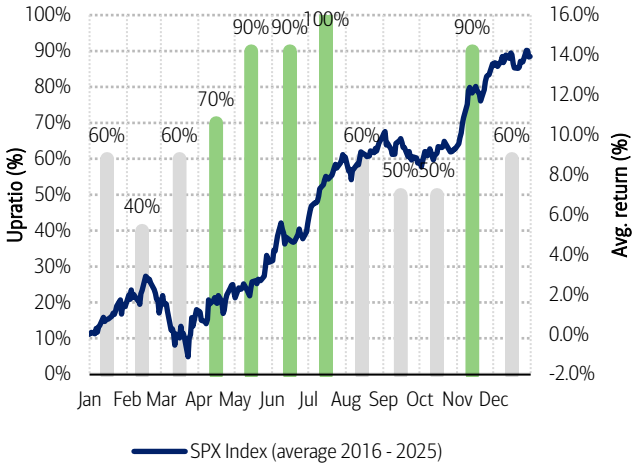


Source: BofA Global Research, Bloomberg

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Exhibit 137: S&P 500 over the last ten years.

The SPX has tended to rally in April-July with July up 100% of the time. Supportive up ratio at year end i.e. November.



Source: BofA Global Research, Bloomberg

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Seasonality Heatmap

Average monthly macro trends

In this section we recap average trend in macro assets by week and month. This includes the DXY, gold, oil, US 10Y yield and S&P 500.

January favors DXY and SPX

In January, the DXY and SPX are supported. The DXY was up most often in week 4, rising 76% of time by about 0.45%. DXY was up in week 1 about 66% of time by 0.39%. SPX was up in January 63% of time by 1.23% with support in weeks 1-2. US 10Y yield tended to decline in week 2 65% of time by -2bps.

February modestly favors higher US 10Y yield

In February, there were few strong up ratios. US 10Y yield was up in February 62% of the time by about 6bps. On average, oil and gold were up.

March favors higher oil, and SPX strength toward month end

In March, oil prices were up 64% of the time by about 1.80%. Support for SPX may be seen later in the month such as weeks 2-4 when it was up 64% of the time by 0.59%.

Exhibit 138: Q1 macro trends. January favors DXY and SPX. February favors higher US 10Y yield. March favors higher oil.
Average percent change in prices for SPX, DXY, commodities and basis points for US 10Y yield by week(s) and months. Green highlights a higher bias for prices and lower for yields. Red highlights a lower bias for prices and higher for yields.

Average return and up ratio		Week 1	Week 2	Week 3	Week 4	Week 1-2	Week 1-3	Week 2-3	Week 2-4	Week 3-4	Month
January	DXY	0.39% (66%)	0.16% (57%)	-0.02% (48%)	0.45% (76%)	0.56% (64%)	0.56% (60%)	0.14% (63%)	0.51% (61%)	0.38% (52%)	0.99% (63%)
	Gold	0.42% (57%)	0.85% (51%)	-0.07% (55%)	-0.14% (43%)	1.31% (51%)	1.08% (57%)	0.65% (55%)	0.61% (52%)	0.29% (48%)	1.55% (52%)
	Oil	0.18% (57%)	-0.65% (48%)	0.36% (52%)	0.09% (47%)	-0.56% (52%)	-0.28% (60%)	-0.37% (48%)	0.09% (56%)	0.00% (47%)	0.07% (52%)
	US 10Y	1.15 (53%)	-1.83 (35%)	2.53 (54%)	-2.64 (49%)	-0.70 (44%)	1.83 (51%)	0.70 (49%)	-1.68 (47%)	-0.08 (42%)	0.13 (54%)
	SPX	0.36% (58%)	0.20% (57%)	0.04% (53%)	0.07% (54%)	0.58% (62%)	0.62% (60%)	0.24% (56%)	0.10% (56%)	-0.06% (54%)	1.23% (63%)
February	DXY	0.08% (50%)	-0.31% (44%)	0.03% (52%)	0.06% (50%)	-0.23% (44%)	-0.20% (46%)	-0.27% (46%)	0.14% (54%)	0.13% (54%)	-0.09% (50%)
	Gold	0.11% (45%)	0.25% (61%)	0.18% (53%)	-0.65% (42%)	0.36% (47%)	0.55% (49%)	0.44% (61%)	0.16% (50%)	-0.25% (38%)	0.47% (50%)
	Oil	-0.68% (40%)	0.30% (55%)	0.42% (55%)	-0.29% (53%)	-0.37% (48%)	0.07% (50%)	0.78% (55%)	-0.51% (53%)	-0.38% (47%)	0.54% (60%)
	US 10Y	3.43 (57%)	2.17 (54%)	0.37 (51%)	-5.09 (38%)	5.60 (52%)	5.97 (56%)	2.54 (56%)	-4.16 (52%)	-7.01 (41%)	6.26 (62%)
	SPX	-0.04% (59%)	0.21% (59%)	-0.03% (49%)	-0.33% (47%)	0.17% (57%)	0.14% (52%)	0.19% (52%)	0.16% (60%)	-0.10% (58%)	-0.08% (52%)
March	DXY	0.12% (63%)	0.06% (44%)	0.03% (56%)	0.03% (57%)	0.18% (54%)	0.21% (57%)	0.10% (46%)	0.08% (48%)	-0.01% (54%)	0.19% (52%)
	Gold	-0.05% (63%)	-0.39% (55%)	0.09% (61%)	0.23% (56%)	-0.45% (41%)	-0.36% (45%)	-0.30% (49%)	-0.08% (47%)	0.36% (47%)	0.05% (44%)
	Oil	0.18% (55%)	-0.07% (60%)	0.17% (62%)	1.02% (64%)	0.19% (57%)	0.52% (62%)	0.28% (62%)	0.67% (56%)	0.83% (56%)	1.80% (64%)
	US 10Y	2.39 (60%)	-0.52 (46%)	0.39 (46%)	1.56 (50%)	1.87 (54%)	2.25 (51%)	-0.14 (51%)	-0.03 (46%)	0.43 (50%)	3.04 (57%)
	SPX	-0.12% (53%)	0.19% (57%)	-0.03% (50%)	0.13% (48%)	0.20% (61%)	0.18% (58%)	0.24% (62%)	0.59% (64%)	0.21% (56%)	0.52% (61%)

Source: BofA Global Research, Bloomberg. Note: End year is 2025 while start of period is in brackets: DXY (1972), XAU (1974), CL1 R:05_0_N (1984), USGG10YR (1963), SPX (1928).

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April favors higher SPX, at times higher US 10Y yield and oil

In April, the SPX was up 64% of the time by about 1.22%. US 10Y yield tended to rise in the second half of April, such as weeks 3-4 up about 69% of the time by about 7bps. Oil tended to rally in week 2-4 up 73% of the time by about 2.55%.

May up ratios more neutral, spots of strength for DXY and oil

The month of May saw few strong up ratios overall. During the month, oil tended to be higher in weeks 1-2 about 64% of time by 2.36%. The DXY was up in week 2 about 69% of the time on average by 0.34%.

June up ratios mostly neutral, some oil strength

In June, monthly up ratios were not significant. However during the month oil tended to rally in weeks 2-4 about 74% of the time by 1.73%. In Week 2, the US 10Y yield declined about 73% of the time by about 4bps.

Exhibit 139: Q2 macro trends. April sees higher SPX, possibly US 10Y yield & oil. May and June less significant, intramonth moves favor oil.
Average percent change in prices for SPX, DXY, commodities and basis points for US 10Y yield by week(s) and months. Green highlights a higher bias for prices and lower for yields. Red highlights a lower bias for prices and higher for yields.

Average return and up ratio		Week 1	Week 2	Week 3	Week 4	Week 1-2	Week 1-3	Week 2-3	Week 2-4	Week 3-4	Month
April	DXY	-0.09% (44%)	-0.14% (44%)	-0.19% (50%)	0.09% (51%)	-0.23% (44%)	-0.41% (50%)	-0.33% (43%)	-0.05% (56%)	0.04% (62%)	-0.35% (43%)
	Gold	0.75% (61%)	0.04% (49%)	0.20% (49%)	0.32% (44%)	0.80% (55%)	1.01% (59%)	0.23% (55%)	0.39% (53%)	0.06% (33%)	0.68% (54%)
	Oil	0.33% (52%)	1.79% (62%)	-0.47% (50%)	0.53% (53%)	2.08% (60%)	1.54% (57%)	1.09% (57%)	2.55% (73%)	1.14% (60%)	2.78% (60%)
	US 10Y	-0.36 (44%)	-0.96 (54%)	5.56 (62%)	2.46 (62%)	-1.32 (54%)	4.24 (60%)	4.60 (60%)	6.65 (62%)	6.91 (69%)	7.22 (56%)
	SPX	0.39% (58%)	0.44% (62%)	0.30% (58%)	-0.11% (57%)	0.84% (64%)	1.17% (58%)	0.75% (66%)	1.01% (61%)	0.41% (54%)	1.22% (64%)
May	DXY	-0.07% (46%)	0.34% (69%)	0.03% (57%)	0.03% (59%)	0.27% (59%)	0.30% (61%)	0.37% (59%)	0.35% (52%)	-0.01% (50%)	0.48% (57%)
	Gold	0.09% (53%)	0.39% (53%)	0.09% (43%)	-0.18% (48%)	0.48% (53%)	0.55% (53%)	0.46% (53%)	0.13% (50%)	-0.12% (50%)	0.11% (48%)
	Oil	1.19% (62%)	1.01% (62%)	-0.09% (40%)	0.14% (53%)	2.36% (64%)	2.37% (52%)	0.93% (50%)	1.56% (58%)	0.68% (53%)	2.36% (50%)
	US 10Y	1.51 (52%)	0.13 (48%)	-0.39 (54%)	-1.26 (45%)	1.64 (56%)	1.25 (52%)	-0.26 (48%)	-1.96 (47%)	-1.89 (42%)	-0.91 (54%)
	SPX	0.39% (59%)	-0.52% (36%)	-0.25% (49%)	0.26% (61%)	-0.14% (50%)	-0.38% (54%)	-0.77% (46%)	-0.42% (55%)	0.12% (57%)	0.08% (60%)
June	DXY	0.08% (46%)	-0.02% (43%)	0.06% (54%)	-0.33% (41%)	0.05% (48%)	0.11% (52%)	0.03% (44%)	-0.52% (36%)	-0.36% (38%)	-0.16% (46%)
	Gold	-0.17% (43%)	-0.04% (55%)	0.03% (55%)	-0.17% (44%)	-0.20% (49%)	-0.18% (43%)	-0.02% (47%)	-0.25% (50%)	-0.22% (39%)	-0.31% (42%)
	Oil	0.06% (43%)	0.27% (48%)	0.05% (52%)	1.44% (65%)	0.31% (52%)	0.31% (48%)	0.24% (45%)	1.73% (74%)	1.84% (61%)	1.35% (55%)
	US 10Y	1.39 (51%)	-3.99 (27%)	2.52 (46%)	0.75 (47%)	-2.60 (43%)	-0.07 (46%)	-1.47 (44%)	-1.29 (44%)	4.17 (58%)	-2.04 (49%)
	SPX	0.43% (57%)	-0.03% (52%)	-0.02% (50%)	0.13% (54%)	0.40% (51%)	0.39% (61%)	-0.05% (50%)	-0.12% (48%)	-0.12% (48%)	0.82% (58%)

Source: BofA Global Research, Bloomberg. Note: End year is 2025 while start of period is in brackets: DXY (1972), XAU (1974), CL1 R:05_0_N (1984), USGG10YR (1963), SPX (1928).

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July favors higher oil, lower DXY, sometimes higher SPX

In July, oil prices were up 67% of the time by about 0.51% with strength in week 2 most evident. The DXY tended to decline later in the month such as weeks 2-4. It was up just 35% of the time. The SPX rallied 70% of the time in week 1 by about 0.76%.

August up ratios mostly neutral, intramonth oil strength, DXY weakness

In August most up ratios were near 50%. However, in weeks 2-4 the DXY was lower 65% of the time by about -0.24% and oil was up 72% of the time by about 1.31%.

September up ratios mostly neutral, intramonth favors gold and oil

In September, most up ratios were near 50%. However intramonth strength was seen in gold during weeks 3-4 rising 70% of the time by about 1.29%. Also, oil prices rose in week 4 71% of the time by about 1.55%.

Exhibit 140: Q3 macro trends. July favors oil. August and September low conviction, intramonth favored oil and gold.
Average percent change in prices for SPX, DXY, commodities and basis points for US 10Y yield by week(s) and months. Green highlights a higher bias for prices and lower for yields. Red highlights a lower bias for prices and higher for yields.

Average return and up ratio		Week 1	Week 2	Week 3	Week 4	Week 1-2	Week 1-3	Week 2-3	Week 2-4	Week 3-4	Month
July	DXY	0.13% (57%)	-0.14% (41%)	-0.24% (39%)	-0.06% (43%)	-0.01% (48%)	-0.25% (50%)	-0.38% (43%)	-0.63% (35%)	-0.40% (35%)	-0.16% (50%)
	Gold	0.27% (61%)	0.19% (55%)	0.29% (57%)	0.21% (49%)	0.48% (57%)	0.76% (61%)	0.48% (59%)	0.80% (58%)	0.32% (51%)	0.56% (54%)
	Oil	-0.32% (45%)	0.99% (67%)	-0.92% (45%)	0.18% (47%)	0.61% (57%)	-0.28% (40%)	0.05% (48%)	0.83% (53%)	-0.28% (53%)	0.51% (67%)
	US 10Y	0.20 (41%)	-1.60 (43%)	2.58 (54%)	-1.31 (50%)	-1.40 (48%)	1.18 (51%)	0.98 (48%)	-1.85 (46%)	0.73 (41%)	-0.21 (44%)
	SPX	0.76% (70%)	0.52% (59%)	-0.27% (50%)	0.21% (54%)	1.28% (67%)	1.00% (62%)	0.26% (58%)	0.44% (56%)	-0.14% (57%)	1.68% (61%)
August	DXY	0.23% (61%)	0.07% (52%)	-0.27% (43%)	0.14% (52%)	0.31% (59%)	0.03% (50%)	-0.20% (43%)	-0.24% (35%)	-0.23% (39%)	0.21% (50%)
	Gold	0.32% (63%)	0.16% (45%)	0.67% (57%)	-0.17% (43%)	0.51% (57%)	1.16% (49%)	0.83% (47%)	0.99% (52%)	0.78% (55%)	1.38% (58%)
	Oil	-0.44% (50%)	0.29% (52%)	0.24% (52%)	0.68% (69%)	-0.08% (40%)	0.23% (48%)	0.55% (52%)	1.31% (72%)	0.81% (64%)	1.73% (55%)
	US 10Y	2.61 (57%)	-3.37 (43%)	-1.97 (38%)	1.95 (48%)	-0.76 (60%)	-2.73 (51%)	-5.34 (40%)	-3.80 (46%)	0.19 (44%)	-3.27 (51%)
	SPX	0.04% (49%)	0.14% (59%)	0.29% (56%)	0.18% (54%)	0.19% (54%)	0.50% (57%)	0.44% (59%)	0.63% (57%)	0.52% (55%)	0.68% (58%)
September	DXY	0.19% (52%)	-0.01% (48%)	-0.13% (48%)	-0.16% (50%)	0.19% (50%)	0.06% (43%)	-0.14% (46%)	-0.41% (45%)	-0.42% (48%)	-0.30% (43%)
	Gold	0.27% (49%)	-0.04% (51%)	0.53% (65%)	0.63% (59%)	0.25% (47%)	0.80% (53%)	0.50% (69%)	1.07% (59%)	1.29% (70%)	1.73% (58%)
	Oil	-0.20% (55%)	0.98% (64%)	-0.66% (48%)	1.55% (71%)	0.79% (52%)	0.13% (45%)	0.32% (55%)	0.61% (61%)	0.50% (61%)	1.45% (50%)
	US 10Y	-0.30 (49%)	1.00 (54%)	0.00 (49%)	-0.64 (38%)	0.70 (57%)	0.70 (54%)	1.00 (52%)	2.11 (53%)	-0.88 (47%)	-1.99 (51%)
	SPX	-0.18% (49%)	-0.19% (56%)	-0.31% (49%)	-0.57% (42%)	-0.37% (50%)	-0.69% (48%)	-0.53% (52%)	-1.02% (48%)	-0.74% (46%)	-1.12% (45%)

Source: BofA Global Research, Bloomberg. Note: End year is 2025 while start of period is in brackets: DXY (1972), XAU (1974), CL1 R:05_0_N (1984), USGG10YR (1963), SPX (1928).

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October up ratios mostly neutral, intramonth may support US 10Y yield

In October, up ratios were mostly around 50%, or a lower conviction month. Intramonth, such as week 1-2, the US 10Y yield rose about 3bps 62% of the time. The SPX rose in week 1 65% of the time by 0.46%.

November favors lower oil and higher S&P 500

In November, oil prices tended to decline about 64% of the time by about -2.97%. In weeks 2-4, oil was down 83% of the time by about -3.72%. The S&P 500 tended to be higher in week 1 and week 4 with the month ending up about 1.00% 62% of the time.

December favors higher S&P 500 and lower DXY

In December, the S&P 500 was up 72% of the time by about 1.27% and the DXY was down 63% of the time by about -0.86%. Intramonth shows support for oil, such as weeks 2-3 up 69% of the time by about 1.15%.

Exhibit 141: Q4 macro trends. October trends show low convictions, possibly higher yield in week 1-2 and SPX in week 1. In November, oil was lower and SPX higher. In December, DXY was lower and SPX higher.

Average percent change in prices for SPX, DXY, commodities and basis points for US 10Y yield by week(s) and months. Green highlights a higher bias for prices and lower for yields. Red highlights a lower bias for prices and higher for yields.

Average return and up ratio		Week 1	Week 2	Week 3	Week 4	Week 1-2	Week 1-3	Week 2-3	Week 2-4	Week 3-4	Month
October	DXY	-0.11% (41%)	-0.10% (43%)	0.21% (50%)	0.04% (59%)	-0.20% (48%)	0.01% (50%)	0.11% (48%)	0.19% (59%)	0.28% (61%)	0.00% (46%)
	Gold	0.42% (55%)	0.21% (59%)	-0.46% (51%)	-0.11% (39%)	0.62% (53%)	0.17% (49%)	-0.23% (57%)	-0.62% (41%)	-0.77% (43%)	0.22% (48%)
	Oil	-0.60% (40%)	0.73% (52%)	-1.20% (43%)	-0.50% (40%)	0.08% (45%)	-1.12% (48%)	-0.51% (50%)	-0.92% (40%)	-1.83% (37%)	-2.08% (43%)
	US 10Y	1.23 (60%)	1.49 (51%)	1.73 (57%)	-1.09 (51%)	2.72 (62%)	4.46 (60%)	3.23 (54%)	1.98 (53%)	0.28 (51%)	-1.34 (54%)
	SPX	0.46% (65%)	0.00% (61%)	-0.14% (53%)	0.12% (51%)	0.45% (64%)	0.34% (57%)	-0.14% (53%)	-0.15% (54%)	0.04% (57%)	0.56% (59%)
November	DXY	0.06% (54%)	0.14% (50%)	0.14% (59%)	-0.08% (47%)	0.20% (52%)	0.35% (52%)	0.28% (56%)	0.21% (53%)	0.20% (61%)	0.23% (54%)
	Gold	0.24% (59%)	-0.29% (49%)	0.62% (59%)	-0.34% (47%)	-0.05% (51%)	0.56% (55%)	0.32% (57%)	0.13% (47%)	0.27% (44%)	0.92% (56%)
	Oil	-0.52% (40%)	-1.44% (33%)	0.04% (52%)	-1.83% (40%)	-1.92% (33%)	-1.83% (48%)	-1.41% (50%)	-3.72% (17%)	-2.21% (30%)	-2.97% (36%)
	US 10Y	-1.14 (43%)	-1.71 (44%)	-3.66 (40%)	-2.48 (38%)	-2.85 (44%)	-6.51 (35%)	-5.36 (33%)	-8.05 (40%)	-6.56 (40%)	-6.99 (44%)
	SPX	0.71% (69%)	-0.19% (46%)	-0.21% (55%)	0.63% (64%)	0.51% (59%)	0.31% (56%)	-0.38% (54%)	0.17% (60%)	0.28% (59%)	1.00% (62%)
December	DXY	0.02% (56%)	-0.05% (50%)	-0.19% (37%)	-0.47% (28%)	-0.02% (50%)	-0.21% (46%)	-0.23% (46%)	-0.81% (32%)	-0.71% (28%)	-0.86% (37%)
	Gold	-0.42% (35%)	-0.12% (55%)	0.44% (67%)	1.22% (70%)	-0.55% (45%)	-0.11% (47%)	0.32% (59%)	1.75% (66%)	1.88% (73%)	1.12% (54%)
	Oil	-1.71% (43%)	0.87% (57%)	0.33% (62%)	1.05% (64%)	-0.96% (50%)	-0.62% (55%)	1.15% (69%)	1.85% (64%)	1.04% (56%)	0.48% (52%)
	US 10Y	3.75 (63%)	-1.55 (55%)	-0.31 (50%)	-0.48 (55%)	2.20 (53%)	1.88 (53%)	-1.86 (53%)	-3.57 (47%)	-3.08 (45%)	0.32 (51%)
	SPX	0.21% (62%)	-0.18% (45%)	0.21% (65%)	0.56% (69%)	0.01% (55%)	0.21% (64%)	0.03% (59%)	0.38% (60%)	0.76% (65%)	1.27% (72%)

Source: BofA Global Research, Bloomberg. Note: End year is 2025 while start of period is in brackets: DXY (1972), XAU (1974), CL1 R:05_0_N (1984), USGG10YR (1963), SPX (1928).

BofA GLOBAL RESEARCH

U.S. Presidential Cycle

- 2026 is the second year (Y2) of the U.S. Presidential Cycle (USPC), a four-year cycle. Since it is President Trump's second term (T2), some may consider it the sixth year.
- The typical stride of a four-year cycle suggests year 2 is the weakest, year 1 is less weak, both underperforming all years. Year 3 tends to be the strongest with year 4 right behind it, both outperforming all years. In a four-year cycle, the SPX was up 76% of the time (29 of 38), by 29.5% on average and 24% median. (Exhibit 142)
- There is a divergence between year 2 and year 6. The average trend in term 1 year 2 is the weakest (-1.2% on average and up just 44% of the time) while term 2 year 2, or year 6, is strong (+11.9% on average and up 85% of the time). See Exhibit 143. President Cleveland's term 2 year 2 underperformed like a term 1 year 2.

S&P 500 and the US Presidential Cycle

The S&P 500 tended to be strongest in year 3, 4 and 6 of the U.S. Presidential Cycles. In fact, in year 4 and year 6 since 1872, the S&P 500 was up 85% of the time with average and median gains in the 10-13% area. Year 4 also showed the lowest standard deviation.

Exhibit 142: S&P 500 tends to underperform in Year 2 and 1 while outperforming in year 3 and 4.
S&P 500 U.S. Presidential Cycle since 1872

	Average	Median	Up ratio	Std Dev
Year 1 & 5	5.3%	7.4%	62% (24 / 39)	19.7%
Year 2 & 6	3.3%	2.0%	58% (22 / 38)	17.1%
Year 3 & 7	10.7%	13.1%	71% (27 / 38)	19.1%
Year 4 & 8	6.8%	9.0%	72% (28 / 39)	16.5%
Four-year term	29.5%	24%	76% (29 / 38)	46.3%
All years	6.5%	7.55%	66% (101 / 154)	18.1%

Source: BofA Global Research, Bloomberg, Fineon

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Exhibit 143: Since 2026 is the second year of Trumps second term, or year 6, note this year tended to outperform.
S&P 500 U.S. Presidential Cycle since 1872

	Average	Median	Up ratio	Std Dev
Year 1	9.0%	9.1%	68% (17 / 25)	16.8%
Year 2	-1.2%	-2.9%	44% (11 / 25)	15.9%
Year 3	13.3%	17.6%	72% (18 / 25)	20.4%
Year 4	10.5%	11.0%	85% (22 / 26)	13.0%
Term 1 Year	7.9%	7.4%	67% (68 / 101)	17.3%
Year 5	-1.4%	-4.9%	50% (7 / 14)	23.1%
Year 6	11.9%	13.6%	85% (11 / 13)	16.6%
Year 7	5.5%	3.5%	69% (9 / 13)	16.0%
Year 8	-0.6%	-2.3%	46% (6 / 13)	20.6%
Term 2 Year	3.8%	8.5%	62% (33 / 53)	19.6%
Term 1 & 2	6.5%	7.5%	66% (101 / 154)	18.2%

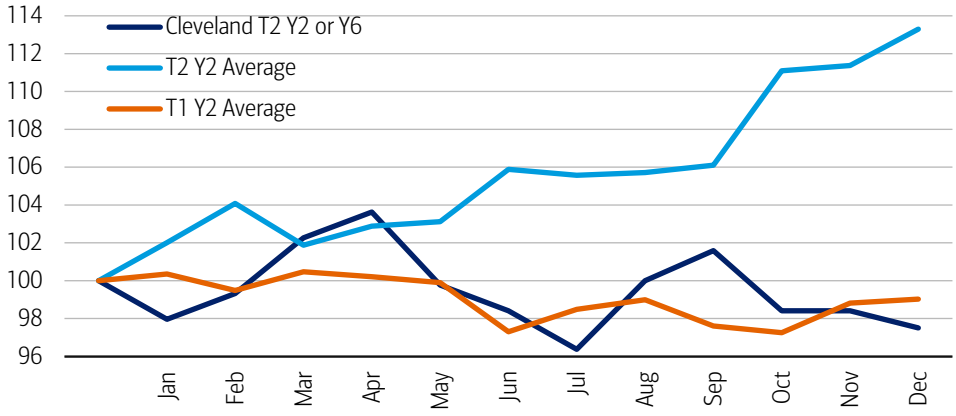
Source: BofA Global Research, Bloomberg, Fineon

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Is 2026 year 2 or year 6 of USPC? Your guess is as good as mine.

Only once has a US President served a second term non-consecutively. In 1893-1897, President Cleveland served a second non-consecutive term, or term 2 year 2 (T2 Y2). Then, the S&P 500 traded sideways and finished down about -2.5%. This was in line with how year 2 of term 1 trades. However, a consecutive term 2 year 2 is quite bullish.

Exhibit 144: Cleveland's second year 2 (year 6) correlates more to a term 1 year 2, not term 2 year 2
S&P500 trend of Cleveland's second year 2 vs average term 1 year 2 vs average term 2 year 2



Source: BofA Global Research, Bloomberg, Fineon

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S&P 500 returns for all US Presidencies back to 1872

The table below shows all the Presidential terms going back to 1872.

Exhibit 145: All Presidential terms going back to 1872
US Presidents and S&P 500 returns by year of term

President	Party	Term	Years	Year 1 Return	Year 2 Return	Year 3 Return	Year 4 Return	Full term Return	Ran for reelection	Reelected
Ulysses S. Grant	Republican	1	1869-1872				6.81%		Yes	Yes
Ulysses S. Grant	Republican	2	1873-1876	-12.75%	2.81%	-4.10%	-17.95%	-29.41%	No	
Rutherford B. Hayes	Republican	1	1877-1880	-9.37%	6.13%	42.96%	18.69%	63.19%	No	
James A. Garfield / Chester A. Arthur	Republican	1	1881-1884	2.98%	-2.89%	-8.51%	-18.84%	-25.74%	No	
Grover Cleveland	Democrat	1	1885-1888	19.83%	8.46%	-6.56%	-2.47%	18.45%	Yes	No
Benjamin Harrison	Republican	1	1889-1892	3.50%	-13.53%	17.61%	1.85%	7.20%	Yes	No
Grover Cleveland	Democrat	1	1893-1896	-19.96%	-2.49%	0.47%	-2.31%	-23.41%	No	
William McKinley	Republican	1	1897-1900	12.56%	18.95%	6.55%	14.12%	62.80%	Yes	Yes
William McKinley / Theodore Roosevelt	Republican	1	1901-1904	15.72%	1.26%	-18.39%	25.57%	20.09%	Yes	Yes
Theodore Roosevelt	Republican	2	1905-1908	15.64%	3.14%	-33.23%	37.44%	9.45%	No	
William Howard Taft	Republican	1	1909-1912	14.06%	-12.14%	0.66%	2.96%	3.88%	Yes	No
Woodrow Wilson	Democrat	1	1913-1916	-14.29%	-8.58%	28.98%	3.38%	4.48%	Yes	Yes
Woodrow Wilson	Democrat	2	1917-1920	-30.61%	16.44%	13.97%	-24.51%	-30.49%	No	
Warren G. Harding / Calvin Coolidge	Republican	1	1921-1924	7.43%	20.90%	-1.46%	18.76%	51.98%	Yes	Yes
Calvin Coolidge	Republican	1	1925-1928	23.25%	5.72%	30.91%	37.88%	135.20%	No	
Herbert Hoover	Republican	1	1929-1932	-11.91%	-28.48%	-47.04%	-15.19%	-71.71%	Yes	No
Franklin D. Roosevelt	Democrat	1	1933-1936	46.62%	-5.99%	41.51%	27.83%	149.36%	Yes	Yes
Franklin D. Roosevelt	Democrat	2	1937-1940	-38.64%	25.33%	-5.43%	-15.32%	-38.42%	Yes	Yes
Franklin D. Roosevelt	Democrat	2	1941-1944	-17.86%	12.43%	19.45%	13.80%	25.53%	Yes	Yes
Harry Truman	Democrat	1	1945-1948	30.72%	-11.85%	0.00%	-0.66%	14.47%	Yes	Yes
Harry Truman	Democrat	2	1949-1952	10.25%	21.78%	16.46%	11.81%	74.84%	No	
Dwight D. Eisenhower	Republican	1	1953-1956	-6.65%	45.02%	26.40%	2.62%	75.60%	Yes	Yes
Dwight D. Eisenhower	Republican	2	1957-1960	-14.31%	38.06%	8.48%	-2.97%	24.51%	No	
John F. Kennedy / Lyndon B. Johnson	Democrat	1	1961-1964	23.13%	-11.81%	18.89%	12.97%	45.84%	Yes	Yes
Lyndon B. Johnson	Democrat	1	1965-1968	9.06%	-13.09%	20.09%	7.66%	22.55%	No	
Richard M. Nixon	Republican	1	1969-1972	-11.36%	0.10%	10.79%	15.63%	13.66%	Yes	Yes
Richard Nixon / Gerald Ford	Republican	2	1973-1976	-17.37%	-29.72%	31.55%	19.15%	-8.97%	Yes	No
Jimmy Carter	Democrat	1	1977-1980	-11.50%	1.06%	12.31%	25.77%	26.34%	Yes	No
Ronald Reagan	Republican	1	1981-1984	-9.73%	14.76%	17.27%	1.40%	23.19%	Yes	Yes
Ronald Reagan	Republican	2	1985-1988	26.33%	14.62%	2.03%	12.40%	66.06%	No	
George H.W. Bush	Republican	1	1989-1992	27.25%	-6.56%	26.31%	4.46%	56.89%	Yes	No
Bill Clinton	Democrat	1	1993-1996	7.06%	-1.54%	34.11%	20.26%	70.01%	Yes	Yes
Bill Clinton	Democrat	2	1997-2000	31.01%	26.67%	19.53%	-10.14%	78.24%	No	
George W. Bush	Republican	1	2001-2004	-13.04%	-23.37%	26.38%	8.99%	-8.21%	Yes	Yes
George W. Bush	Republican	2	2005-2008	3.00%	13.62%	3.53%	-38.49%	-25.47%	No	
Barack Obama	Democrat	1	2009-2012	23.45%	12.78%	0.00%	13.41%	57.90%	Yes	Yes
Barack Obama	Democrat	2	2013-2016	29.60%	11.39%	-0.73%	9.54%	56.98%	No	
Donald Trump	Republican	1	2017-2020	19.42%	-6.24%	28.88%	16.26%	67.77%	Yes	No
Joe Biden	Democrat	1	2021-2024	26.89%	-19.44%	24.23%	23.31%	56.59%	No	No
Donald Trump	Republican	2	2025-2028	16.39%						

Source: BofA Global Research, Bloomberg, Fineon

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Decennial Cycle

- The Decennial Cycle is a seasonality framework that compares cross-decade historical performance patterns according to the last digit of the calendar year.
- While traditionally used to analyze the S&P 500 as there are more observations (15-16), we begin tracking macro assets, too, with as few as 5 observations.
- **SPX** best year ends in 5, worst in 0. **Oil's** best year ends in 9, worst in 8. **Gold's** best year unclear, worst ends in 1. **DXY** best year ends in 1, worst ends in 7. **US10Y yield** tended to rise in years ending in 3 and declined in years ending in 0, 1 and 2.

What is the Decennial Cycle?

Also known as the 10-year cycle, the Decennial Cycle observes cyclical tendencies for years ending in the same digit. We compare average return, median return, % time up and standard deviation of returns. Like many forms of technical or statistical analysis, the Decennial Cycle should not be used as a standalone investment strategy. Rather it is a piece of evidence to consider and becomes more powerful when paired with other technical tools and techniques. Many technicians combine the Decennial Cycle with seasonal trends, the four-year US Presidential Cycle and other cycle-based studies. Sometimes multiple cycles align to suggest greater confidence in a tendency.

SPX: Years ending in 6 are average, lower vol

On a total return basis since 1871 for years ending in 6, the S&P 500 was up 80% of the time (12 out of 15) by an average of 9.55%, median 11.14%, with a standard deviation of 12.86%. In comparison to the average and median statistics of grouped years, returns were slightly lower, up ratios slightly higher and standard deviation of returns lower. In conclusion, years ending in 6 had the sixth best average return, fifth best median return, tied for third best up ratio and the lowest standard deviation. This suggests an average year with lower volatility. In Y2 of the US Presidential Cycle, the takeaways are similar. In USPC Y2s ending in 6, the S&P was up about 71% of the time by an average of 6.69%, median 11.14%. Its standard deviation mirrored all year 6s at 11.43%.

SPX best year ends in a 5, worst in a 0

Years ending in five saw the S&P 500 total return index up 100% of the time for an average gain of 23.88%, median 29.55% and standard deviation of 15.14%. Years ending in zero were the weakest, up just 53% of the time with an average gain of 3.89%, median 0.48% and deviation of 18.23%. Even years faced year 2 of the US Presidential Cycle, which is often a weaker year, where years ending in 0, 2 and 4 underperformed.

Exhibit 146: SPX in years ending in 6 were avg years w/lower volatility. Years ending in 5 and 8 were the best, ending in 0 and 7 the worst.
SPX Decennial year average total returns (1874-2025)

Decennial Year	Average return	Median return	% Up	Standard deviation	Observations
1	8.31%	14.33%	73.33%	20.14%	15
2	6.72%	7.79%	75.00%	14.69%	16
3	9.98%	7.12%	56.25%	20.67%	16
4	10.47%	10.09%	75.00%	18.68%	16
5	23.88%	29.55%	100.00%	15.14%	15
6	9.55%	11.14%	80.00%	12.86%	15
7	2.28%	5.17%	53.33%	22.02%	15
8	16.84%	16.61%	86.67%	21.67%	15
9	16.49%	18.35%	80.00%	15.60%	15
0	3.89%	0.48%	53.33%	18.23%	15
Averages	10.81%	11.27%	73.20%	18.73%	

Source: BofA Global Research, GFD

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Exhibit 147: In years ending in 6 that were year 2 of the USPC, the SPX had average year w/lower vol
SPX Decennial year average total returns (1874-2025) for USPC Y2

Decennial Year	Average return	Median return	% Up	Standard deviation	Observations
2	3.56%	3.55%	62.50%	18.90%	8
4	5.77%	1.68%	62.50%	22.30%	8
6	6.69%	11.14%	71.43%	11.43%	7
8	20.91%	24.40%	87.50%	15.50%	8
0	0.46%	-3.10%	42.86%	18.21%	7
Averages	7.68%	6.64%	65.79%	18.29%	

Source: BofA Global Research, GFD

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Crude oil: Years ending in 6 lean bullish

While we have just four observations to consider, crude oil tended to outperform during years ending in 6 with average volatility. It rose 75% of the time (3 out of 4) by +11.46% on average. Its median return was also higher than all years. Years ending in 9 were best, while years ending in 8 were the worst.

Exhibit 148: Oil rose more than avg most of the time during years ending in 6. It was up +11.5% 75% of the time (vs +7.55% / 61% all years).
Crude oil Decennial year average returns (1984-2025)

Decennial Year	Average return	Median return	% Up	Standard deviation
1	1.11%	-8.91%	50.00%	40.15%
2	14.72%	4.35%	75.00%	28.93%
3	-6.66%	-3.25%	50.00%	15.86%
4	0.48%	0.10%	60.00%	31.59%
5	4.92%	4.83%	50.00%	29.28%
6	11.46%	16.30%	75.00%	34.52%
7	7.71%	2.78%	50.00%	37.68%
8	-26.71%	-28.27%	25.00%	23.41%
9	62.85%	56.20%	100.00%	40.04%
0	7.41%	9.92%	75.00%	21.40%
Averages	7.55%	4.23%	60.98%	34.92%

Source: BofA Global Research, Bloomberg

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Gold: Years ending in 6 are a touch bullish

Since 1920, gold in years ending in 6 rose on average 4.11%, median 0.01% with a lower standard deviation of returns or 9.63%. Gold's best average year ends in 9, but up ratio was low. Gold's worst year ends in 1.

Exhibit 149: During years ending in 6, gold underperformed all years. It rose 4.11% on avg with neutral bias. In all years, it rose an avg of 6%.
Gold Decennial year average returns (1920-2025)

Decennial Year	Average return	Median return	% Up	Standard deviation
1	-1.45%	-0.06%	27.27%	12.17%
2	8.04%	0.29%	54.55%	15.53%
3	9.31%	3.19%	63.64%	24.86%
4	10.17%	0.03%	54.55%	25.00%
5	-0.67%	0.24%	60.00%	10.69%
6	4.11%	0.01%	50.00%	9.63%
7	6.94%	0.02%	50.00%	15.74%
8	4.40%	0.13%	60.00%	13.96%
9	15.11%	-0.07%	30.00%	40.66%
0	6.53%	0.29%	70.00%	12.50%
Averages	6.26%	0.04%	51.92%	19.89%

Source: BofA Global Research, Bloomberg

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DXY: Mixed in years ending in 6, best ends in 1, worst 7

In years ending in six, the DXY trend is unclear. It was up 60% of the time (3 out of 5) but average return was down while median was up. Years ending in 1 tended to be up while years ending in 7 tended to be down.

Exhibit 150: The US dollar trended softer in years ending in 6. It fell -3.15% on avg 60% of the time. In all years, it trended flat with neutral bias.
US dollar index Decennial year average returns (1968-2025)

Decennial Year	Average return	Median return	% Up	Standard deviation
1	3.81%	3.91%	83.33%	7.88%
2	2.77%	3.85%	50.00%	9.42%
3	-1.06%	-0.89%	50.00%	9.35%
4	2.41%	1.04%	50.00%	10.45%
5	1.09%	6.39%	60.00%	12.70%
6	-3.15%	1.01%	60.00%	8.77%
7	-6.08%	-8.31%	20.00%	11.42%
8	0.52%	2.27%	66.67%	7.21%
9	0.65%	0.02%	50.00%	4.09%
0	-0.67%	0.30%	50.00%	7.02%
Averages	0.17%	0.33%	54.39%	8.68%

Source: BofA Global Research, Bloomberg

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US 10Y yield mixed in years ending in 6

In years ending in six, the US 10Y yield trend is unclear. US 10Y yield was up 50% of the time but avg gain was -23bps while median gain was +8bps. Years ending in zero, one and two tended to see yield decline while years ending in three tended to see yield rise.

Exhibit 151: Years ending in 6 saw US 10Y yield decline more than all years. Yield fell by -23bps with neutral bias. Median return was +8bps.
US 10Y yield Decennial year average returns (1962-2025)

Decennial Year	Average return	Median return	% Up	Standard deviation
1	-21.82	-33.55	33.33%	116.0
2	-34.59	-6.59	33.33%	198.0
3	42.92	42.95	85.71%	77.7
4	30.24	7.00	57.14%	91.5
5	-61.78	13.56	66.67%	138.0
6	-23.18	8.25	50.00%	95.2
7	37.87	46.56	50.00%	98.0
8	-8.64	27.84	66.67%	115.5
9	72.50	140.23	66.67%	134.8
0	-33.76	-77.38	33.33%	132.1
Averages	1.16	11.51	54.84%	120.51

Source: BofA Global Research, Bloomberg

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Benner Cycle

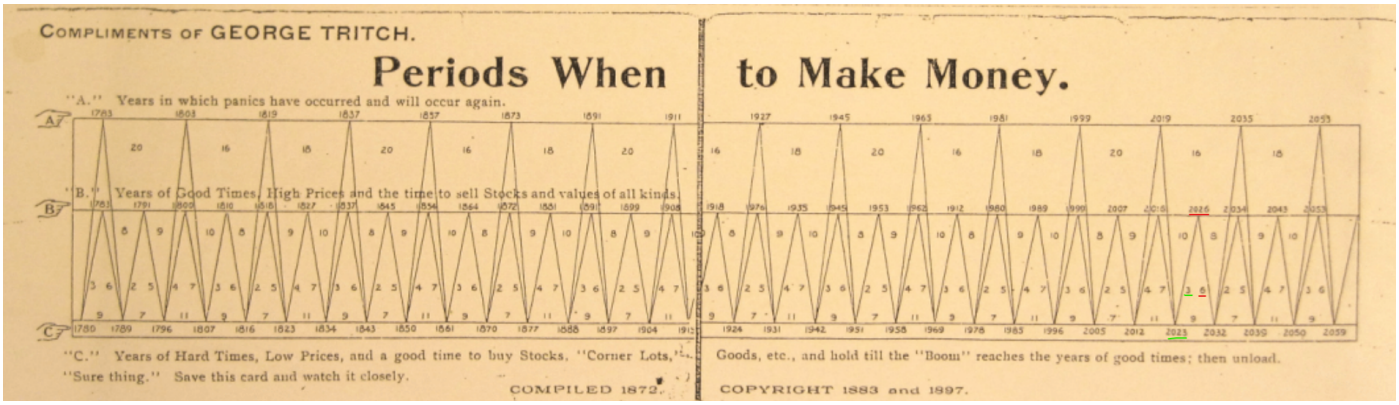
2026 is the last good year

- 2026 is projected to be the final "good year" for asset prices within the current Benner Cycle framework. The suggests sticking with risk assets in 2024-2026 and increasing caution / prudence from 2027-2032.
- Since 1887 the "good years" outperformed the "hard years." Average annual SPX returns were 11.73% vs 4.45%, median 12.35% vs 3.00%, standard deviation 16.7% vs 18.7%, time up 82% vs 59%, max gain 47% vs 45%, max loss -33% vs -47%.
- In FICC markets, oil and gold tended to outperform in good years vs hard years. The DXY tended to underperform in good years vs hard years. US 10Y yield tended, modestly, to rise less in good years than hard years.

Benner Cycle summary

Sam Benner was a farmer and commodity enthusiast in the 1800s. He founded the Benner Cycle where he described a pattern of long-, medium- and short-term cycles. There is a 54-year cycle and a 27-year cycle. The 54-year cycle includes recurring 20-, 16- and 18-year cycles. We are presently in a 16-year cycle from 2019-2035. The 27-year cycle includes recurring 7-, 9- and 11-year cycles. We are presently in a 9-year cycle from 2023-2032. The 9-year cycle is comprised of three "good" years and six "hard" years. 2024-2026 is a cycle of good years. 2026 is estimated to be the last good year and, according to the description, years of "good times, high prices and the time to sell stocks and valuations of all kinds." 2027-2032 is noted as hard year cycle and, according to the description, years of "Hard times, low prices and a good time to buy stocks."

Exhibit 152: Extended Benner Cycle shows 2026 is the last of three good years (Middle row B)
Benner's 54-year cycle comprised of a 20-, 16- and 18-year cycle. And the 27-year cycle comprised of 7-, 9- and 11-year cycles



Source: Sam Benner, George Tritch, Tritch Hardware, BofA Global Research

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Trends in "Good years" outperform the "Hard years"

Exhibit 153 shows that good years outperform hard years across all metrics and favors a constructive case for 2026. Exhibit 154 suggests that buying during the last hard year and selling one year after the last good year outperformed selling at the end of the last good year. In other words, investors who averaged in during 2023 may outperform if exiting in 2027 rather than in 2026.

Exhibit 153: SPX in good year cycles outperform bad year cycles
Brenner Cycle good years vs bad year annualized measures 1878-2025

	Good years	Hard years
Average	11.73%	4.45%
Median	12.35%	3.00%
Standard deviation	16.68%	18.72%
% time up	82.00%	58.76%
Max	47%	45%
Min	-33%	-47%

Source: BofA Global Research, Bloomberg, Fineon Inc

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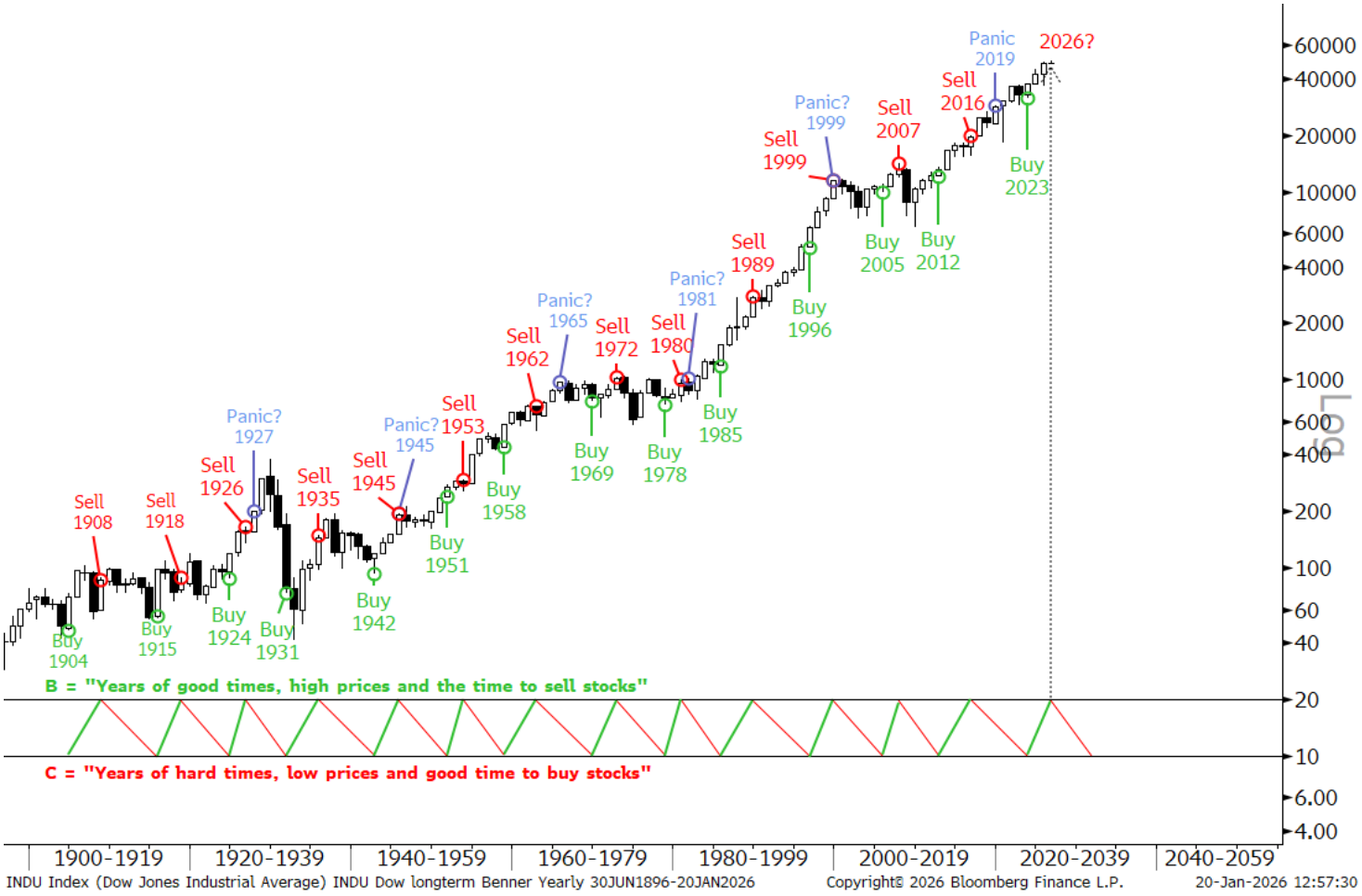
Exhibit 154: Selling SPX a year after last good year better than selling last good year
Benner Cycle study, 1877-2016.

	Buy last hard and sell last good yr	Buy last hard and sell last good +1yr	Buy last hard and sell last good +2yr
Average	34.91%	47.03%	50.64%
Median	28.84%	37.95%	74.87%
Std Dev	36.42%	35.40%	46.90%
% time up	81.25%	100%	75%
Min	-19.40%	0.62%	21.47%
Max	99.50%	110.38%	123.98%

Source: BofA Global Research, Bloomberg, Fineon Inc.

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Exhibit 155: Buy years and sell years from the Benner Cycle shown to visualize periods of investing aggressively and conservatively, respectively.
Dow Jones Industrial Average with Benner Cycle points plotted.



Source: BofA Global Research, Bloomberg

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Crude oil trends in "Good years" outperform the "Hard years"

Exhibit 156 shows that crude oil in good years outperformed hard years in terms of average return. However, oil's up ratio was modestly lower during good years, and it was more volatile (higher standard deviation). Exhibit 154 suggests that buying during the last hard year and selling two years after the last good year outperformed selling at the end of the last good year and selling one year after the last good year.

Exhibit 156: Crude oil average return in good year cycles outperformed hard year cycle. However, oil volatility was higher and up ratio weaker.
Benner Cycle good years vs hard year annualized measures. 1878-2025

	Good years	Hard years
Average	7.56%	5.81%
Median	0.00%	0.67%
Standard deviation	35.31%	28.73%
% time up	45.10%	51.55%
Max	151.06%	115.67%
Min	-45.55%	-59.09%

Source: BofA Global Research, Bloomberg, Fineon Inc

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Exhibit 157: Buying the last hard year cycle and selling 2 yrs after the last good year yielded higher returns but also higher volatility.
Benner Cycle study, 1877-2016.

	Buy last hard and sell last good yr	Buy last hard and sell last good +1yr	Buy last hard and sell last good +2yr
Average	16.66%	24.88%	40.93%
Median	6.07%	5.51%	4.98%
Std Dev	61.04%	62.63%	101.56%
% time up	56.25%	56.25%	56.25%
Min	-63.17%	-65.62%	-52.78%
Max	154.78%	161.10%	285.73%

Source: BofA Global Research, Bloomberg, Fineon Inc

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US dollar index trends in "Good years" underperform "Hard years"

Exhibit 158 indicates that the US dollar tended to modestly underperform during Benner Cycle good years when compared to hard years. During both good and hard years, DXY saw roughly flat returns and neutral up ratios. The dollar tended to rise more when bought during the last hard year of the Benner Cycle and sold two years following the last good year (Exhibit 159).

Exhibit 158: DXY tended to perform worse during good years than in hard years. It saw lower returns and was up less frequently.
Benner Cycle good years vs hard year annualized measures. 1971-2025.

	Good years	Hard years
Average	-0.31%	0.21%
Median	0.33%	0.75%
Standard deviation	9.23%	8.90%
% time up	52.63%	55.56%
Max	13.08%	15.82%
Min	-17.56%	-18.49%

Source: BofA Global Research, Bloomberg, Fineon Inc

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Exhibit 159: Buying the last Benner Cycle hard year and selling 2 years after the last good year produced the best average and median returns.
Benner Cycle study, 1969-2016.

	Buy last hard and sell last good yr	Buy last hard and sell last good +1yr	Buy last hard and sell last good +2yr
Average	-2.95%	-1.65%	2.46%
Median	-6.15%	0.55%	4.32%
Std Dev	18.60%	24.90%	27.35%
% time up	33.33%	50.00%	50.00%
Min	-30.33%	-36.98%	-36.31%
Max	21.01%	25.93%	32.31%

Source: BofA Global Research, Bloomberg, Fineon Inc

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Gold trends in "Good years" outperform "Hard years"

Gold exhibited leadership during Benner Cycle good years. It tended to rise by an average of 11% about 52% of the time versus 6% about 55% of the time hard years. Gold delivered the best performance when it was bought during the last hard year and sold 2 years after the last good year. It gained by 51% about 67% of the time.

Exhibit 160: Gold performed better during Benner Cycle good years. It returned 11.2% on average about 52% of the time versus 6% on avg 55% of the time during hard years.
Benner Cycle good years vs hard year annualized measures. 1941-2025.

	Good years	Hard years
Average	11.20%	5.97%
Median	2.76%	0.28%
Standard deviation	29.56%	17.89%
% time up	51.72%	55.36%
Max	126.55%	72.16%
Min	-28.03%	-31.57%

Source: BofA Global Research, Bloomberg, Fineon Inc

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Exhibit 161: Gold saw the best returns on average when bought during the last hard year of the Benner Cycle then sold 2 years after the last good year. It was up 51.38% on avg 67% of the time.
Benner Cycle study. 1942-2016.

	Buy last hard and sell last good yr	Buy last hard and sell last good +1yr	Buy last hard and sell last good +2yr
Average	29.86%	36.43%	51.38%
Median	4.73%	7.71%	12.26%
Std Dev	72.14%	67.75%	105.06%
% time up	66.67%	66.67%	66.67%
Min	-27.52%	-28.24%	-29.70%
Max	206.06%	145.00%	295.47%

Source: BofA Global Research, Bloomberg, Fineon Inc

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US 10Y yield trends in "Good years" underperformed "Hard years"

Benner Cycle good and hard years did not have a material impact on the US 10Y yield. Its average and median returns were roughly flat. US 10Y yield tended to see larger declines from the last hard year to the last good year.

Exhibit 162: Benner Cycle good vs hard years did not have a significant impact on US 10Y yield performance. However good years saw modestly lower rises than hard years.
Benner Cycle good years vs hard year annualized measures. 1878-2025.

	Good years	Hard years
Average	0.002	0.2
Median	1.0	2.3
Standard deviation	73.7	85.2
% time up	51.0%	50.5%
Max	210.0	236.0
Min	-177.0	-362.0

Source: BofA Global Research, Bloomberg, Fineon Inc

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Exhibit 163: US 10Y yield tended to decline from the end of the hard year cycle through the end of the good year cycle
Benner Cycle study, 1789-2016.

	Buy yield (sell price) in last hard year and sell yield (buy price) in last good yr	Buy yield (sell price) in last hard year and sell yield (buy price) in last good +1yr	Buy yield (sell price) in last hard year and sell yield (buy price) in last good +2yr
Average	-62.52	-53.62	-50.88
Median	-13.03	-27.64	-28.62
Std Dev	278.61	313.78	294.45
% time up	34.6%	38.5%	42.3%
Min	-1356.82	-1454.12	-1348.69
Max	312.33	554.00	438.00

Source: BofA Global Research, Bloomberg, Fineon Inc

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January Trifecta

- The January Trifecta (JT) is a three-part seasonal stock market indicator created by the Stock Trader's Almanac, Yale Hirsch and Jeff Hirsch. It combines three January-related signals including the Santa Claus Rally (SCR), the First Five Days (FFD) and the January Barometer.

- The SCR measures performance of the last five trading days of the year plus the first two of January. The FFD measures performance of the first five trading days of the year. The JB measures the performance of the entire month of January. The JT is all three combined.

- We review what each asset did after each part of the JT and all together for the S&P 500, DXY, Oil, Gold and US10Y yield. An up JT tended to precede a higher SPX, DXY and US10Y yield. A down JT favors lower oil and higher gold.

January Trifecta combinations

We define a January Trifecta as having occurred when the SCR, FFD and JB are all positive or all negative. The macro results of a positive or negative JT are as follows with other notable outcomes mentioned.

SPX: A positive JT is bullish, beware if SCR up while FFD and Jan down.

A positive JT led to the rest of the year being up 83% of the time, or 34 of 41 times, with strong average and median returns. Only four times was the SCR down, FFDs down and Jan up. However each time the rest of year was up more than JT years. We could call this the "January Override" as January reverses the down SCR and FFD bias. Beware if the SCR is up, FFDs down and Jan down. In the past, average and median returns for the rest of the year were negative and up just 36% of time. An all-negative JT preceded weaker returns for the rest of the year and lower up ratio, just 63% of the time or 5 of 8.

Exhibit 164: An all-positive January Trifecta is bullish for the S&P 500, all-negative JT a headwind. SCR up, FFD down, Jan down = Bearish. SCR down, FFD up, Jan up = Bullish.
S&P 500 January Trifecta combinations from 1928-2025

Criteria	Average Feb - Dec return	Median Feb - Dec return	% time up	Standard Deviation	Max	Min	Count
Santa rally up, first five days up and Jan up	10.20%	12.07%	83%	17.85%	43.66%	-49.85%	41
Santa rally down, first five days down and Jan down	0.20%	6.09%	63%	16.83%	16.81%	-34.48%	8
Santa rally down, first five days up and Jan up	9.99%	11.38%	71%	10.21%	22.92%	-4.64%	7
Santa rally up, first five days down and Jan up	-0.69%	2.09%	56%	20.38%	25.40%	-32.69%	9
Santa rally up, first five days up and Jan down	6.65%	7.49%	71%	18.03%	35.02%	-22.15%	14
Santa rally up, first five days down and Jan down	-1.11%	-5.95%	36%	19.67%	47.58%	-29.00%	11
Santa rally down, first five days up and Jan down	-2.28%	2.45%	67%	9.16%	3.54%	-12.85%	3
Santa rally down, first five days down and Jan up	16.64%	19.45%	100%	7.11%	21.38%	6.31%	4
Overall	6.44%	8.67%	71%	17.62%	47.58%	-49.85%	97

Source: BofA Global Research, Bloomberg

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DXY: A positive JT is bullish, beware if SCR up while FFD and Jan down.

A positive JT resulted in Feb-Dec rising 63% of the time (10 of 16) on average by 0.44%, median 3.24%. A negative JT showed stronger average and median returns but occurred just six times. Only twice has the SCR been up, FFD down and Jan up, however the DXY was higher at the end of the year both times. The "January Override" may also apply to the DXY. If the SCR was down, FFDs down and January up, average and median returns for the rest of the year were up and the index was up 6 of 9 times or 67% of the time.

Exhibit 165: When JT all positive, rest of year leans positive for the US dollar index. When JT all negative, rest of year leans positive, too.
DXY January Trifecta combinations from 1971-2025

Criteria	Average Feb - Dec return	Median Feb - Dec return	% time up	Standard Deviation	Max	Min	Count
Santa rally up, first five days up and Jan up	0.44%	3.24%	63%	8.87%	11.78%	-19.16%	16
Santa rally down, first five days down and Jan down	2.32%	3.58%	50%	6.55%	8.26%	-5.97%	6
Santa rally down, first five days up and Jan up	-1.42%	-3.42%	43%	7.99%	8.64%	-10.49%	7
Santa rally up, first five days down and Jan up	4.09%	4.09%	100%	2.09%	5.57%	2.61%	2
Santa rally up, first five days up and Jan down	-5.43%	-6.75%	25%	4.51%	1.05%	-9.27%	4
Santa rally up, first five days down and Jan down	-4.01%	-3.35%	50%	5.54%	0.85%	-10.19%	4
Santa rally down, first five days up and Jan down	-6.10%	-6.20%	50%	8.79%	3.14%	-14.75%	6
Santa rally down, first five days down and Jan up	1.23%	4.21%	67%	8.09%	9.92%	-15.27%	9
Overall	-0.82%	0.75%	56%	7.90%	11.78%	-19.16%	54

Source: BofA Global Research, Bloomberg

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Oil: A negative JT tended to be bearish for oil

When JT was all positive, there was little noteworthy bias. When the JT was negative, oil was down 5 of 7 times with lower returns in Feb-Dec. Lack of observations a challenge, however, when SCR was down, FFDs up and Jan up, oil was up two of two times. When SCR up, FFDs down and Jan down, oil was up 3 of 4 times. When SCR was down, FFDs down and Jan up, the rest of year was down 3 of 4 times (No January override).

Exhibit 166: All-positive JT did not impact oil. All-negative JT saw crude lower 5 of 7 times, weaker Feb-Dec returns. Other scenarios lacked observations.
Oil January Trifecta combinations from 1990-2025

Criteria	Average Feb - Dec return	Median Feb - Dec return	% time up	Standard Deviation	Max	Min	Count
Santa rally up, first five days up and Jan up	9.03%	-4.00%	50%	40.71%	100.78%	-30.77%	10
Santa rally down, first five days down and Jan down	0.85%	-11.23%	29%	43.80%	65.08%	-45.36%	7
Santa rally down, first five days up and Jan up	21.78%	21.78%	100%	6.87%	26.64%	16.92%	2
Santa rally up, first five days down and Jan up	9.58%	6.31%	67%	14.46%	25.40%	-2.95%	3
Santa rally up, first five days up and Jan down	7.76%	9.30%	50%	43.46%	60.16%	-51.39%	6
Santa rally up, first five days down and Jan down	25.52%	8.79%	75%	44.04%	90.40%	-5.90%	4
Santa rally down, first five days up and Jan down	-	-	-	-	-	-	-
Santa rally down, first five days down and Jan up	-7.84%	-4.24%	25%	15.80%	7.20%	-30.06%	4
Overall	7.94%	-1.00%	50%	36.21%	100.78%	-51.39%	36

Source: BofA Global Research, Bloomberg

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Gold: All-negative JT favored higher gold for rest of year 71% of time

When the JT was positive, gold was up the rest of the year 59% of the time. When the JT was all negative, gold was up 71% of the time. Low observation count for other scenarios. When the SCR was down, FFDs up and Jan up, gold was down 2 of 2 times. When the SCR was up, FFDs down and Jan up, gold was up the rest of year 4 of 4 times.

Exhibit 167: Gold after a positive JT leaned higher the rest of the year. However a negative JT was more supportive from a % time up and Std Dev perspective.
Gold January Trifecta combinations from 1976-2025

Criteria	Average Feb - Dec return	Median Feb - Dec return	% time up	Standard Deviation	Max	Min	Count
Santa rally up, first five days up and Jan up	6.11%	6.98%	59%	18.79%	54.35%	-23.44%	17
Santa rally down, first five days down and Jan down	6.02%	4.94%	71%	14.73%	28.68%	-17.32%	14
Santa rally down, first five days up and Jan up	-5.68%	-5.68%	0%	2.95%	-3.59%	-7.77%	2
Santa rally up, first five days down and Jan up	47.03%	25.41%	100%	48.19%	119.08%	18.23%	4
Santa rally up, first five days up and Jan down	7.31%	0.65%	60%	14.29%	31.47%	-3.51%	5
Santa rally up, first five days down and Jan down	-11.86%	-10.59%	50%	14.99%	1.49%	-27.77%	4
Santa rally down, first five days up and Jan down	18.06%	18.06%	100%	-	18.06%	18.06%	1
Santa rally down, first five days down and Jan up	0.87%	-4.69%	33%	10.38%	12.85%	-5.54%	3
Overall	7.50%	4.06%	62%	22.84%	119.08%	-27.77%	50

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

US 10Y Yield: A positive JT favors higher yield the rest of the year

When JT was all positive, yield was up the rest of the year 73% of the time (11 of 15) on average by 16bps, median 38bps. When the JT was all negative, trend bias was unclear.

Remaining combinations have few observations, however, SCR down, FFDs up and Jan down favored lower yield. Also, SCR down, FFDs down and Jan up favored higher yield.

Exhibit 168: US 10Y yield after an all-positive JT tended to rise the rest of the year 73% of time, or 11 of 15.
US 10Y yield January Trifecta combinations from 1963-2025

Criteria	Average Feb - Dec return	Median Feb - Dec return	% time up	Standard Deviation	Max	Min	Count
Santa rally up, first five days up and Jan up	16	38	73%	140	210	-375	15
Santa rally down, first five days down and Jan down	3	9	50%	98	168	-138	8
Santa rally down, first five days up and Jan up	-12	-2	43%	12	179	-183	7
Santa rally up, first five days down and Jan up	-8	9	50%	49	27	-78	4
Santa rally up, first five days up and Jan down	-101	-115	25%	94	26	-201	4
Santa rally up, first five days down and Jan down	3	-2	50%	124	218	-219	10
Santa rally down, first five days up and Jan down	-13	-37	29%	89	118	-131	7
Santa rally down, first five days down and Jan up	42	25	63%	86	169	-99	8
Overall	0.34	8.63	52%	111.68	218.00	-375.30	63

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Part 1: The Santa Claus Rally (SCR)

The Santa Claus Rally and macro markets

The Santa Claus Rally (SCR) suggests that the last five trading days of the prior year through the first two trading days of the new year may indicate future trend for the month, quarter or first half of the new year. This may be due to tax-loss harvesting reversing, holiday optimism, less trading and portfolio rebalancing.

SPX: A negative SCR favors soft 1Q, bullish 1H. Positive SCR supportive.

The SCR has been negative 22 times since 1928. January and 1Q have lower up ratios than all periods, especially 1Q, so a negative SCR is initially a headwind. However the S&P 500 was up by the end of 1H 68% of the time. When the SCR was positive, average trend and up ratios were modestly supportive of typical S&P 500 uptrend bias.

Exhibit 169: For the S&P 500, a negative SCR favors underperformance in 1Q and a rebound by end 1H. A positive SCR is modestly supportive of price.
S&P 500 after a negative and positive Santa Claus Rally from 1928-2025 and vs all periods

	Santa Claus Rally negative?			Santa Claus Rally positive?			All periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	1.14%	0.47%	1.40%	0.59%	1.23%	3.75%	0.71%	1.06%	3.22%
Median	0.66%	-0.61%	3.84%	0.79%	2.05%	5.05%	0.79%	1.78%	4.42%
% time up	55%	45%	68%	60%	63%	63%	59%	59%	64%
Std dev	4.23%	8.49%	12.32%	4.78%	8.32%	13.02%	4.64%	8.32%	12.84%
Max	9.15%	16.56%	16.58%	12.04%	18.42%	52.59%	12.04%	18.42%	52.59%
Min	-4.74%	-22.66%	-41.32%	-10.95%	-20.10%	-23.03%	-10.95%	-22.66%	-41.32%
Count	22	22	22	75	75	75	97	97	97

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

SPX after SCR in Y2 USPC: A negative SCR is also a headwind in Jan, 1Q, 1H

While year 2 of the USPC is a weaker period into midterms, if the SCR was negative, price was down in Q1 4 of 4 times with an average decline of -9.63%, median -5.67%. When positive, up ratios and returns improved, but still underperformed.

Exhibit 170: For the SPX in year 2 of the USPC, a negative SCR was bearish for the S&P 500, but just four observations. A positive SCR preceded a 1H decline
S&P 500 after a negative and positive Santa Claus Rally in year 2 of the U.S. Presidential cycle from 1928-2025 and vs all periods

	Santa Claus Rally negative?			Santa Claus Rally positive?			All Y2 periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	-0.96%	-9.63%	-1.55%	-0.17%	1.65%	-2.30%	-0.30%	-0.23%	-2.17%
Median	-1.22%	-5.67%	-1.35%	0.38%	1.96%	-0.93%	0.31%	-1.88%	-0.93%
% time up	50%	0%	50%	55%	55%	40%	54%	46%	42%
Std dev	3.40%	8.74%	6.35%	5.39%	7.42%	12.86%	5.06%	8.60%	11.91%
Max	3.15%	-4.52%	5.19%	12.04%	18.42%	18.95%	12.04%	18.42%	18.95%
Min	-4.57%	-22.66%	-8.70%	-9.03%	-11.88%	-23.03%	-9.03%	-22.66%	-23.03%
Count	4	4	4	20	20	20	24	24	24

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

DXY: Negative SCR modestly favors 1Q, positive SCR modestly favors January

When the DXY saw a negative Santa Claus rally, it tended to be up modestly in 1Q about 61% of the time by an average of 1.39%, median 1.09%. When the SCR was positive, DXY was up in January 62% of the time by an average of 0.77% and median of 0.75%.

Exhibit 171: For the DXY, a negative SCR modestly favored an up 1Q and a positive SCR modestly favored an up January.
DXY after a negative and positive Santa Claus Rally from 1972-2025 and vs all periods

	Santa Claus Rally negative?			Santa Claus Rally positive?			All periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	0.49%	1.39%	1.67%	0.77%	0.05%	-0.21%	0.63%	0.75%	0.77%
Median	0.44%	1.09%	0.53%	0.75%	-0.41%	-0.77%	0.65%	0.57%	0.02%
% time up	57%	61%	54%	62%	46%	46%	59%	54%	50%
Std dev	2.58%	5.21%	7.81%	2.07%	3.95%	6.11%	2.33%	4.66%	7.04%
Max	6.84%	11.11%	20.83%	4.19%	7.99%	9.75%	6.84%	11.11%	20.83%
Min	-4.11%	-8.30%	-10.59%	-3.58%	-8.24%	-14.85%	-4.11%	-8.30%	-14.85%
Count	28	28	28	26	26	26	54	54	54

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Oil: A negative SCR favors oil more than a positive SCR

In all periods, oil tended to rise in Q1 and H1. However, a negative SCR period shows oil tended to be higher more often in January, 1Q and 1H. Average change, median change and up ratios after a negative SCR were stronger than after a positive SCR. For example, after a negative SCR, oil was up 77% of the time (10 of 13 times) at the end of 1Q by an average of 5.46%, median 6.59%.

Exhibit 172: Oil after a negative SCR tended to outperform, such as up in 1Q 77% of the time or 10 of 13 times.
Crude oil after a negative and positive Santa Claus Rally from 1990-2025 and vs all periods

	Santa Claus Rally negative?			Santa Claus Rally positive?			All periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	1.92%	5.46%	12.01%	-0.94%	3.00%	8.68%	0.09%	3.89%	9.88%
Median	3.54%	6.59%	12.16%	-1.72%	2.43%	5.18%	1.60%	4.85%	11.04%
% time up	69%	77%	69%	43%	65%	57%	53%	69%	61%
Std dev	7.17%	12.53%	17.96%	8.32%	21.13%	26.34%	7.94%	18.33%	23.43%
Max	10.96%	26.17%	34.36%	14.50%	39.78%	60.88%	14.50%	39.78%	60.88%
Min	-15.46%	-22.96%	-19.31%	-18.22%	-67.52%	-37.72%	-18.22%	-67.52%	-37.72%
Count	13	13	13	23	23	23	36	36	36

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Gold: If a negative SCR, gold lower in Jan. If positive SCR, gold up in Jan and 1Q.

History suggests when gold was down in the SCR period, it was also lower in January and may face headwinds in 1Q and 1H. However when gold was up in the SCR period, gold prices were higher in January 70% of the time and in 1Q 67% of the time. Both its average and median returns outperformed all periods.

Exhibit 173: A negative SCR suggests lower gold in January and, to a lesser extent, in 1Q / 1H. A positive SCR was bullish for gold in January and 1Q.
Gold after a negative and positive Santa Claus Rally from 1976-2025 and vs all periods

	Santa Claus Rally negative?			Santa Claus Rally positive?			All periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	-0.26%	-0.36%	-0.64%	1.83%	1.66%	3.25%	0.99%	0.85%	1.69%
Median	-0.33%	-0.47%	0.35%	2.61%	3.39%	4.62%	0.98%	2.56%	1.64%
% time up	40%	50%	55%	70%	67%	60%	58%	60%	58%
Std dev	4.06%	7.28%	8.85%	5.17%	8.46%	12.82%	4.83%	8.00%	11.46%
Max	8.34%	11.91%	14.40%	10.02%	18.31%	26.94%	10.02%	18.31%	26.94%
Min	-7.54%	-20.69%	-21.31%	-15.48%	-22.00%	-28.91%	-15.48%	-22.00%	-28.91%
Count	20	20	20	30	30	30	50	50	50

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

US 10Y Yield: Down SCR favored higher yield ending 1H, up SCR favored up 1Q

When US 10Y yield was down in the SCR period, yield tended to be higher ending the 1H of the following year 70% of the time (21 of 30 times). When yield rose in the SCR period, 1Q was up 67% of the time or 21 of 33 occurrences.

Exhibit 174: When US 10Y yield declined in the SCR period, yield tended to be higher ending 1H of next year 70% of the time. If yield rose in the SCR period, it tended to be higher ending 1Q.

US 10Y yield after a negative and positive Santa Claus Rally from 1963-2025 and vs all periods

	Santa Claus Rally negative?			Santa Claus Rally positive?			All periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	-1	-0.1	18	-1	16	7	-1	8	12
Median	0	3	22	-2	8	0	-1	6	19
% time up	50%	57%	70%	42%	67%	48%	46%	62%	59%
Std dev	25	58	71	25	53	82	25	56	76
Max	53	98	171	53	204	209	53	204	209
Min	-47	-169	-171	-43	-80	-167	-47	-169	-171
Count	30	30	30	33	33	33	63	63	63

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Part 2: The First Five Days

The outcome of the First Five Trading Days (FFD) of the year may indicate future trend for the month, quarter or first half of the year. In this section, we review macro market trends if the FFDs were positive or negative in all years and in year 2 of the USPC.

SPX: A down FFDs may precede a 1Q and 1H rebound. Up FFDs favors up 1H.

When the FFDs of the year were negative, the S&P 500's average and median returns outperformed in January and 1Q compared to when the FFDs were up. However when the FFDs were up, the S&P 500 was higher ending the 1H 68% of the time.

Exhibit 175: A down FFDs is bullish in January, 1Q and slightly less bullish 1H

S&P 500 trends after the first five trading days in January from 1928-2025

	Trend if FFDs down			Trend if FFDs up			Trend all periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	1.18%	2.36%	2.37%	0.35%	0.42%	3.67%	0.63%	1.07%	3.23%
Median	0.47%	1.90%	2.42%	0.45%	1.23%	4.79%	0.46%	1.25%	4.48%
% time up	58%	61%	61%	57%	62%	68%	57%	61%	65%
Std dev	4.57%	8.40%	10.96%	4.10%	8.04%	13.67%	4.26%	8.17%	12.78%
Max	14.45%	19.16%	20.61%	9.91%	19.02%	53.02%	14.45%	19.16%	53.02%
Min	-6.54%	-14.15%	-20.79%	-9.22%	-24.91%	-48.31%	-9.22%	-24.91%	-48.31%
Count	33	33	33	65	65	65	98	98	98

Source: BofA Global Research, Bloomberg

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SPX after FFDs in Y2 of USPC: If FFDs were up, weakness lies ahead.

An up FFDs in Y2 of U.S. Presidential Cycle tended to precede a weak Jan, 1Q and 1H.

Exhibit 176: S&P 500 tended to decline if the FFDs were up in Y2 of USPC

S&P 500 trends after the first five trading days in January in year 2 of the U.S. Presidential Cycle from 1928-2025

	Trend if FFDs down			Trend if FFDs up			Trend all Y2 periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	1.87%	4.62%	-1.08%	-1.05%	-2.84%	-2.23%	0.17%	0.27%	-1.75%
Median	0.59%	1.26%	-1.69%	-0.31%	-2.22%	-1.31%	-0.23%	-0.39%	-1.31%
% time up	60%	60%	50%	36%	43%	43%	46%	50%	46%
Std dev	5.35%	8.86%	14.24%	4.31%	7.71%	10.02%	4.88%	8.86%	11.68%
Max	14.45%	18.08%	20.61%	4.97%	8.06%	17.17%	14.45%	18.08%	20.61%
Min	-3.45%	-6.35%	-20.79%	-8.26%	-24.91%	-21.54%	-8.26%	-24.91%	-21.54%
Count	10	10	10	14	14	14	24	24	24

Source: BofA Global Research, Bloomberg

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DXY: FFDs up or down shows little trend bias

Exhibit 177: The outcome of the FFDs of the year showed little trend bias for DXY. However, if FFDs were down, average and median returns were more positive than if FFDs were up.
DXY trends after the first five trading days in January from 1971-2025

	Trend if FFDs down			Trend if FFDs up			Trend all periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	1.05%	1.39%	1.12%	0.07%	0.04%	0.20%	0.46%	0.58%	0.57%
Median	0.70%	1.66%	0.03%	0.04%	0.30%	0.40%	0.40%	0.54%	0.17%
% time up	55%	59%	50%	52%	52%	58%	53%	55%	55%
Std dev	2.42%	4.68%	6.72%	2.26%	4.60%	6.89%	2.35%	4.64%	6.78%
Max	5.83%	10.00%	19.69%	5.47%	8.77%	13.04%	5.83%	10.00%	19.69%
Min	-3.06%	-7.36%	-8.79%	-4.05%	-8.25%	-14.86%	-4.05%	-8.25%	-14.86%
Count	22	22	22	33	33	33	55	55	55

Source: BofA Global Research, Bloomberg

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Oil: FFDs down and oil up 78% of time in Jan. If FFDs up, buy Jan dip for up 1Q.

When oil was down in the FFDs of January, it tended to rebound by the end of January, rising 78% of the time (14 of 18) by an average of 2.29%, median 4.81%. When FFDs were up, the rest of Jan may dip before a bullish 1Q, which saw oil rise 78% of the time.

Exhibit 178: When FFDs were down, oil tended to be up in January. If FFDs were up, January tended to be down, 1Q and 1H tended to be up.
Oil trends after the first five trading days in January from 1990-2025

	Trend if FFDs down			Trend if FFDs up			Trend all periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	2.29%	1.53%	7.64%	-0.58%	7.51%	13.79%	0.86%	4.52%	10.72%
Median	4.81%	8.52%	4.20%	-0.96%	5.06%	16.94%	2.65%	5.33%	11.26%
% time up	78%	67%	50%	39%	78%	72%	58%	72%	61%
Std dev	7.89%	20.66%	25.83%	7.09%	13.08%	21.55%	7.53%	17.31%	23.65%
Max	12.04%	21.23%	67.60%	11.72%	28.23%	47.59%	12.04%	28.23%	67.60%
Min	-20.72%	-65.64%	-34.12%	-12.44%	-23.33%	-25.62%	-20.72%	-65.64%	-34.12%
Count	18	18	18	18	18	18	36	36	36

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Gold: When the FFDs were up, rest of January up 72% of time

When gold was up in the first five days, it tended to be up in the rest of January 72% of the time by an average of 1.89%, median 2.49%.

Exhibit 179: If gold was up in the FFDs, it tended to be higher for the rest of January more so than if the FFDs were down.
Gold trend after the first five trading days from 1975-2025

	Trend if FFDs down			Trend if FFDs up			Trend all periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	0.40%	2.35%	1.21%	1.89%	-0.36%	2.18%	1.13%	1.02%	1.69%
Median	0.73%	2.40%	2.45%	2.49%	0.38%	3.88%	1.01%	1.40%	2.76%
% time up	65%	58%	62%	72%	60%	56%	69%	59%	59%
Std dev	4.57%	6.22%	11.34%	3.98%	8.96%	10.86%	4.32%	7.73%	11.01%
Max	10.07%	13.66%	24.38%	7.92%	17.34%	24.09%	10.07%	17.34%	24.38%
Min	-12.33%	-11.08%	-26.27%	-6.08%	-20.05%	-20.67%	-12.33%	-20.05%	-26.27%
Count	26	26	26	25	25	25	51	51	51

Source: BofA Global Research, Bloomberg

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US 10Y Yield: If FFDs down, 1H up 67% of time. If FFDs up, Jan down & Q1 up.

When US 10Y yield was down in the first five days of the year, yield tended to be higher 67% of the time ending 1H. If FFDs were up, there was a modest tendency to see yield move lower by the end of January and then rise ending 1Q.

Exhibit 180: When US 10Y yield was up in the first five days of the year, it tended to be higher in 1Q
US 10Y yield trend after the first five trading days from 1963-2025

	Trend if FFDs down			Trend if FFDs up			Trend all periods		
	January	1Q	1H	January	1Q	1H	January	1Q	1H
Average	-2	8	24	-2	6	-1	-2	7	11
Median	-1	6	27	-5	15	-3	-3	13	14
% time up	47%	57%	67%	36%	61%	45%	41%	59%	56%
Std dev	20	46	76	26	64	73	23	56	75
Max	44	109	212	56	207	125	56	207	212
Min	-35	-115	-125	-67	-174	-176	-67	-174	-176
Count	30	30	30	33	33	33	63	63	63

Source: BofA Global Research, Bloomberg

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Part 3: January Barometer – So goes Jan, so goes the year?

SPX: Up Jan says rest of year up 79% of time. Down Jan a headwind.

It is said that if January is up, it bodes well for the year (at least for the S&P 500), and conversely if down. We find this adage to be true. If January was down, the rest of Q1 and H1 saw negative returns with lower up ratios. If January was up, the rest of Q1 was up 66% of time, 1H was up 75% of the time and the year up 79% of the time.

Exhibit 181: When SPX down in Jan, rest of Q1 and H1 struggled. When SPX up in Jan, the year gets better and better.
S&P 500 January Barometer from 1928-2025

	If January down...			If January up...			All periods		
	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec
Average	-0.19%	-1.67%	3.09%	0.75%	5.05%	8.99%	0.40%	2.51%	6.76%
Median	-0.49%	-0.16%	2.45%	1.04%	4.65%	11.78%	0.82%	3.77%	9.21%
% time up	43%	49%	59%	66%	75%	79%	57%	65%	71%
Std dev	6.05%	12.18%	18.28%	6.37%	10.40%	17.32%	6.24%	11.52%	17.83%
Max	10.63%	15.70%	47.58%	12.81%	57.20%	43.66%	12.81%	57.20%	47.58%
Min	-19.87%	-44.21%	-34.48%	-20.49%	-13.34%	-49.85%	-20.49%	-44.21%	-49.85%
Count	37	37	37	61	61	61	98	98	98

Source: BofA Global Research, Bloomberg

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SPX J.B. in Y2 USPC: Year 2 weakness feeds through JB theory

When SPX was down in January in year 2 of the USPC, the rest of Q1 (Feb-Mar) bounced back 70% of the time. But the rest of the 1H (Feb-Jun) was up just 30% of the time (or down 70%). So, a down January in Y2 is headwind with choppy scenarios. When January was up in year 2 of USPC, up ratios and returns struggled in Feb-Mar and Feb-Jun, but to a lesser degree than if January was down. However, the Feb-Dec period saw stronger average and median returns likely due to a strong Q4 rally.

Exhibit 182: In year 2 of the USPC, a down January leads to up Feb-Mar but down Feb-Jun. An up January saw less-weak average and median returns until year end when the Feb-Dec returns went strongly positive.
S&P 500 January Barometer from 1928-2025 in Y2 of USPC only.

	If January down...			If January up...			All Y2 periods		
	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec
Average	1.58%	-6.17%	-0.86%	-0.95%	1.15%	5.67%	0.10%	-1.90%	2.95%
Median	1.28%	-9.95%	4.02%	-0.95%	-0.76%	10.60%	1.00%	-3.88%	8.04%
% time up	70%	30%	60%	50%	43%	57%	58%	38%	58%
Std dev	4.47%	11.09%	16.88%	9.12%	10.03%	21.42%	7.51%	10.89%	19.54%
Max	8.90%	9.97%	17.11%	12.81%	18.44%	37.96%	12.81%	18.44%	37.96%
Min	-7.01%	-20.47%	-29.00%	-20.49%	-12.18%	-32.69%	-20.49%	-20.47%	-32.69%
Count	10	10	10	14	14	14	24	24	24

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

DXY J.B.: So goes January, so goes the DXY

When DXY was down in January, it also tended to be down at the end of March and June.
When DXY was up in January, it tended to be up ending June and December.

Exhibit 183: If DXY down in Jan, it tended to favor a down Feb-Mar and Feb-Jun. If DXY was up in Jan, it favored an up Feb-Jun and Feb-Dec.
DXY January Barometer from 1971-2025

	If January down...			If January up...			All periods		
	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec
Average	-0.62%	-1.91%	-3.24%	0.55%	1.32%	0.48%	0.10%	0.08%	-0.94%
Median	-0.83%	-1.60%	-3.67%	0.68%	0.98%	3.33%	0.29%	0.26%	0.65%
% time up	38%	33%	43%	59%	65%	62%	51%	53%	55%
Std dev	4.10%	6.53%	7.17%	3.17%	5.26%	8.07%	3.56%	5.93%	7.88%
Max	12.21%	17.11%	8.26%	8.64%	13.09%	11.78%	12.21%	17.11%	11.78%
Min	-7.19%	-13.88%	-14.75%	-6.03%	-11.73%	-19.16%	-7.19%	-13.88%	-19.16%
Count	21	21	21	34	34	34	55	55	55

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Oil J.B.: An up January modestly favors an up Feb-Jun

Exhibit 184: Oil trend after an up or down January showed little conviction. At best, an up January favors higher oil by end of Jun 63% of time
Crude oil January Barometer from 1990-2025

	If January down...			If January up...			All periods		
	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec
Average	2.14%	11.17%	9.09%	4.70%	8.58%	6.91%	3.49%	9.80%	7.94%
Median	4.20%	8.08%	-5.90%	0.89%	8.70%	0.95%	3.53%	8.39%	-1.00%
% time up	59%	59%	47%	58%	63%	53%	58%	61%	50%
Std dev	20.60%	27.24%	42.09%	10.49%	18.56%	31.19%	15.89%	22.76%	36.21%
Max	35.06%	67.68%	90.40%	31.45%	51.29%	100.78%	35.06%	67.68%	100.78%
Min	-60.28%	-23.84%	-51.39%	-10.58%	-24.74%	-30.77%	-60.28%	-24.74%	-51.39%
Count	17	17	17	19	19	19	36	36	36

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Gold J.B.: After a down Jan, gold might bounce back in Feb-Mar

Exhibit 185: When gold was down in January, it tended to rally back modestly in Feb-March 64% of the time.
Gold January Barometer from 1975-2025

	If January down...			If January up...			All periods		
	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec
Average	1.28%	-0.69%	2.85%	-1.34%	1.66%	10.90%	-0.05%	0.51%	6.95%
Median	1.43%	0.56%	1.84%	-0.83%	-0.11%	7.29%	1.34%	-0.08%	3.20%
% time up	64%	52%	64%	42%	46%	58%	53%	49%	61%
Std dev	6.05%	9.77%	16.05%	7.75%	9.62%	27.79%	7.03%	9.67%	22.94%
Max	12.55%	14.91%	31.47%	11.62%	18.74%	119.08%	12.55%	18.74%	119.08%
Min	-17.31%	-25.80%	-27.77%	-24.27%	-16.72%	-23.44%	-24.27%	-25.80%	-27.77%
Count	25	25	25	26	26	26	51	51	51

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

US 10Y Yield J.B.: An up Jan favors higher yield

An up January tended to lead to higher yield ending March and June 71% of the time.

Exhibit 186: When yield was up in January, it tended to be higher ending March and June 71% of time.
US 10Y Yield January Barometer from 1963-2025

	If January down...			If January up...			All periods		
	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec	Feb-Mar	Feb-Jun	Feb-Dec
Average	6	5	-15	12	21	13	9	14	0
Median	7	-7	-20	9	25	28	8	10	9
% time up	59%	45%	41%	71%	71%	62%	65%	59%	52%
Std dev	42	79	106	50	61	116	46	69	112
Max	110	220	218	151	124	210	151	220	218
Min	-84	-138	-219	-171	-173	-375	-171	-173	-375
Count	29	29	29	34	34	34	63	63	63

Source: BofA Global Research, Bloomberg

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Don't sell in May!

- It is commonly suggested that the equity market cycle favors selling in May to reduce exposure to risk assets ahead of a mid-year/summer period when market participants take vacations, liquidity declines and drawdowns may worsen.
- When we review 6-, 3- and 1-month average trend change, we do not see this. In fact, we could argue that seasonality favors buying in May and selling in July/August.
- While May-Oct does have the lowest average 6-month return, it is back-end-loaded to the Aug-Oct period. Maybe we should say "sell in August and buy Halloween."

May-Oct: Lowest average & median returns since 1928

When assessing the validity of "Sell in May and Go Away", we use S&P 500 data since 1928. Seasonality shows that May-October is the weakest 6-month period of the year. May-October is up 66% of the time with average return of 2.4% (3.3% median). November-April is the strongest 6-month period of the year on average with the highest up ratio of 70.1%. March-August was also strong. (Exhibit 187**Error! Reference source not found.**).

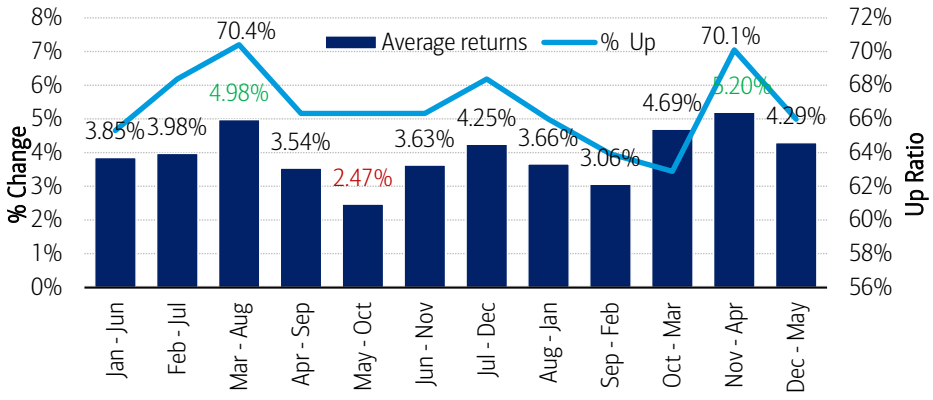
Exhibit 187: S&P 500 - 6-month seasonal returns – 1928 to present
May through October has the worst average and median S&P 500 returns of any 6-month period

6-month period	Average return	Median return	% of time up	Average correction	Median correction
January-June	3.85%	5.01%	65.3%	-7.60%	-5.92%
February-July	3.98%	4.56%	68.4%	-7.90%	-4.85%
March-August	4.98%	4.94%	70.4%	-7.58%	-5.39%
April-September	3.54%	3.94%	66.3%	-7.79%	-4.29%
May-October	2.47%	3.35%	66.33%	-8.67%	-5.21%
June-November	3.63%	4.58%	66.3%	-8.31%	-5.24%
July-December	4.25%	5.52%	68.4%	-8.28%	-4.93%
August-January	3.66%	4.95%	66.0%	-8.68%	-5.68%
September-February	3.06%	5.00%	63.9%	-8.56%	-5.29%
October-March	4.69%	5.92%	62.9%	-7.65%	-4.94%
November-April	5.20%	5.39%	70.10%	-6.76%	-4.11%
December-May	4.29%	4.63%	66.0%	-7.28%	-4.60%
Averages	3.96%	4.79%	66.70%	-7.88%	-4.93%

Source: BofA Global Research, Bloomberg

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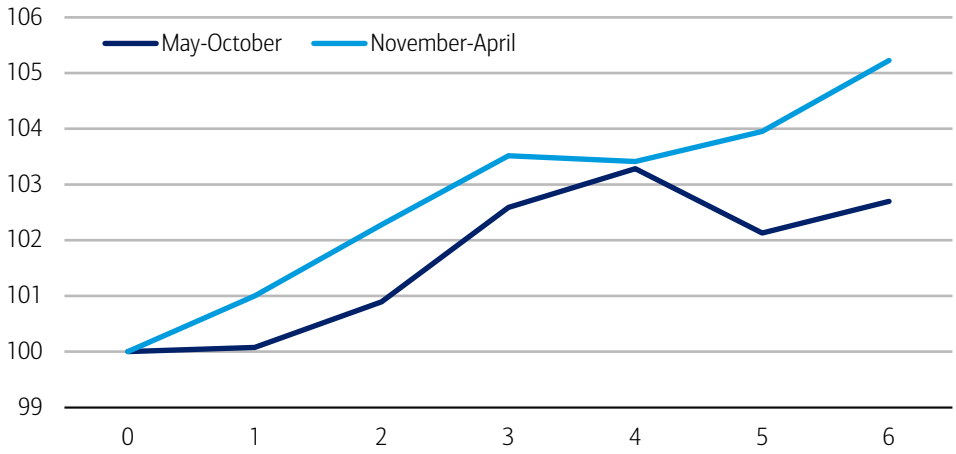
Exhibit 188: S&P 500 - 6-month seasonality returns – 1928 to present
May-October has the weakest average S&P 500 return of 2.47% while November-April has the best average 6-month period return of 5.2%. March-August was also strong.



Source: BofA Global Research, Bloomberg

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Exhibit 189: Average trend from Nov-Apr outperforms May-Oct
Average trend in May-Oct vs Nov-Apr since 1928



Source: BofA Global Research, Bloomberg

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But May-October is back-end-loaded

The average 3-month return for all 3-month periods going back to 1928 is 2.01%. The second strongest 3-month period of the year is June-August with a 3.3% average return, which suggests a "summer rally". The weakest 3-month period of the year is August-October, which returns -0.02% on average. Both 3-month periods are between May and October. This suggests that the weak May-October period is back-end-loaded and often a sell in July/August rather than a sell in May pattern.

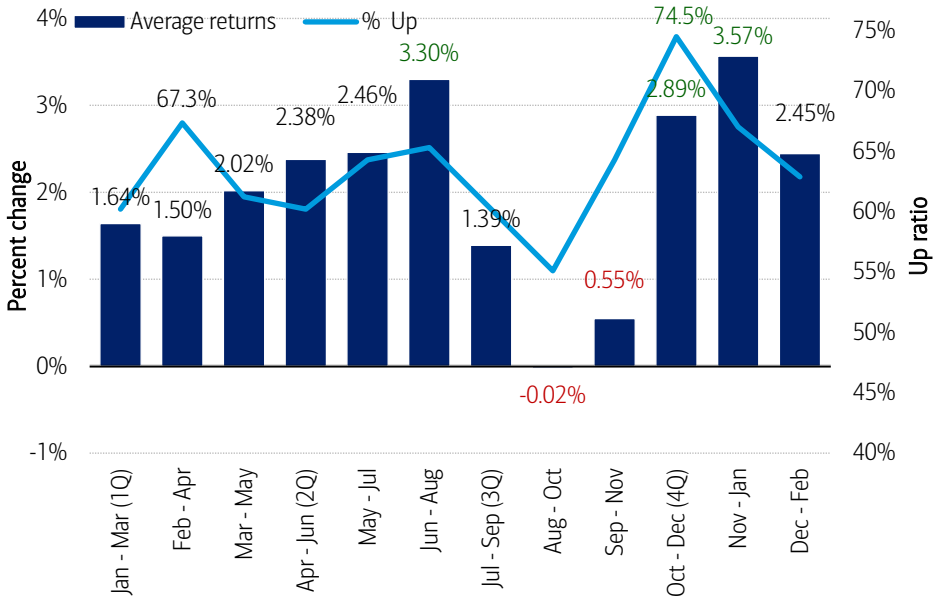
Exhibit 190: S&P 500 - 3-month average trend favors long Nov-Jan and Jun-Aug, not Aug-Oct.
S&P 500 - 3-month seasonal returns – 1928 to present

3-month period	Average return	Median return	% of time up	Average correction	Median correction
Jan - Mar (1Q)	1.64%	2.10%	60.20%	-5.28%	-3.74%
Feb - Apr	1.50%	2.28%	67.35%	-5.51%	-3.87%
Mar - May	2.02%	2.61%	61.22%	-5.64%	-2.97%
Apr - Jun (2Q)	2.38%	2.24%	60.20%	-5.34%	-3.60%
May - Jul	2.46%	2.04%	64.29%	-5.35%	-2.97%
Jun - Aug	**3.30%**	**2.45%**	**65.31%**	**-5.08%**	**-3.36%**
Jul - Sep (3Q)	1.39%	2.91%	60.20%	-5.39%	-2.92%
Aug - Oct	-0.02%	0.90%	55.10%	-7.38%	-5.31%
Sep - Nov	0.55%	2.45%	64.29%	-7.40%	-4.87%
Oct - Dec (4Q)	2.89%	4.63%	74.49%	-5.66%	-3.52%
Nov - Jan	3.57%	4.17%	67.01%	-4.56%	-2.25%
Dec - Feb	2.45%	2.50%	62.89%	-4.58%	-3.63%
Averages	2.01%	2.49%	63.54%	-5.59%	-3.59%

Source: BofA Global Research, Bloomberg

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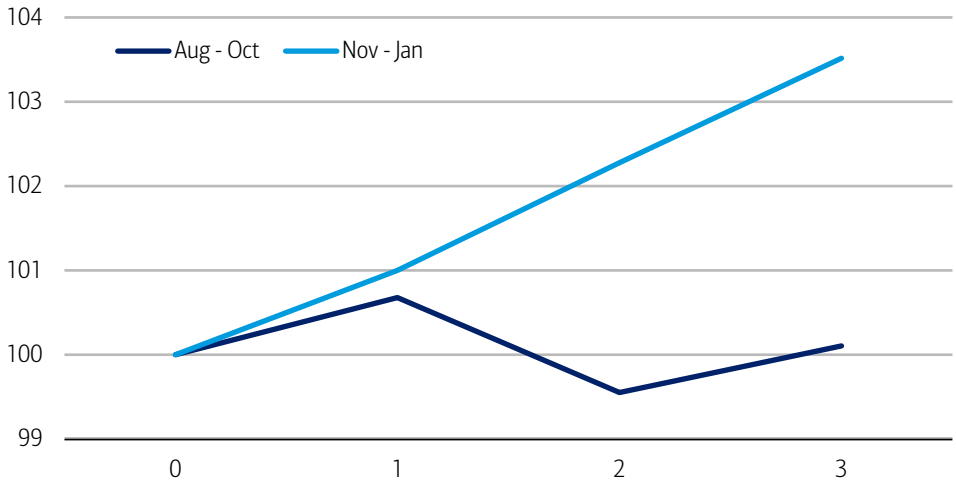
Exhibit 191: S&P 500 - 3-month seasonal returns – 1928 to present
The historical pattern below suggests buying in May and then selling in July/August, rather than selling in May



Source: BofA Global Research, Bloomberg

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Exhibit 192: Aug-Oct underperforms Nov-Jan. September is a weak month.
Average trend in Aug-Oct vs Nov-Jan



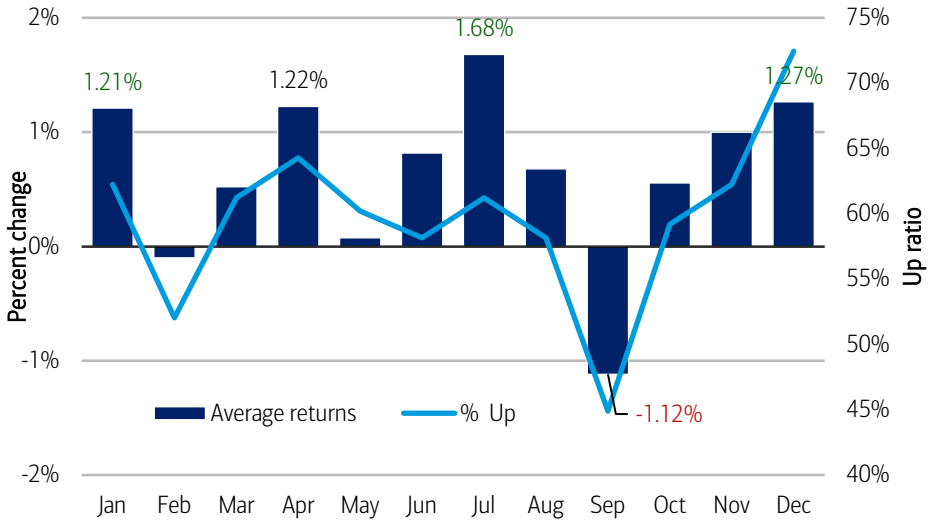
Source: BofA Global Research, Bloomberg

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Monthly seasonality says buy Feb and Sep dips

The ebbs and flows of monthly SPX seasonality suggest buying into weakness in February and September and selling into strength in April, July/August and December/January. It is not "sell in May and go away" but rather "buy in May and sell in July/August".

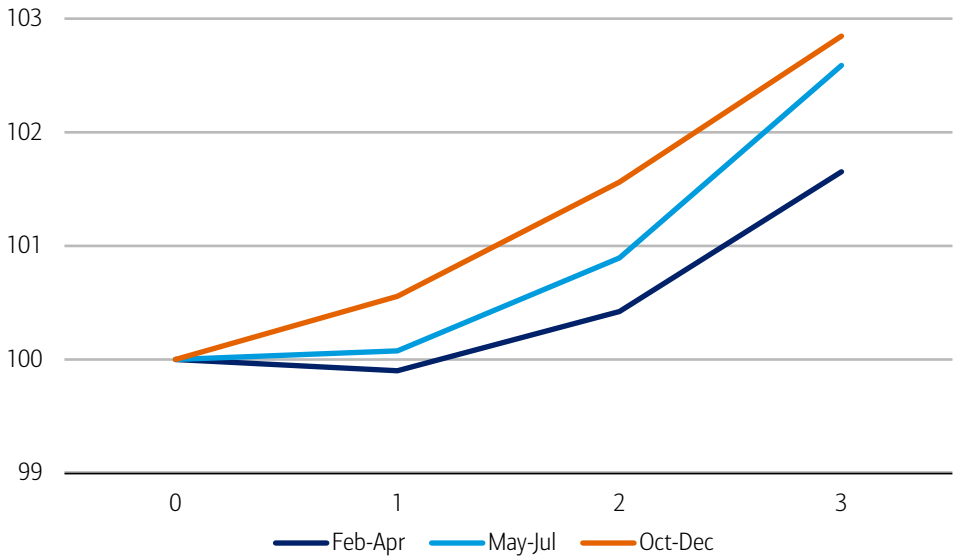
Exhibit 193: S&P 500 - 1-month seasonality returns – 1928 to present
Monthly SPX seasonality suggest buying weakness in February and September and selling strength in April, July/August and December/January.



Source: BofA Global Research, Bloomberg

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Exhibit 194: Average trend in Oct-Dec outperforms May-Jul and Feb-Apr
Comparison of three average trend periods



Source: BofA Global Research, Bloomberg

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Dow Theory

Dow Theory is among the older indicators in technical analysis. Charles Dow and Edward Jones developed the Dow Theory in the late 1800s. These gentlemen should sound familiar. Together, Mr. Dow and Mr. Jones founded Dow Jones and Company, which created the Dow Jones Industrial Average (DJIA). Although the Dow Theory is considered an indicator for technical analysis, Charles Dow believed that the stock market provided a reliable measure of overall business conditions and used it as an economic barometer.

What is the Dow Theory

Investors who are familiar with the Dow Theory see it as the interplay between the trends of the DJIA and the Dow Jones Transportation Average (DJTA). When both trends are rising (higher highs and higher lows), Dow Theory is bullish and suggests a primary uptrend for the US equity market. When both trends are declining (lower highs and lower lows), Dow Theory is bearish and suggests a primary downtrend. The other condition is when the DJIA and the DJTA are not moving either up or down together, which leads to non-confirmation of either a bullish or bearish trend. This may or may not precede a definitive reversal of the primary trend. In our view, Dow Theory is much more than that, and its application should not be limited to just the DJIA and DJTA.

The major tenets of Dow Theory

Dow Theory has several major tenets: The market discounts everything, the market has three trends, primary trends have three phases, the indices mush confirm each other, volume must confirm the trend and trends persist until definitively reversed. We discuss these tenets below.

The market (price action) discounts everything

A cornerstone of technical analysis is that market action discounts everything. This suggests that that technical analysis and the Dow Theory work in line with the efficient market hypothesis, which states that asset prices reflect all available information. Big beats or big misses from corporate earnings reports, out of consensus economic data releases, geopolitics, wars and natural calamities can cause dislocations in the market, but even these unexpected events get quickly absorbed into price action.

In practice, prices likely reflect actual fundamental and macro conditions as well as retail and institutional investor sentiment, opinions, and narratives surrounding these fundamental and macro themes. Sometimes noise from algorithmic and high frequency trading can have a short-lived impact on price action, but these factors most likely do not impact the longer-term trend of the market and Dow Theory application.

There are three market trends: Primary, secondary and minor

A series of higher highs and higher lows represents an uptrend, while a series of lower highs and lower lows represents a downtrend. Dow Theory states that the market has three trends: primary, secondary and minor.

- **Primary trends** are what the Dow Theory focuses on. These trends reflect the "tide" of the market and typically last a year or more.
- **Secondary trends** are "waves" that either continue or represent corrections within the primary trend. When these intermediate-term trends correct the primary trend, they tend to last from three weeks to a quarter and can retrace from one-third to two-thirds of the previous trend. Fibonacci levels can also guide investors during these corrective phases.
- **Minor trends** provide "ripples" within secondary trends that usually last less than three weeks. Investors who follow Dow Theory view these trends as market noise.

Primary trends have three phases

In the Dow Theory, primary uptrends and primary downtrends tend to occur in three phases. The three phases for a primary uptrend (aka bull market) are accumulation, public participation (mark-up) and excess. The three phases for a primary downtrend (aka bear market) are distribution, public participation (mark-down) and panic. Combining the phases for bullish and bearish primary trends show the market cycle. For example, panic moves into accumulation as a downtrend ends and transitions to an uptrend, and excess shifts into distribution as an uptrend ends and changes into a downtrend.

The three phases for a primary uptrend

- **Accumulation** starts when sentiment is bearish and the prevailing economic or fundamental narratives are negative, but the primary market downtrend has discounted most or all the bad news. Accumulation reflects informed and contrarian fundamental buying. This phase may also represent some technical buying from those who identified bullish divergences (e.g., improving breadth) on the most recent primary bear market lows. In summary, the accumulation phase represents buying, or covering shorts, when all seems bleak, and nobody wants to buy.

- **Public participation (mark-up)** occurs when technical trend followers as well as institutional and retail investors buy into a rally as fundamental and economic news improves. A big mark-up in the equity market tends to occur during this phase, which is generally the longest phase of primary bull markets.

- **Excess** represents the climactic and often euphoric buying phase for investors that happens near the end of a primary uptrend. Some of these investors bought during the public participation phase, while others have missed the rally and have just started buying since the fundamental and economic news flow remains mostly bullish. During this phase, some of the more informed investors who bought in the accumulation phase take a contrarian bearish view and begin to sell to a public that does not want to sell or initiate short positions.

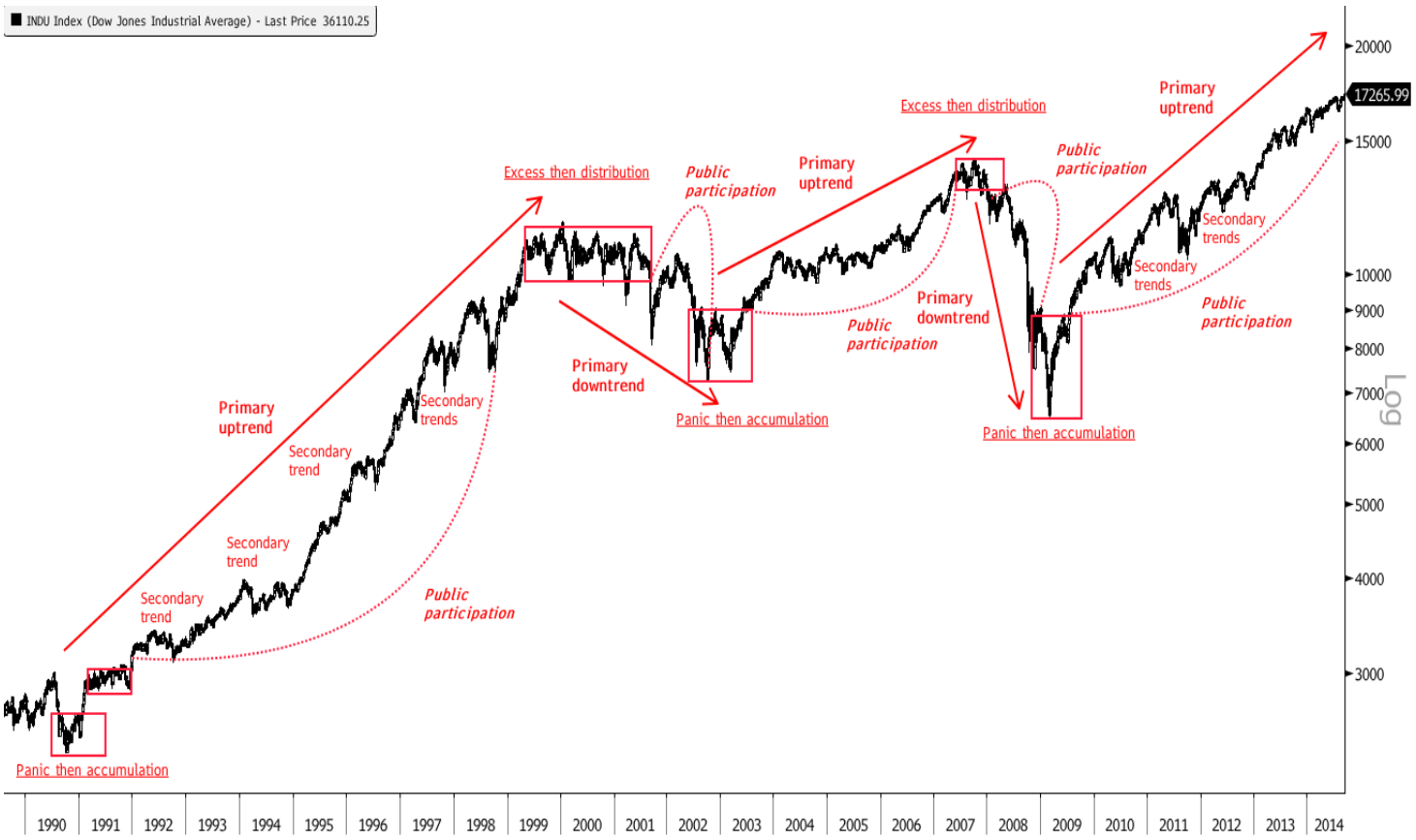
The three phases for a primary downtrend

- **Distribution** happens when economic and fundamental narratives remain positive, but public excesses (aka speculation) have gotten to the point of no marginal buyers for equities. Informed fundamental investors and technical traders, some of whom bought in the accumulation phase, sell or short the market. Astute technical traders may spot bearish divergences (e.g., weakening breadth) and start to sell out long equity positions and/or short the market.

- **Public participation (mark-down)** occurs when technical trend followers and other investors sell into a downtrend as fundamental and economic news deteriorates. A big mark-down in the equity market tends to occur during this phase, which is generally the longest phase of primary bear markets. Short positions typically increase during this phase as well.

- **Panic** is the climactic and often despondent selling phase for investors that happens near the end of a primary downtrend. Some of these investors sold during the public participation phase, while others missed earlier opportunities to sell and have just begun selling since the fundamental and economic new flows remains predominantly bearish. During this phase, some of the more informed investors who sold in the distribution phase take a contrarian bullish view and begin to buy stocks, or cover shorts, in an environment in which no investors want to buy.

Exhibit 195: Dow Jones Industrial Average
The cycle goes on. Primary uptrend: Accumulation, public participation and excess. Primary downtrend: Distribution, public participation and panic.



Source: BofA Global Research, Bloomberg

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The averages must confirm each other

The Dow Theory has four positions: A confirmed primary uptrend, a non-confirmed primary uptrend, a confirmed downtrend and a non-confirmed downtrend. We consider this a market cycle. Primary uptrends are bullish. Non-confirmed uptrends set up corrections within an ongoing bull market but can sometimes precede a shift into a confirmed primary downtrend. Primary downtrends are bearish. Non-confirmed downtrends set up corrections within an ongoing bear market but can sometimes precede a shift into a confirmed primary uptrend.

- **A confirmed primary uptrend (bull market)** occurs when both the DJIA and DJTA are trending higher and moving to new highs together.
- **A non-confirmed primary uptrend (negative divergence)** happens when both the DJIA and DJTA are trending higher, but one index begins to lag the other. The primary bull market is intact but with a non-confirmation (aka negative divergence) within the primary uptrend. Secondary downtrends, or corrections, against the primary uptrend often emerge when this bearish divergence occurs. However, these non-confirmations can also precede a transition to a primary downtrend.
- **A confirmed primary downtrend (bear market)** occurs when both the DJIA and DJTA are trending lower and moving to new lows together.
- **A non-confirmed primary downtrend (positive divergence)** happens when both the DJIA and DJTA are trending lower, but one index begins to improve relative to the other. The primary bear market is intact but with a non-confirmation (aka positive divergence) within the primary downtrend. Secondary uptrends against the

primary downtrend often appear when this bullish divergence occurs. However, these non-confirmations can also herald the switch to a primary downtrend.

Volume must confirm the trend

Since Dow Theory signals are based only on daily closing prices, this tenet on volume provides only a secondary indicator. Price is king and most important. The theory is that volume should expand in the direction of the major trend. This means that a primary bull market should see volume expand on rallies and diminish as prices drop in secondary corrections. The opposite should happen in primary bear markets. In practice, volume has taken on the role of a sentiment indicator. Rallies have tended to grind higher on lower volume (calmness) and declines have generally happened on higher volume (urgency).

A trend is in effect until it gives definitive signals that it has reversed

It can be difficult to distinguish between a secondary correction in an existing trend and the first move of a new trend in the opposite direction. However, the good news is that many indicators highlighted in this primer can help investors navigate these shifts in market trends. In strict Dow Theory terms, we will focus on price trends as defined by peaks and troughs on the DJIA and DJTA using daily closing price charts.

- **A shift from a primary bull market to a primary bear market** happens when either both the DJIA and DJTA form a series of lower highs and lower lows or when both indices break their most recent higher lows after a bearish non-confirmation.
- **A shift from a primary bear market to a primary bull market** happens when either both the DJIA and DJTA form a series of higher lows and higher highs or when both indices break their most recent lower highs after a bullish non-confirmation.

Exhibit 196: Dow Jones Industrial Average (top) and Dow Jones Transportation Average (bottom): Daily closing price chart
Shifts in the Dow Theory from 2007 into 2009.



Source: BofA Global Research, Bloomberg

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Dow Theory is a trend confirmation signal

Definitive shifts to primary uptrends and primary downtrends in the Dow Theory are not early technical signals, but rather trend confirmation signals that can happen well after the market low or market peak. Most confirmed primary uptrend Dow Theory signals happen moving from the accumulation phase to the public participation phase in a primary uptrend. Most confirmed primary downtrend Dow Theory signals occur as the distribution phase shifts to the public participation, or mark-down, phase in a primary downtrend.

Other applications of Dow Theory

The Dow Theory has come a long way since Charles Dow monitored the movements of the Dow Jones Rail Average and Dow Jones Industrial Average over 100 years ago. The rails shipped raw materials to industrial manufacturers and shipped finished products from the industrials, so watching the interplay between these two economically sensitive indices made sense. Today, there are many other indices to watch to gauge the health of the equity market, which is a leading indicator of the economy.

Small caps vs large caps

Non-confirmation signals, or lower highs, for the Russell 2000 (RTY) against the higher highs on the SPX preceded the US equity market corrections in late 2018 and early 2020. The RTY bottomed ahead of the SPX, forming a tactical bullish divergence on the spike low for US equities in March 2020.

Exhibit 197: S&P 500 (top) and Russell 2000 (bottom): Daily closing price chart
Non-confirmation signals, or lower highs, for the Russell 2000 (RTY) against the higher highs on the SPX preceded the US equity market corrections in late 2018 and early 2020. The RTY bottomed ahead of the SPX, forming a tactical bullish divergence on the spike low for US equities in March 2020.



Source: BofA Global Research, Bloomberg

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Capital goods vs transports

Using S&P 500 GICs sectors and industry group charts also can help gauge the health of the US equity market. Turns in the cyclical groups of capital goods and transports can confirm bullish and bearish turns for the US equity market.

Exhibit 198: S&P 500 Capital Goods Index (top) and S&P 500 Transportation Index (bottom): Daily closing price chart
Turns in the cyclical groups of capital goods and transports can confirm bullish and bearish turns for the US equity market.



Source: BofA Global Research, Bloomberg

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Market Breadth

Advance-decline lines

The cumulative advance-decline (A-D) line measures the extent of participation, or breadth, of a rally or decline for a market index. A rising A-D line means improving market breadth, while a declining A-D line reflects weakening market breadth. Standard A-D lines include the NYSE stocks and S&P 500 A-D lines, but investors can create A-D lines for other indices, asset classes, sectors and even portfolios.

A-D line calculation: Previous day's value of A-D line + current day's net advances (advances – declines)

Exhibit 199: Sample calculation of an advance-decline line
The advance-decline line is a cumulative measure of net advancing issues

Day	Advances	Declines	Net	Cumulative A-D
1	1200	800	400	400
2	1400	700	700	1100
3	1600	400	1200	2300
4	900	1100	-200	2100

Source: BofA Global Research, Bloomberg

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How to use the A-D line in uptrends

A rally for an index with a rising or strengthening A-D line suggests a broad rally. Strong breadth confirms the rally, which is bullish and suggests that the uptrend can continue. When the A-D line begins to lag and does not confirm new highs for an index within an uptrend, breadth has narrowed with fewer stocks advancing and more stocks starting to decline, even as the underlying index has continued to rally. This decrease in upside participation provides a negative divergence and is a sign that the uptrend for the market index may be coming to an end.

Negative (or bearish) divergence: The price trend is moving higher but a technical indicator begins to fall and does not confirm the move higher in the price trend. Higher highs in price and lower highs in the indicator reflect a bearish divergence. This is a sign that the price uptrend may be coming to an end.

Exhibit 200: NYSE Comp (top) and NYSE stocks advance-decline line (bottom): Daily chart
Broadening rally and bullish confirmation from late 2020 into mid 2021. A narrowing rally and bearish divergences from mid 2021 into January 2022. Confirmation of the 2022 downtrend into October.



Source: BofA Global Research, Bloomberg

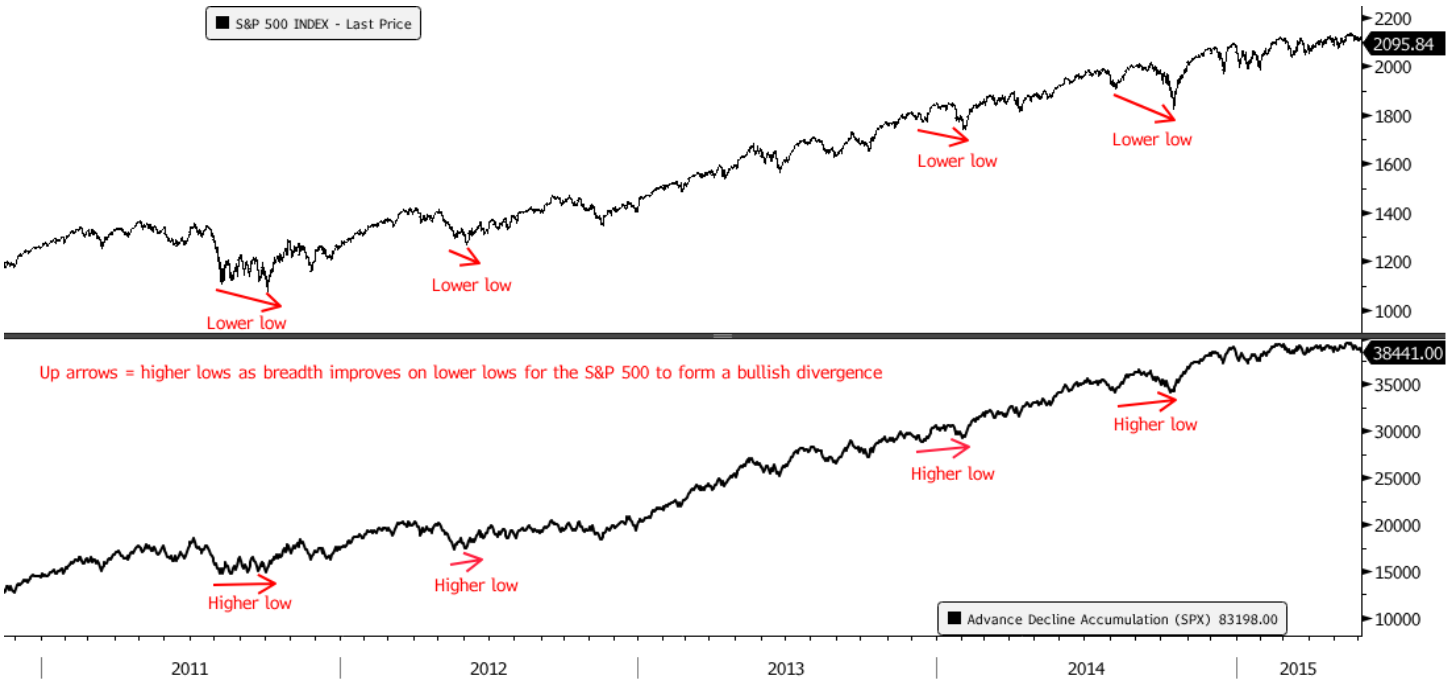
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How to use the A-D line in downtrends and corrections

A decline for an index with a falling or weakening A-D line suggests a broad decline. Weak breadth confirms the decline, which is bearish and suggests that the downtrend can continue. When the A-D line begins to improve and does not confirm new lows for an index within a downtrend, or a correction, breadth has improved with fewer stocks declining and more stocks starting to rise, even as the underlying index has continued to fall. This decrease in downside participating provides a positive divergence and is a sign that the downtrend for the market index may be coming to an end.

Positive (or bullish) divergence: The price trend is moving lower but a technical indicator begins to rise and does not confirm the move lower in the price trend. Lower lows in price and higher lows in the indicator reflect a bullish divergence. This is a sign that the price downtrend may be coming to an end.

Exhibit 201: S&P 500 (top) and S&P 500 advance-decline line (bottom): Daily chart
Breadth confirmed the 2011 and 2012 corrections until the A-D line improved on higher lows. The A-D line formed higher lows on the 2014 corrections on the SPX.



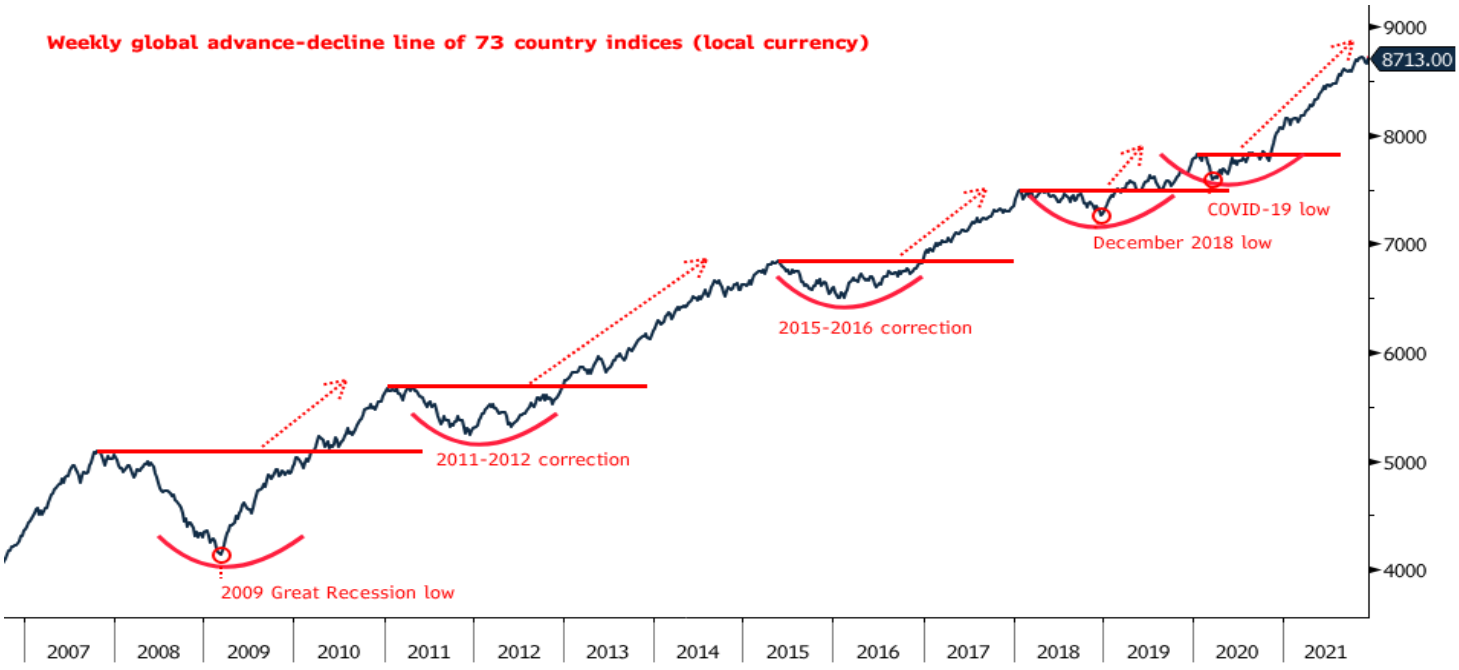
Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Global breadth: Weekly global A-D line of 73 country indices

The weekly global A-D line of 73 country indices measures global equity market breadth. A rising global A-D line means strong breadth at the global index level. A falling global A-D line reflects weak global breadth. Upside breakouts for this A-D line confirm equity bull markets, while breakdowns confirm equity market corrections or bear markets.

Exhibit 202: Weekly global A-D line of 73 country indices
The weekly global A-D line of 73 country indices measures global equity market breadth at the index level.



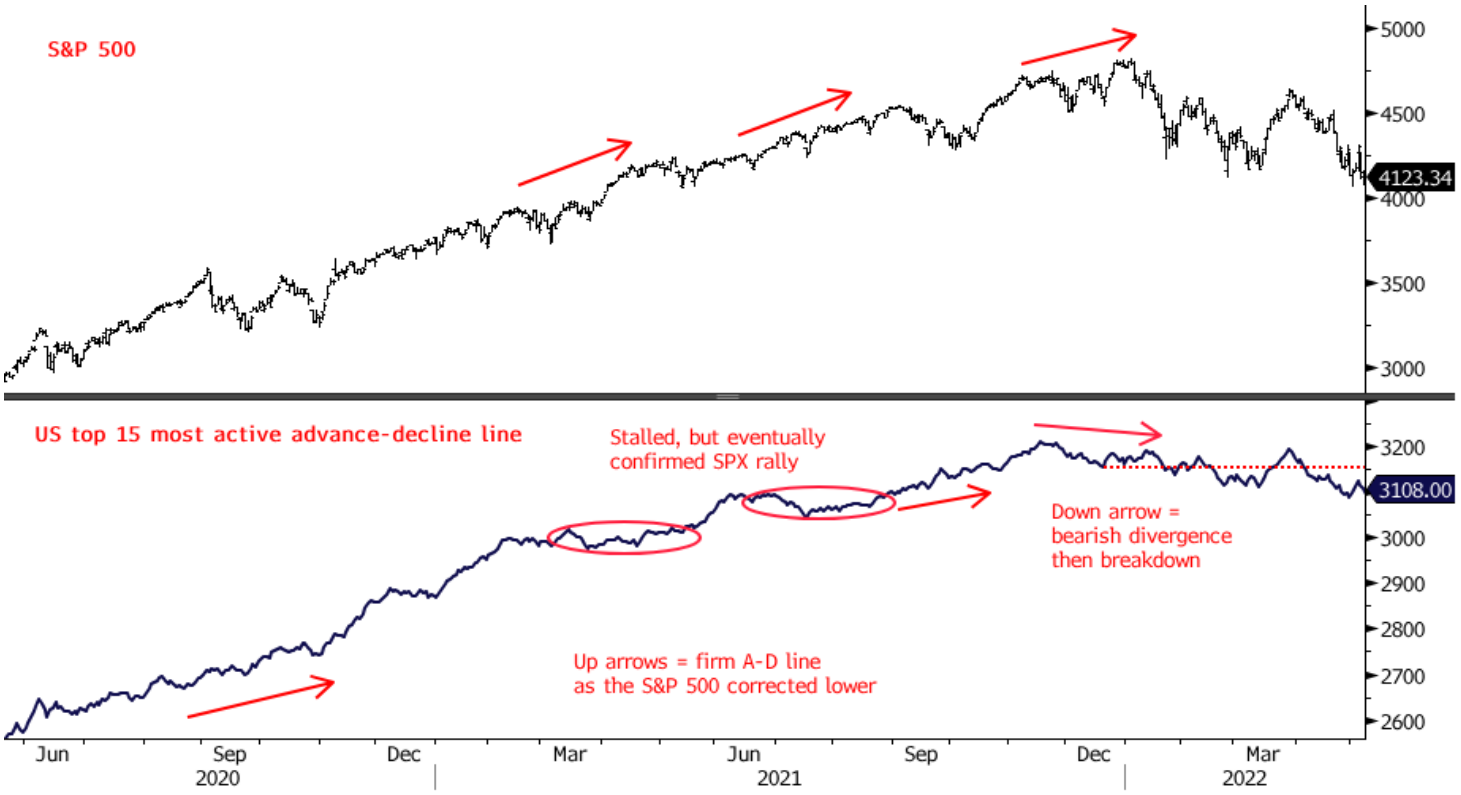
Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

US top 15 most active stocks A-D line

The US top 15 most active A-D line measures the breadth of the top 15 most actively traded US stocks by share volume, excluding those stocks with a market cap below $500 million. This is a big money indicator. When the most active A-D line is rising, the breadth for the most heavily traded stocks is strong. In addition, to positive breadth, this also suggests big money accumulation, or buying, of US stocks. This scenario is bullish. When the most active A-D line is declining, breadth is weak and reflects big money distribution, or selling, of the most heavily traded stocks in the US. This is bearish.

Exhibit 203: S&P 500 (top) and US top 15 most active advance-decline line (bottom): Daily chart
The most active A-D line measures the breadth of the top 15 most actively traded US stocks by share volume.



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Do not use the NASDAQ Comp A-D line to measure market breadth

The A-D line for the NASDAQ Comp has a downward bias and should not be used to measure market breadth. Listing requirements are not as stringent on the NASDAQ as they are on the NYSE. Many IPOs get listed only to fail, and this is more likely to happen on the NASDAQ than it is on the NYSE. Stocks that go from "IPO" to "zero" provide this downward bias, which renders the NASDAQ Comp A-D line useless, for the most part, as a market breadth indicator.

Exhibit 204: NASDAQ Comp (top) and the NASDAQ Comp advance-decline line (bottom): Daily chart

The A-D line for the NASDAQ Comp has a downward bias and should not be used to measure market breadth. Stocks that go from "IPO" to "zero" provide this downward bias, which renders the NASDAQ Comp A-D line useless, for the most part, as a market breadth indicator.



Source: BofA Global Research, Bloomberg

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Beware of using the SPW vs SPX as a breadth indicator

Investors often complain about the narrowness of the US equity market when the larger stocks are stronger than the smaller stocks in the SPX. This concentration shows up in the S&P 500 equal weighted (SPW) relative to the SPX ratio (SPW/SPX). When this ratio is rising, the average stock is beating the largest stocks in the index. When it is falling, the average stock is lagging the largest stocks. Investors view a rising SPW/SPX as bullish for market breadth and a declining ratio as bearish for market breadth. Following this view means that investors would have missed the 1994-2000 and 2016-2020 rallies and would have gotten bullish on the SPX when the SPW/SPX ratio bottomed near the 2000 peak for US equities. SPW/SPX also rose during the 2022 correction.

SPW versus SPX measures concentration of alpha, not market breadth

When the SPW is lagging the SPX, the "average" stock is not generating alpha relative to the SPX. In this environment, a smaller group of large and mega cap stocks (aka "FANG" or the "Magnificent Seven") provide leadership and stronger relative performance. If most stocks are rising, but the largest stocks are outperforming, the SPX A-D line is trending higher as the SPW vs SPX ratio is declining. In other words, market breadth (A-D line) and concentration of leadership (SPW/SPX) are two different indicators. When SPW drops vs the SPX, the US equity market can stay bullish if the rally is confirmed by the A-D line. The market becomes bearish when the A-D line begins to weaken along with the SPW/SPX ratio as it did in early 2000 and in late 2021.

Exhibit 205: S&P 500 (SPX) relative to the S&P 500 Equal Weight Index (SPW) (top) and the S&P 500 advance-decline line (bottom): Daily chart
When the SPW is lagging the SPX, the "average" stock is not generating alpha relative to the SPX. In this environment, a smaller group of large and mega cap stocks (aka "FANG" or the "Magnificent Seven") provide leadership and stronger relative performance. If most stocks are rising, but the largest stocks are outperforming, the SPX A-D line is trending higher as the SPW vs SPX ratio is declining. In other words, market breadth (A-D line) and concentration of leadership (SPW/SPX) are two different indicators. One example: The big SPX rally from 1994 to 2000. The SPW lagged the SPX the entire time. The SPX A-D line rose until 1998 prior to a narrowing of market breadth into the 2000 peak for US equities.



Source: BofA Global Research, Bloomberg

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Exhibit 206: S&P 500 (top) and S&P 500 equal-weighted index relative to the S&P 500 (SPW/SPX) (bottom): Daily chart 2008 to present

The SPW vs. SPX ratio is not market breadth indicator. The SPX can rally significantly even when the SPW versus the SPX is declining, such as during the 2016 into early 2020 period.



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 207: S&P 500 (top) and S&P 500 equal-weighted index relative to the S&P 500 (SPW/SPX) (bottom): Daily chart 1990 to 2008

The SPW vs. SPX ratio is not market breadth indicator. Investors using the SPW/SPX ratio to measure market breadth would have missed the 1994-2000 rally and bought the 2000-2002 bear market.



Source: BofA Global Research, Bloomberg

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Arms Index

The Arms Index was developed by Richard Arms in 1967. Many traders refer to the Arms as TRIN, which stands for short-term trading index. We calculate this market breadth indicator by dividing the advance vs decline ratio by the up volume vs down volume ratio. We use this on the NYSE, but TRIN can be created for any index with breadth and up/down volume statistics.

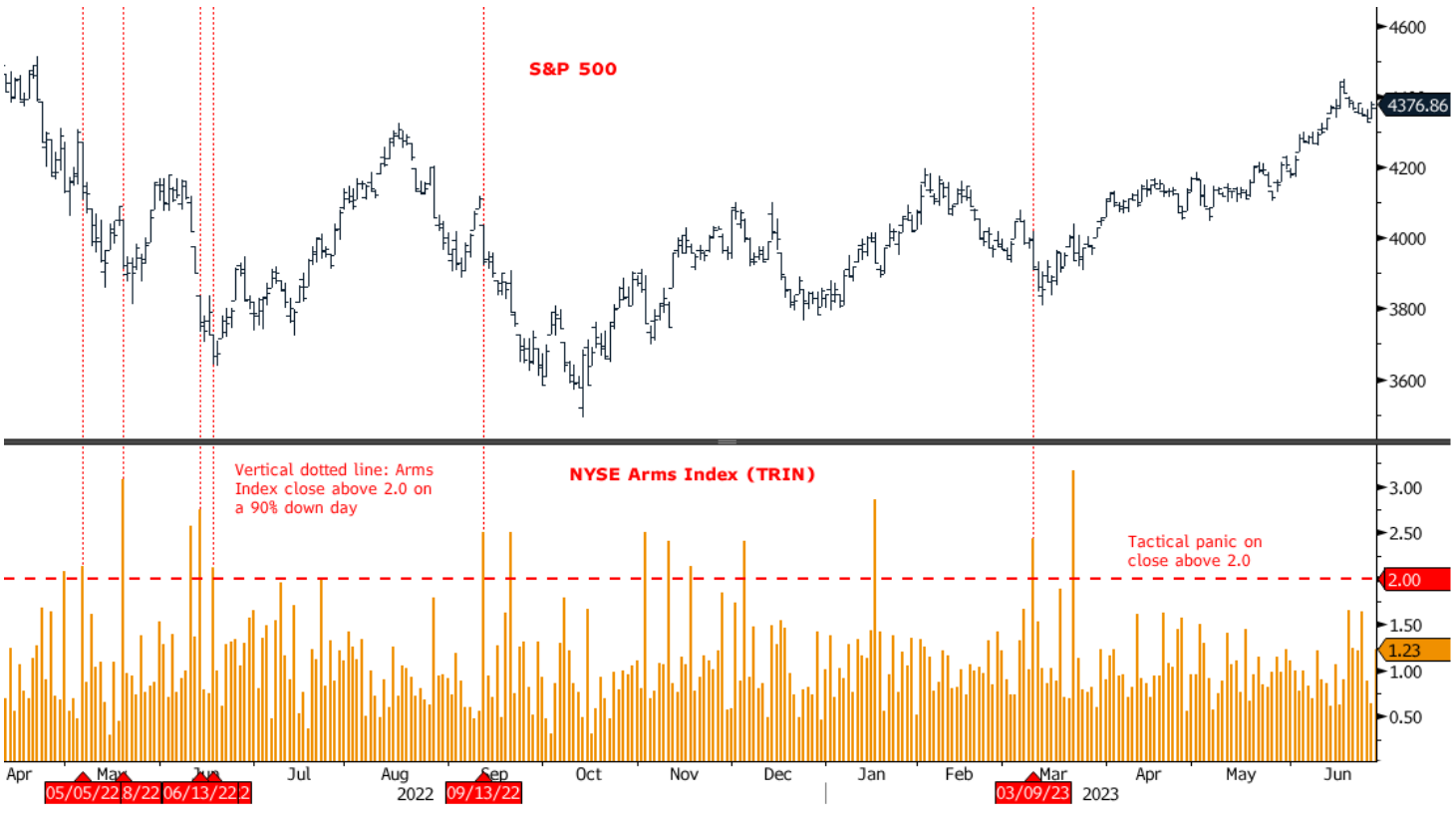
> **Arms / TRIN calculation**: (advancing stocks / declining stocks) / (volume in advancing stocks / volume in declining stocks)

Since the Arms index is a ratio of the A-D breadth ratio and the A-D volume ratio, a reading below 1.0 means that the A-D volume ratio is greater than the A-D breadth ratio. In theory, rallies for the market index should have low TRIN readings as the volume ratio tends to be higher than the breadth ratio under this scenario. A decline for the index tends to have a higher reading for the breadth ratio vs the volume ratio, which puts the TRIN above 1.0. This means that the Arms Index tends to move inversely to the direction of the market.

Readings above 2.0 often climatic for the Arms Index

When the NYSE Arms Index moves above 2.0, the volume ratio is much weaker than the breadth ratio. This can provide a tactical panic signal, especially when it is on a closing price basis and coincides with a 90% down day on the NYSE (see the previous section for 90% up and down days).

Exhibit 208: S&P 500 (top) and NYSE Arms (TRIN) (bottom): Daily chart
When the NYSE Arms closes above 2.0, it can provide a tactical panic signal, especially when it coincides with a 90% down day on the NYSE.



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Percentage of stocks above 10, 50 and 200-day MAs

The percentage of an index's stocks above 10-day, 50-day or 200-day moving averages (MAs) is another way to measure market breadth. A high or rising percentage of stocks above a moving average shows stronger breadth and broadening participation. A low or falling percentage of stocks above a moving average reflects weaker breadth and narrowing participation.

Like the A-D line, the percentage of stocks above a MA also generates bullish and bearish divergences as an advance warning for a change in the price trend of the underlying index. In addition, the percentage of stocks above MAs indicators also provide oversold and overbought signals.

Percentage of stocks above 10-day MAs is a short-term measure of breadth

Exhibit 209: S&P 500 (top) with the percentage of S&P 500 stocks above 10-day moving averages (bottom): Daily chart
The percentage of stocks above 10-day MAs provides a short-term, or tactical, measure of breadth and shows bullish and bearish divergences as well as oversold and overbought conditions.

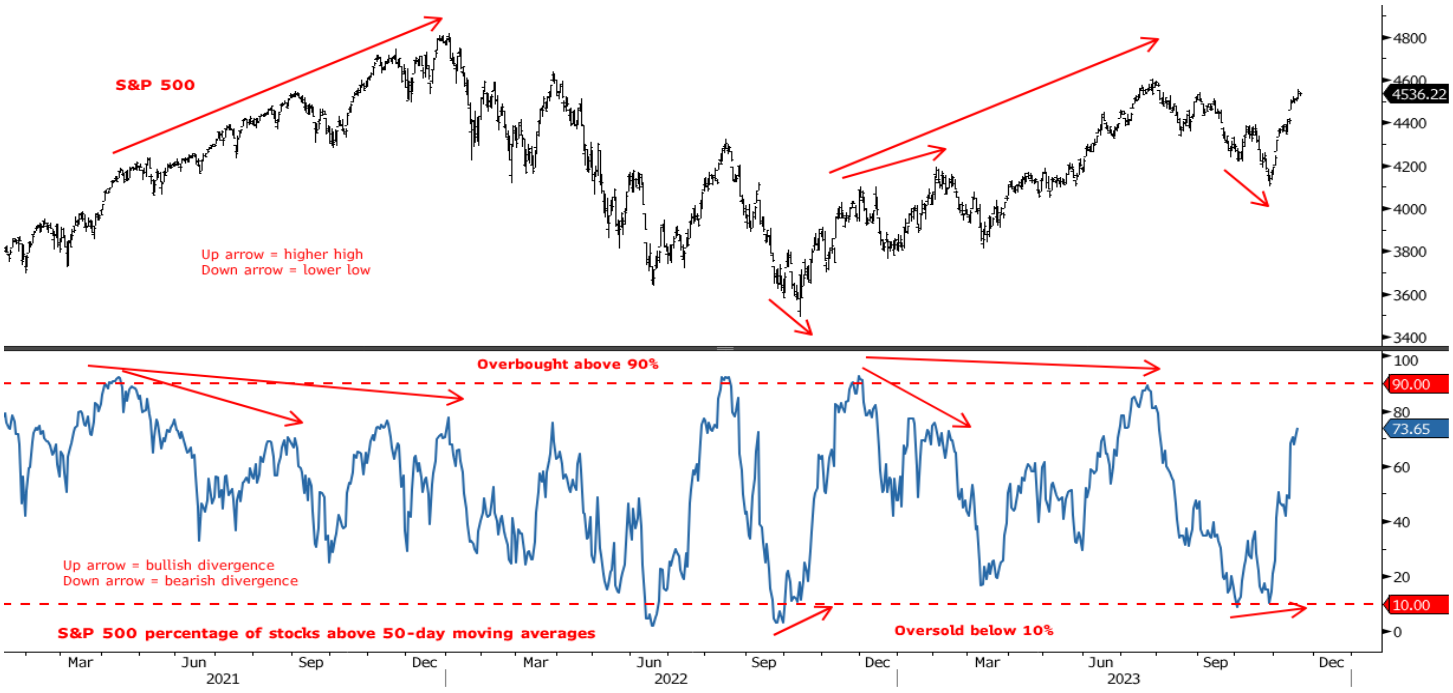


Source: BofA Global Research, Bloomberg

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Percentage of stocks above 50-day MAs gauges short-to-medium-term breadth

Exhibit 210: S&P 500 (top) with the percentage of S&P 500 stocks above 50-day moving averages (bottom): Daily chart
The percentage of stocks above 50-day MAs offers a short-to-intermediate-term measure of market breadth. This indicator also shows bullish and bearish divergences as well as oversold and overbought conditions.



Source: BofA Global Research, Bloomberg

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Percentage of stocks above 200-day MAs is a longer-term breadth indicator

Exhibit 211: S&P 500 (top) with the percentage of S&P 500 stocks above 200-day moving averages (bottom): Daily chart
The percentage of stocks above 200-day MAs is an intermediate-to-long-term breadth measure. This indicator also shows bullish and bearish divergences as well as oversold and overbought conditions.



Source: BofA Global Research, Bloomberg

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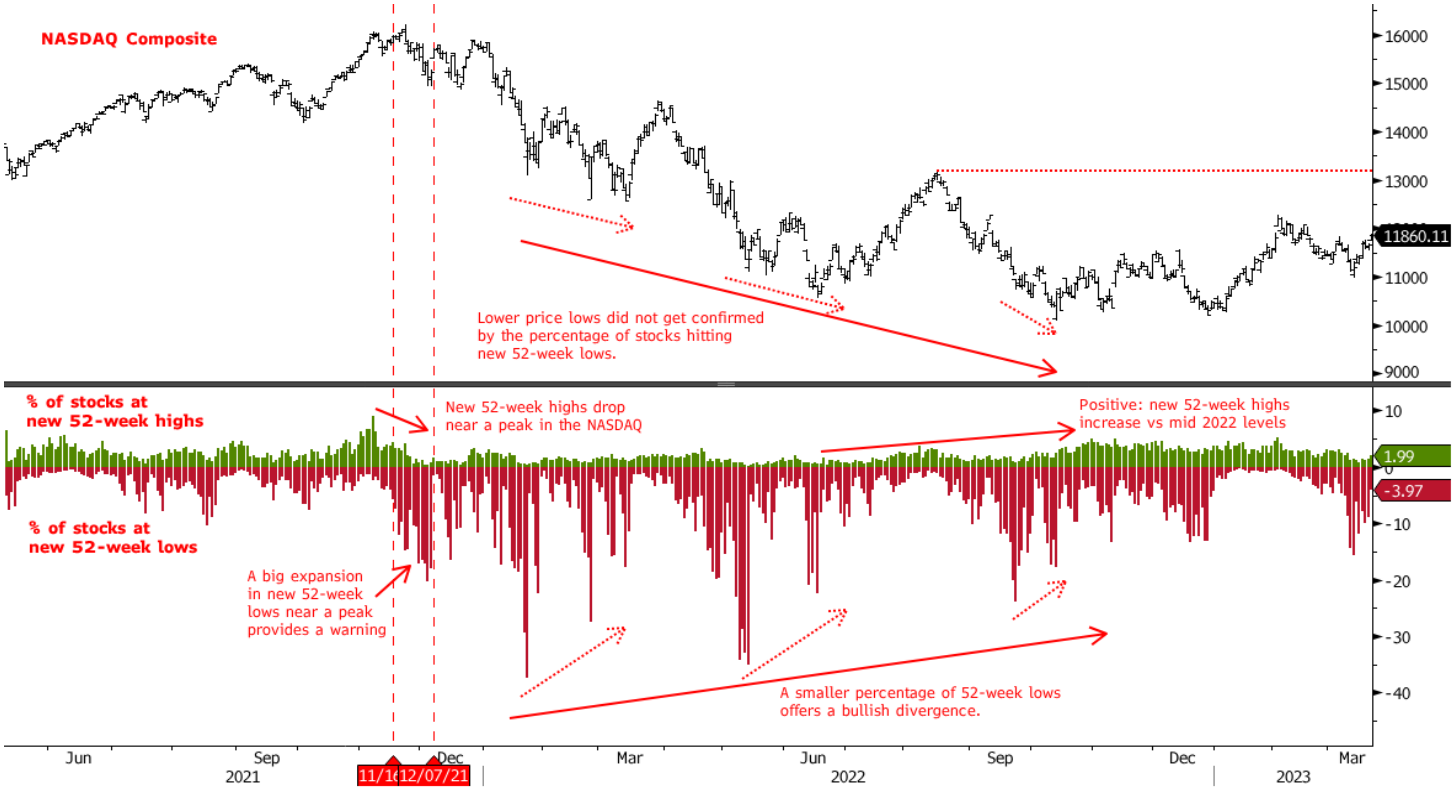
New highs and new lows

The percentage of stocks hitting new highs and the percentage of stocks reaching new lows offer additional insights into market breadth. We will focus on the 52-week highs and 52-week lows in this primer, but some investors may find that other timeframes, such as 24 or 4 weeks for new highs and new lows, are also useful for measuring market breadth.

When an index is reaching new highs in an uptrend, an increasing percentage of stocks achieving new highs suggests a broad rally and confirms the highs on the index. When an index is hitting new lows in a downtrend, an increasing percentage of stocks moving to new lows indicates a broad decline and confirms the lows on the index. Like the breadth indicators highlighted above, new highs and new lows can generate divergences that precede a change in the price trend of an index.

Another signal is when the percentage of stocks at new 52-week lows increases significantly on a minor pullback from a high on an index. This suggests that breadth is diminishing under the surface. This happened to the NASDAQ Comp in late 2021. New 52-week highs also declined as new 52-week lows expanded.

Exhibit 212: NASDAQ Comp (top) with the percentage of NASDAQ stocks at new 52-week highs and new 52-week lows (bottom): Daily chart
The percentage of stocks at new 52-week highs and the percentage of stocks at new 52-week lows can provide some important market breadth signals.



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

10-day MA of new 52-week highs minus new 52-week lows

Taking the 10-day moving average of new 52-week highs minus new 52-week lows shows the breadth battle between the bulls and bears. When above zero, this indicator is bullish. When below zero, it is bearish. A move above zero can confirm an uptrend, while a move below zero can confirm a downtrend. Like other breadth indicators, this indicator can generate divergences to provide a warning ahead of index turning points.

Spike lows below -100 in the 10-day MA of new 52-week highs minus new 52-week lows on the SPX provided oversold readings in early 2016, late 2018 and early 2020. Movements for this indicator above zero in early 2016, early 2019 and early 2020 (post-COVID low) confirmed stronger rallies. This breadth indicator held above or near zero to keep those rallies intact throughout 2016-2017, 2019 and 2020-2021. Spike highs above 100 for this breadth indicator in early 2018 and mid 2021 preceded bearish divergences for the SPX in late 2018 and early 2022, respectively. A January-February bearish divergence for the 10-day MA of new 52-week highs minus new 52-week lows preceded the COVID-19 correction.

Exhibit 213: S&P 500 (top) with the 10-day moving average of new 52-week highs minus new 52-week lows (bottom): Daily chart
Taking the 10-day moving average of new 52-week highs minus new 52-week lows shows the breadth battle between the bulls and bears.



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

NYSE 90% up days

- After an NYSE 90% up day, S&P 500 tended to be higher 10-80 trading days later about 78% of the time.
- In FICC after an NYSE 90% up day, the average and median trends in US 10Y yield and DXY were lower, but hit ratios weren't as consistent. Crude oil tended to rise 50-80 trading days later about 62-68% of the time.
- After a NYSE 90% down day with the Arms index above 2.0, the S&P 500 tended to be higher 70-76% of the time over the following 10-80 trading days.

NYSE 90% up day defined

A 90% up day occurs when the percentage of advancing stocks and up volume on an index (NYSE) exceed 90%. This strong breadth and accumulation (up volume) day suggests an urgency to buy stocks and confirms rallies from important market lows. The tendency is for stronger forward returns after this breadth signal. We review the behavior of the S&P 500 and other macro assets following prior NYSE 90% up days.

A NYSE 90% up day was bullish for the S&P 500

A 90% up day was bullish for the S&P 500. Strong up ratios were observed 10-80 trading days later (average 78%). SPX was higher (70-82%) by an average of 4.00% and median 4.31%. The strongest biases were observed 40 to 80 trading days later when SPX rose 79-82% of the time. The average return during this period was 5.49% while the median return averaged 5.83%. For all periods, SPX tended to rise between 69-71% of the time (average 71%) with average return 2.21%, median return 3.20%.

Exhibit 214: S&P 500 returns and up ratios after NYSE 90% up day tended to be bullish with price rising as much as 82% of the time 70 trading days later.
Average and median returns and up ratios following NYSE 90% up day

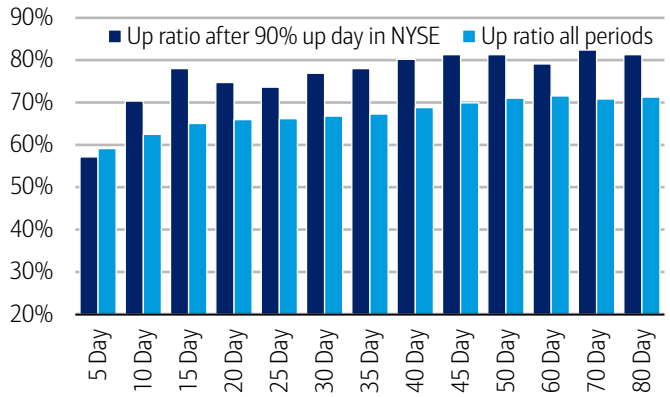
Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	57%	70%	78%	75%	74%	77%	78%	80%	81%	81%	79%	82%	81%
Up	52	64	71	68	67	70	71	73	74	74	72	75	74
Down	39	27	20	23	24	21	20	18	17	17	19	16	17
Average	0.13%	1.06%	2.20%	2.42%	2.58%	3.23%	3.55%	4.48%	4.71%	4.88%	5.52%	6.31%	7.06%
Median	0.50%	1.38%	2.29%	2.89%	3.11%	3.38%	3.64%	4.40%	4.51%	4.81%	5.93%	7.37%	7.96%
Standard deviation	3.44%	3.80%	3.87%	4.79%	5.37%	5.16%	5.81%	6.45%	6.54%	7.43%	8.01%	8.82%	8.87%
Min	-17.43%	-15.39%	-8.65%	-9.80%	-15.21%	-15.10%	-15.40%	-11.43%	-18.60%	-23.25%	-17.44%	-20.58%	-17.80%
Max	8.13%	12.02%	13.73%	14.38%	20.11%	17.22%	21.41%	27.78%	23.80%	27.17%	30.97%	27.89%	31.77%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Average return and up ratio after a NYSE 90% up day outperformed

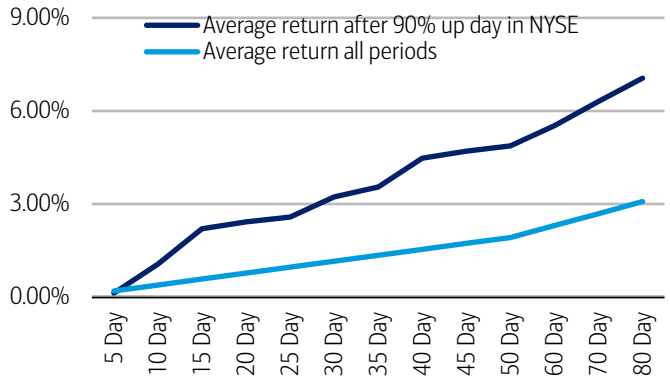
Exhibit 215: SPX up ratio after NYSE 90% up day was greater than the up ratio of all periods
Up ratio after 90% up day vs all periods



Source: BofA Global Research, Bloomberg

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Exhibit 216: SPX average returns after an NYSE 90% up day outperformed all periods.
Average trend after NYSE 90% up day vs all periods



Source: BofA Global Research, Bloomberg

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US 10Y yield saw modest declines after a NYSE 90% up day

The US 10Y yield tended to decline, modestly, following a NYSE 90% up day. Yield was lower for all observed periods. Average US 10Y yield declines ranged from -2bps to -9bps. Median declines ranged from -2bps to -5bps. Up ratios were less consistent. Yield tended to be lower 10, 20, 25 and 45 trading days later 62-64% of the time.

Exhibit 217: US 10Y yield average and median returns and up ratios after NYSE 90% up day
Average and median US 10Y yield returns and up ratios following NYSE 90% up day

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	41%	38%	42%	36%	38%	40%	41%	45%	37%	42%	45%	44%	43%
Up	37	35	38	33	35	36	37	41	34	38	41	40	39
Down	54	56	53	58	56	55	54	50	57	53	50	51	52
Average	-4	-5	-4	-6	-9	-8	-8	-6	-6	-4	-2	-2	-2
Median	-3	-4	-3	-4	-5	-4	-5	-2	-4	-4	-3	-3	-5
Standard deviation	16	19	25	28	34	33	39	39	43	46	47	48	47
Min	-84	-87	-130	-125	-177	-167	-164	-166	-178	-181	-150	-166	-126
Max	31	57	64	52	72	56	69	85	94	109	106	97	89

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

US dollar index saw modest declines after a NYSE 90% up day

The DXY tended to decline, modestly, following a NYSE 90% up day. Average and median trends were consistently negative, however % up ratios were not consistently weak. There was a tendency for DXY to be lower 20-30 trading days after a 90% up day.

Exhibit 218: US dollar index (DXY) average and median returns after an NYSE 90% up day were negative, however % up ratio was not consistently weak.
Average and median US dollar index returns and up ratios following NYSE 90% up day

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	45%	40%	40%	37%	36%	38%	42%	35%	41%	40%	45%	46%	44%
Up	41	36	36	34	33	35	38	32	37	36	41	42	40
Down	50	55	55	57	58	56	53	59	54	55	50	50	51
Average	-0.20%	-0.33%	-0.51%	-0.45%	-0.49%	-0.59%	-0.73%	-0.81%	-1.04%	-1.16%	-0.85%	-0.95%	-1.09%
Median	-0.24%	-0.41%	-0.57%	-0.62%	-0.56%	-0.74%	-0.76%	-0.74%	-1.00%	-0.86%	-0.69%	-0.46%	-0.83%
Standard Deviation	1.50%	1.92%	2.11%	2.66%	2.80%	3.07%	3.42%	3.51%	3.58%	3.92%	4.31%	4.41%	4.78%
Min	-5.17%	-5.70%	-4.79%	-5.79%	-8.59%	-6.95%	-7.50%	-7.28%	-7.32%	-8.93%	-10.60%	-9.85%	-10.45%
Max	2.93%	6.18%	5.47%	6.30%	7.75%	6.13%	7.58%	7.27%	8.37%	8.73%	11.96%	8.80%	9.88%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Gold showed some tendency to rise after a NYSE 90% day

Gold's average and median trends after an NYSE 90% up day were positive. However the % up ratio was not consistently above 60%. There may be strength 15-30 and 60-80 trading days after as corresponding up ratios were stronger versus other periods.

Exhibit 219: Gold average and median returns after an NYSE 90% up day were positive, but up ratios were not consistently strong.
Average and median gold returns and up ratios following NYSE 90% up day

Ticker: XAU Curncy					Start Year: 2006								
Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	62%	58%	64%	62%	57%	63%	60%	59%	58%	60%	66%	65%	65%
Up	56	53	58	56	52	57	55	54	53	55	60	59	59
Down	35	38	34	35	39	35	36	37	38	36	31	32	32
Average	0.59%	0.70%	0.95%	1.40%	1.36%	1.65%	1.80%	1.74%	2.28%	2.50%	3.16%	3.77%	4.22%
Median	0.62%	0.63%	1.49%	1.72%	1.26%	2.11%	2.24%	1.37%	2.05%	1.97%	1.74%	2.54%	3.16%
Standard Deviation	2.74%	4.18%	4.45%	4.93%	4.96%	5.36%	6.48%	6.49%	6.66%	6.91%	7.25%	8.41%	8.90%
Min	-6.57%	-12.28%	-13.15%	-10.85%	-11.41%	-11.30%	-12.51%	-11.66%	-11.42%	-8.12%	-11.66%	-13.41%	-15.11%
Max	7.44%	10.82%	9.71%	14.38%	15.82%	14.32%	19.68%	16.42%	19.37%	24.51%	24.18%	32.34%	29.89%

Source: BofA Global Research, Bloomberg

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Crude oil tended to be higher 50-80 trading days after a NYSE 90% up day

Crude oil tended to trade higher after a 90% up day 50-80 trading days later. Average and median returns were also positive beginning 50 trading days later at about 4% and rising to almost 9% after 80 trading days.

Exhibit 220: WTI crude oil average and median returns and up ratios after NYSE 90% up day
Average and median crude oil returns and up ratios following NYSE 90% up day

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	52%	53%	54%	58%	57%	64%	56%	59%	59%	62%	65%	66%	68%
Up	47	48	49	53	52	58	51	54	54	56	59	60	62
Down	44	43	43	38	39	33	40	37	37	35	32	31	29
Average	-0.29%	-0.01%	0.08%	0.77%	1.12%	2.18%	2.33%	3.31%	3.96%	4.45%	6.75%	8.49%	9.00%
Median	0.12%	0.45%	1.14%	1.77%	1.69%	2.31%	1.47%	2.99%	2.47%	3.95%	7.38%	6.33%	7.87%
Standard Deviation	6.07%	9.32%	10.65%	11.73%	11.81%	11.65%	13.15%	15.31%	15.60%	17.49%	19.75%	21.05%	20.90%
Min	-22.89%	-55.64%	-52.68%	-42.61%	-37.28%	-32.87%	-42.16%	-48.18%	-42.48%	-51.99%	-48.64%	-42.76%	-50.52%
Max	12.43%	18.09%	25.23%	35.09%	28.53%	29.37%	31.28%	42.98%	51.84%	55.81%	62.64%	72.20%	79.95%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

*NYSE 90% down day defined

A 90% down day occurs when the percentage of both declining stocks and down volume on an index exceeds 90%. This weak breadth and distribution (down volume) day suggests an urgency to sell stocks.

NYSE 90% down day points to weak breadth and volume internals

In theory, a 90% down day seems bearish and sometimes it can be, especially if it occurs near a market peak. However, the data show positive SPX returns after a 90% down day for NYSE stocks, which suggests that 90% down days can often provide a bearish capitulation or washout signal that is contrarian bullish on a tactical basis. In the past, the SPX was higher 66-70% of the time 10-80 trading days later.

Exhibit 221: S&P 500 after a NYSE 90% down day – price tended to be up 66-70% of the time 10-80 trading days later.
S&P 500 forward trend after 162 NYSE 90% down days

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	62%	67%	66%	66%	70%	67%	67%	67%	70%	70%	69%	69%	67%
Up	100	108	107	107	113	109	108	108	114	114	112	112	108
Down	62	54	55	55	49	53	54	54	48	48	50	50	54
Average	0.72%	0.77%	1.17%	1.19%	2.07%	2.16%	2.57%	2.57%	3.30%	3.73%	4.30%	4.28%	4.88%
Median	0.88%	1.70%	2.35%	2.43%	3.34%	3.05%	4.22%	4.23%	4.80%	5.52%	5.46%	5.88%	6.76%
Standard deviation	4.31%	5.86%	6.95%	7.92%	8.08%	8.88%	8.67%	9.00%	9.89%	9.80%	10.59%	10.95%	12.31%
Min	-10.55%	-16.85%	-19.68%	-23.27%	-26.43%	-28.82%	-26.58%	-26.30%	-30.25%	-29.84%	-28.89%	-28.87%	-29.71%
Max	19.11%	16.43%	22.02%	22.24%	25.49%	27.40%	26.91%	28.56%	36.14%	29.61%	38.14%	35.33%	35.85%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

90% down day with the Arms Index above 2.0 provides a better bullish signal

When a NYSE 90% down day occurs with the daily NYSE Arms Index (aka TRIN) closing above 2.0 (see the next page for more on TRIN), it suggests tactical panic and can provide a better contrarian bullish climactic tactical signal for US equities.

Exhibit 222: SPX returns after a 90% down day with daily ARMS above 2.0 are stronger than 90% down days with the ARMS closing below 2.0
NYSE stocks 90% down day with daily TRIN closing above 2.0 (103 observations) and S&P returns

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	64%	70%	70%	69%	71%	69%	69%	71%	76%	77%	73%	75%	70%
Up	66	72	72	71	73	71	71	73	78	79	75	77	72
Down	37	31	31	32	30	32	32	30	25	24	28	26	31
Average	0.88%	1.35%	1.90%	1.78%	2.46%	2.58%	2.91%	3.38%	3.88%	4.49%	4.54%	4.35%	4.90%
Median	0.99%	2.20%	2.72%	2.86%	3.51%	3.23%	4.55%	4.63%	5.25%	6.04%	7.08%	5.93%	7.05%
Standard deviation	3.62%	4.73%	5.14%	5.72%	6.69%	7.10%	7.25%	7.07%	7.79%	7.76%	8.49%	9.07%	10.54%
Min	-10.55%	-16.85%	-13.82%	-15.16%	-26.43%	-23.48%	-26.58%	-22.86%	-26.84%	-26.32%	-27.54%	-28.87%	-29.71%
Max	11.45%	9.25%	17.42%	14.02%	19.21%	20.07%	21.30%	22.02%	28.96%	29.61%	27.43%	34.01%	28.56%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

NYSE Breadth Thrust

- An NYSE breath thrust signals strong participation in a stock market rally. A healthy rally typically sees many stocks rising together rather than just major constituents.
- A breadth thrust occurs when the 10-day exponential average of advancing issues divided by advancing issues plus declining issues (Advances / (Advances + Declines)) moves from a weak breadth reading of below 0.40 to a stronger breadth reading of above 0.615 within ten trading days. Dr. Martin Zweig developed the breadth thrust.
- After a breadth thrust, the S&P 500 tended to outperform over short- and medium-term outlooks. The US 10Y yield and the DXY initially rose within 20 trading days after. Oil was down 5 of 7 times in ten trading days. Gold was higher 67% of the time in 15-, 30- and 35-trading days.

S&P 500 after a breadth thrust

There have been 32 breadth thrust signals since 1930. In the past, the S&P 500 tended to be higher in or after 130 days at least 75% of the time. The signal has been noticeably stronger since the 1939. The S&P 500 rose 100% of the time, or 17 of 17 times, in 130 trading days.

Exhibit 223: The 32nd signal was confirmed on April 24, 2025. The last 16 signals preceded higher prices in 130 and 190 days. Prior to 1939, the signal was not as consistent.
SPX after breadth thrusts since 1931

Signal	Entry date	Signal date	30-days return	65-days return	130-days return	190-days return	250-days return
1	12/22/1930	1/3/1931	7.0%	3.0%	-9.0%	-39.1%	-49.0%
2	10/28/1931	11/6/1931	-26.6%	-32.5%	-49.5%	-31.1%	-38.1%
3	1/4/1932	1/13/1932	-5.2%	-30.5%	-44.0%	-19.0%	-17.8%
4	11/2/1932	11/11/1932	-13.9%	-20.2%	26.8%	46.9%	30.9%
5	12/24/1932	1/6/1933	-17.0%	2.9%	42.7%	19.1%	47.4%
6	3/30/1933	4/10/1933	37.0%	83.5%	40.8%	54.0%	68.5%
7	6/1/1934	6/9/1934	-10.9%	-13.3%	-9.2%	-19.3%	-0.3%
8	9/17/1934	9/27/1934	0.5%	3.6%	-2.6%	14.1%	26.3%
9	10/29/1934	11/8/1934	-0.1%	-1.2%	8.4%	28.0%	41.7%
10	6/28/1937	7/6/1937	5.0%	-19.0%	-26.5%	-43.0%	-26.7%
11	12/27/1937	1/6/1938	-2.4%	-15.8%	4.6%	12.3%	14.4%
12	3/30/1938	4/9/1938	-2.4%	19.0%	31.4%	26.9%	8.4%
13	4/29/1938	5/9/1938	-1.6%	17.5%	25.6%	16.0%	6.2%
14	9/24/1938	10/1/1938	4.8%	2.3%	-15.1%	-9.7%	2.2%
15	12/21/1938	12/31/1938	-5.4%	-18.5%	-13.5%	-2.2%	-2.4%
16	6/30/1939	7/12/1939	-5.7%	10.3%	5.0%	5.2%	-13.7%
17	11/29/1943	12/8/1943	3.3%	5.0%	9.1%	8.4%	12.3%
18	2/25/1949	3/8/1949	0.5%	-6.4%	4.2%	10.3%	14.6%
19	7/12/1950	7/21/1950	5.5%	14.2%	23.2%	26.0%	25.6%
20	6/27/1962	7/11/1962	3.6%	-1.2%	12.0%	19.1%	21.3%
21	10/23/1962	11/5/1962	7.2%	13.4%	20.3%	20.0%	26.5%
22	11/19/1971	12/3/1971	6.8%	12.2%	10.5%	14.9%	20.0%
23	9/30/1974	10/10/1974	-2.3%	2.7%	25.0%	36.4%	24.3%
24	12/23/1974	1/3/1975	15.3%	14.5%	34.1%	21.6%	27.0%
25	8/9/1982	8/20/1982	7.5%	18.8%	32.4%	43.5%	44.6%
26	3/5/2009	3/18/2009	9.9%	16.0%	34.9%	39.6%	46.0%
27	10/3/2011	10/12/2011	-3.8%	7.2%	14.1%	12.1%	19.4%
28	10/8/2013	10/18/2013	3.2%	4.8%	7.2%	13.9%	6.8%
29	9/28/2015	10/8/2015	3.4%	-3.7%	3.3%	6.9%	7.3%
30	12/21/2018	1/7/2019	9.2%	13.3%	18.2%	13.5%	26.9%
31	10/23/2023	11/3/2023	8.8%	14.7%	19.8%	22.0%	31.4%
32	4/10/2025	4/24/2025	9.4%	16.2%	25.6%		
		Average	1.6%	4.1%	9.7%	11.8%	14.6%
		Median	3.3%	4.2%	11.3%	14.1%	19.4%
		% time up	59.4%	65.6%	75.0%	77.4%	77.4%

Source: BofA Global Research, Bloomberg, Global Financial Data (GFD)

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S&P 500 index following a NYSE breadth thrust

SPX exhibited a strong tendency to rise after a NYSE breadth thrust. Bullish hit ratios were observed between 5-25, 35-40, 50 and 70 trading days later – the S&P 500 was higher between 66-72% of the time. Stronger average and median gains tended to occur after 25 days. Average and median returns and hit ratios were positive for all periods.

Exhibit 224: A NYSE breadth thrust was bullish for the S&P 500 index. Strong hit ratios were observed 5-25, 35-40, 50, and 70 trading days later. Average and median returns and up ratios were positive for all periods.
SPX trend and up ratios following NYSE breadth thrust signal

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	72%	69%	66%	69%	72%	59%	69%	66%	56%	69%	63%	66%	59%
Up	23	22	21	22	23	19	22	21	18	22	20	21	19
Down	9	10	11	10	9	13	10	11	14	12	12	11	13
Average	0.33%	0.76%	1.23%	1.69%	2.28%	1.58%	3.17%	4.44%	4.11%	4.82%	4.37%	3.84%	4.29%
Median	0.77%	1.31%	3.07%	3.45%	2.71%	3.27%	2.74%	2.67%	1.58%	3.45%	4.09%	4.86%	7.65%
Standard deviation	2.58%	5.56%	8.15%	8.49%	9.75%	10.83%	12.65%	13.17%	14.08%	15.03%	19.64%	18.21%	18.24%
Min	-7.16%	-11.46%	-14.84%	-16.38%	-27.90%	-26.64%	-28.35%	-27.27%	-20.79%	-22.32%	-31.02%	-31.48%	-32.27%
Max	3.90%	18.04%	29.36%	28.44%	35.32%	37.00%	47.40%	56.27%	60.09%	61.77%	81.19%	61.62%	58.41%

Source: BofA Global Research, Bloomberg
Note: SPX index data since 1931

BofA GLOBAL RESEARCH

US 10Y yield following NYSE breadth thrust

US 10Y yield tended to move higher after a NYSE breadth thrust. Bullish hit ratios occurred 10-20 trading days later. A bearish hit ratio occurred at 25 days. At 10 days, yield was up 69% of the time (+3bps avg/median). At 15-20 trading days, yield was higher 77% of the time by an average of +7bps and +4bps, median +3 and +5bps, respectively. At 25 days, yield was lower 69% of the time by -1bp (avg), -2bps (median).

Exhibit 225: Following a breadth thrust, US 10Y yield tended to be up 10-20 trading days later 69-77% of the time. 25 days later, yield was lower 69% of the time. No other strong hit ratios were observed.
US 10Y yield trend and up ratios following NYSE breadth thrust signal

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	54%	69%	77%	77%	31%	38%	46%	62%	46%	46%	46%	38%	46%
Up	7	9	10	10	4	5	6	8	6	6	6	5	6
Down	7	4	3	3	9	8	8	6	8	7	7	8	7
Average	4	3	7	4	-1	-5	-9	-9	-9	-4	0	-3	-7
Median	0	3	3	5	-2	-4	-1	1	-1	-1	-2	-2	-6
Standard deviation	17	8	17	14	19	27	54	55	50	66	73	66	61
Min	-9	-8	-22	-25	-30	-52	-171	-168	-137	-178	-178	-177	-160
Max	53	23	39	23	41	57	63	59	66	121	141	115	88

Source: BofA Global Research, Bloomberg
Note: USGG10YR Index data since 1962

BofA GLOBAL RESEARCH

US dollar index following a NYSE breadth thrust

A NYSE breadth thrust does not appear to have a material impact on the US dollar. No strong hit ratios were observed for the DXY. Average and median returns were muted.

Exhibit 226: A NYSE breadth thrust did not seem to impact the US dollar index. No strong up ratios were observed and avg / median returns were modest.
DXY trend and up ratios following NYSE breadth thrust signal

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	36%	55%	64%	64%	55%	55%	45%	45%	45%	45%	45%	55%	45%
Up	4	6	7	7	6	6	5	5	5	5	5	6	5
Down	7	5	4	4	5	5	6	6	6	6	6	5	6
Average	-0.09%	0.25%	0.30%	0.35%	0.56%	0.26%	-0.34%	-0.59%	-0.46%	-0.43%	-0.34%	-0.52%	-0.88%
Median	-0.27%	0.70%	0.08%	0.59%	1.09%	0.02%	-0.77%	-0.47%	-2.24%	-1.58%	-0.65%	0.59%	-1.13%
Standard Deviation	0.65%	0.99%	1.67%	1.56%	1.97%	2.62%	2.67%	2.66%	3.57%	3.64%	3.85%	3.71%	3.34%
Min	-0.99%	-1.05%	-2.16%	-1.75%	-2.01%	-2.85%	-3.56%	-4.09%	-4.77%	-4.92%	-6.13%	-7.19%	-6.24%
Max	0.88%	1.33%	2.63%	2.75%	3.62%	4.25%	4.70%	3.53%	4.92%	5.20%	5.54%	4.70%	3.35%

Source: BofA Global Research, Bloomberg
Note: US dollar index (DXY) data since 1971

BofA GLOBAL RESEARCH

Crude oil following a NYSE breadth thrust

Crude oil tended to decline 5-25 trading days following a NYSE breadth thrust signal. The only strong bias occurred 10 days later. Oil tended to be lower 71% of the time by an average of -1.59% and median -4.59%. Crude tended to be higher 30-70 trading days later, but with weaker bias.

Exhibit 227: Oil tended to decline following a NYSE breadth thrust. It was lower 71% of the time by an avg of -1.59%, median -4.59% 10 trading days later.
Crude oil trend and up ratios following NYSE breadth thrust signal

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	43%	29%	43%	43%	43%	57%	57%	57%	57%	57%	57%	57%	43%
Up	3	2	3	3	3	4	4	4	4	4	4	4	3
Down	4	5	4	4	4	3	3	3	3	3	3	3	4
Average	-0.91%	-1.59%	-0.45%	-0.41%	-0.79%	0.75%	5.88%	3.95%	2.41%	6.72%	8.01%	6.24%	3.82%
Median	-3.22%	-4.59%	-1.86%	-2.53%	-3.19%	2.85%	15.97%	9.11%	3.69%	8.82%	4.01%	3.77%	-0.67%
Standard Deviation	5.36%	6.57%	7.11%	8.89%	12.27%	12.12%	13.94%	14.80%	18.27%	21.15%	25.26%	27.46%	20.69%
Min	-6.17%	-8.19%	-7.02%	-9.28%	-15.54%	-15.60%	-15.62%	-19.14%	-27.94%	-27.05%	-27.23%	-42.65%	-34.70%
Max	7.91%	9.25%	9.87%	11.89%	19.90%	17.81%	17.35%	19.88%	24.85%	35.60%	47.32%	41.43%	27.39%

Source: BofA Global Research, Bloomberg
Note: CL1 R:05_0_N Comdty data since 1986

BofA GLOBAL RESEARCH

Gold following a NYSE breadth thrust

Gold tended to rise following a NYSE breadth thrust. Strong tendencies occurred 15, 30 and 35 trading days later – it was higher 67% of the time. At 15 trading days, gold's average gain was +1.05%, at 30 TDs -0.34%, at 35 days +1.44%. Strong hit ratios were also observed 70-80 trading days later. Gold was lower 78% of the time at 70 days. It was down 67% of the time 80 trading days later.

Exhibit 228: Gold tended to rise after a breadth thrust. It was higher 67% of the time 15, 30 and 35 trading days later. At days 70-80, it tended to be lower.
Gold trend and up ratios following NYSE breadth thrust signal

Days after	5 days	10 days	15 days	20 days	25 days	30 days	35 days	40 days	45 days	50 days	60 days	70 days	80 days
% Up Ratio	56%	44%	67%	56%	56%	67%	67%	56%	44%	56%	44%	22%	33%
Up	5	4	6	5	5	6	6	5	4	5	4	2	3
Down	4	5	3	4	4	3	3	4	5	4	5	7	6
Average	0.87%	2.94%	1.05%	1.12%	-0.23%	-0.34%	1.44%	0.96%	-0.98%	-0.37%	-0.53%	-0.36%	0.65%
Median	0.19%	-0.01%	0.61%	1.03%	0.29%	0.26%	2.98%	0.64%	-0.36%	0.83%	-0.12%	-1.12%	-0.72%
Standard Deviation	3.67%	7.59%	5.61%	4.75%	4.51%	3.76%	5.83%	5.46%	5.20%	5.65%	3.57%	5.00%	5.53%
Min	-3.29%	-1.55%	-6.53%	-5.41%	-5.55%	-4.99%	-6.62%	-6.75%	-8.59%	-7.82%	-5.65%	-4.33%	-4.31%
Max	8.15%	22.64%	13.58%	9.77%	7.05%	5.75%	11.25%	10.80%	8.31%	10.54%	4.33%	12.29%	13.52%

Source: BofA Global Research, Bloomberg
Note: XAU Curncy data since 1975

BofA GLOBAL RESEARCH

VIX spikes and macro trends

What happens after the VIX spikes above 45?

In April 2025, volatility struck financial markets as risk assets sold off due to tariff headlines and so-called "Liberation Day." The VIX spiked above 45. This has happened twelve times since 1997. We review how macro markets react after such a VIX spike.

SPX: After the VIX spiked > 45, SPX tended to bounce back the following week but remained volatile for up to 8 weeks. Thereafter an uptrend began. In the past, the SPX was up 11 of 12 times or 91.7% of the time 16 weeks after the spike. (Exhibit 229)

US 10Y Yield: Yield tended to decline more and more often than all periods and was down 11 of 12 times after seven weeks. (Exhibit 230)

Oil: Tended to decline on average, median, and underperformed all periods. For example, eight weeks later oil was up just 25% of the time. (Exhibit 231)

DXY: Tended to trade sideways for up to seven weeks and then outperformed, especially 11-13 weeks later. (Exhibit 232)

Gold: Price was higher 83% of the time at one week and 75% of the time at 4 weeks and 13 weeks. However hit ratio in between was a little less significant. (Exhibit 233)

Exhibit 229: S&P 500 after a VIX spike above 45 tended to bounce back one week late. However SPX could decline again within 6-8 weeks to form a bottom. Then price tended to outperform all periods, up as much as 91.7% of the time (11 of 12) 16 weeks after the spike.
S&P 500 trends after the VIX spiked above 45

Date	1-wk	2-wk	3-wk	4-wk	5-wk	6-wk	7-wk	8-wk	9-wk	10-wk	11-wk	12-wk	13-wk	14-wk	15-wk	16-wk
10/31/1997	1.41%	1.50%	5.30%	4.46%	7.56%	4.24%	3.52%	2.39%	6.61%	1.43%	5.13%	4.70%	7.18%	10.70%	11.53%	13.08%
9/4/1998	3.61%	4.74%	7.28%	2.95%	1.08%	8.47%	9.94%	12.81%	17.16%	15.59%	19.47%	22.43%	20.83%	19.77%	21.99%	25.91%
9/21/2001	7.78%	10.93%	13.03%	11.15%	14.37%	12.57%	16.00%	17.90%	19.11%	17.98%	19.93%	16.29%	18.54%	20.21%	21.40%	18.62%
7/26/2002	1.34%	6.54%	8.90%	10.32%	7.41%	4.82%	4.33%	-0.87%	-2.99%	-6.13%	-2.05%	3.70%	5.25%	5.64%	4.91%	6.68%
10/3/2008	-18.20%	-14.44%	-20.24%	-11.87%	-15.31%	-20.55%	-27.22%	-18.47%	-20.30%	-19.97%	-19.23%	-20.60%	-15.23%	-19.00%	-22.66%	-24.32%
5/21/2010	0.16%	-2.10%	0.36%	2.74%	-1.00%	-5.99%	-0.89%	-2.10%	1.38%	1.28%	3.12%	-0.78%	-1.47%	-2.12%	1.55%	2.01%
8/12/2011	-4.69%	-0.17%	-0.41%	-2.09%	3.16%	-3.60%	-4.02%	-1.98%	3.88%	5.04%	9.02%	6.31%	7.21%	3.13%	-1.71%	5.55%
8/28/2015	-3.40%	-1.40%	-1.55%	-2.89%	-1.89%	1.31%	2.22%	4.34%	4.55%	5.55%	1.72%	5.04%	5.09%	5.17%	1.18%	0.84%
2/9/2018	4.30%	4.88%	2.74%	6.38%	5.06%	-1.19%	0.81%	-0.58%	1.40%	1.93%	1.92%	1.67%	4.13%	3.57%	3.89%	4.39%
2/28/2020	0.61%	-8.23%	-21.98%	-13.97%	-15.76%	-5.56%	-2.70%	-3.98%	-4.18%	-0.83%	-3.06%	0.04%	3.05%	8.11%	2.95%	4.86%
8/9/2024	3.93%	5.43%	5.69%	1.20%	5.27%	6.71%	7.37%	7.61%	8.81%	9.74%	8.68%	7.20%	12.19%	9.85%	11.70%	12.88%
4/4/2025	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%	5.70%
Average	**0.21%**	**1.12%**	**0.40%**	**1.17%**	**1.31%**	**0.58%**	**1.26%**	**1.90%**	**3.43%**	**3.11%**	**4.20%**	**4.31%**	**6.04%**	**5.89%**	**5.20%**	**6.35%**
Median	**1.34%**	**1.50%**	**2.74%**	**2.74%**	**3.16%**	**1.31%**	**2.22%**	**-0.58%**	**3.88%**	**1.93%**	**3.12%**	**4.70%**	**5.25%**	**5.64%**	**3.89%**	**5.55%**
% time up	**75.0%**	**58.3%**	**66.7%**	**66.7%**	**66.7%**	**58.3%**	**66.7%**	**50.0%**	**75.0%**	**75.0%**	**75.0%**	**83.3%**	**83.3%**	**83.3%**	**83.3%**	**91.7%**
All periods	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	0.18%	0.36%	0.53%	0.71%	0.88%	1.06%	1.23%	1.41%	1.58%	1.76%	1.94%	2.11%	2.29%	2.47%	2.65%	2.84%
Median	0.28%	0.63%	0.89%	1.20%	1.40%	1.58%	1.76%	1.95%	2.05%	2.31%	2.64%	2.84%	3.14%	3.34%	3.52%	3.63%
% time up	56.7%	60.5%	61.6%	62.2%	63.2%	64.4%	65.9%	65.7%	66.7%	68.5%	69.0%	70.1%	69.5%	69.7%	69.6%	69.7%
Spread to all	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	0.03%	0.76%	-0.13%	0.47%	0.42%	-0.48%	0.02%	0.49%	1.84%	1.35%	2.26%	2.19%	3.75%	3.42%	2.55%	3.51%
Median	1.05%	0.87%	1.85%	1.54%	1.76%	-0.27%	0.46%	-2.52%	1.83%	-0.38%	0.48%	1.86%	2.12%	2.30%	0.36%	1.92%
% time up	18.26%	-2.17%	5.04%	4.51%	3.51%	-6.12%	0.75%	-15.69%	8.25%	6.55%	6.02%	13.23%	13.82%	13.59%	13.70%	21.98%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 230: US 10Y yield after the VIX spiked above 45 tended to decline more often than usual, such as up just 8.3% of the time (1 of 12) at seven weeks. Yield tended to decline more on average versus all periods.
US 10Y yield trends after the VIX spiked above 45

Date	1-wk	2-wk	3-wk	4-wk	5-wk	6-wk	7-wk	8-wk	9-wk	10-wk	11-wk	12-wk	13-wk	14-wk	15-wk	16-wk
10/31/1997	7.10	4.80	-2.00	4.30	8.30	-10.30	-9.50	-9.30	-18.40	-41.80	-29.90	-14.40	-32.60	-21.40	-34.70	-28.80
9/4/1998	-18.30	-31.80	-45.30	-73.30	-22.90	-56.90	-31.00	-40.80	-7.30	-20.30	-19.90	-20.50	-38.60	-39.80	-45.10	-16.50
9/21/2001	-10.30	-18.70	-2.20	-7.00	-16.20	-33.30	-38.80	15.20	29.60	6.10	47.60	49.70	39.30	42.20	43.70	17.50
7/26/2002	-9.13	-12.67	-5.73	-14.83	-24.08	-37.04	-47.65	-59.67	-72.33	-71.71	-60.35	-27.64	-29.03	-37.72	-52.59	-35.30
10/3/2008	26.74	32.68	8.25	34.99	18.98	13.18	-40.57	-68.31	-89.94	-103.26	-148.00	-147.13	-123.43	-121.31	-128.43	-98.59
5/21/2010	5.43	-3.56	-0.33	-1.84	-13.01	-26.09	-18.59	-31.64	-24.36	-33.27	-42.13	-56.63	-62.73	-59.32	-54.09	-44.62
8/12/2011	-19.25	-6.49	-26.91	-33.65	-20.69	-42.14	-33.94	-17.84	-0.71	-3.56	6.19	-22.21	-19.84	-24.44	-29.13	-22.17
8/28/2015	-5.63	0.76	-4.71	-1.84	-18.78	-9.26	-14.73	-9.41	-3.86	14.45	8.51	8.16	3.94	8.86	-5.37	2.33
2/9/2018	2.37	1.48	1.31	4.26	-0.67	-3.77	-11.23	-7.77	-2.45	10.90	10.56	9.85	11.83	20.47	8.01	5.10
2/28/2020	-38.63	-18.83	-30.32	-47.40	-55.38	-42.95	-50.69	-54.78	-53.68	-46.55	-50.58	-48.95	-49.60	-25.35	-44.52	-45.49
8/9/2024	-5.72	-14.08	-3.64	-23.18	-28.85	-19.85	-18.92	2.74	16.05	14.30	30.01	44.38	36.45	49.96	46.04	22.87
4/4/2025	49.52	33.06	24.10	31.40	38.42	48.27	51.67	40.61	51.13	40.44	38.08	28.26	35.14	41.50	42.12	39.35
Average	**-1.32**	**-2.78**	**-7.29**	**-10.67**	**-11.24**	**-18.35**	**-22.00**	**-20.08**	**-14.69**	**-19.52**	**-17.49**	**-16.43**	**-19.10**	**-13.86**	**-21.17**	**-17.03**
Median	**-5.67**	**-5.02**	**-2.92**	**-4.42**	**-17.49**	**-22.97**	**-24.96**	**-13.63**	**-5.58**	**-11.93**	**-6.85**	**-17.45**	**-24.44**	**-22.92**	**-31.92**	**-19.34**
% time up	**41.7%**	**41.7%**	**25.0%**	**33.3%**	**25.0%**	16.7%	8.3%	**25.0%**	**25.0%**	**41.7%**	**50.0%**	**41.7%**	**41.7%**	**41.7%**	**33.3%**	**41.7%**
All periods	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	-0.14	-0.29	-0.45	-0.64	-0.82	-0.99	-1.18	-1.36	-1.55	-1.75	-1.96	-2.16	-2.35	-2.54	-2.73	-2.92
Median	-0.55	-0.88	-0.93	-1.62	-2.04	-2.19	-2.77	-2.62	-2.63	-2.68	-3.06	-2.31	-2.70	-2.45	-3.41	-3.80
% time up	47.8%	47.4%	48.2%	47.3%	46.6%	47.1%	45.6%	46.9%	47.0%	47.4%	47.3%	47.7%	48.1%	47.5%	47.7%	46.5%
Spread to all	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	-1.17	-2.49	-6.84	-10.04	-10.42	-17.35	-20.82	-18.72	-13.14	-17.78	-15.54	-14.27	-16.74	-11.32	-18.44	-14.10
Median	-5.12	-4.15	-1.99	-2.79	-15.45	-20.78	-22.20	-11.01	-2.94	-9.26	-3.80	-15.15	-21.74	-20.48	-28.51	-15.54
% time up	-6.11%	-5.77%	-23.18%	-13.93%	-21.58%	-30.43%	-37.28%	-21.86%	-21.98%	-5.71%	2.74%	-6.00%	-6.39%	-5.88%	-14.33%	-4.86%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 231: Oil after the VIX spiked above 45 tended to decline especially 6-8 weeks later. Oil was up just 25% of the time at eight weeks. Trends after the VIX spiked underperformed all periods.
Oil trends after the VIX spiked above 45

Date	1-wk	2-wk	3-wk	4-wk	5-wk	6-wk	7-wk	8-wk	9-wk	10-wk	11-wk	12-wk	13-wk	14-wk	15-wk	16-wk
10/31/1997	-1.47%	-0.38%	-6.26%	-9.16%	-11.24%	-13.61%	-12.05%	-13.66%	-17.31%	-21.11%	-21.68%	-25.33%	-18.36%	-20.78%	-24.00%	-22.96%
9/4/1998	-1.71%	6.17%	7.95%	7.20%	-0.07%	-3.02%	-3.70%	-1.17%	-4.93%	-6.99%	-11.65%	-18.71%	-23.44%	-26.05%	-22.82%	-23.03%
9/21/2001	-9.78%	-13.79%	-13.36%	-14.29%	-15.17%	-22.29%	-14.44%	-29.26%	-26.99%	-25.14%	-26.68%	-24.95%	-24.45%	-21.41%	-16.75%	-24.22%
7/26/2002	1.13%	1.21%	7.42%	7.87%	9.19%	11.57%	12.32%	12.43%	15.07%	11.61%	10.66%	11.53%	1.92%	2.22%	-2.86%	-3.88%
10/3/2008	-17.23%	-23.17%	-31.67%	-27.77%	-34.98%	-39.24%	-46.82%	-42.02%	-56.53%	-50.70%	-54.88%	-59.83%	-50.64%	-56.51%	-54.65%	-50.50%
5/21/2010	5.61%	2.10%	5.34%	11.74%	12.59%	3.00%	8.64%	9.05%	12.76%	12.72%	15.22%	7.64%	5.40%	7.32%	6.51%	9.15%
8/12/2011	-3.48%	-0.01%	1.25%	2.18%	3.28%	-6.48%	-7.24%	-2.81%	1.66%	2.37%	9.30%	10.40%	15.94%	14.39%	13.34%	18.25%
8/28/2015	1.84%	-1.30%	-0.44%	1.06%	0.71%	9.75%	5.53%	-1.37%	3.03%	-2.06%	-9.91%	-7.34%	-7.76%	-11.61%	-21.23%	-20.26%
2/9/2018	3.97%	7.35%	3.46%	4.80%	5.42%	11.28%	9.70%	4.83%	13.83%	15.54%	15.03%	17.77%	19.43%	20.56%	14.66%	11.17%
2/28/2020	-7.77%	-29.11%	-49.44%	-51.94%	-36.68%	-49.15%	-44.08%	-62.15%	-55.81%	-44.73%	-34.05%	-25.71%	-20.71%	-11.64%	-18.99%	-11.01%
8/9/2024	-1.69%	-2.62%	-4.28%	-11.93%	-10.66%	-7.60%	-11.27%	-3.20%	-1.67%	-10.61%	-6.59%	-9.57%	-8.41%	-12.78%	-7.29%	-11.50%
4/4/2025	-0.79%	4.34%	1.66%	-5.97%	-1.56%	-0.03%	-0.74%	-1.94%	4.18%	17.73%	19.12%	5.69%	8.08%	10.42%	6.55%	5.11%
Average	**-2.62%**	**-4.10%**	**-6.53%**	**-7.18%**	**-6.60%**	**-8.82%**	**-8.68%**	**-10.94%**	**-9.39%**	**-8.45%**	**-8.01%**	**-9.87%**	**-8.58%**	**-8.82%**	**-10.63%**	**-10.31%**
Median	**-1.58%**	**-0.20%**	**0.41%**	**-2.45%**	**-0.82%**	**-4.75%**	**-5.47%**	**-2.37%**	**0.00%**	**-4.52%**	**-8.25%**	**-8.45%**	**-8.08%**	**-11.62%**	**-12.02%**	**-11.26%**
% time up	**33.3%**	**41.7%**	**50.0%**	**50.0%**	**41.7%**	33.3%	33.3%	25.0%	**50.0%**	**41.7%**	**41.7%**	**41.7%**	**41.7%**	**41.7%**	**33.3%**	**33.3%**
All periods	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	0.19%	0.38%	0.56%	0.76%	0.95%	1.15%	1.35%	1.54%	1.74%	1.94%	2.14%	2.33%	2.51%	2.70%	2.87%	3.05%
Median	0.44%	0.60%	0.61%	0.95%	1.09%	1.23%	1.43%	1.85%	2.17%	2.22%	2.13%	2.29%	2.23%	2.42%	2.83%	3.02%
% time up	53.7%	53.6%	54.2%	53.5%	54.4%	54.4%	55.8%	55.9%	56.8%	56.7%	56.6%	56.2%	56.8%	56.4%	56.7%	57.2%
Spread to all	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	-2.81%	-4.48%	-7.10%	-7.94%	-7.55%	-9.97%	-10.02%	-12.48%	-11.14%	-10.39%	-10.15%	-12.20%	-11.09%	-11.52%	-13.50%	-13.35%
Median	-2.02%	-0.79%	-0.20%	-3.41%	-1.91%	-5.97%	-6.90%	-4.22%	-2.17%	-6.74%	-10.37%	-10.74%	-10.31%	-14.04%	-14.85%	-14.28%
% time up	-20.37%	-11.98%	-4.16%	-3.53%	-12.72%	-21.06%	-22.48%	-30.87%	-6.84%	-15.00%	-14.95%	-14.55%	-15.12%	-14.78%	-23.34%	-23.85%

Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

DXY after a VIX spike above 45

Exhibit 232: DXY after a VIX spike > 45 tended to trade sideways for up to seven weeks, potentially bottoming, and then strengthening 11-13 weeks later.
DXY trends after the VIX spiked above 45.

Date	1-wk	2-wk	3-wk	4-wk	5-wk	6-wk	7-wk	8-wk	9-wk	10-wk	11-wk	12-wk	13-wk	14-wk	15-wk	16-wk
10/31/1997	-0.41%	0.53%	1.00%	2.23%	3.37%	3.22%	3.03%	3.00%	4.66%	5.53%	5.59%	2.93%	5.31%	4.06%	4.57%	4.89%
9/4/1998	-1.73%	-1.52%	-1.85%	-2.88%	-5.19%	-6.09%	-5.12%	-4.60%	-3.51%	-2.51%	-2.43%	-1.40%	-3.35%	-4.81%	-4.24%	-3.37%
9/21/2001	0.43%	-0.12%	0.61%	1.74%	2.32%	1.36%	1.93%	3.09%	3.91%	2.84%	2.79%	1.78%	3.60%	3.83%	3.02%	3.29%
7/26/2002	0.06%	1.47%	0.04%	1.28%	0.35%	0.22%	1.36%	0.86%	0.86%	0.93%	0.60%	1.79%	1.23%	-0.50%	-1.98%	-1.47%
10/3/2008	3.34%	2.61%	7.63%	6.62%	6.97%	7.54%	9.81%	7.72%	8.48%	4.15%	1.23%	0.72%	1.90%	2.93%	4.85%	6.59%
5/21/2010	1.30%	3.35%	2.50%	0.38%	-0.07%	-1.11%	-1.67%	-3.38%	-3.41%	-4.49%	-5.81%	-2.84%	-2.71%	-2.87%	-3.83%	-3.13%
8/12/2011	-0.80%	-1.06%	0.20%	3.46%	2.67%	5.22%	5.29%	5.52%	2.70%	2.40%	0.62%	3.16%	3.13%	4.63%	6.81%	5.39%
8/28/2015	0.13%	-0.95%	-1.29%	0.17%	-0.29%	-1.35%	-1.63%	1.06%	0.87%	3.19%	3.01%	3.60%	4.07%	2.34%	1.52%	2.70%
2/9/2018	-1.48%	-0.62%	-0.56%	-0.39%	-0.23%	-1.11%	-0.52%	-0.37%	-0.71%	-0.14%	1.22%	2.35%	2.32%	3.53%	4.21%	4.11%
2/28/2020	-2.22%	0.63%	4.77%	0.24%	2.49%	1.38%	1.68%	2.29%	0.97%	1.63%	2.31%	1.76%	0.22%	-1.22%	-0.83%	-0.52%
8/9/2024	-0.65%	-2.34%	-1.39%	-1.90%	-1.96%	-2.34%	-2.67%	-0.60%	-0.24%	0.35%	1.09%	1.11%	1.81%	3.44%	4.28%	2.52%
4/4/2025	-2.84%	-3.68%	-3.45%	-2.91%	-2.61%	-1.87%	-3.80%	-3.59%	-3.72%	-4.70%	-4.19%	-5.46%	-5.67%	-5.02%	-4.41%	-5.22%
Average	**-0.41%**	**-0.14%**	**0.68%**	**0.67%**	**0.65%**	**0.42%**	**0.64%**	**0.92%**	**0.91%**	**0.77%**	**0.50%**	**0.79%**	**0.99%**	**0.86%**	**1.17%**	**1.32%**
Median	**-0.53%**	**-0.37%**	**0.12%**	**0.31%**	**0.14%**	**-0.45%**	**0.42%**	**0.96%**	**0.87%**	**1.28%**	**1.15%**	**1.77%**	**1.85%**	**2.63%**	**2.27%**	**2.61%**
% time up	**41.7%**	**41.7%**	**58.3%**	**66.7%**	**50.0%**	**50.0%**	**50.0%**	**58.3%**	**58.3%**	**66.7%**	**75.0%**	**75.0%**	**75.0%**	**58.3%**	**58.3%**	**58.3%**
All periods	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	0.01%	0.03%	0.04%	0.05%	0.07%	0.08%	0.09%	0.10%	0.12%	0.13%	0.14%	0.15%	0.16%	0.18%	0.19%	0.20%
Median	0.03%	0.03%	0.06%	0.07%	0.10%	0.12%	0.15%	0.07%	0.19%	0.20%	0.17%	0.14%	0.18%	0.16%	0.17%	0.25%
% time up	51.0%	50.7%	51.8%	51.3%	51.5%	51.6%	51.5%	50.6%	52.3%	52.5%	51.5%	51.5%	51.7%	52.0%	52.2%	52.3%
Spread to all	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	-0.42%	-0.17%	0.64%	0.62%	0.59%	0.34%	0.55%	0.81%	0.79%	0.64%	0.36%	0.64%	0.82%	0.69%	0.98%	1.12%
Median	-0.55%	-0.40%	0.06%	0.25%	0.04%	-0.57%	0.27%	0.89%	0.68%	1.09%	0.98%	1.63%	1.68%	2.48%	2.10%	2.36%
% time up	-9.30%	-9.02%	6.51%	15.41%	-1.48%	-1.60%	-1.54%	7.71%	6.00%	14.22%	23.46%	23.46%	23.35%	6.34%	6.11%	6.00%

Source: BofA Global Research, Bloomberg

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Exhibit 233: Gold after the VIX spiked above 45 tended to outperform. Price was higher 83% of the time at one week and 75% of the time at 4 weeks and 13 weeks. Hit ratios in between those weeks were less significant. The spread of average and median trend after signals vs all periods tended to be positive.
Gold trends after the VIX spiked above 45.

Date	1-wk	2-wk	3-wk	4-wk	5-wk	6-wk	7-wk	8-wk	9-wk	10-wk	11-wk	12-wk	13-wk	14-wk	15-wk	16-wk
10/31/1997	-0.39%	-2.60%	-2.09%	-4.64%	-7.45%	-9.18%	-7.19%	-5.39%	-7.45%	-10.53%	-6.84%	-3.80%	-2.89%	-4.27%	-3.72%	-4.56%
9/4/1998	2.41%	1.60%	2.54%	4.78%	3.45%	4.57%	2.09%	1.99%	2.21%	3.14%	3.21%	3.26%	1.95%	1.39%	0.77%	-0.21%
9/21/2001	0.62%	-0.15%	-2.40%	-4.05%	-4.84%	-3.95%	-4.80%	-5.70%	-6.36%	-5.85%	-5.80%	-4.46%	-4.56%	-5.11%	-4.29%	-1.34%
7/26/2002	1.19%	3.56%	3.36%	1.09%	2.96%	5.43%	4.31%	6.06%	5.33%	6.09%	4.28%	3.00%	3.33%	5.27%	5.86%	5.53%
10/3/2008	1.74%	-6.25%	-12.07%	-13.37%	-11.84%	-11.19%	-4.20%	-2.10%	-9.47%	-1.58%	0.32%	4.04%	4.77%	2.23%	0.91%	7.68%
5/21/2010	3.20%	3.65%	4.21%	6.77%	6.68%	2.94%	2.94%	1.35%	1.02%	0.33%	2.41%	3.26%	4.32%	5.16%	5.90%	5.85%
8/12/2011	6.03%	4.61%	7.80%	6.23%	3.70%	-5.15%	-7.04%	-6.23%	-3.76%	-6.00%	-0.19%	0.46%	2.40%	-1.30%	-3.62%	-0.01%
8/28/2015	-1.04%	-2.26%	0.50%	1.11%	0.44%	2.02%	3.85%	2.72%	0.75%	-3.86%	-4.38%	-4.90%	-6.72%	-4.16%	-5.19%	-5.94%
2/9/2018	2.30%	0.92%	0.46%	0.55%	-0.18%	2.33%	0.67%	1.24%	2.24%	1.50%	0.56%	-0.16%	0.20%	-1.79%	-1.09%	-1.77%
2/28/2020	5.56%	-3.52%	-5.49%	2.68%	2.21%	7.00%	6.13%	9.08%	7.24%	7.38%	9.96%	9.40%	9.12%	6.27%	9.15%	9.98%
8/9/2024	3.15%	3.34%	2.96%	2.72%	6.02%	7.84%	9.33%	9.14%	9.27%	11.93%	13.01%	12.55%	10.42%	5.43%	11.72%	8.71%
4/4/2025	6.56%	9.50%	9.26%	6.66%	9.44%	5.44%	10.51%	8.26%	8.96%	12.97%	10.87%	7.77%	9.84%	10.45%	10.26%	9.84%
Average	**2.61%**	**1.03%**	**0.75%**	**0.88%**	**0.88%**	**0.68%**	**1.38%**	**1.70%**	**0.83%**	**1.29%**	**2.28%**	**2.53%**	**2.68%**	**1.63%**	**2.22%**	**2.81%**
Median	**2.35%**	**1.26%**	**1.52%**	**1.89%**	**2.59%**	**2.63%**	**2.52%**	**1.67%**	**1.62%**	**0.92%**	**1.48%**	**3.13%**	**2.86%**	**1.81%**	**0.84%**	**2.76%**
% time up	**83.3%**	**58.3%**	**66.7%**	**75.0%**	**66.7%**	**66.7%**	**66.7%**	**66.7%**	**66.7%**	**58.3%**	**66.7%**	**66.7%**	**75.0%**	**58.3%**	**58.3%**	**50.0%**
All periods	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	0.15%	0.30%	0.45%	0.61%	0.76%	0.92%	1.07%	1.23%	1.38%	1.55%	1.71%	1.87%	2.03%	2.19%	2.35%	2.52%
Median	0.19%	0.18%	0.21%	0.20%	0.32%	0.27%	0.47%	0.59%	0.53%	0.64%	0.86%	0.96%	1.16%	1.37%	1.48%	1.61%
% time up	53.9%	53.0%	53.0%	52.1%	52.8%	52.1%	54.0%	54.4%	53.4%	54.4%	56.4%	56.8%	57.6%	58.2%	58.3%	58.0%
Spread to all	**1-wk**	**2-wk**	**3-wk**	**4-wk**	**5-wk**	**6-wk**	**7-wk**	**8-wk**	**9-wk**	**10-wk**	**11-wk**	**12-wk**	**13-wk**	**14-wk**	**15-wk**	**16-wk**
Average	2.46%	0.73%	0.30%	0.27%	0.12%	-0.24%	0.31%	0.48%	-0.55%	-0.25%	0.58%	0.67%	0.65%	-0.56%	-0.13%	0.30%
Median	2.17%	1.08%	1.31%	1.70%	2.27%	2.36%	2.04%	1.08%	1.09%	0.27%	0.62%	2.16%	1.70%	0.44%	-0.64%	1.15%
% time up	29.46%	5.31%	13.70%	22.89%	13.87%	14.56%	12.62%	12.28%	13.30%	3.89%	10.28%	9.88%	17.42%	0.18%	0.01%	-8.04%

Source: BofA Global Research, Bloomberg

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Margin Debt

- Margin debt is the amount of money borrowed by market participants to purchase securities. It can be considered a measure of risk appetite.

- Margin debt is reported by FINRA. When the year-over-year margin debt growth rate becomes stretched, it can imply that risk appetite, or lack thereof, may be stretched. We discuss the margin debt YoY thresholds that imply peak or trough potential.

- Margin debt data can be found here: https://www.finra.org/rules-guidance/key-topics/margin-accounts/margin-statistics

What is margin debt?

Margin debt is the amount of money market participants borrow from brokerages to purchase more securities than they could with cash. Another word for margin debt is "leverage." The Financial Industry Regulatory Authority (FINRA) tracks this borrowing across member firms and publishes it monthly. Increases in margin debt balances may be observed when market participants feel more confident. Lower borrowing can indicate uncertainty or even concern.

Borrowing on margin can help generate larger returns. However, it can also be risky. For bullish participants, gains grow faster when the stock market goes up, but losses can also grow faster if the market falls. If the value of a margin account drops too far, the brokerage that issued the loan can demand repayment or even liquidate securities without the purchaser's agreement. This is known as a margin call. Margin calls or forced sales can make prices fall even further, creating a negative feedback loop.

Margin debt as a predictive indicator

Because rising or falling margin debt can impact how market participants act, it can be treated as a sentiment gauge and measure of potential market risk. In the past, rapid margin debt growth (increased borrowing to buy more securities) has occurred toward the end of bull markets. Conversely, fast declines in margin debt can precede market bottoms. It should not be used in isolation but can be a complementary tool when paired with other technical analysis frameworks and indicators.

S&P 500 vs. Margin debt YoY

Margin debt YoY change is frothy, but not at big bubble highs

We conclude YoY Margin Debt reaching/exceeding about 40% is a headwind for risk assets with increased risk of a dip. However MD YoY rising above 60% may signal a major peak in a key theme driving risk assets. See Exhibit 234.

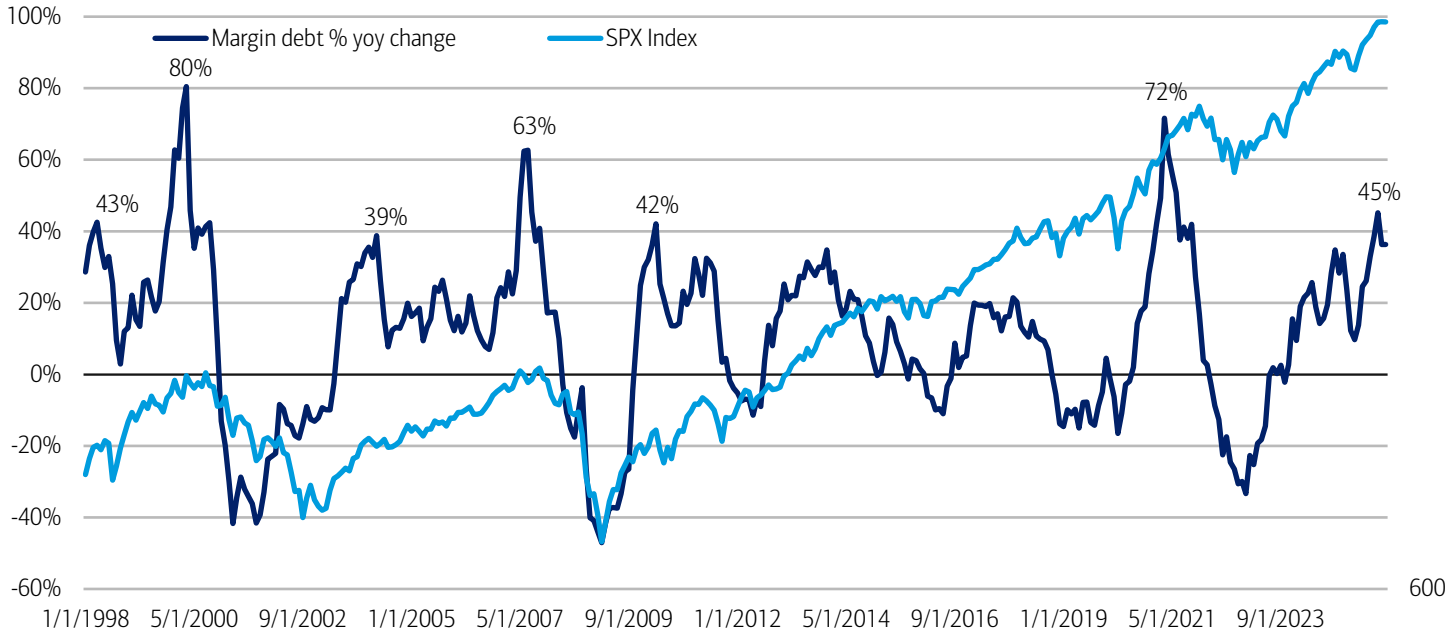
Margin debt jetted past $1 trillion in 2H25. However the amount of debt is less relevant than the rate of change. The 12-month growth rate – or year-over-year change (YoY) – reached 45% in 2025 and ended the year at 36%. This is a high level akin to 1998 (43%), 2004 (39%) and 2010 (42%). However, it is not stretched to outlier levels like those seen prior to market peaks in 2000 (dot-com), in 2007 (GFC) and 2021 (war on inflation).

In March 2000, YoY margin debt reached 80% and turned down. Through August 2000 the SPX's uptrend slowed forming a top with a peak at 1,553. Then a downturn followed as the dot-com bubble popped.

In June 2007, YoY margin debt reached 63%. Through October 2007, the SPX uptrend slowed forming a top with a peak at 1,576. Then a downturn followed with the onset of the Global Financial Crisis.

In March 2021, YoY margin debt reached 72% under the guise of extremely loose monetary and fiscal policy in the pandemic recovery. The S&P 500 trended higher through year-end, peaking in early January 2022 at 4,819. Then a downturn followed as tighter monetary policy was necessary to combat high inflation.

Exhibit 234: Margin debt YoY change in 2025 reached as high as 45%. This is frothy like that of 2010, 2004 and 1998 when dips followed. It is not as high as levels seen prior to the dot-com (80%), GFC (63%) and pandemic recovery (72%) peaks that preceded more persistent downturns.
YoY % change in margin debt with S&P 500 Index overlaid



Source: BofA Global Research, FINRA, Bloomberg

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When margin debt reaches 60, a SPX peak or top may be near

In the past, the S&P 500 tended to peak and decline after margin debt rose above 60%. Following a margin debt print above 60%, the SPX tended to be lower 67% of the time, but just 2 out of 3 occurrences, since 1998. The S&P was up 100% of the time at the 3-month mark, which may correspond to a last gasp rally during the start of a bear market. See Exhibit 235.

Exhibit 235: SPX forward returns and up ratios from the month margin debt moved above 60%
SPX forward returns from the month the level first reached 60

Days after	1-month	2-month	3-month	6-month	9-month	12-month	18-month	24-month
% Up Ratio	33%	33%	100%	33%	33%	33%	0%	33%
Up	1	1	3	1	1	1	0	1
Down	2	2	0	2	2	2	3	2
Average	-1.02%	-1.04%	3.91%	1.70%	1.91%	-3.65%	-22.11%	-19.09%
Median	-3.20%	-1.95%	2.00%	-1.00%	-2.23%	-10.14%	-16.66%	-21.86%
Standard deviation	5.50%	6.46%	3.70%	5.86%	16.39%	15.50%	15.80%	21.28%
Min	-5.09%	-7.00%	1.56%	-2.33%	-12.02%	-14.86%	-39.92%	-38.85%
Max	5.24%	5.82%	8.17%	8.42%	19.97%	14.03%	-9.75%	3.43%

Source: BofA Global Research, FINRA, Bloomberg
Note: Data start year 1998. If multiple consecutive signals occur, only the first signal is taken.

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Margin debt above 40 was bullish for SPX in the immediate term, then weak

Historically, the SPX tended to move a little higher 1-2 months following a margin debt rise above 40%. One month after it was up 80% of the time (4 of 5 occurrences since 1998). However it tended to deliver mixed returns with neutral bias in the months that followed. It tended to be weakest 18 months later, down 60% of the time by -6.60% on average. See Exhibit 236.

Exhibit 236: SPX forward returns and up ratios from the month margin debt moved above 40%
SPX forward returns from the month the level first reached 40

Days after	1-month	2-month	3-month	6-month	9-month	12-month	18-month	24-month
% Up Ratio	80%	60%	40%	60%	40%	40%	40%	60%
Up	4	3	2	3	2	2	2	3
Down	1	2	3	2	3	3	3	2
Average	1.07%	-0.08%	-1.63%	-0.60%	0.10%	0.32%	-6.60%	-0.15%
Median	2.39%	0.72%	-3.68%	2.74%	-5.99%	-6.03%	-7.25%	4.41%
Standard deviation	3.72%	2.35%	7.65%	9.41%	16.72%	20.33%	29.42%	26.97%
Min	-5.39%	-2.99%	-12.24%	-13.36%	-16.16%	-23.61%	-47.74%	-30.76%
Max	3.94%	2.74%	6.83%	8.37%	21.83%	23.48%	27.33%	30.23%

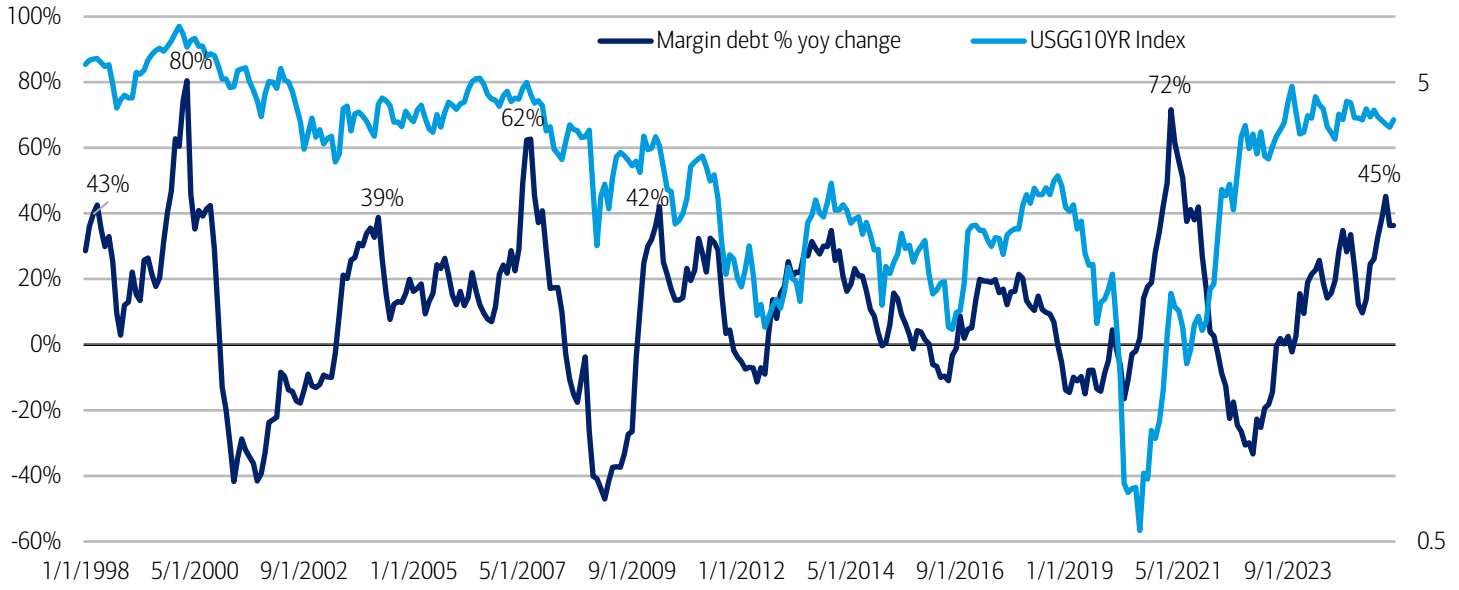
Source: BofA Global Research, FINRA, Bloomberg
Note: Data start year 1998. If multiple consecutive signals occur, only the first signal is taken.

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US 10Y yield vs. Margin debt YoY

After peaks in YoY margin debt in 2000, 2007 and 2021, the US 10Y yield tended to decline. However, the turn down after the peak in 2021 was temporary. In 1998, 2004 and 2009 when it reached 43%, 39% and 42%, respectively, yield went down, sideways and down.

Exhibit 237: Prior to 2021, margin debt spikes tended to coincide w/ US 10Y yield peaks. However in 2021, yield dipped and then rose to higher highs again.
YoY % change in margin debt with US 10Y yield overlaid



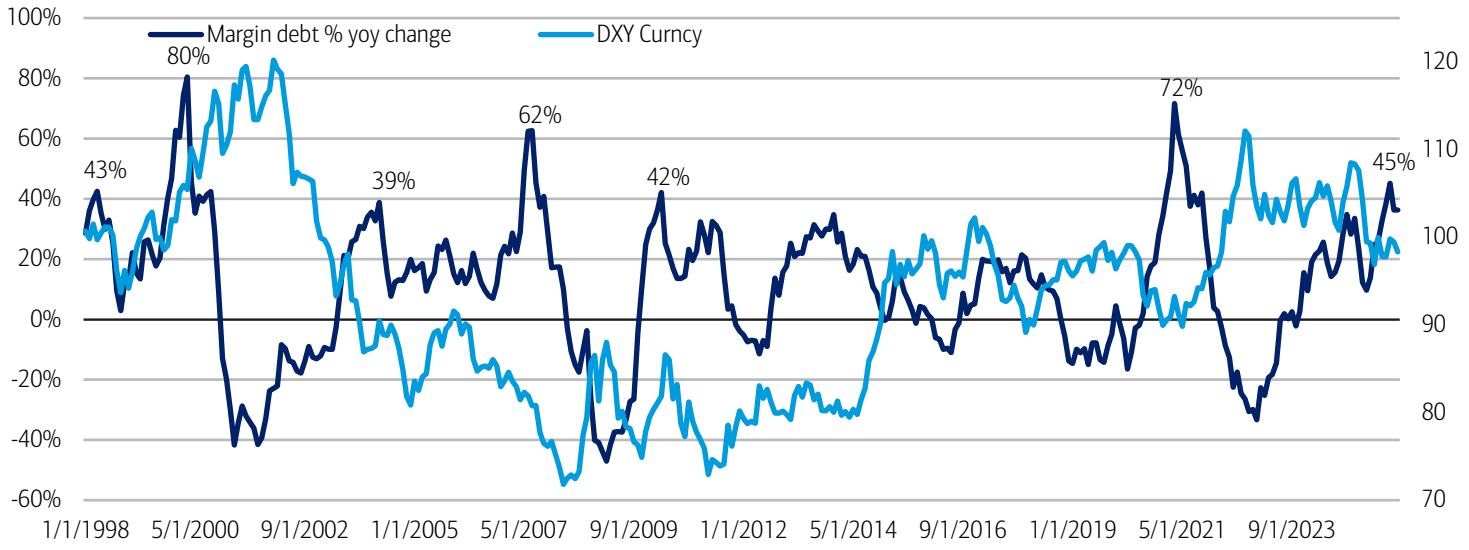
Source: BofA Global Research, FINRA, Bloomberg

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US dollar index margin debt YoY

When YoY margin debt peaked, we observe random trends in the DXY thereafter. In 1998, the DXY declined. In 2000, the DXY uptrend continued. In 2004, the DXY declined and rallied back again. In 2007, the DXY continued to decline. In 2009 the DXY peaked and declined. In 2021, the DXY bottomed and rallied.

Exhibit 238: Peaks in YoY margin debt do not appear to have much relevance to the direction of the DXY.
YoY % change in margin debt with US dollar index (DXY) overlaid



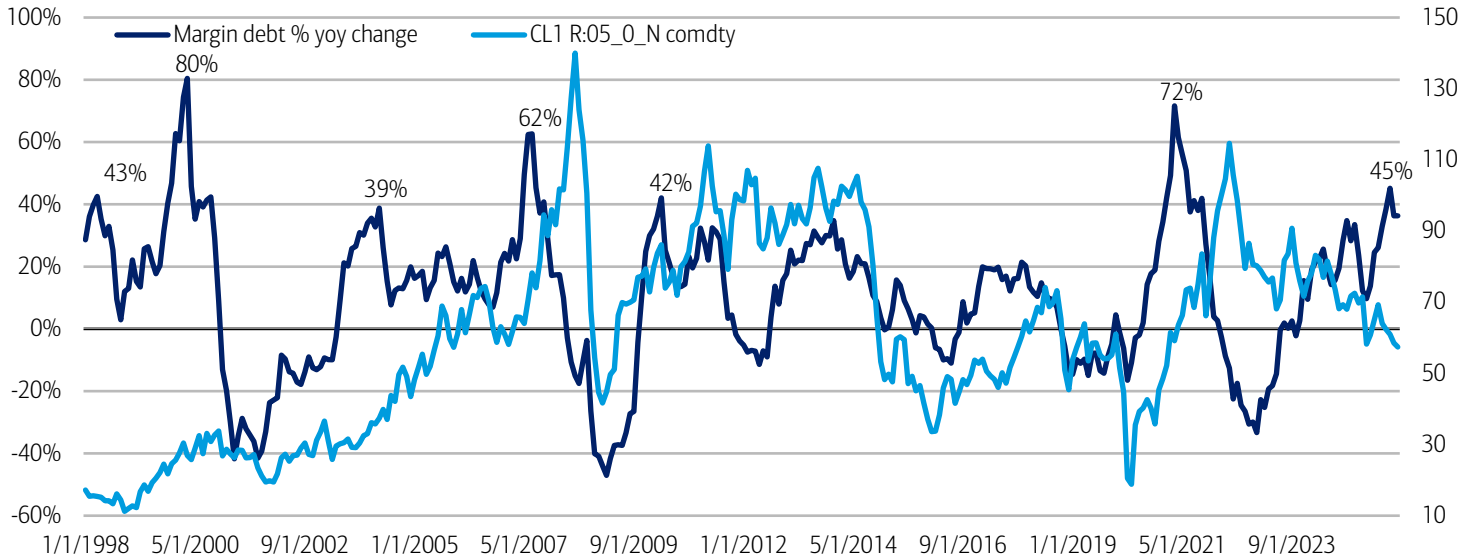
Source: BofA Global Research, FINRA, Bloomberg

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Crude oil margin debt YoY

When YoY margin debt peaked, we tend to see oil strength follow. When YoY margin debt peaked in 1998, oil was near a key low and rallied. When it peaked at 80% in 2000, oil continued a bit higher and then rolled over. In 2004 at 39%, oil prices rallied. In 2007 at 62%, oil prices rallied. In 2009 at 42%, oil prices dipped and rallied. In 2021 at 72%, oil prices also rallied.

Exhibit 239: When YoY margin debt is high or stretched, we observe some tendency for oil prices to find support and rally.
YoY % change in margin debt with WTI crude oil future (CL) overlaid



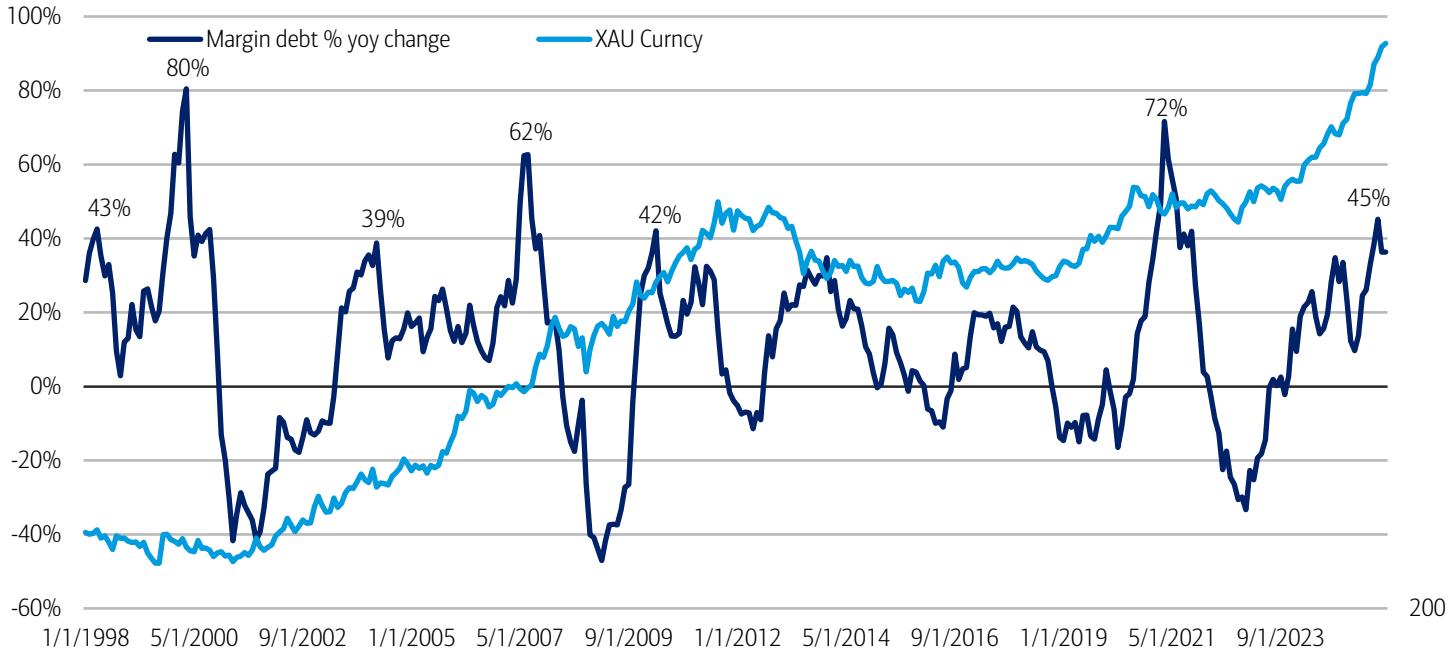
Source: BofA Global Research, FINRA, Bloomberg

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Gold margin debt YoY

When YoY margin debt peaked, we see gold tending to trade sideways or possibly higher. In 1998, gold went lower. In 2001, gold made a higher low and then rose. In 2004, it went sideways and then higher. In 2007 it also went sideways and higher. In 2009 it continued higher. In 2021 it went sideways.

Exhibit 240: Gold after a peak in YoY margin debt was either sideways or higher.
YoY % change in margin debt with gold (XAU) overlaid



Source: BofA Global Research, FINRA, Bloomberg

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Digital Assets & Technical Analysis

- Many technical strategies have proven valuable in timing entry into and exit from digital asset markets such as bitcoin.
- In this section we recap a few technical methods that have proven valuable in recent years including support and resistance levels, trend lines, a head and shoulders top pattern, Fibonacci and the Relative Strength Index (RSI).
- This section is meant to offer more examples of technical theories discussed in this primer.

Bitcoin had a trend line in 2016-17 that broke in 2018

Trend lines imply trend direction and signal change

A trend line is a guide to the overall direction of the market in an upward, sideways or downward direction. It is present for four reasons:

1. Estimate support in an uptrend as price corrects the larger trend. Support is a level where buyers outnumber sellers resulting in prices turning up.
2. Estimate resistance in downtrends during relief rallies. Resistance is a level where sellers begin to outweigh buyers, which turns prices lower.
3. Create a parallel channel line based on 1 and 2. A support line in an uptrend can be used to estimate the top of a channel. The resistance line in a downtrend can be used to estimate the bottom of a channel.
4. When a trend line is broken, the trend has changed.

Exhibit 241: XBT/USD trend line examples - illustrative example
Green arrows show support, red arrows resistance, a channel formed in 2015-2017, breakouts tend to retest lines and those levels can be traded such as mid-2017 and 4Q218.



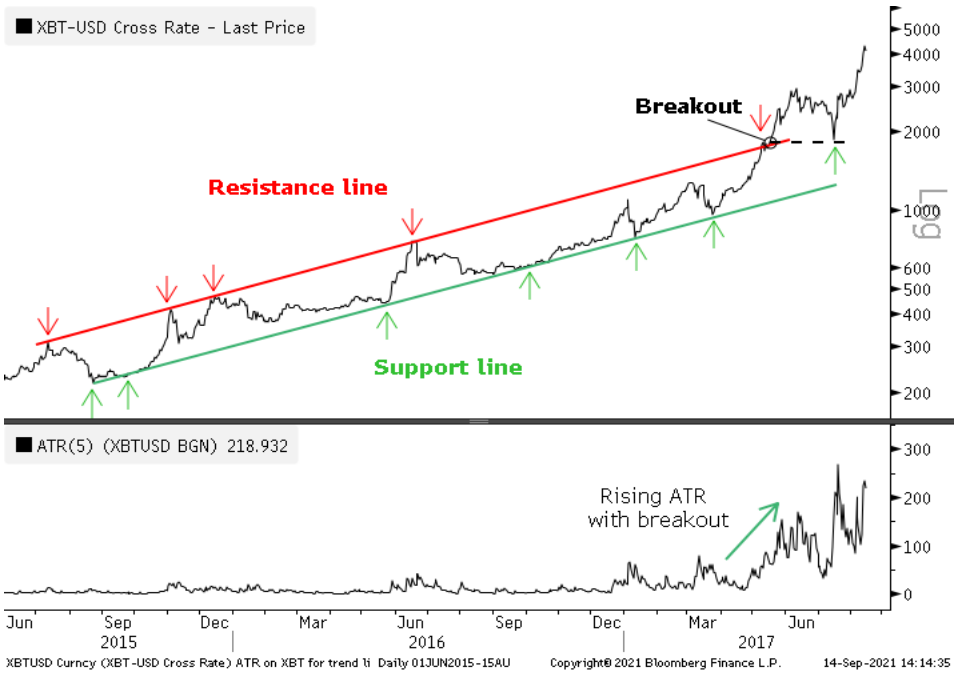
Source: BofA Global Research, Bloomberg

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When is a trend line broken?

A variety of characteristics help to differentiate good trend lines from not so good ones. There are also a few ways to define a trend line as broken. For more on this, please see page 11 in the Terrific Trend Lines section and Exhibit 242 as another supporting example.

Exhibit 242: XBT/USD with Average True Range indicator (ATR) - illustrative example
When price rallied above the resistance line, the ATR was rising and confirmed the breakout higher.



Source: BofA Global Research, Bloomberg

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Projecting price after a qualified trend line break

There are a few ways to calculate targets after price breaks a support or resistance line. The most straight forward way is to take the height of a prior trend and add it to the breakout point. In Exhibit 243 below, we take the height of the channel and add it to the breakout point to estimate an initial target. If price broke below the support line, we would take the height of the channel and subtract it from the breakdown point to estimate an initial target.

Exhibit 243: XBT/USD - Measured move example
The height of the channel added to the breakout point approximated a target that was reached. The breakout point also offered support in the correction following the target being reached.



Source: BofA Global Research, Bloomberg

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Bitcoin had a head and shoulders top in 2021

The head and shoulders pattern is a popular chart pattern that usually implies the prior trend is changing direction. In the section called "A well massaged head and shoulders" we discussed the inputs to this pattern. Below is another example.

The basics of a head and shoulders top

The only way there is a head and shoulders top pattern is if there is an uptrend to reverse. If there is not an uptrend prior to the formation then there is not a trend to correct or change. These are the basics inputs to a top pattern.

1. **Three peaks and two troughs**. The first peak is the left shoulder, the second the head and the third the right shoulder. The head is always the highest high. The troughs separate the peaks from one another and help create the neckline.
2. **Estimate a neckline by drawing a trend line after the second trough forms**. In a top pattern, the neckline should have a slightly positive slope or at least be flat. This is important because price will be in the earlier phase of a downtrend when the neckline breaks, making the sell entry point higher.
3. **The right shoulder should make a lower high than the head**. It is also ideal for the right shoulder to be similar or smaller than the left shoulder.
4. **A decisive move down through the neckline** with greater than average volume should occur to solidify the pattern.
5. **The neckline may be tested as resistance on a light volume rally** and offers a second opportunity to sell. An appropriate stop is often the top of the right shoulder.

Exhibit 244: XBT/USD – Head and shoulders top - illustrative example
The May 2021 decline was preempted by a head and shoulders top pattern and target reached.



Source: BofA Global Research, Bloomberg

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Calculate targets with measured moves and Fibonacci

There are three ways to calculate targets for a head and shoulders top. The first is to use a measured move projection of the height of the pattern subtracted from the breakdown point. This can be done by drawing/coping an arrow as shown in Exhibit 244. A second way to estimate multiple targets is to use a Fibonacci Extension annotation as shown in Exhibit 245. Measure the distance from the top of the head to the neckline and subtract that from the breakdown point. This will also estimate multiple target levels where the 100% line is equal to the arrow method. Shallow and deeper levels can also be estimated. A third way is to find the low and high of the uptrend preceding the head and

shoulders top. Calculating a Fibonacci retracement on this distance. For more on Fibonacci, please see section Finding Fibonacci.

Exhibit 245: Calculating a target with Fibonacci retracements and extensions
Fibonacci extension measuring downside target levels after a head and shoulders top



Source: BofA Global Research, Bloomberg

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Bitcoin had a head and shoulders bottom in 2022-2023

Exhibit 246: A head and shoulders base on XBTUSD
A head and shoulders base on XBTUSD in 2022-2023 estimated upside to 35,000, which was reached.



Source: BofA Global Research, Bloomberg

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Bitcoin has tended to respond to RSI conditions

Measure and trade with momentum

The Relative Strength Index (RSI), which is the most popular technical indicator used on Bloomberg, measures the momentum of a trend by comparing the average size of the up periods to the average size of the down periods over a specified timeframe, such as 14 days. Common thought suggests if momentum reaches an extreme then a reversal, or at least a consolidation, will occur. While this may be the short-term outcome, sometimes the trend resumes until momentum finally depletes with a bullish or bearish divergence. In this primer, we will review RSI and the signals it can generate.

The essence of the RSI calculation and levels

If price went up $1.00 for 10 of the last 14 days and down $0.25 four of the last 14 days, then we know price went up on average much more than down. However, in financial markets, the up/down relationship is rarely this clear cut. This type of market may be described as overbought and the opposite as oversold. Both mean the market moved too far too fast. RSI is normalized to fall between 0 – 100. So when it is above 70, the trend is described as overbought. When it is below 30, it is described as oversold.

The facts and myths of overbought and oversold

The terms overbought and oversold mean the market may have risen or fallen too much, especially in the short term. Exiting or reversing a position solely because the market is deemed overbought or oversold may be the right trade for a short period of time but could be the wrong trade for the intermediate-term trend. One way of filtering this is to incorporate support and resistance. In theory, momentum complements a breakout similar to greater than average volume. If price and momentum break up through resistance, the expected overbought dip could be buyable. Similarly, if price and momentum break down through support, a bounce may be sold.

Momentum trades vs momentum trends

In a trend-less market, also referred to as a sideways or range-bound trend, RSI rarely moves through overbought or oversold readings. When price reaches the high end of the range and RSI is equal to or less than its prior peaks, price can be sold. If price were to breakout to new highs and RSI is still not overbought, then the breakout lacks confirmation from RSI and may soon reverse. However, when price and RSI break out together, a new uptrend is under way, and longs could be held or added to in a dip.

Price vs RSI divergences signal a trend is changing

When prices make lower lows and the RSI indicator makes higher lows, a bullish divergence is forming. Conversely, when price makes higher highs and RSI makes lower highs, a bearish divergence is forming. Think of price as a car and RSI as the foot on the accelerator. If a car is going uphill and the foot is pushing the accelerator, then the car is moving uphill. However, if the pressure on the accelerator starts to decrease, the car will be climbing the hill at a slower speed (with less momentum). Eventually the car makes it to the top of the hill, or the peak of the uptrend, and the foot comes completely off the pedal. If the bearish divergence is correct, all that's left is for the car to go down the hill.

Adjust overbought and oversold levels for trend direction

In the below chart, another important aspect of using RSI is shown. When markets are in a downtrend, naturally they should be oversold more often than not. When markets are in uptrends, they should be overbought more often. When a clearly defined trend is present, it usually makes sense to adjust the overbought and oversold levels. In uptrends, consider using 40-80 and in downtrends 20-60. If price is in a downtrend and reverses so RSI fails at 60, it is still in a downtrend. If RSI moves through 60, the downtrend may be near an end. The same goes for an uptrend and the 40 level.

Please see the section called "Relative Strength Index is the most popular" for more.

Exhibit 247: RSI is the most popular technical indicator (excluding moving averages)
Historical example from March 2020 to May 2021 using RSI to confirm trend breakouts, trend tops and oversold/overbought moves.



Source: BofA Global Research, Bloomberg

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Futures positioning data

- The Commitment of Traders report issued by the Commodity Futures Trading Commission provides a summary of how a variety of market participants are positioned in the commodity and financial futures market, i.e. long or short.

- When market participants get too long or short, there is a greater risk of a correction or trend change. Sometimes this can occur very fast and can be useful to combine with the aforementioned technical strategies.

CFTC Commitment of Traders (COT) report

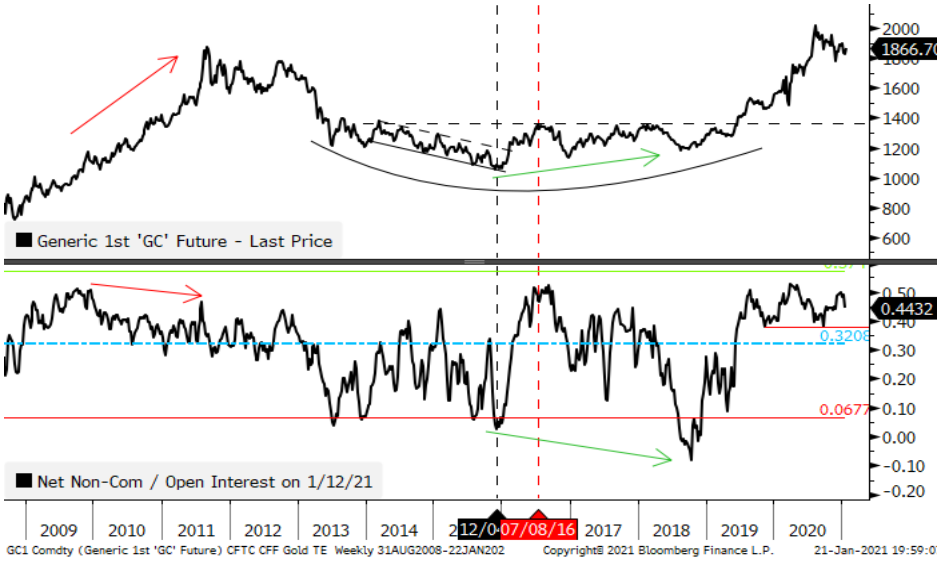
The Commitments of Traders (COT) report provides weekly positioning and flow data. More specifically, COT reports are a breakdown of each Tuesday's open interest for markets in which 20 or more traders hold positions equal to or above the reporting levels established by the Commodity Futures Trading Commission (CFTC)[1]. As of this writing, information is released weekly on Fridays at 3:30pm EST for as-of Tuesday data. Sometimes release dates move due to holidays.

"Speculator" positions often used as contrarian indicator

The legacy COT report divides data into two major categories: commercial and non-commercial. The commercial category represents hedgers while the non-commercial represents speculators. The speculators net position, normally measured as a percentage of the open interest, is often used as a contrarian sentiment indicator. When readings are extremely bullish (above 2 standard deviations for example), it is contrarian bearish, and vice versa. Just like any sentiment indicator, it is best combined with price trend and technical indicators.

In December of 2015, gold prices bottomed while positioning was the shortest in many years. A strong reversal/breakout and uptrend followed to take price from just below $1,100/oz to $1,400/oz. Positioning was stretched long and reversed lower. After trading rangebound for two years, the market found itself very short – even more so than in 2015 but at higher price levels. We view this as a positive divergence in conjunction with a large rounded base forming, suggesting significant upside.

Exhibit 248: Gold future (top), Net non-commercial as % of open interest (bottom) with a mean and +/- 2 standard deviation lines.
Divergence between price trend and positioning trend can indicate trend change coming.



Source: BofA Global Research, Bloomberg (G392)

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[1] Source: http://www.cftc.gov/Marketreports/CommitmentsofTraders/index.htm

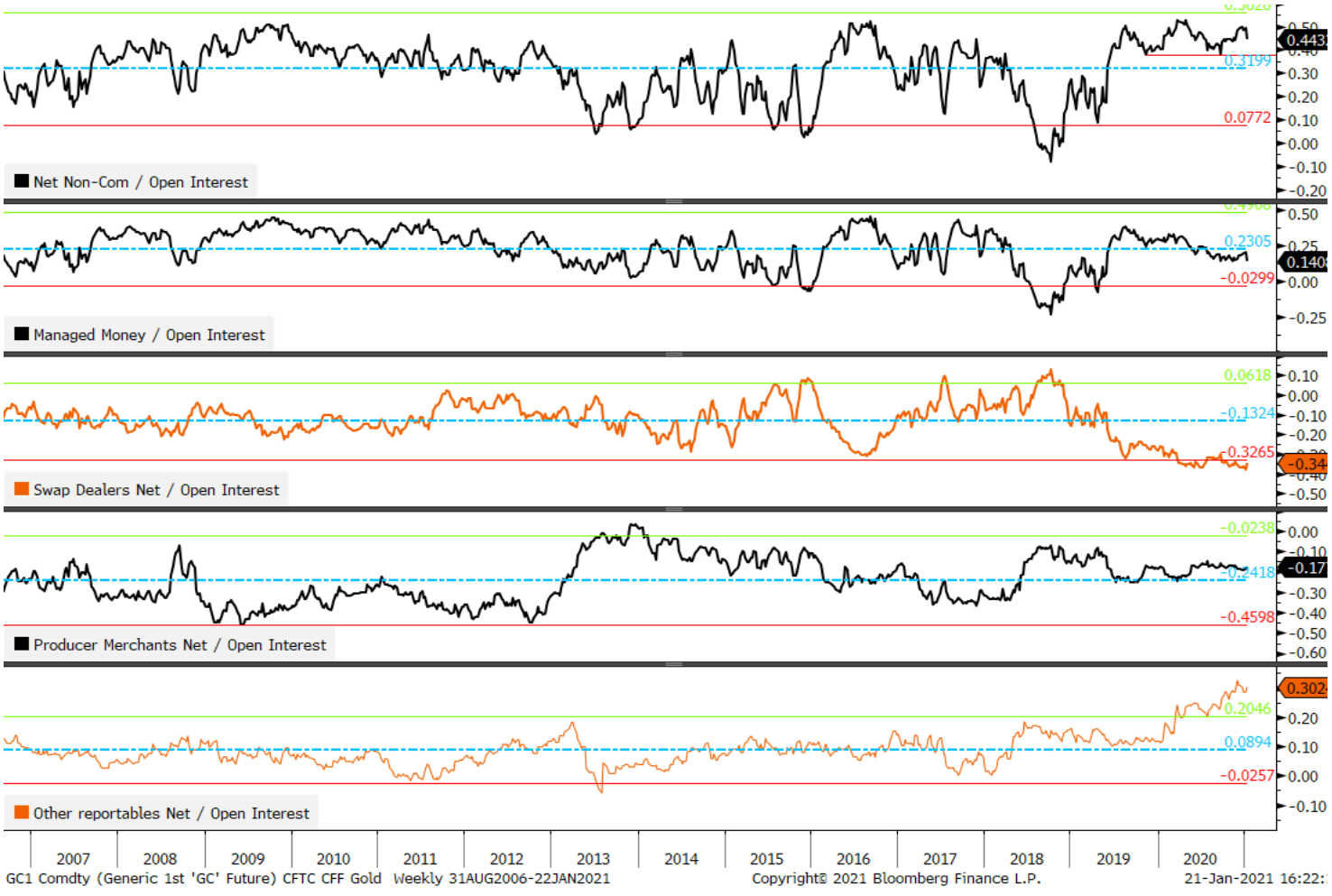
Disaggregated positioning data reveal market dynamics

To improve transparency of the most vulnerably positioned asset and/or market player, the CFTC publishes a disaggregated COT report. The disaggregated commodity report (CFF) divides data into four categories: Producer, Swap Dealers, Managed Money and Other Reportables. In financial futures the four categories are Dealer, Asset Manager, Leveraged Funds and Other Reportables.

In our technical work we tend to find net-noncommercial and the following three positioning datasets most useful in warning of a trend reversal. The latter three tend to have a relatively larger size in the market or build large positions in a trend faster than most. Adverse or surprise outcomes often lead to a dramatic shift and a need to exit. This can cause a strong trend reversal.

CFF Managed Money = CTAs, CPOs or funds identified by CFTC
TFF Asset Manager = institutional investors such as mutual funds
TFF Leveraged Funds = HFs, CTAs, CPOs or funds identified by CFTC

Exhibit 249: Net non-commercials, managed money, swap dealers, producer merchants and other reportables
Comparing a variety of positioning data



Source: BofA Global Research, Bloomberg

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Trading Market Distributions

- Market profile charts compare price at time distributions to identify important levels of support, resistance and targets.
- The current and historical point of controls and value area lines are two of the most important levels.
- Trading between double distributions is a popular strategy amongst market profile practitioners.

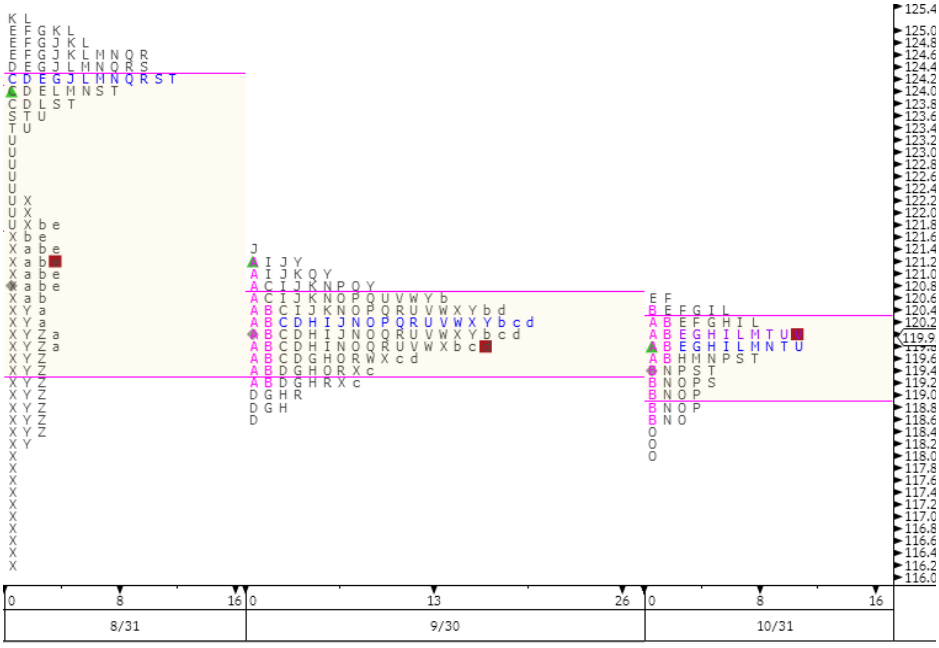
A market profile shows price distribution

A market profile is the distribution of prices to identify a market's fair value price, trading patterns, and support and resistance levels. It is used to determine if a market is trending or range-bound. It is founded on the principle that prices move slowly through time (range-bound) and fast through price (trending). One of its goals is to help traders catch trending moves and stay out of choppy, range-bound moves by offering key levels that define each market and signal when that market is changing.

A profile reduces dependency on time, focuses on price

The chart below consists of three distributions, each representing one month of trading. It is composed of letters that represent a time period. A letter can represent a specified number of minutes, a trading day, week, or even month. In the example below, an "A" represents the first day of the month, and each set of letters is a one-month distribution of prices. The magenta letters highlight the first two periods of a market profile and are called the initial balance. The magenta lines define 70% of the distribution. The basics of the chart suggest that when price trades above the initial balance or value area line, it may trend higher. When price trades below either, it may trend lower.

Exhibit 250: Three-month market profile Exhibit
Price action represented in a market profile Exhibit for three months



Source: BofA Global Research, Bloomberg
*The 8/31 profile does not have any A's because the first day of the month was a Saturday and there was no trading.

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Green triangles, gray diamonds, and red squares

In a default market profile chart, many symbols are present to identify important points. Most important is the point of control (POC). This is the longest horizontal row in a market profile distribution that shows the price level most often reached. It shows *frequency* at price or *time* at price, *not volume* at price. Historical POCs create key levels of support, resistance and targets for trends. Type MKTP<GO> on Bloomberg to see the chart in its default colors.

Exhibit 251: Market profile symbols and colors
Market profile symbols explained



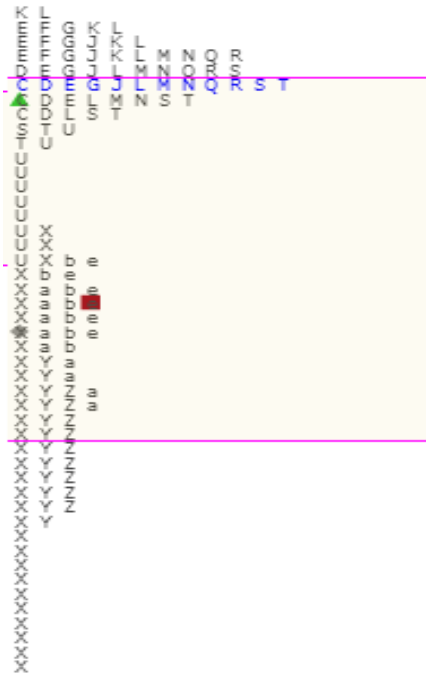
Source: BofA Global Research

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A collapsed bar chart is a market profile

Each letter in a distribution represents the high/low range for a given period. The market picture chart collapses the ranges in real time to form the distribution. The three images below compare a market profile to a split profile and to a bar chart. The split profile and bar chart look alike. Now, imagine the chart collapsing the bars to the left, forming a distribution of prices resulting in a market profile.

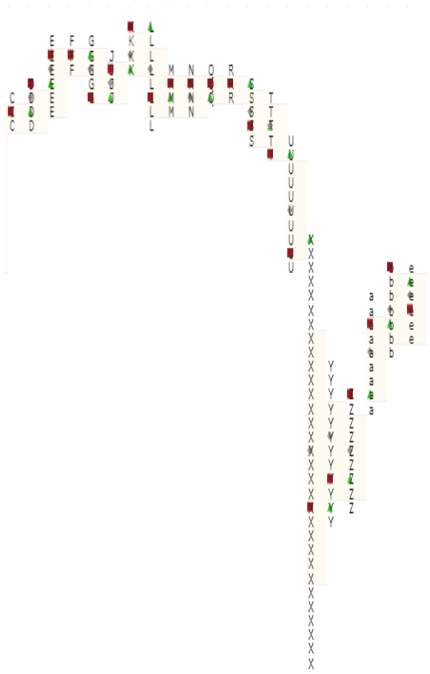
Exhibit 252: Profile distribution
A distribution with a "P" shape



Source: BofA Global Research, Bloomberg

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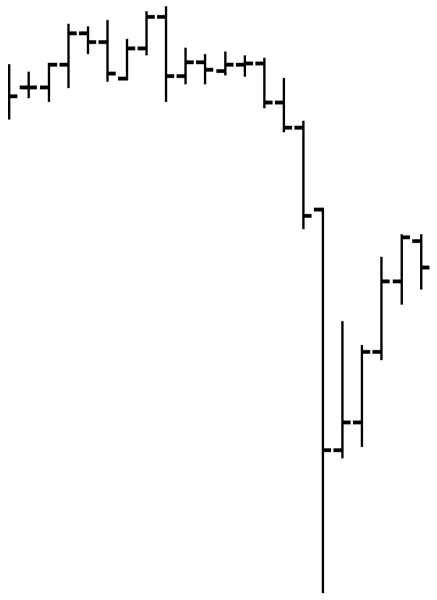
Exhibit 253: Split profile with each day separated
The "P" shape split apart by time...



Source: BofA Global Research, Bloomberg

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Exhibit 254: Looks like a bar Exhibit
...looks just like a bar Exhibit.



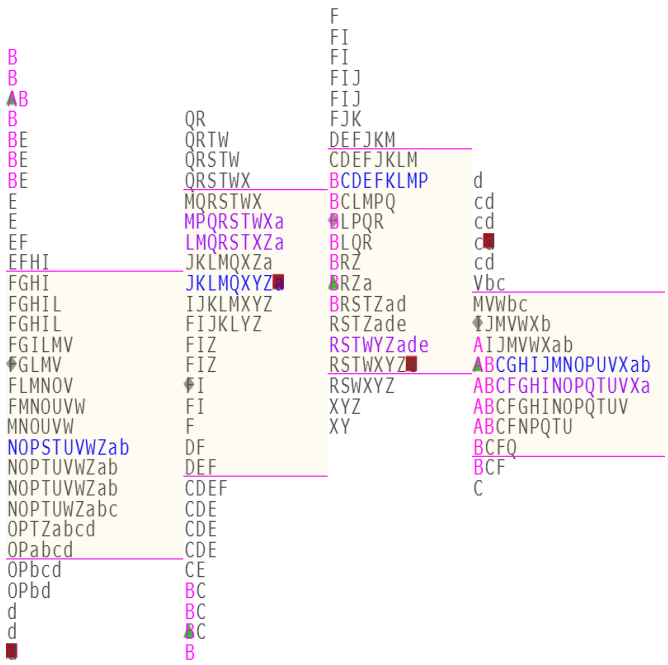
Source: BofA Global Research, Bloomberg

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Normal, uptrend, downtrend and double distributions

There are many ways price can be distributed over time. In market profile analysis, we differentiate between normal, uptrend, downtrend and double distributions.

Exhibit 255: Four types of profiles
Distribution examples



Source: BofA Global Research, Bloomberg

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The image to the left presents four distribution types:

- The first profile is a downtrend where the price opened at the highs, forming the initial balance. Soon thereafter, it traded below the initial balance and began its downtrend, closing at the lows of the month.
- The second profile is an uptrend, where the price opened and held the lows of the prior month and began trading higher above the initial balance. Price traded as high as the open of the prior distribution, formed its point of control at the value area of the prior month, and closed there.
- The third profile is a double distribution, which shows the level where sellers sold the price lower and buyers bid prices higher, forming the gap. Price closed at the lower value area line.
- The fourth is a relatively normal distribution with a POC in the center of the value area equal to the prior month's close, emphasizing the importance of that level. Toward the end of the month, price rallied to begin filling the gap in the third profile and failed at the third profile's POC.

Applying market profile analysis

The items that comprise a market profile can be used to describe market activity and generate buy and sell signals. They enable the analyst to do the following:

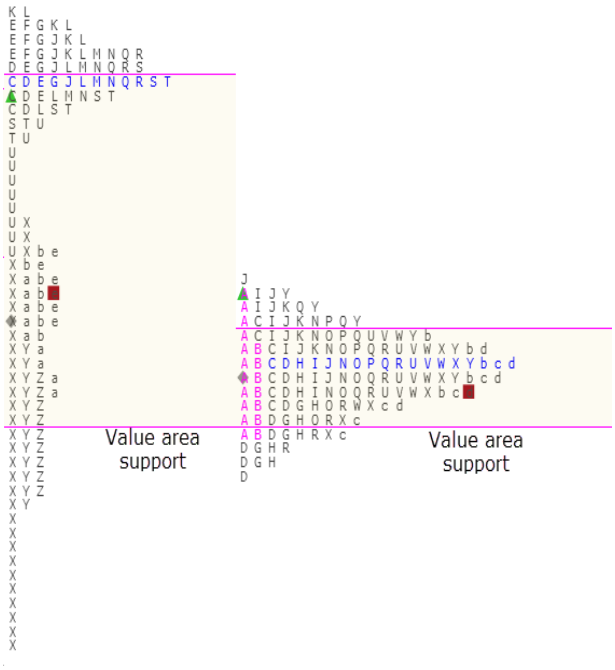
- Understand where price is in relation to current and historical POCs, value areas, initial balance lines, highs, lows, closes and midpoints. Use these to find breakouts and trade in the direction of the move and to identify key levels to forecast price.
- Identify a double distribution and trade with an expectation the gap may be filled. Create targets based on the double distribution value area and POCs.
- Consider a reversion trade when the market moves too far too fast in one direction and finds support or resistance at a key level. This is sometimes called a single print tail or retracement trade.
- Split and join profiles to separate trending and range-bound price action to identify early breakouts, targets, and support and resistance levels.
- Analyze multiple distributions historically to see if the same defined components of a market profile appear at or near the same level. For example, if over the course of 10 distributions, multiple POCs, value area levels and symbols overlap.

Example 1

Exhibit 256 depicts an uptrend distribution followed by a normal distribution. The two profiles have equal lower value areas and suggest support at that level (119.30). If price started trading below 119.30, it would be interpreted as bearish.

Joining these two distributions together creates the distribution in Exhibit 257. The double distribution has a more pronounced POC at 120.20. A bullish signal could be interpreted if price decisively traded above the POC and midpoint with anticipation of an uptrend to fill the gap in the distribution. The first target could be the top of the value area (T1) and a second could be the longest point of the upper distribution (T2).

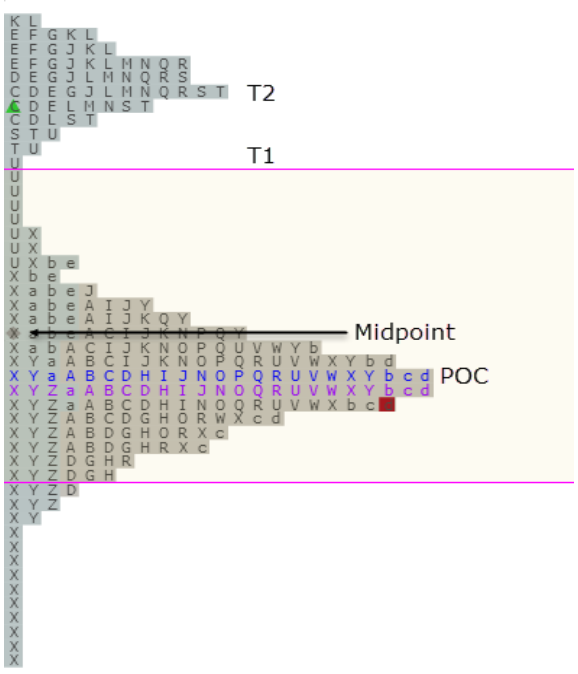
Exhibit 256: Two distributions
Two distributions can make a double distribution



Source: BofA Global Research, Bloomberg

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Exhibit 257: Joined distributions
Two distributions can make a double distribution



Source: BofA Global Research, Bloomberg

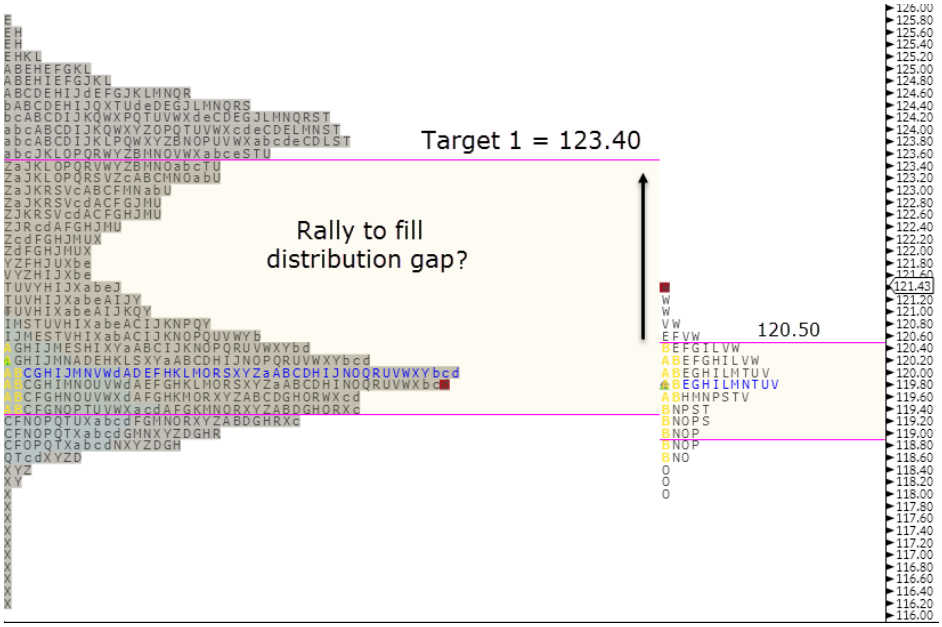
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Example 2

The distribution on the left joins six months of data to form a large double distribution. The POC is on the lower portion of the distribution at 120, with value area lines equal to 119.30 and 123.40.

The current distribution broke above its value area high level and the initial balance high of 120.50. This suggests a new trend may be beginning with a target of 123.40.

Exhibit 258: Multi-month distribution compared to the current month
When price starts filling a gap in a double distribution



Source: BofA Global Research, Bloomberg

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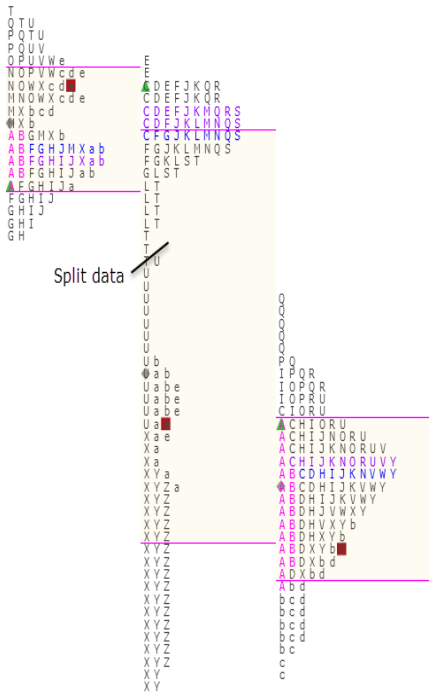
Example 3

In Exhibit 259, the first distribution is a small double distribution, the second formed an abnormal trending distribution, and the third is a much more normal distribution. To define the important levels in this scenario, we should first split the top of the middle distribution from the rest. This forms Exhibit 260. Then we can join the two left distributions together and the two right distributions together. The result is Exhibit 261.

Exhibit 261 suggests the market is normally distributed and range-bound. Price is trading below the POC and above the lower value area line. If the price traded below the value area, it would be bearish. If price traded above the value area, then it would be bullish. Otherwise, reversion to the POC is anticipated in a normal distribution.

When a new distribution begins, we can look to the most recent normal distribution for key levels. This includes the POC at the center of it and the two value area lines.

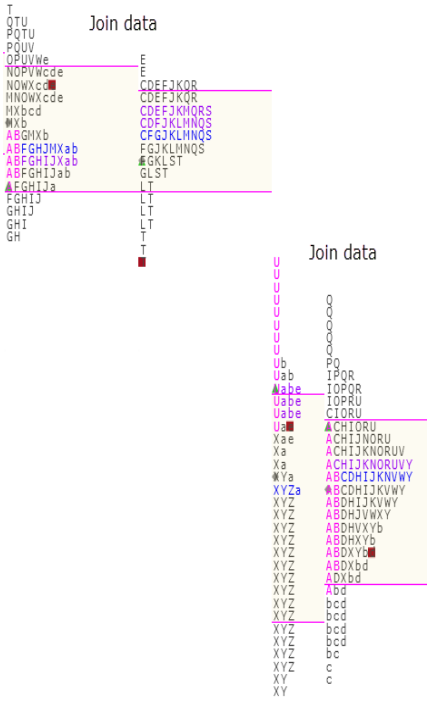
Exhibit 259: Three distributions
Splitting a distribution



Source: BofA Global Research, Bloomberg

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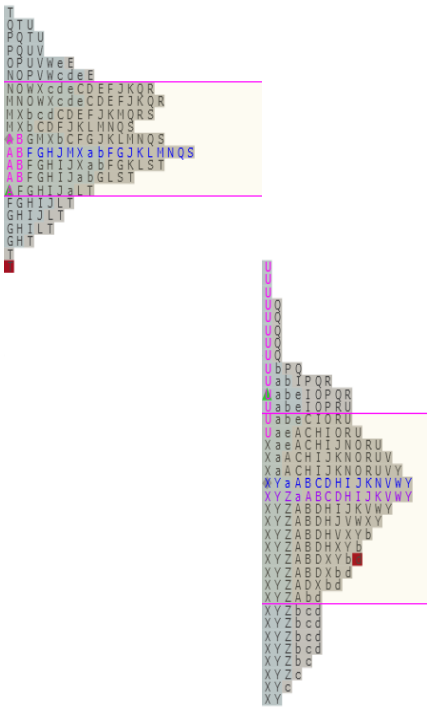
Exhibit 260: Splitting the first
After splitting a distribution



Source: BofA Global Research, Bloomberg

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Exhibit 261: Joining the partial distributions
Joining relevant distributions



Source: BofA Global Research, Bloomberg

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Market profile accreditation and more information

Market profile was created by J. Peter Steidlmayer while working at the Chicago Board of Trade. More information can be found in his book published in 2002 called Steidlmayer on Markets: Trading with Market Profile. The Chicago Mercantile Exchange also hosts content on their website about market profile: (http://www.cmegroup.com/education/interactive/marketprofile/).

Appendix: Acronyms & abbreviations

Exhibit 262: Common acronyms and abbreviations
This list may not be comprehensive and is subject to change

Acronym/Abbreviation	Definition	Category	Acronym/Abbreviation	Definition	Category
1H	First Half	Time Period	INR	Indian Rupee	FX
2H	Second Half	Time Period	Jan	January	Time Period
1Q / Q1	First Quarter	Time Period	JPY	Japanese Yen	FX
2Q / Q2	Second Quarter	Time Period	Jul	July	Time Period
3Q / Q3	Third Quarter	Time Period	Jun	June	Time Period
4Q / Q4	Fourth Quarter	Time Period	KRW	South Korean Won	FX
A/D	Accumulation/Distribution Line	Technical Analysis	LATAM (LatAm)	Latin America	Geography
Apr	April	Time Period	M (m)	Month	Time Period
ATH	All-Time High	Statistical	MA	Moving Average	Technical Analysis
ATR	Average True Range	Technical Analysis	MACD	Moving Average Convergence Divergence	Technical Analysis
AUD	Australian Dollar	FX	MD	US Memorial Day	Other
Aug	August	Time Period	MFI	Money Flow Index	Technical Analysis
Avg (avg)	Average	Statistical	MXN	Mexican Peso	FX
BB	Bollinger Bands	Technical Analysis	NDX	NASDAQ-100 Index	Equities
BCOM	Bloomberg Commodity Index	Commodity	NFP	Non-farm Payrolls	Economics
BoE	Bank of England	Economics	NOK	Norwegian Krone	FX
BofA	Bank of America	Other	Nov	November	Time Period
BoJ	Bank of Japan	Economics	NYE	New Years Eve	Time Period
bp(s)	Basis point(s)	Statistical	NZD	New Zealand Dollar	FX
BRIC	Brazil, Russia, India, China	Geography	OBV	On-Balance Volume	Technical Analysis
BRL	Brazilian Real	FX	Pd (pd)	Period	Time Period
CAD	Canadian Dollar	FX	PLN	Polish Zloty	FX
CCI	Commodity Channel Index	Technical Analysis	PMI	Purchasing Managers Index	Economics
CHF	Swiss Franc	FX	Q	Quarter	Time Period
CL1	WTI Crude Oil (front-month contract)	Commodity	QE	Quantitative Easing	Economics
CLP	Chilean Peso	FX	QoQ	Quarter-on-quarter (-over-)	Statistical
CME	Chicago Mercantile Exchange	Other	ROC	Rate of Change	Technical Analysis
CMF	Chaikin Money Flow	Technical Analysis	RSI	Relative Strength Index	Technical Analysis
CMT	Chartered Market Technician	Other	RTY	Russell 2000 Index	Equities
CNH	Chinese Yuan (offshore)	FX	SD	Standard Deviation	Statistical
CNY	Chinese Yuan (onshore)	FX	SEK	Swedish Krona	FX
COP	Colombian Peso	FX	SGD	Singapore Dollar	FX
CPI	Consumer Price Index	Economics	SMA	Simple Moving Average	Technical Analysis
CTA	Commodity Trading Advisor	Other	SPX	S&P 500	Equities
D (d)	Day	Time Period	Stoch	Stochastic Oscillator	Technical Analysis
Dec	December	Time Period	TSY(s)	Treasury(ies)	Fixed Income/Rates
DMI	Directional Movement Index	Technical Analysis	UK	United Kingdom	Geography
DXY	US Dollar Index	FX	US	United States	Geography
EM	Emerging Markets	Geography	US 10Y	US 10-Year Treasury	Fixed Income/Rates
EMA	Exponential Moving Average	Technical Analysis	US Urate	US Unemployment Rate	Economics
EMBI	Emerging Market Bond Index	Fixed Income/Rates	USD	US Dollar	FX
EMEA	Europe, Middle East, Africa	Geography	USGG10YR	US 10Y Yield	Fixed Income/Rates
EOM	End of Month	Time Period	USPC	US Presidential Cycle	Other
EOY	End of Year	Time Period	UST	US Treasuries	Fixed Income/Rates
EU	European Union	Economics	USUR	US Unemployment Rate	Economics
EUR	Euro	FX	Vs (vs)	Versus	Statistical
EWT	Elliott Wave Theory	Technical Analysis	VWAP	Volume Weighted Average Price	Technical Analysis
Feb	February	Time Period	Wk (wk)	Week	Time Period
Fed	Federal Reserve	Economics	Y (y)	Year	Time Period
Fib	Fibonacci	Technical Analysis	Y1	Year 1	Time Period
FOMC	Federal Open Market Committee	Economics	Y2	Year 2	Time Period
G10	Group of 10 (major economies)	Geography	Y3	Year 3	Time Period
GBP	British Pound Sterling	FX	Y4	Year 4	Time Period
GDP	Gross Domestic Product	Economics	YE	Year-end	Time Period
Golden Cross	50d SMA crosses above 200d SMA	Technical Analysis	YOY (YoY)	Year-on-year (-over-)	Statistical
H&S	Head and shoulders	Technical Analysis	Yr (yr)	Year	Time Period
HKD	Hong Kong Dollar	FX	YTD	Year-to-date	Statistical
HUF	Hungarian Forint	FX	YTG	Year-to-go	Statistical
INDU	Dow Jones Industrial Average	Equities	ZAR	South African Rand	FX

Source: BofA Global Research

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